National Aeronautics and Space Administration

SMALL BUSINESS INNOVATION RESEARCH (SBIR) & SMALL BUSINESS TECHNOLOGY TRANSFER (STTR)

Fiscal Year 2019 General Solicitation

Amended: February 12, 2019
Amended: March 4, 2019

Opening Date: February 5, 2019
Closing Date: March 29, 2019

Fiscal Year 2019 SBIR/STTR Solicitation
Noteworthy Changes

EHB Modernization

This is the second year proposals will be submitted in the modernized SBIR/STTR Electronic Handbook (EHB). It was modernized with the goal of providing a seamless user experience by building an intuitive, user-friendly and integrated digital service. We have continued to make refinements over the past year to continue to provide clearer instructions throughout the process. Over time, the new interface should improve efficiency, eliminate duplication of effort and overall make your proposal submission process simpler.

Research Topics for SBIR and STTR

The STTR subtopics will appear in an integrated list with the SBIR subtopics again this year. They will be clearly marked as STTR subtopics so that offerors will know that the additional Research Institution (RI) partnership is required before submitting a proposal. This will assist Firms in seeing related subtopics across both programs.

"Pointers" to Assist You in Finding the Appropriate Subtopic

Subtopic pointers are used to indicate subtopics that are asking for related technologies. Where applicable, these pointers will appear in the subtopic headers to assist proposers with identifying those related subtopics that potentially seek related technologies for different customers or applications. Pointers in conjunction with the focus area listings of subtopics will make it easier for proposers to find all subtopics that may be of interest.

CCRPP Is Back For 2019

The Civilian Commercialization Readiness Pilot Program (CCRPP) is an additional funding opportunity available to small businesses, with the purpose of accelerating the transition of SBIR and STTR funded technologies to commercialization. The funding is a combination of additional SBIR/STTR Program investment and NASA or non-NASA entity investment. The program will match between $500,000 and $1 million of external investment. The primary objective of the NASA CCRPP is an infusion or commercialization, not an incremental improvement in technology maturation alone. Technology maturation without infusion or commercialization will not be accepted for CCRPP. For additional information, please see https://sbir.nasa.gov/content/post-phase-ii-initiatives#CCRPP.

Moon to Mars Campaign

NASA is implementing a program for the exploration and utilization of the Moon followed by missions to Mars and other destinations, called the Moon to Mars campaign (see https://www.nasa.gov/feature/nasa-unveils-sustainable-campaign-to-return-to-moon-on-to-mars and https://www.nasa.gov/topics/moon-to-mars). An early element of the exploration campaign is the delivery of payloads to the Moon for scientific study and the advancement of technology capabilities to support sustained lunar surface operations. In December 2018, the agency announced the selection of nine U.S. companies who are now eligible to bid on NASA delivery services for payloads to the lunar surface through Commercial Lunar Payload Services (CLPS) contracts (https://www.nasa.gov/press-release/nasa-announces-new-partnerships-for-commercial-lunar-payload-delivery-services).

Due to this emerging new commercial payload delivery service, a highlight in this year's solicitation is that a number of the subtopics are encouraging proposers to consider the potential for developing a lunar payload as part of their technology development effort. Where appropriate, the technology development project may consider a lunar payload as a deliverable by the end of Phase II (or perhaps in a post Phase II effort). While not all proposals from those subtopics are expected to produce a payload as their deliverable, suitable payloads which are developed may be eligible (through subsequent competitive selection) for delivery to the lunar surface at no cost. However, selection for an SBIR award will not guarantee selection for a future lunar payload flight opportunity.

The 18 subtopics that highlight the lunar payload opportunity include, but are not limited to, in situ resource utilization, thermal management, energy storage, rad hard high performance computing, cryogenic fluid management, various lunar sensors, solar arrays, and coordination of space vehicle swarms. The CLPS companies are expected in the coming months to produce payload user guides (with some ready now and perhaps all such user guides available by time this SBIR/STTR solicitation's Phase I projects are awarded in summer 2019). The payload user guides will provide insights into size, power, interfaces, and similar constraints that should be helpful to payload developers. In the meantime, some preliminary payload constraints are provided within the relevant subtopic descriptions.

To learn more about the NASA's interest in lunar payload development refer to a recent NASA ROSES solicitation: (https://nspires.nasaprs.com/external/viewrepositorydocument/cmdocumentid=649890/solicitationId=%7B2D390C4D-39F9-E880-34C8-C07DC523698E%7D/viewSolicitationDocument=1/C.28%20LSITP%20Amend%2034.pdf).

STMD:
Z1.03 – Kilowatt-Class Energy Conversion for Small Fission Reactors
Z1.04 – Long Duration Lunar Energy Storage and Discharge
Z2.01 – Spacecraft Thermal Management
Z4.01 – MISSE Experiments
Z4.02 – In-Space Sub-Modular Assembly
Z5.05 – Lunar Rover technologies for In-Situ Resource Utilization and Exploration Missions
Z6.01 – High Performance Space Computing Technology
Z7.04 – Lander Systems Technologies
Z8.03 – Low Cost Radiation Hardened Integrated Circuit Technology
Z10.01 – Cryogenic Fluid Management
Z12.02 – Payloads for Lunar Resources: Volatiles

SMD:
S1.06 – Particles and Fields Sensors & Instrument Enabling Technologies
S1.07 – In Situ Instrument/Technologies for Lunar and Planetary Science
S3.01 – Power Generation and Conversion
S3.03 – Power Electronics and Management, and Energy Storage
S4.04 – Extreme Environments Technology

HEOMD:
H5.01 – Lunar Surface Solar Array Structures

STTR:
T4.03 – Coordination and Control of Swarms of Space Vehicles

Travel in Phase I
Due to the limited budget and short period of performance in Phase I contracts, travel is highly discouraged. If the purpose of the meeting cannot be accomplished via videoconference or teleconference, the proposer must provide rationale for the trip in the proposal budget form. All travel must be approved by the Contracting Officer.

Updated Certifications
The certifications collected at time of proposal, time of award, and during the lifecycle have been revised to match those required in the SBIR and STTR Policy Directives located at https://www.sbir.gov/. These certifications will look similar to those you may have seen in the past from NASA's SBIR and STTR programs, but with some updated language. You will see one set of certifications twice. Once at time of proposal and again at time of award. The purpose

of presenting these certifications at time of proposal is to speed up the award timeline by preparing you for what will be asked of your company by the Contracting Officer at time of award.

Request for Information (RFI)

After last year's solicitation period, the SBIR/STTR Program released an RFI asking for feedback on our existing subtopics. The inputs received were reviewed and provided to the Subtopic Managers for consideration as they updated or developed new subtopics for this year's solicitations. We hope that you see some of your inputs incorporated.

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1. Program Description

1.1 Introduction

This document includes instructions for two NASA program solicitations with separate subtopics under which small business concerns (SBCs) are invited to submit proposals: the Small Business Innovation Research (SBIR) Program and the Small Business Technology Transfer (STTR) Program. While the SBIR and STTR subtopics appear in an integrated list in Chapter 9, each subtopic will indicate its program of origin. Program background information, eligibility requirements for participants, information on the three program phases, and information for submitting responsive proposals are contained herein. The fiscal year 2019 Solicitation period for Phase I proposals begins February 5, 2019 and ends March 29, 2019.

The NASA SBIR and STTR Programs do not fund proposals solely directed towards system studies, market research, routine engineering, development of existing product(s), proven concepts, or modifications of existing products without substantive innovation.

It is anticipated that some SBIR and STTR Phase I proposals will be selected for negotiation of firm-fixed-price contracts approximately during the month of June 2019. Historically, 24 percent of SBIR Phase I proposal submissions receive awards, while 35 percent of STTR Phase I proposals receive awards. About 41 percent of the completed Phase I projects receive funding for Phase II development.

Under this Solicitation NASA will not accept more than 10 proposals to either program from any one firm (20 total) in order to ensure the broadest participation of the small business community. NASA does not plan to award more than 5 SBIR contracts and 2 STTR contracts (7 total) to any offeror. See section 3.1.

Proposals including all relevant documentation must be submitted online via the Proposal Submissions Electronic Handbook at http://sbir.nasa.gov. Unsolicited proposals will not be accepted.

1.2 Program Management and Alignment

The Space Technology Mission Directorate (STMD) provides overall policy direction for implementation of the NASA SBIR/STTR Programs. The NASA SBIR/STTR Program Management Office, which operates the programs in conjunction with NASA mission directorates and centers, is hosted at the NASA Ames Research Center. NASA Shared Services Center (NSSC) provides the overall procurement management for the programs.

For the SBIR Program, NASA research and technology areas to be solicited are identified annually by the agency's mission directorates. The directorates identify high priority research problems and technology needs for their respective programs and projects. The range of problems and technologies is broad, and the list of topics and subtopics vary in content from year to year to maintain alignment with current interests.

The STTR Program is aligned with the priorities of NASA's Space Technology Roadmaps, as well as the associated core competencies of the NASA centers. Again, the range of technologies is broad, and the list of topics and subtopics vary in content from year to year to maintain alignment with current interests.

For information regarding the mission directorates and the NASA centers see section 7.1.

For details on the research subtopic descriptions by Focus Area please see section 9.

1.3 Three-Phase Program

Both the SBIR and STTR Programs are divided into three funding and development stages. These three phases are described in detail on the NASA SBIR/STTR website: http://sbir.nasa.gov/content/nasa-sbirsttr-basics.

Phase I and II

Maximum value and period of performance for Phase I and Phase II contracts:

Phase I Contracts	**SBIR**	**STTR**
Maximum Contract Value	$125,000	$125,000
Period of Performance	6 months	13 months
Phase II Contracts	**SBIR**	**STTR**
Maximum Contract Value	$750,000	$750,000
Maximum Period of Performance	24 months	24 months

Post-Phase II Opportunities for Continued Technology Development

The NASA SBIR/STTR Program has two initiatives for supporting its small business partners beyond the basic Phase II award. These are the Phase II Extended (Phase II-E) contract option and the Civilian Commercialization Readiness Pilot Program (CCRPP) contract.

Please refer to http://sbir.nasa.gov/content/post-phase-ii-initiatives for eligibility, application deadlines, matching requirements and further information.

Phase III

Phase III is the commercialization of innovative technologies, products and services resulting from either a Phase I or Phase II contract. This includes further development of technologies for transition into NASA programs, other government agencies or the private sector. Phase III contracts are funded from sources other than the SBIR and STTR programs and may be awarded without further competition.

Please refer to http://sbir.nasa.gov/content/post-phase-ii-initiatives for Phase III information.

1.4 Availability of Funds

All Phase I, Phase II and post-Phase II awards are subject to availability of funds. NASA has no obligation to make any specific number of awards based on this solicitation, and may elect to make several or no awards in any specific technical topic or subtopic.

1.5 Eligibility Requirements

1.5.1 Small Business Concern

Only firms qualifying as SBCs, as defined here: http://sbir.nasa.gov/content/nasa-sbirsttr-program-definitions, are eligible to participate in these programs. Socially and economically disadvantaged and women-owned SBCs are particularly encouraged to propose.

1.5.2 Place of Performance

Research/Research & Development (R/R&D) must be performed in the United States (See: http://sbir.nasa.gov/content/nasa-sbirsttr-program-definitions). However, based on a rare and unique circumstance (for example, if a supply, material or other item or project requirement is not available in the United States), NASA may allow a particular portion of the research or R&D work to be performed or obtained in a country outside of the United States. Proposals must clearly indicate if any work will be performed outside the United States, including subcontractor performance. Prior to award, approval by the Contracting Officer for such specific condition(s) must be in writing.

Note: NASA will not approve purchases from or work with countries that appear on the list of Designated Countries. For reference, please visit https://oiir.hq.nasa.gov/nasaecp/ - Designated Countries List

1.5.3 Principal Investigator (PI) Employment Requirement.

The primary employment of the Principal Investigator (PI) shall be with the SBC under the SBIR Program, while under the STTR Program, either the SBC or Research Institution (RI) shall employ the PI. Primary employment means that more than 50% of the PI's total employed time (including all concurrent employers, consulting and self-employed time) is spent with the SBC or RI at time of award and during the entire period of performance. Primary employment with a small business concern precludes full-time employment at another organization. If the PI does not currently meet these primary employment requirements, then the offeror must explain how these requirements will be met if the proposal is selected for contract negotiations that may lead to an award. Co-Principal Investigators are not allowed.

Note: NASA considers a full-time workweek to be nominally 40 hours and we consider a 19.9-hour or more workweek elsewhere to be in conflict with this rule. In rare occasions, minor deviations from this requirement may be necessary; however, any minor deviation must be approved in writing by the Contracting Officer after consultation with the NASA SBIR/STTR Program Manager/Business Manager.

Requirements	**SBIR**	**STTR**
Primary Employment	PI shall be primarily employed with the SBC	PI shall be primarily employed with the RI or SBC
Employment Certification	The offeror must certify in the proposal that the primary employment of the PI will be with the SBC at the time of award and during the conduct of the project	The offeror must certify in the proposal that the primary employment of the PI will be with the SBC or the RI at the time of award and during the conduct of the project
Co-PIs	Not Allowed	Not Allowed
Misrepresentation of Qualifications	Shall result in rejection of the proposal or termination of the contract	Shall result in rejection of the proposal or termination of the contract
Substitution of PIs	Requires a prior approval from NASA	Requires a prior approval from NASA

1.5.4 Restrictions on Venture Capital-owned Businesses

At the current time, small businesses owned in majority part by multiple venture capital operating companies, hedge funds or private equity firms are not eligible to submit proposals to the NASA SBIR/STTR Solicitation.

1.5.5 Joint Ventures and Limited Partnerships

Both joint ventures and limited partnerships are permitted, provided the entity created qualifies as an SBC in accordance with the definition of an SBC here: http://sbir.nasa.gov/content/nasa-sbirsttr-program-definitions. A statement of how the workload will be distributed, managed and charged should be included in the proposal. A copy or comprehensive summary of the joint venture agreement or partnership agreement should be included on the Technical Proposal upload page.

1.5.6 Required Benchmark Transition Rate

The Transition Rate requirement applies only to SBIR and STTR Phase I applicants that have received more than 20 Phase I awards over the past five fiscal years, excluding the most recently completed fiscal year. To be eligible for a NASA Phase I SBIR or STTR award, the applicant must meet the Phase I to Phase II Transition Rate Benchmark as required by the SBIR/STTR Reauthorization Act of 2011 and extended by the Fiscal Year 2017 National Defense Authorization Act. Companies with more than 20 Phase I awards during the past five years can view their Transition Rate if they log onto their Company Registry account at www.sbir.gov. For these companies, the benchmark establishes a minimum number of Phase II awards the company must have received for a given number of Phase I awards received during the five-year time period in order to be eligible to receive a new Phase I award.

On June 1 of each year, the SBA assesses SBIR/STTR awardees using SBIR and STTR award information across all federal agencies reported on www.sbir.gov to determine if they meet the benchmark requirements. Companies that failed to meet the transition rate benchmark on June 1, 2018 are not eligible to submit a Phase I proposal during the period June 1, 2018 through May 31, 2019. Companies were notified by the SBA if they failed to meet the benchmark.

More information on the Transition Rate requirements is available at https://www.sbir.gov/faqs/performance-benchmarks.

1.6 NASA Technology Available (TAV) for SBIR/STTR Use

Offerors have the option of using technology developed by NASA (Technology Available (TAV)) related to the subtopic to which they are proposing. NASA has over 1400 patents available for licensing in its portfolio, including many sensors and materials related patents. NASA has over 1000 available software codes/tools listed in its Software Catalog (software.nasa.gov). While NASA scientists and engineers conduct breakthrough research that leads to innovations, the range of NASA's effort does not extend to commercial product development in any of its intramural research areas. Additional work is often necessary to exploit these NASA technologies (TAVs) for either infusion or commercial viability and likely requires innovation on behalf of the private sector. These technologies can be searched via the NASA Technology Transfer Portal, http://technology.nasa.gov, and may be a NASA owned patent and/or computer software. Use of a TAV requires a patent license or Software Usage Agreement from NASA. TAVs are available for use during both Phase I and Phase II award periods, including any extensions. NASA provides these technologies "as is" and makes no representation or guarantee that additional effort will result in infusion or commercial viability.

Whether or not a firm proposes the use of a NASA patent or computer software within their proposed effort will not in any way be a factor in the selection for award.

Use of NASA Software

If an offeror intends to use NASA software, a Software Usage Agreement (SUA), on a non-exclusive, royalty-free basis, is necessary, and the clause at 48 C.F.R. 1852.227-88, “Government-Furnished Computer Software and Related

Technical Data," will apply to the contract. A Software Usage Agreement (SUA) shall be requested from the appropriate NASA Center Software Release Authority (SRA), after contract award.

Use of NASA Patent

All offerors submitting proposals including the use of a NASA patent must submit an application for a non-exclusive, royalty-free evaluation license. Once a firm has identified a patent to license in the NASA patent portfolio (http://technology.nasa.gov), there is a link on the patent webpage that says "Click Here to License this Technology". Firms will be directed to NASA's Automated Licensing System (ATLAS) to finalize their license with the appropriate field center technology transfer office. The completed evaluation license application must be uploaded on the EHB Proposal Certifications page. Such grant of non-exclusive evaluation license will be set forth in the successful offeror's SBIR/STTR contract. The evaluation license will automatically terminate at the end of the SBIR/STTR contract. License applications will be treated in accordance with federal patent licensing regulations as provided in 37 CFR Part 404.

In addition to an evaluation license, if the proposed work includes the making, using or selling of products or services incorporating a NASA patent, successful awardees will be given the opportunity to negotiate a non-exclusive commercialization license or, if available, an exclusive commercialization license to the NASA patent. Commercialization licenses are also provided in accordance with 37 CFR Part 404.

An SBIR/STTR awardee that has been granted a non-exclusive, royalty-free evaluation license to use a NASA patent under the SBIR/STTR award may, if available and on a non-interference basis, also have access to NASA personnel knowledgeable about the NASA patent. Licensing Executives located at the appropriate NASA field center will be available to assist awardees requesting information about a patent that was identified in the SBIR/STTR contract and, if available and on a non-interference basis, provide access to the inventor or surrogate for the purpose of knowledge transfer.

Note: Access to the inventor for the purpose of knowledge transfer will require the requestor to enter into a Non-Disclosure Agreement (NDA), or other agreement, such as a Space Act Agreement. The awardee may be required to reimburse NASA for knowledge transfer activities. For Phase I proposals this is a time-consuming process and is not recommended.

1.7 I-Corps ™

The NASA SBIR/STTR Program is partnering with the National Science Foundation (NSF) to offer the NSF Innovation Corps Program (I-Corps ™) (hereinafter I-Corps). I-Corps focuses on educating teams on how to translate technologies from the laboratory into the marketplace. Participation in I-Corps will require selected contractors to conduct either 30 interviews (shortened version for the SBIR Program) or 100 interviews (full version for the STTR Program) to enable contractors to understand the commercial potential of their ideas. Selected contractors will be awarded training grants, separate from their Phase I contract, that must be completed prior to the conclusion of Phase I contracts. The program is described further at http://sbir.nasa.gov/content/I-Corps. The application process for I-Corps is described in Section 3.3.6. NASA will conduct an abbreviated competition for I-Corps after it selects offerors for Phase I SBIR and STTR contracts. NASA anticipates awarding a total of approximately 42 grants to SBIR and STTR Phase I contractors. The amount of funding is up to $25,000 for the full I-Corps Program for STTR firms, and up to $10,000 for the shortened version for SBIR firms.

1.8 Technical and Business Assistance

In accordance with the Small Business Act, NASA may authorize the recipient of a Phase II SBIR/STTR award to purchase technical and business assistance services through one or more outside vendors. The offeror may also seek

business-related services aimed at improving its commercialization success from an entity, such as a public or private organization or an agency of or other entity established or funded by a State that facilitates or accelerates the commercialization of technologies or assists in the creation and growth of private enterprises that are commercializing technology

These services may, as determined appropriate, include access to a network of non-NASA scientists and engineers engaged in a wide range of technologies, assistance with product sales, intellectual property protections, market research, market validation, and development of regulatory plans and manufacturing plans, or access to technical and business literature available through online databases, for the purpose of assisting such concerns in:

1. Making better technical decisions concerning such projects.
2. Solving technical problems which arise during the conduct of such projects.
3. Minimizing technical risks associated with such projects.
4. Commercializing new commercial products and processes resulting from such projects, including intellectual property protections.

If you are interested in proposing the use of a vendor for technical and business assistance, you must complete the Technical and Business Assistance section located under Other Direct Costs (ODCs) in the Proposal Budget form. You must provide the vendor name and contact information, the proposed amount not to exceed $5,000 per year per Phase II project and a detailed explanation of the services to be provided. You must also upload a price quote from the vendor including their DUNS number. Technical and business assistance does not count toward the maximum award size of your Phase II. Approval of technical and business assistance is not guaranteed and is subject to review by the Contracting Officer.

A description of any technical and business assistance obtained under this section and the benefits and results of the technical or business assistance provided will be a required deliverable of your Phase II contract.

Note: SEC. 854(c) of H.R.5515 - John S. McCain National Defense Authorization Act for Fiscal Year 2019 revised requirements for Technical and Business Assistance, including the amount agencies may make available. NASA is maintaining a maximum of $5,000 per year per Phase II project while it evaluates how to implement an expanded program for the future. Commercialization assistance does not count toward the maximum award size of your Phase II.

1.9 NASA Mentor-Protégé Program (MPP)

The purpose of the NASA Mentor-Protégé Program (MPP) is to provide incentives to NASA contractors, performing under at least one active approved subcontracting plan negotiated with NASA, to assist protégés in enhancing their capabilities to satisfy NASA and other contract and subcontract requirements. The NASA MPP, established under the authority of Title 42, U.S.C., 2473(c)(1) and managed by the Office of Small Business Programs (OSBP), includes an Award Fee Pilot Program. Under the Award Fee Pilot Program, a mentor is eligible to receive an award fee at the end of the agreement period based upon the mentor's performance of providing developmental assistance to an active SBIR/STTR Phase II contractor in a NASA Mentor-Protégé agreement (MPA).

The evaluation criterion is based on the amount and quality of technology transfer and business development skills that will increase the protégé's Technology Readiness Levels (TRLs). TRLs measure technology readiness on a scale of 1 to 9. A mentor should attempt to raise the TRL of the protégé and outline the goals and objectives in the MPA and the award fee plan. A separate award fee review panel set up by NASA OSBP will use the semiannual reports, annual reviews and the award fee plan in order to determine the amount of award fee given at the end of the performance period of the agreement.

For more information on the Mentor-Protégé Program please visit: http://www.osbp.nasa.gov/mpp/index.html.

1.10 NASA Procurement Ombudsman Program

The NASA Procurement Ombudsman Program is available under this solicitation as a procedure for addressing concerns and disagreements. The clause at NASA FAR Supplement (NFS) 1852.215-84 ("Ombudsman") is incorporated into this solicitation.

The cognizant ombudsman is:

William Roets
Deputy Assistant Administrator for Procurement
Office of Procurement
NASA Headquarters
Washington, DC 20546-0001
Telephone: 202-358-4483
Fax: 202-358-3082
Email: agency-procurementombudsman@nasa.gov

1.11 General Information

1.11.1 Means of Contacting NASA SBIR/STTR Program

1. NASA SBIR/STTR Website: http://sbir.nasa.gov

2. Help Desk: The NASA SBIR/STTR Help Desk can answer any questions regarding clarification of proposal instructions and any administrative matters. The Help Desk may be contacted by:

    a. Email: sbir@reisystems.com
    b. Telephone: 301-937-0888 between 9:00 a.m.-5:00 p.m. (Mon.-Fri., Eastern Time)
    c. The requestor must provide the name and telephone number of the person to contact, the organization name and address, and the specific questions or requests.

3. NASA SBIR/STTR Program Manager: Specific information requests that could not be answered by the Help Desk should be emailed to:

    ARC-SBIR-PMO@mail.nasa.gov

1.11.2 Questions about this Solicitation

To ensure fairness, questions relating to the intent and/or content of research topics in this Solicitation cannot be addressed during the open solicitation period. Only questions requesting clarification of proposal instructions and administrative matters will be addressed.

The cut-off date and time for receipt of Phase I solicitation procurement related questions and answers is 5:00 p.m. Eastern, March 22, 2019.

The cut-off date and time for receipt of Phase II solicitation procurement related questions and answers is seven calendar days prior to the end of the Phase I contract.

1.11.3 NASA Electronic Handbook (EHB)

NASA uses the EHB for all proposal submissions. See section 6 for more information.

1.12 Definitions

A comprehensive list of definitions related to the SBIR and STTR Programs is available at: http://sbir.nasa.gov/content/nasa-sbirsttr-program-definitions. These definitions include those from the SBIR and STTR policy directives, as well as terms specific to NASA. Offerors are strongly encouraged to review these prior to submitting a proposal.

2. Certifications and Other Proposal Information

2.1 SBA Firm Registry

SBA maintains and manages a Company Registry at http://www.sbir.gov to track ownership and affiliation requirements for all companies applying to the SBIR/STTR Program. The SBIR and STTR Policy Directives require each small business concern (SBC) applying for a Phase I or Phase II award to register in the Company Registry prior to submitting an application. A PDF document with the SBC registration information is available for download by the SBC upon successful registration. This PDF document must be saved by the SBC for inclusion in applications submitted to SBIR/STTR agencies. All SBCs must report and/or update ownership information to SBA prior to each SBIR/STTR application submission or if any information changes prior to award.

In the NASA SBIR/STTR Proposal Submissions Electronic Handbook (EHB), the SBC must provide their unique SBC Control ID that gets assigned by SBA upon completion of the Company Registry registration, as well as upload the PDF document validating their registration. This information is submitted to NASA via the Firm Certifications form and is applicable across all proposals submitted by the SBC for that specific solicitation.

2.2 System for Award Management (SAM) Registration

Offerors should be aware of the requirement to register in SAM prior to contract award.

Note: To avoid a potential delay in contract award, offerors are required to register prior to submitting a proposal. To be eligible for SBIR/STTR awards firms must be registered under the applicable NAICS code. SBIR/STTR Phase I and II awards use NAICS codes 541713 or 541715. Offerors that are not registered should consider applying for registration immediately upon receipt of this solicitation. Offerors and contractors may obtain information on SAM registration and annual confirmation requirements at https://www.sam.gov/SAM or by calling (866) 606-8220. In the past, registration and updates to registration have required a notarized letter with a processing period of several weeks.

SAM is the primary repository for contractor information required for the conduct of business with NASA. It is maintained by the Department of Defense. To be registered in SAM, all mandatory information, which includes the Data Universal Numbering System (DUNS) or DUNS+4 number, and a Commercial and Government Entity (CAGE) code, must be validated in SAM.

- The DUNS number is a 9-digit number assigned by Dun and Bradstreet Information Services to identify unique business entities. The DUNS+4 is similar, but includes a 4-digit suffix that may be assigned by a parent (controlling) business concern. To obtain a DUNS number please follow instructions at http://www.dnb.com.
- The CAGE code is assigned by the Defense Logistics Information Service (DLIS) to identify a commercial or government entity. If an SBC does not have a CAGE code, one will be assigned during the SAM registration process.

Note: It is recommended to list Purpose of Registration as "All Awards" on your SAM Registration.

2.3 FAR Certifications

SAM contains required certifications offerors may access at https://www.acquisition.gov/browsefar as part of required registration (see FAR 4.1102). Offerors must complete these certifications to be eligible for award.

Offerors should be aware that SAM requires all offerors provide representations and certifications electronically via the website, and to update the representations and certifications as necessary, but at least annually, to keep them current, accurate and complete. NASA will not enter into any contract wherein the Contractor is not compliant with the requirements stipulated herein.

2.3.1 52.222-37 Employment Reports on Special Disabled Veterans, Veterans of the Vietnam-Era and Other Eligible Veterans

In accordance with Title 38, United States Code, Section 4212(d), the U.S. Department of Labor (DOL) Veterans' Employment and Training Service (VETS) collects and compiles data on the Federal Contractor Program Veterans' Employment Report (VETS-4212 Report) from federal contractors and subcontractors who receive federal contracts that meet the threshold amount of $100,000. The VETS-4212 reporting cycle begins annually on August 1 and ends September 30. Any federal contractor or prospective contractor that has been awarded or will be awarded a federal contract with a value of $100,000 or greater must have a current VETS-4212 report on file. Please visit the DOL VETS-4212 website at http://www.dol.gov/vets/programs/fcp/main.htm. NASA will not enter into any contract wherein the firm is not compliant with the requirements stipulated herein.

2.4 Firm Certifications

Offerors must complete the Firm Certifications section of the Proposal Submissions Electronic Handbook, answering Yes or No to certifications as applicable.

Firms should carefully read each of the certification statements. The federal government relies on the information to determine whether the business is eligible for a SBIR or STTR Program award. A similar certification will be used to ensure continued compliance with specific program requirements during the life of the funding agreement. The definitions for the terms used in this certification are set forth in the Small Business Act, SBA regulations (13 C.F.R. Part 121), the SBIR and STTR Policy Directives and any statutory and regulatory provisions referenced in those authorities.

If the Contracting Officer believes that the business may not meet certain eligibility requirements at the time of award, they are required to file a size protest with the U.S. Small Business Administration (SBA), who will determine eligibility. At that time, SBA will request further clarification and supporting documentation in order to assist in the eligibility determination. Additionally, the Contracting Officer may request further clarification and supporting documentation regarding eligibility to determine whether a referral to SBA is required.

2.5 NASA Clauses

The following NASA clauses are necessary to implement restrictions in NASA appropriations. Offerors must comply with these clauses to be eligible for award.

1852.203-71 Requirement to Inform Employees of Whistleblower Rights (Aug 14)

(a) The Contractor shall inform its employees in writing, in the predominant native language of the workforce, of contractor employee whistleblower rights and protections under 10 U.S.C. 2409, as described in subpart 1803.9 of the NASA FAR Supplement.
(b) The Contractor shall include the substance of this clause, including this paragraph (b), in all subcontracts.

1852.225-72 Restriction on funding Activity with China – Representation.

(a) (a) Definition - “China” or “Chinese-owned” means the People’s Republic of China, any company owned by the People’s Republic of China or any company incorporated under the laws of the People’s Republic of China.
(b) (b) Public Laws 112-10, Section 1340(a) and 112-55, Section 536, restrict NASA from contracting to participate, collaborate, or coordinate bilaterally in any way with China or a Chinese-owned company with funds appropriated on or after April 25, 2011. Contracts for commercial and non-developmental items are excepted from the prohibition as they constitute purchase of goods or services that would not involve participation, collaboration, or coordination between the parties.
(c) (c) Representation. By submission of its offer, the offeror represents that the offeror is not China or a Chinese-owned company.

2.6 False Statements

Note: Knowingly and willfully making any false, fictitious or fraudulent statements or representations may be a felony under the Federal Criminal False Statement Act (18 U.S.C. Sec 1001), punishable by a fine and imprisonment of up to five years in prison. The Office of the Inspector General (OIG) has full access to all proposals submitted to NASA.

Pursuant to NASA policy, any company representative who observes crime, fraud, waste, abuse, or mismanagement or receives an allegation of crime, fraud, waste, abuse or mismanagement from a federal employee, contractor, grantee, contractor or grantee employee, or any other source will report such observation or allegation to the OIG. NASA contractor employees and other individuals are also encouraged to report crime, fraud, waste and mismanagement in NASA's programs to the OIG. The OIG offers several ways to report a complaint:

NASA OIG Hotline
1-800-424-9183
TDD: 1-800-535-8134

NASA OIG Cyber Hotline
http://oig.nasa.gov/cyberhotline.html

NASA OIG Headquarters
Office of Investigations fax number: 202-358-3914

Or by mail:
NASA Office of Inspector General
P.O. Box 23089
L'Enfant Plaza Station
Washington, DC 20026

2.7 Software Development Standards

Offerors proposing projects involving the development of software may be required to comply with the requirements of NASA Procedural Requirements (NPR) 7150.2A, “NASA Software Engineering Requirements” which are available online at http://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=7150&s=2.

2.8 Human and/or Animal Subject

Offerors should be aware of the requirement that an approved protocol by a NASA Review Board is required if the proposed work includes human or animal subject. An approved protocol shall be provided to the Contracting Officer prior to the initiation of any human and/or animal subject research. Offerors shall identify the use of human or animal subject in the Proposal Certifications form. For additional information, contact the NASA SBIR/STTR Program Management Office at ARC-SBIR-PMO@mail.nasa.gov. Reference 14 CFR 1230 and 1232.

Note: Due to the complexity of the approval process, use of human and/or animal subjects is not allowed for Phase I contracts.

2.9 HSPD-12

Firms that require access to federally controlled facilities or access to a federal information system (federally controlled facilities and federal information system are defined in FAR 2.101(b)(2)) for six consecutive months or more must adhere to Homeland Security Presidential Directive 12 (HSPD-12), "Policy for a Common Identification Standard for Federal Employees and Contractors", and Federal Information Processing Standards Publication (FIPS PUB) Number 201, "Personal Identity Verification (PIV) of Federal Employees and Contractors," which require agencies to establish and implement procedures to create and use a government-wide secure and reliable form of identification no later than October 27, 2005. See: https://nvlpubs.nist.gov/nistpubs/FIPS/NIST.FIPS.201-2.pdf. In accordance with the FAR clause 52.204-9, "Personal Identity Verification of Contractor Personnel," which states in part that the contractor shall comply with the requirements of this clause and shall ensure that individuals needing such access shall provide the personal background and biographical information requested by NASA.

Note: Additional information regarding PIV credentials can be found at https://csrc.nist.gov/Projects/PIV.

3. Proposal Preparation Instructions and Requirements

3.1 Fundamental Considerations

Multiple Proposal Submissions

Each proposal submitted must be based on a unique innovation, must be limited in scope to just one subtopic and shall be submitted only under that one subtopic within each program. An offeror shall not submit more than 10 proposals to each of the SBIR or STTR Programs (20 total). An offeror may submit more than one unique proposal to the same subtopic; however, an offeror shall not submit the same (or substantially equivalent) proposal to more than one subtopic. Submitting substantially equivalent proposals to several subtopics may result in the rejection of all such proposals. In order to enhance SBC participation, NASA does not plan to select more than 5 SBIR proposals and 2 STTR proposals from any one offeror under this solicitation (7 total).

3.2 Proprietary Information in the Proposal Submission

If proprietary information is provided by an applicant in a proposal, which constitutes a trade secret, proprietary commercial or financial information, confidential personal information or data affecting the national security, it will be treated in confidence to the extent permitted by law. This information must be clearly marked by the applicant as confidential proprietary information. NASA will treat in confidence pages listed as proprietary during the review step of proposal submission.

Information contained in unsuccessful proposals will remain the property of the applicant. However, the government will retain copies of all proposals.

3.2.1 Release of Certain Proposal Information

In submitting a proposal, the offeror agrees to permit the government to disclose publicly the information contained in the Contact Information form, Proposal Summary form and Briefing Chart. Other proposal data is considered to be the property of the offeror, and NASA will protect it from public disclosure to the extent permitted by law including requests submitted under the Freedom of Information Act (FOIA).

3.3 Phase I Proposal Requirements

3.3.1 General Requirements

A competitive proposal will clearly and concisely: (1) describe the proposed innovation relative to the state of the art; (2) address the scientific, technical, and commercial merit and feasibility of the proposed innovation, and its relevance and significance to NASA interests as described in section 9 of this solicitation; and (3) provide a preliminary strategy that addresses key technical, market and business factors pertinent to the successful development, demonstration of the proposed innovation, and its transition into products and services for NASA mission programs, the commercial aerospace industry, and other potential markets and customers.

3.3.2 Format Requirements

Note: The government administratively screens all proposals and reserves the right to reject any proposal that does not conform to the following formatting requirements. Offerors that repeatedly violate solicitation formatting instructions are at higher risk of rejection for nonconformance on subsequent SBIR/STTR proposals.

Page Limitations and Margins

Note: Technical proposal uploads with any page(s) going over the required page limit will not be accepted.

A Phase I technical proposal shall not exceed a total of 19 standard 8 1/2 x 11 inch (21.6 x 27.9 cm) pages. Proposals uploaded with more than 19 pages will prompt a warning which will prevent the completed proposal from being submitted. Each page shall be numbered consecutively at the bottom. Margins shall be 1.0 inch (2.5 cm). The space allocated to each part of the technical content will depend on the project chosen and the offeror's approach. The additional forms required for proposal submission will not count against the 19-page limit.

Type Size
No type size smaller than 10 point shall be used for text or tables, except as legends on reduced drawings. Proposals prepared with smaller font sizes may be rejected without consideration.

Header/Footer Requirements
Header must include firm name, proposal number and project title. Footer must include the page number and proprietary markings if applicable. Margins can be used for header/footer information.

Classified Information
NASA does not accept proposals that contain classified information.

Project Title
The proposal project title shall be concise and descriptive of the proposed effort. The title should not use acronyms or words like "Development of" or "Study of." The NASA research subtopic title must not be used as the proposal title.

Proposal Package

Each complete proposal package submitted shall contain the following items:

1. Proposal Contact Information (**3.3.3.1**)
2. Proposal Certifications, electronically endorsed. (**3.3.3.2**)
3. Proposal Summary (must not contain proprietary data). (**3.3.3.3**)
4. Proposal Budget (including letters of availability for facilities and subcontractors/consultants, if applicable) (**3.3.3.4**)
5. Technical Proposal - 10 parts in the order specified in section 3.3.4, and not to exceed 19 pages **(both SBIR and STTR)**, including all graphics, with a table of contents. (**3.3.4**)
6. Research Agreement between the SBC and RI **(STTR only)**. (**3.3.5**)
7. Briefing Chart (must not contain proprietary data). (**3.3.7**)
8. NASA Evaluation License Application, only if TAV is being proposed. (**1.6**)
9. I-Corps Opt-In Form (**3.3.6**)

Note: Letters expressing general technical interest or letters of funding support commitments (for Phase I) are not required or desired and will not be considered during the review process. However, if submitted, such letter(s) will count against the page limit.

Firm Level Forms

In addition to the above items, each offeror must submit the following firm level forms, which must be filled out once during each submission period and are applicable to all firm proposal submissions:

1. Firm Certifications (**3.3.8**)
2. Audit Information (**3.3.9**)
3. Prior Awards Addendum (**3.3.10**)
4. Commercial Metrics Survey (**3.3.11**)

Previews of all forms and certifications are available via the NASA SBIR/STTR Firm Library, located at: http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html.

Note: Website references, relevant technical papers, product samples, videotapes, slides or other ancillary items will not be considered during the review process.

3.3.3 Forms

All form submissions shall be completed electronically, and do not count towards the 19-page limit for the technical proposal.

3.3.3.1 Proposal Contact Information

A sample Contact Information form is provided in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. The offeror shall provide complete information for each contact person and submit the form as required in section 6.

Note: Contact Information is public information and may be disclosed.

3.3.3.2 Proposal Certifications

A sample Proposal Certifications form is provided in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. The offeror shall provide complete information for each item and submit the form as required in section 6.

3.3.3.3 Proposal Summary

A sample Proposal Summary form is provided in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. The offeror shall provide complete information for each item and submit the form as required in section 6.

Note: The Proposal Summary, including the Technical Abstract, is public information and may be disclosed. Do not include proprietary information in this form.

3.3.3.4 Proposal Budget

A sample of the Proposal Budget form is provided in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. The offeror shall complete the Proposal Budget following the instructions provided with the sample form. The total requested funding for the Phase I effort shall not exceed $125,000. Contextual help is provided on the electronic budget form for additional explanation. Information shall be submitted to explain the offeror's plans for use of the requested funds to enable NASA to determine whether the proposed price is fair and reasonable.

Note: The government is not responsible for any monies expended by the firm before award of any contract.

In addition, the following additional uploads, must be submitted in the Proposal Budget form, as applicable:

Proposal Requirements for Use of Federal Services, Facilities or Equipment:
In cases where an offeror seeks to use NASA or another federal department or agency services, equipment or facilities, the offeror shall provide the following:

1. Statement, signed by the appropriate government official at the effected federal department or agency, verifying that the resources should be available during the proposed period of performance.
2. Signed letter on company letterhead from the contractor's Small Business Official explaining why the SBIR/STTR research project requires the use of federal services, equipment or facilities, including data that verifies the absence of non-federal facilities or personnel capable of supporting the research effort, a statement confirming that the facility proposed is not a federal laboratory, if applicable, and the associated cost estimate.

Note: Use of federal laboratories/facilities for Phase I contracts is highly discouraged. Approval for use of federal facilities and labs, for a Phase I proposal, requires Program Executive approval during negotiations if selected for award.

See Part 8 of the Technical Proposal for additional information on use of federal facilities.

Use of Subcontractors and Consultants:
Subject to the restrictions set forth below, the SBC may establish business arrangements with other entities or individuals to participate in performance of the proposed R/R&D effort. Subcontractors' and consultants' work has the same place of performance restrictions as stated in section 1.5.2.

Note:

1. ***Offerors should list consultants by name and specify, for each, the number of hours and hourly costs.***
2. ***Breakdown of subcontractor budget should mirror the SBC's own breakdown in the Proposal Budget form and include breakdowns of direct labor, other direct costs, profit, as well as indirect rate agreements.***
3. ***A signed letter of commitment is required for each subcontractor and/or consultant. For educational institutions, the letter must be from the institution's Office of Sponsored Programs.***

The following restrictions apply to the use of subcontracts/consultants and the formula below must be used in preparing budgets with subcontractors/consultants:

SBIR Phase I Subcontracts/Consultants	**STTR Phase I Subcontracts/Consultants**
The proposed subcontracted business arrangements, including consultants, must not exceed 33 percent of the research and/or analytical work [as determined by the total cost of the proposed subcontracting effort (to include the appropriate OH and G&A) in comparison to the total effort (total contract price including cost sharing, if any, less profit if any)].	A minimum of 40 percent of the research or analytical work must be performed by the proposing SBC and minimum of 30 percent must be performed by the RI. Any subcontracted business effort other than that performed by the RI, shall not exceed 30 percent of the research and/or analytical work [as determined by the total cost of the subcontracting effort (to include the appropriate OH and G&A) in comparison to the total effort (total contract price including cost sharing, if any, less profit if any)].

Example:	Total price to include profit	- $99, 500
	Profit	- $3,000
	Total price less profit	- $99,500 - $3,000 = $96,500
	Subcontractor cost	- $29,500
	G&A	- 5%
	G&A on subcontractor cost	- $29,500 x 5% = $1,475
	Subcontractor cost plus G&A	- $29,500 + $1,475 = $30,975
	Percentage of subcontracting effort – subcontractor cost plus G&A / total price less profit	**- $30,975 / $96,500 = 32.1%**

For an SBIR Phase I this is acceptable since it is below the limitation of 33%.
For an STTR Phase I, where there is a subcontract with a company other than the RI, this is unacceptable since it is above the 30% limitation.

See Part 9 of the Technical Proposal for additional information on the use of Subcontractors and Consultants.

Travel in Phase I
Travel is highly discouraged during the performance of a Phase I contract. If the purpose of the meeting cannot be accomplished via videoconference or teleconference, the contractor must provide rationale for the trip in the proposal budget form. All travel must be approved by the Contracting Officer.

3.3.4 Technical Proposal

This part of the submission should not contain any budget data and must consist of all ten (10) parts listed below in the given order. All ten parts of the technical proposal must be numbered and titled. Parts that are not applicable must be included and marked "Not Applicable." A proposal omitting any part will be considered non-responsive to this solicitation and may be rejected during administrative screening. The required table of contents is provided below:

Phase I Table of Contents
Part 1: Table of Contents……..……Page 1
Part 2: Identification and Significance of the Innovation
Part 3: Technical Objectives
Part 4: Work Plan
Part 5: Related R/R&D
Part 6: Key Personnel and Bibliography of Directly Related Work
Part 7: Potential Future Applications and Relationship with Future R/R&D
Part 8: Facilities/Equipment
Part 9: Subcontracts and Consultants
Part 10: Related, Essentially Equivalent and Duplicate Proposals and Awards

Part 1: Table of Contents
The technical proposal shall begin with a brief table of contents indicating the page numbers of each of the parts of the proposal.

Part 2: Identification and Significance of the Proposed Innovation
Succinctly describe:

- The proposed innovation.
- The relevance and significance of the proposed innovation to an interest, need or needs, within a subtopic described in section 9.
- The proposed innovation relative to the state of the art.

Part 3: Technical Objectives
State the specific objectives of the Phase I R/R&D effort including the technical questions posed in the subtopic description that must be answered to determine the feasibility of the proposed innovation.

Note: All offerors submitting proposals who are planning to use NASA Intellectual Property (IP) must describe their planned developments with the IP. The NASA Evaluation License Application should be added as an attachment in the Proposal Certifications form (see section 1.6).

Part 4: Work Plan

Include a detailed description of the Phase I R/R&D plan to meet the technical objectives. The plan should indicate what will be done, where it will be done and how the R/R&D will be carried out. Discuss in detail the methods planned to achieve each task or objective. Task descriptions, schedules, resource allocations, estimated task hours for each key personnel and planned accomplishments including project milestones shall be included. Offerors should ensure that the estimated task hours provided in the work plan for key personnel are consistent with the hours reported in the Proposal Budget form. If the offeror is a joint venture or limited partnership, a statement of how the workload will be distributed, managed and charged should be included here.

STTR: In addition, the work plan will specifically address the percentage and type of work to be performed by the SBC and the RI. The plan will provide evidence that the SBC will exercise management direction and control of the performance of the STTR effort, including situations in which the PI may be an employee of the RI.

Part 5: Related R/R&D
Describe significant current and/or previous R/R&D that is directly related to the proposal including any conducted by the PI or by the offeror. Describe how it relates to the proposed effort and any planned coordination with outside sources. The offeror must persuade reviewers of his or her awareness of key recent R/R&D conducted by others in the specific subject area.

Part 6: Key Personnel and Bibliography of Directly Related Work
Identify all key personnel involved in Phase I activities whose expertise and functions are essential to the success of the project. Provide bibliographic information including directly related education and experience.

The PI is considered key to the success of the effort and must make a substantial commitment to the project. The following requirements are applicable:

> **Functions:** The functions of the PI are: planning and directing the project, leading it technically and making substantial personal contributions during its implementation, serving as the primary contact with NASA on the project and ensuring that the work proceeds according to contract agreements. Competent management of PI functions is essential to project success. The Phase I proposal shall describe the nature of the PI's activities and the amount of time that the PI will personally apply to the project. The amount of time the PI proposes to spend on the project must be acceptable to the Contracting Officer.

Qualifications: The qualifications and capabilities of the proposed PI and the basis for PI selection are to be clearly presented in the proposal. NASA has the sole right to accept or reject a PI based on factors such as education, experience, demonstrated ability and competence, and any other evidence related to the specific assignment.

Eligibility: This part shall also establish and confirm the eligibility of the PI, and indicate the extent to which other proposals recently submitted or planned for submission in Fiscal Year 2019 and existing projects commit the time of the PI concurrently with this proposed activity. Any attempt to circumvent the restriction on PIs working more than half time for an academic or a nonprofit organization by substituting an ineligible PI will result in rejection of the proposal. However, for an STTR the PI can be primarily employed by either the SBC or the RI. Please see section 1.5.3 for further explanation.

Part 7: Potential Future Applications and Relationship with Future R/R&D

Discuss the anticipated results of the proposed investigation if this project is successful (through Phase I & II). Discuss the significance of the Phase I effort in relation to its foundation for Phase II and potential follow-on development, applications and potential for commercialization.

Commercialization is defined as (1) the transition of technology into products and services for NASA mission programs (2) other government agencies and (3) non-government markets. Discuss (1) potential and targeted applications of the technology and paths to infusion in future NASA missions and/or programs (2) possible applications in other government agencies and (3) potential non-government markets. Provide a preliminary commercialization strategy that addresses key technical, market and business factors pertinent to the successful development of the proposed innovation, and its transition into products and services for NASA mission programs, other government agencies and non-government markets. If feasible, discuss potential barriers to entry, potential competitors, risks, and existing and projected commitments for Phase III funding, investment, sales, licensing, and other indicators of commercial potential and feasibility.

Please also discuss your experience and record in technology commercialization, as applicable.

Note: Companies with no SBIR/STTR awards or fairly recent awards will not be penalized under past performance for the lack of past SBIR/STTR commercialization.

Part 8: Facilities/Equipment

Offerors must describe the necessary instrumentation and facilities to be used to perform the proposed work. Offerors must ensure the resources are adequate and address any reliance on external sources, such as government furnished equipment or facilities. In cases where an offeror seeks to use NASA or another federal department or agency services, equipment or facilities, the offeror shall describe in this part why the use of government furnished equipment or facilities is necessary. See section 3.3.3.4 and 5.13 for additional requirements when proposing use of federal facilities. The narrative description of facilities and equipment should support the proposed approach and documentation in the Proposal Budget form.

Note: Use of federal laboratories/facilities for Phase I contracts is highly discouraged. Approval for use of federal facilities and labs, for a Phase I proposal, requires Program Executive approval during negotiations if selected for award.

Part 9: Subcontracts and Consultants

The offeror must describe all subcontracting or other business arrangements and identify the relevant organizations and/or individuals with whom arrangements are planned. The expertise to be provided by the entities must be described

in detail, as well as the functions, services and number of hours. Offerors are responsible for ensuring that all organizations and individuals proposed to be utilized are actually available for the time periods proposed. Subcontract costs shall be documented in the Subcontractors/Consultants section of the Proposal Budget form and supporting documentation should be uploaded for each (appropriate documentation is specified in the form). The narrative description of subcontracts and consultants in the technical proposal should support the proposed approach and documentation in the Proposal Budget form.

Part 10: Related, Essentially Equivalent, and Duplicate Proposals and Awards

WARNING – While it is permissible with proper notification to submit identical proposals or proposals containing a significant amount of essentially equivalent work for consideration under numerous federal program solicitations, it is unlawful to enter into funding agreements requiring essentially equivalent work.

If an applicant elects to submit identical proposals or proposals containing a significant amount of essentially equivalent work under other federal program solicitations, a statement must be included in each such proposal indicating:

1. The name and address of the agencies to which proposals were submitted or from which awards were received.
2. Date of proposal submission or date of award.
3. Title, number and date of solicitations under which proposals were submitted or awards received.
4. The specific applicable research topics for each proposal submitted or award received.
5. Titles of research projects.
6. Name and title of principal investigator or project manager for each proposal submitted or award received.

Offerors are at risk for submitting essentially equivalent proposals and therefore are strongly encouraged to disclose these issues to the soliciting agency to resolve the matter prior to award.

A summary of essentially equivalent work information, as well as related research and development on proposals and awards is also required on the Proposal Certifications form (if applicable).

3.3.5 Research Agreement (Applicable for STTR proposals only)

STTR: The Research Agreement (different from the Allocation of Rights Agreement, see: http://sbir.nasa.gov/content/nasa-sbirsttr-program-definitions) is a single-page document electronically submitted and endorsed by the SBC and RI. A model agreement is provided, or firms can create their own custom agreement. The Research Agreement should be submitted as required in section 6.

All STTR Phase I proposals must provide sufficient information to convince NASA that the proposed SBC/RI cooperative effort represents a sound approach for converting technical information resident at the RI into a product or service that meets a need described in a Solicitation research topic.

3.3.6 Applications to I-Corps

Firms proposing to this solicitation will be allowed to also propose participation in the SBIR/STTR I-Corps Program using the following submittal process. I-Corps awards will be made separately from the Phase I contract as a training grant.

3.3.6.1 Step 1: Opt-In Form

Phase I SBIR/STTR offerors must complete a short I-Corps Opt-In Form as part of their Phase I proposal submission. Representations in the form will determine an offeror's eligibility to participate in I-Corp. The form also asks that offerors provide a brief summary explaining the value of I-Corps to their companies. In the event a large number of offerors express interest, the government reserves the right to limit the number of offerors invited to submit I-Corps proposals based upon the government's assessment of the initial summary statements.

3.3.6.2 I-Corps Proposal

To be qualified to submit an I-Corps proposal: 1) offerors must have submitted the I-Corps Opt-In Form as part of their Phase I proposals; 2) offerors must be qualified to participate in I-Corps and 3) offerors must be selected for a Phase I award. Participating offerors must form a team composed of three main members: The Principal Investigator, the Entrepreneurial Lead and the Mentor, as described in http://sbir.nasa.gov/content/I-Corps. The I-Corps proposal shall follow the same format requirements as the SBIR/STTR Phase I proposal, shall be limited to six pages and shall include the following sections in order to be considered complete:

- I-Corps Team and Commercialization Plan (limited to five pages).
    - I-Corps Team: Biographical sketches of I-Corps team members and their commitment to participate in I-Corps (limited to one page per team member).
    - Commercialization Plan (limited to one page). This shall include:
        - Composition and roles (Principal Investigator, Entrepreneurial Lead and Mentor) of the team members proposing to undertake the commercialization feasibility research.
        - Building off the commercialization information provided in the Phase I proposal, include an additional, brief description of the potential non-NASA commercial impacts of the project, what types of customer discovery the firm hopes to accomplish through I-Corps and what steps the company will take to move the project closer to commercialization.
- I-Corps Proposal Budget (limited to one page).
    - Capped at $10,000 for each SBIR team and $25,000 for each STTR team.
    - Only recovery of certain direct costs associated with participation in I-Corps is allowed, no recovery of indirect costs is allowed.
    - The budget should include the following five components:
        - Maximum of $5,500 for Entrepreneurial Lead stipend (no stipend for the Principal Investigator or I-Corps Mentor)
        - An estimate for the travel costs associated with team member participation in required kick-off and close out/lessons learned meetings (i.e., airfare, per diem costs). Suggested limit is $5,500 per team.
        - Costs for workshop registration fees that will be paid to the instruction service (logistics) providers. This is expected to be $4,500 per team.
        - Estimated costs for travel associated with the three team members traveling as a group to conduct customer interviews (30 interviews for SBIR participants and 100 interviews for STTR participants). Suggested limits are $2,550 for SBIR teams and $10,000 for STTR teams.

The I-Corps proposal will be due one week after formal notification that the firm has been selected for negotiation of a Phase I SBIR or STTR contract. The firm shall submit their I-Corps proposal into the Proposal Submissions EHB, which shall be re-opened for those firms which have met the three qualifications identified above.

Note: Proposals for I-Corps have separate page limitations outside the page limitations for the technical proposal.

3.3.7 Briefing Chart

The one-page briefing chart is required to assist in the ranking and advocacy of proposals prior to selection and contains the following sections with summary information:

- Identification and Significance of Innovation
- Technical Objectives
- Proposed Deliverables
- NASA Applications
- Non-NASA Applications
- Graphic

It shall not contain any proprietary data or ITAR restricted data. An electronic form will be provided during the submissions process.

Note: The briefing chart is public information and may be disclosed. Do not include proprietary information in this form.

3.3.8 Firm Certifications

Firm certifications that are applicable across all proposal submissions submitted to this solicitation must be completed via the Firm Certifications section of the Proposal Submissions Electronic Handbook. The offeror shall answer Yes or No as applicable. An example of the certifications can be found in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. An electronic form will be provided during the submissions process.

Note: The designated Firm Admin, typically the first person to register your firm, is the only individual authorized to update the certifications.

3.3.9 Audit Information

Although firms are not required to have an approved accounting system, knowledge that a firm has an approved accounting system facilitates NASA's determination that rates are fair and reasonable. To assist NASA, the SBC shall complete the questions in the Audit Information form regarding the firm's rates and upload the federal agency audit report or related information that is available from the last audit. There is a separate "Audit Information" section in the Proposal Budget form that shall also be completed. If your firm has never been audited by a federal agency, then answer "No" to the first question and you do not need to complete the remainder of the form. An electronic form will be provided during the submissions process.

The Contracting Officer will use this Audit Information to assist with negotiations if the proposal is selected for award. The Contracting Officer will advise offerors what is required to determine reasonable cost and/or rates in the event the Audit Information is not adequate to support the necessary determination on rates.

Note: The designated firm admin, typically the first person to register your firm, is the only individual authorized to update the audit information.

3.3.10 Prior Awards Addendum

If the SBC has received more than 15 Phase II awards in the prior 5 fiscal years, submit name of awarding agency, solicitation year, phase, date of award, funding agreement/contract number, and topic or subtopic title for each Phase II. If your firm has received any SBIR or STTR Phase II awards, even if it has received fewer than 15 in the last 5 years, it is still recommended that you complete this form for those Phase II awards your firm did receive. This information will be useful when completing the Commercialization Metrics Survey, and in tracking the overall success of the SBIR and STTR Programs. Any NASA Phase II awards your firm has received will be automatically populated in the electronic form, as well as any Phase II awards previously entered by the SBC during prior submissions (you may update the information for these awards). An electronic form will be provided during the submissions process.

Note: The designated firm admin, typically the first person to register your firm, is the only individual authorized to update the addendum information.

3.3.11 Commercial Metrics Survey

NASA has instituted a comprehensive commercialization survey/data gathering process for firms with prior NASA SBIR/STTR awards. If the SBC has received any Phase III awards resulting from work on any NASA SBIR or STTR awards, provide the related Phase I or Phase II contract number, name of Phase III awarding agency, date of award, funding agreement number, amount, project title and period of performance. The survey will also ask for firm financial, sales and ownership information, as well as any commercialization success the firm has had as a result of SBIR or STTR awards. This information must be updated annually during proposal submission via the EHB. This information allows firms to demonstrate their ability to carry SBIR/STTR research through to achieve commercial success, and allow agencies to track the overall commercialization success of their SBIR and STTR Programs. The survey should be limited to information requested above. An electronic form will be provided during the submissions process.

Note: Information received from SBIR/STTR awardees completing the survey is kept confidential, and will not be made public except in broad aggregate, with no firm-specific attribution.

The Commercialization Metrics Survey is a required part of the proposal submissions process and must be completed via the Proposal Submissions Electronic Handbook. Also, Companies with no SBIR/STTR awards or only fairly recent awards will not be penalized under past performance for the lack of past SBIR/STTR commercialization.

3.3.12 Allocation of Rights Agreement (STTR awards only)

No more than 10 business days after the notification of selection for negotiation, the offeror should provide to the Contracting Officer, a completed **Allocation of Rights Agreement (ARA)**, which has been signed by authorized representatives of the SBC, RI, and subcontractors and consultants, as applicable. The ARA shall state the allocation of intellectual property rights with respect to the proposed STTR activity and planned follow-on research, development and/or commercialization. A sample ARA is available in the NASA SBIR/STTR Firm Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html of this Solicitation.

If the ARA form is completed and available at the time of submission, offers should upload it in in the Proposal Budget form, which will help to expedite contract negotiations.

3.4 Phase II Proposal Requirements

3.4.1 General Requirements

The Phase I contract will serve as a request for proposal (RFP) for the Phase II follow-on project. Phase II proposals are more comprehensive than those required for Phase I. Submission of a Phase II proposal is in accordance with Phase I contract requirements and is voluntary. NASA assumes no responsibility for any proposal preparation expenses.

A competitive Phase II proposal will clearly and concisely (1) describe the proposed innovation relative to the state of the art and the market, (2) address Phase I results relative to the scientific, technical merit and feasibility of the proposed innovation and its relevance and significance to the NASA interests and (3) provide the planning for a focused project that builds upon Phase I results and encompasses technical, market, financial and business factors relating to the development and demonstration of the proposed innovation, and its transition into products and services for NASA mission programs, other government agencies and non-government markets.

3.4.2 Format Requirements

Note: The government administratively screens all proposals and reserves the right to reject any proposal that does not conform to the following formatting requirements.

Page Limitations and Margins

Note: Technical proposal uploads with any page(s) going over the required page limit will not be accepted.

A Phase II technical proposal shall not exceed a total of 46 standard 8 1/2 x 11 inch (21.6 x 27.9 cm) pages. Proposals uploaded with more than 46 pages will prompt a warning which will prevent the completed proposal from being submitted. Each page shall be numbered consecutively at the bottom. Margins shall be 1.0 inch (2.5 cm). The space allocated to each part of the technical content will depend on the project chosen and the offeror's approach. The additional forms required for proposal submission will not count against the 46-page limit.

Type Size
No type size smaller than 10 point shall be used for text or tables, except as legends on reduced drawings. Proposals prepared with smaller font sizes may be rejected without consideration.

Header/Footer Requirements
Header must include firm name, proposal number and project title. Footer must include the page number and proprietary markings if applicable. Margins can be used for header/footer information.

Classified Information
NASA does not accept proposals that contain classified information.

Project Title
The proposal project title shall be concise and descriptive of the proposed effort. The title should not use acronyms or words like "Development of" or "Study of." The NASA research topic title must not be used as the proposal title.

Proposal Package

Each complete proposal package submitted shall contain the following items:

1. Proposal Contact Information. (**3.4.3.1)**
2. Proposal Certifications, electronically endorsed. (**3.4.3.2**)
3. Proposal Summary (must not contain proprietary data). (**3.4.3.3**)
4. Proposal Budget. (**3.4.3.4**)
5. Technical Content - 10 Parts in the order specified in section 3.4.4, not to exceed 46 pages **(for SBIR AND STTR)**, including all graphics, and starting with a table of contents. (**3.4.4**)
6. Research Agreement between the SBC and RI **(STTR only)**. (**3.4.5**)
7. Briefing Chart (must not contain proprietary data). (**3.4.7**)
8. NASA Evaluation License Application, only if TAV is being proposed. (**1.6**)
9. Capital Commitments Addendum Supporting Phase II and Phase III (optional). (**3.4.6**)

Note: Letters expressing general technical interest are not required or desired and will not be considered during the review process. However, if submitted, such letter(s) will count against the page limit. Letters of funding support commitments are allowable for Phase II proposals but will be considered only under Factor 4 - Commercial Potential and Feasibility. Letters of funding support commitments should be submitted as part of the Capital Commitments Addendum.

Firm Level Forms

In addition to the above items, each offeror must submit the following firm level forms, which must be filled out once during each submission period and are applicable to all firm proposal submissions:

1. Firm Certifications (**3.4.8**)
2. Audit Information (**3.4.9**)
3. Prior Awards Addendum (**3.4.10**)
4. Commercial Metrics Survey (**3.4.11**)

Previews of all forms and certifications are available via the NASA SBIR/STTR Firm Library, located at: http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html.

Note: Website references, relevant technical papers, product samples, videotapes, slides or other ancillary items will not be considered during the review process.

3.4.3 Forms

All form submissions shall be done electronically, and do not count towards the 46-page limit.

3.4.3.1 Contact Information

A sample Contact Information form is provided in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. The offeror shall provide complete information for each contact person and submit the form as required in section 6.

Note: Contact Information, is public information and may be disclosed.

3.4.3.2 Proposal Certifications

A sample Proposal Certifications form is provided in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. The offeror shall provide complete information for each item and submit the form as required in section 6.

3.4.3.3 Proposal Summary

A sample Proposal Summary form is provided in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. The offeror shall provide complete information for each item and submit the form as required in section 6.

Note: The Proposal Summary, including the Technical Abstract, is public information and may be disclosed. Do not include proprietary information in this form.

3.4.3.4 Proposal Budget

A sample of the Proposal Budget form is provided in the NASA SBIR/STTR Firms Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. The offeror shall complete the Budget Summary following the instructions provided with the sample form. The total requested funding for the Phase II effort shall not exceed $750,000. Contextual help is provided on the electronic budget form for additional explanation. Information shall be submitted to explain the offeror's plans for use of the requested funds to enable NASA to determine whether the proposed price is fair and reasonable.

Note: The government is not responsible for any monies expended by the firm before award of any contract.

In addition, the following additional uploads, must be submitted in the Proposal Budget form, as applicable:

Proposal Requirements for Use of Federal Services, Facilities or Equipment:
In cases where an offeror seeks to use NASA or another federal department or agency services, equipment or facilities, the offeror shall provide the following:

1. Statement, signed by the appropriate government official at the effected federal department or agency, verifying that the resources should be available during the proposed period of performance. Offerors must upload this letter in the Proposal Budget form.
2. Signed letter on company letterhead from the contractor's Small Business Official explaining why the SBIR/STTR research project requires the use of federal services, equipment or facilities, including data that verifies the absence of non-federal facilities or personnel capable of supporting the research effort, a statement confirming that the facility proposed is not a federal laboratory, if applicable, and the associated cost estimate. Offerors must upload this letter in the Proposal Budget form.

See Part 8 of the Technical Proposal for additional information on use of federal Facilities.

Use of Subcontractors and Consultants:

Subject to the restrictions set forth below, the SBC may establish business arrangements with other entities or individuals to participate in performance of the proposed R/R&D effort.

Note:

1. ***Offerors should list consultants by name and specify, for each, the number of hours and hourly costs.***

2. ***Breakdown of subcontractor budget should mirror the SBC's own breakdown in the Proposal Budget form and include breakdowns of direct labor, other direct costs, profit, as well as indirect rate agreements.***
3. ***A signed letter of commitment is required for each subcontractor and/or consultant. For educational institutions, the letter must be from the institution's office of sponsored programs.***

Subcontractors' and consultants' work has the same place of performance restrictions as stated in section 1.5.2.

The following restrictions apply to the use of subcontracts/consultants and the formula below must be used in preparing budgets with subcontractors/consultants:

SBIR Phase II Subcontracts/Consultants	STTR Phase II Subcontracts/Consultants
The proposed subcontracted business arrangements including consultants, must not exceed 50 percent of the research and/or analytical work [as determined by the total cost of the proposed subcontracting effort (to include the appropriate OH and G&A) in comparison to the total effort (total contract price including cost sharing, if any, less profit if any)].	A minimum of 40 percent of the research or analytical work must be performed by the proposing SBC and minimum of 30 percent must be performed by the RI. Any subcontracted business effort other than that performed by the RI, shall not exceed 30 percent of the research and/or analytical work [(as determined by the total cost of the subcontracting effort (to include the appropriate OH and G&A) in comparison to the total effort (total contract price including cost sharing, if any, less profit if any)].

Example:	Total price to include profit	- $725,000
	Profit	- $21,750
	Total price less profit	- $725,000 - $21,750 = $703,250
	Subcontractor cost	- $250,000
	G&A	- 5%
	G&A on subcontractor cost	- $250,000 x 5% = $12,500
	Subcontractor cost plus G&A	- $250,000 + $12,500 = $262,500
	Percentage of subcontracting effort – subcontractor cost plus G&A / total price less profit	**- $262,500 / $703,250 = 37.3%**

For an SBIR Phase II this is acceptable since it is below the limitation of 50%.

For an STTR Phase II, where there is a subcontract with a company other than the RI, this is unacceptable since it is above 30% limitation.

See Part 9 of the Technical Proposal for additional information on the use of Subcontractors and Consultants.

Milestone Plan

For Phase II, offerors shall submit a proposed quarterly milestone plan with the Proposal Budget form. The milestone plan shall be in accordance with the proposed work plan outlining the work to be accomplished each quarter and the cost proposed associated with each of the quarterly milestones. The cost breakdown shall be similar to the Proposal Budget form for each of the proposed quarterly milestones (i.e., each milestone should include the labor, supplies, travel and profit associated with those tasks to be accomplished that quarter). The proposed cost associated with each quarterly milestone must be realistic for the work to be accomplished but is not required to be equally distributed across each quarter.

3.4.4 Technical Proposal

This part of the submission shall not contain any budget data and must consist of all ten (10) parts listed below in the given order. All ten parts of the technical proposal must be numbered and titled. Parts that are not applicable must be included and marked "Not Applicable." A proposal omitting any part will be considered non-responsive to this Solicitation and may be rejected during administrative screening. The required table of contents is provided below:

Phase II Table of Contents

Part 1: Table of Contents…….……Page 1
Part 2: Identification and Significance of the Innovation and Results of the Phase I Proposal
Part 3: Technical Objectives
Part 4: Work Plan
Part 5: Related R/R&D
Part 6: Key Personnel
Part 7: Phase III Efforts, Commercialization and Business Planning
Part 8: Facilities/Equipment
Part 9: Subcontracts and Consultants
Part 10: Related, Essentially Equivalent and Duplicate Proposals and Awards

Part 1: Table of Contents

The technical proposal shall begin with a brief table of contents indicating the page numbers of each of the parts of the proposal.

Part 2: Identification and Significance of the Innovation and Results of the Phase I Award

Please provide a summary of your Phase I results and building on those results succinctly describe the Phase II proposed work including:

- The proposed innovation.
- The relevance and significance of the proposed innovation to an interest, need or needs within the subtopic.
- The proposed innovation relative to the state of the market, the state of the art and its feasibility.

Please be advised that the evaluators may review the Phase I final technical report to verify accuracy of this summary. However proposers shouldn't rely on this and should include relevant results in the Phase II proposal.

Part 3: Technical Objectives

Define the specific objectives of the Phase II research and technical approach, including the technical questions posed in the subtopic description that must be answered to determine the feasibility of the proposed innovation.

Note: All offerors submitting proposals who are planning to use NASA IP must describe their planned developments with the IP. The NASA Evaluation License Application should be added as an attachment under Proposal Certifications (see section 1.6).

Part 4: Work Plan

Include a detailed description of the Phase II R/R&D plan to meet the technical objectives. The plan should indicate what will be done, where it will be done and how the R/R&D will be carried out. Discuss in detail the methods planned to achieve each task or objective. Task descriptions, schedules, resource allocations, estimated task hours for each key personnel and planned accomplishments including project milestones shall be included. Offerors should ensure that

the estimated task hours provided in the work plan for key personnel are consistent with the hours reported in the Proposal Budget form. If the offeror is a joint venture or limited partnership, a statement of how the workload will be distributed, managed and charged should be included in the proposal.

STTR: In addition, the work plan will specifically address the percentage and type of work to be performed by the SBC and the RI. The plan will provide evidence that the SBC will exercise management direction and control of the performance of the STTR effort, including situations in which the PI may be an employee of the RI.

Part 5: Related R/R&D

Describe significant current and/or previous R/R&D that is directly related to the proposal including any conducted by the PI or by the offeror. Describe how it relates to the proposed effort and any planned coordination with outside sources. The offeror must persuade reviewers of his or her awareness of key recent R/R&D conducted by others in the specific subject area.

Part 6: Key Personnel and Bibliography of Directly Related Work

Identify all key personnel involved in Phase II activities whose expertise and functions are essential to the success of the project. Provide bibliographic information including directly related education and experience.

The PI is considered key to the success of the effort and must make a substantial commitment to the project. The following requirements are applicable:

Functions: The functions of the PI are: planning and directing the project, leading it technically and making substantial personal contributions during its implementation, serving as the primary contact with NASA on the project and ensuring that the work proceeds according to contract agreements. Competent management of PI functions is essential to project success. The Phase II proposal shall describe the nature of the PI's activities and the amount of time that the PI will personally apply to the project. The amount of time the PI proposes to spend on the project must be acceptable to the Contracting Officer.

Qualifications: The qualifications and capabilities of the proposed PI and the basis for PI selection are to be clearly presented in the proposal. NASA has the sole right to accept or reject a PI based on factors such as education, experience, demonstrated ability and competence, and any other evidence related to the specific assignment.

Eligibility: This part shall also establish and confirm the eligibility of the PI, and indicate the extent to which other proposals recently submitted or planned for submission in the year and existing projects commit the time of the PI concurrently with this proposed activity. Any attempt to circumvent the restriction on PIs working more than half time for an academic or a nonprofit organization by substituting an ineligible PI will result in rejection of the proposal. However, for an STTR the PI can be primarily employed by either the SBC or the RI. Please see section 1.5.3 for further explanation.

Note: If the Phase II PI is different than that proposed under the Phase I, please provide rationale for the change.

Part 7: Phase III Efforts, Commercialization and Business Planning

Present a plan for commercialization of the proposed innovation. Commercialization encompasses the transition of technology into products and services for NASA mission programs, other government agencies and non-government markets.

This part should complement, through narrative, the data provided by the applicant in the Commercialization Metrics Survey (CMS), including financial information. The CMS data is intended to support the company's claims about their ability to achieve the proposed innovation's commercialization for firms that have previously received SBIR/STTR awards, and provide a level of confidence regarding the SBC's future viability.

Note: Letters of funding support commitments should be submitted as part of the Capital Commitments Addendum and will be considered only under Factor 4 - Commercial Potential and Feasibility (section 4).

The commercialization plan should address the following areas:

Commercial Potential – Market
Describe (1) market segmentation and analysis, by providing the scope in dollars, and describe the commercial Total Addressable Market (TAM) that is appropriate to the proposed innovation, the segment in dollars of the TAM addressable by the proposed innovation and the projected percentage of the offeror's market share in 2-3 years after entry into the identified market; (2) the proposed innovation in terms of target customers (e.g., NASA, other federal agency, commercial enterprise); and (3) the competitive landscape, by identifying potential competitors by company name within the identified market, discussing the barriers to entry and how many years it would take a competitor to enter this segment in terms of capitalization, technology, and people, and describing how the proposed innovation is different from the potential competition.

Commercial Intent – Plan
Describe (1) the commercial development plan by providing a development timeline to bring technology to market, discuss consultants, incubators and research institutions to achieve the plan; (2) the applicable business model (spin-out, license, OEM) the offeror would use to bring the innovation to market, the channels of distribution (direct sales, distributors, etc.) that would be used in bringing the innovation into the identified market, the pro-forma 2-3 year revenue dollar projections based on the proposed innovation's penetration of the identified market and any follow-on development (long term > 5 years) plans to expand your proposed innovation's market presence; (3) the risks to the commercial development plan and what mitigations, if any, can be taken over a reasonable period of time to lessen the risks and (4) Intellectual Property protection methods, plans or processes within your company.

Commercial Capability – Execution
Describe (1) the current and future company capitalization by discussing the technical, operations/manufacturing and business staff conducting the project; the physical plant, including facilities and the capital equipment, tooling and test equipment used to conduct the investigation; how the innovation will enter into production (i.e., in house or through a licensee) and what changes (if any) will be made to company capitalization for commercialization; (2) existing and future business relationships in terms of any formal Partnerships, Joint Ventures, Licensing Agreements with other companies/organizations and; (3) as applicable, the approach, path to market and revenues from past commercialization(s) resulting from SBIR/STTR awards disclosed in the CMS.

Part 8: Facilities/Equipment
Offerors must describe the necessary instrumentation and facilities to be used to perform the proposed work. Offerors must ensure the resources are adequate and address any reliance on external sources, such as government furnished equipment or facilities. In cases where an offeror seeks to use NASA or another federal department or agency services, equipment or facilities, the offeror shall describe in this part why the use of government furnished equipment or facilities is necessary. See section 3.4.3.4 and 5.13 for additional requirements when proposing use of federal facilities.

Part 9: Subcontracts and Consultants

The offeror must describe all subcontracting or other business arrangements, and identify the relevant organizations and/or individuals with whom arrangements are planned. The expertise to be provided by the entities must be described in detail, as well as the functions, services, number of hours and labor rates. Offerors are responsible for ensuring that all organizations and individuals proposed to be utilized are actually available for the time periods proposed. Subcontract costs shall be documented in the Subcontractors/Consultants section of the Proposal Budget form and supporting documentation should be uploaded for each (appropriate documentation is specified in the form). The narrative description of subcontracts and consultants in the technical proposal should support the proposed approach and documentation in the Proposal Budget form.

Part 10: Related, Essentially Equivalent, and Duplicate Proposals and Awards

WARNING – While it is permissible with proposal notification to submit identical proposals or proposals containing a significant amount of essentially equivalent work for consideration under numerous federal program solicitations, it is unlawful to enter into funding agreements requiring essentially equivalent work.

If an applicant elects to submit identical proposals or proposals containing a significant amount of essentially equivalent work under other federal program solicitations, a statement must be included in each such proposal indicating:

1. The name and address of the agencies to which proposals were submitted or from which awards were received.
2. Date of proposal submission or date of award.
3. Title, number and date of solicitations under which proposals were submitted or awards received.
4. The specific applicable research topics for each proposal submitted for award received.
5. Titles of research projects.
6. Name and title of principal investigator or project manager for each proposal submitted or award received.

Offerors are at risk for submitting essentially equivalent proposals and therefore, are strongly encouraged to disclose these issues to the soliciting agency to resolve the matter prior to award.

A summary of essentially equivalent work information, as well as related research and development on proposals and awards is also required on the Proposal Certifications form (if applicable).

3.4.5 Research Agreement (Applicable for STTR proposals only)

STTR: The Research Agreement (different from the Allocation of Rights Agreement, see: http://sbir.nasa.gov/content/nasa-sbirsttr-program-definitions) is a single-page document electronically submitted and endorsed by the SBC and RI. A model agreement is provided, or firms can create their own custom agreement. The Research Agreement should be submitted as required in section 6.

All STTR Phase I proposals must provide sufficient information to convince NASA that the proposed SBC/RI cooperative effort represents a sound approach for converting technical information resident at the RI into a product or service that meets a need described in a Solicitation research topic.

3.4.6 Capital Commitments Addendum Supporting Phase II and Phase III

Offerors for Phase II contracts are strongly urged to obtain non-SBIR/STTR funding support commitments for follow-on Phase III activities along with additional support of their Phase II effort from viable sources outside of the SBC. In order to be considered valid, the funding support commitments must show that a specific and substantial amount

(≥ $50,000) will be made available to the SBC to provide capital for the Phase III activities and (if applicable) support of their Phase II effort. They must indicate the source, date, and conditions and contingencies under which the funds will be made available. If Phase III activities will be funded internally by the SBC, the offeror should ensure that the financials provided in the Commercial Metric Survey (CMS) are current and accurate.

Evidence of funding support commitments from outside parties must be provided in writing and shall accompany the Phase II proposal. Letters of commitment will specify the level of funding commitments, other sources to be provided and any funding contingencies/conditions, in addition to timing of the funding being made available to the SBC. Expressions of technical interest in the Phase II research or of potential future financial support are not sufficient and will not be considered as capital commitments to the Phase II proposal and Phase III activities. Letters of commitment must be signed by a duly authorized representative of the outside funding source and SBC with the authority to obligate funding.

3.4.7 Briefing Chart

A one-page briefing chart is required to assist in the ranking and advocacy of proposals prior to selection and contains the following sections:

- Identification and Significance of Innovation
- Technical Objectives
- Proposed Deliverables
- NASA Applications
- Non-NASA Applications
- Graphic

The briefing chart shall not contain any proprietary data or ITAR restricted data. An electronic form will be provided during the submissions process.

Note: The briefing chart is public information and may be disclosed. Do not include proprietary information in this form.

3.4.8 Firm Certifications

Firm certifications that are applicable across all proposal submissions submitted to this solicitation must be completed via the Firm Certifications section of the Proposal Submissions Electronic Handbook. The offeror shall answer Yes or No as applicable. An example of the certification can be found in the NASA SBIR/STTR Firm Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html.

Note: The designated firm admin, typically the first person to register your firm, is the only individual authorized to update the certifications.

3.4.9 Audit Information

Although firms are not required to have an approved accounting system, knowledge that a firm has an approved accounting system facilitates NASA's determination that rates are fair and reasonable. To assist NASA, the SBC shall complete the questions in the Audit Information form regarding the firm's rates and upload the federal agency audit report or related information that is available from the last audit. There is a separate "Audit Information" section in the Proposal Budget form that shall also be completed. If your firm has never been audited by a federal agency, then

answer "No" to the first question and you do not need to complete the remainder of the form. An electronic form will be provided during the submissions process.

The Contracting Officer will use this Audit Information to assist with negotiations if the proposal is selected for award. The Contracting Officer will advise offerors what is required to determine reasonable cost and/or rates in the event the Audit Information is not adequate to support the necessary determination on rates.

Note: The designated firm admin, typically the first person to register your firm, is the only individual authorized to update the audit information.

3.4.10 Prior Awards Addendum

If the SBC has received more than 15 Phase II awards in the prior 5 fiscal years, submit name of awarding agency, solicitation year, phase, date of award, funding agreement/contract number, and topic or subtopic title for each Phase II. If your firm has received any SBIR or STTR Phase II awards, even if it has received fewer than 15 in the last 5 years, it is still recommended that you complete this form for those Phase II awards your firm did receive. This information will be useful when completing the Commercialization Metrics Survey, and in tracking the overall success of the SBIR and STTR Programs. Any NASA Phase II awards your firm has received will be automatically populated in the electronic form, as are any Phase II awards previously entered by the SBC during prior submissions (you may update the information for these awards). An electronic form will be provided during the submissions process.

Note: The designated firm admin, typically the first person to register your firm, is the only individual authorized to update the addendum information.

3.4.11 Commercial Metrics Survey (CMS)

NASA has instituted a comprehensive commercialization survey/data gathering process for firms with prior NASA SBIR/STTR awards. If the SBC has received any Phase III awards resulting from work on any NASA SBIR or STTR awards, provide the related Phase I or Phase II contract number, name of Phase III awarding agency, date of award, funding agreement number, amount, project title and period of performance. The survey will also ask for firm financial, sales and ownership information, as well as any commercialization success the firm has had as a result of SBIR or STTR awards. This information must be updated annually during proposal submission via the EHB. This information will allow firms to demonstrate their ability to carry SBIR/STTR research through to achieve commercial success, and allow agencies to track the overall commercialization success of their SBIR and STTR Programs. The survey should be limited to information requested above. An electronic form will be provided during the submissions process.

Note: Information received from SBIR/STTR awardees completing the survey is kept confidential, and will not be made public except in broad aggregate, with no firm-specific attribution.

The Commercialization Metrics Survey is a required part of the proposal submissions process and must be completed via the Proposal Submissions Electronic Handbook. Also, Companies with no SBIR/STTR awards or only fairly recent SBIR/STTR awards will not be penalized under past performance for the lack of past SBIR/STTR commercialization.

3.4.12 Contractor Responsibility Information

No later than 10 business days after the notification of selection for negotiations, the offeror shall provide a signed statement from your financial institution(s), on its letterhead, stating whether or not your firm is in good standing and how long you have been with the institution.

3.4.13 Allocation of Rights Agreement (STTR awards only)

No more than 10 business days after the notification of selection for negotiation, the offeror should provide to the Contracting Officer, a completed **Allocation of Rights Agreement (ARA)**, which has been signed by authorized representatives of the SBC, RI, and subcontractors and consultants, as applicable. The ARA shall state the allocation of intellectual property rights with respect to the proposed STTR activity and planned follow-on research, development and/or commercialization. A sample ARA is available in the NASA SBIR/STTR Firm Library http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html of this Solicitation.

If the ARA form is completed and available at the time of submission, offers should upload it in the Proposal Budget form, which will help to expedite contract negotiations.

4. Method of Selection and Evaluation Criteria

4.1. Access to Proprietary Data by Non-NASA Personnel

4.1.1 Non-NASA Reviewers

In addition to government personnel, NASA, at its discretion and in accordance with 1815.207-71 of the NASA FAR Supplement, may utilize individuals from outside the government with highly specialized expertise not found in the government in the proposal review process. Any decision to obtain an outside evaluation shall take into consideration requirements for the avoidance of organizational or personal conflicts of interest and the competitive relationship, if any, between the prospective contractor or subcontractor(s) and the prospective outside evaluator. Any such evaluation will be under agreement with the evaluator that the information (data) contained in the proposal will be used only for evaluation purposes and will not be further disclosed. Such requests for non-NASA Reviewers must be approved by the NASA SBIR/STTR Program Manager.

4.1.2 Non-NASA Access to Confidential Business Information

In the conduct of proposal processing and potential contract administration, the agency may find it necessary to provide proposal access to other NASA contractor and subcontractor personnel. NASA will provide access to such data only under contracts that contain an appropriate NFS 1852.237-72 Access to Sensitive Information clause that requires the contractors to fully protect the information from unauthorized use or disclosure.

4.2 Phase I Proposals

All proposals will be evaluated and ranked on a competitive basis. Proposals will be initially screened to determine responsiveness. Proposals determined to be responsive to the administrative requirements of this Solicitation and having a reasonable potential of addressing a NASA interest, as evidenced by the technical abstract included in the Proposal Summary form, will be technically evaluated by NASA personnel to determine the most promising technical and scientific approaches. Each proposal will be reviewed on its own merit. NASA is under no obligation to fund any proposal or any specific number of proposals in a given subtopic. It also may elect to fund several or none of the proposed approaches to the same subtopic.

4.2.1 Evaluation Process

Proposals shall provide all information needed for complete evaluation. Evaluators will not seek additional information. NASA scientists and engineers will perform evaluations. Also, qualified experts outside of NASA (including industry, academia and other government agencies) may assist in performing evaluations as required to determine or verify the merit of a proposal. Offerors should not assume that evaluators are acquainted with the firm, key individuals, or with any experiments or other information. Any pertinent references or publications should be noted in Part 5 of the technical proposal.

4.2.2 Phase I Evaluation Criteria

NASA intends to select for award those proposals that offer the most advantageous research and development, deliver technological innovation that contribute to NASA's missions, provide societal benefit and grow the US economy. NASA will give primary consideration to the scientific and technical merit and feasibility of the proposal and its benefit to NASA interests. Each proposal will be evaluated and scored on its own merits using the factors described below:

Factor 1: Scientific/Technical Merit and Feasibility
The proposed R/R&D effort will be evaluated on whether it offers a clearly innovative and feasible technical approach to the described NASA problem area. Proposals must clearly demonstrate relevance to the subtopic as well as one or more NASA mission and/or programmatic needs. Specific objectives, approaches and plans for developing and verifying the innovation must demonstrate a clear understanding of the problem and the current state of the art. The degree of understanding and significance of the risks involved in the proposed innovation must be presented.

Factor 2: Experience, Qualifications and Facilities
The technical capabilities and experience of the PI, project manager, key personnel, staff, consultants and subcontractors, if any, are evaluated for consistency with the research effort and their degree of commitment and availability. The necessary instrumentation or facilities required must be shown to be adequate and any reliance on external sources, such as government furnished equipment or facilities, addressed (section 3.3.4, part 8).

Factor 3: Effectiveness of the Proposed Work Plan
The work plan will be reviewed for its comprehensiveness, effective use of available resources, labor distribution and the proposed schedule for meeting the Phase I objectives. The methods planned to achieve each objective or task should be discussed in detail. Please see Factor 5 for price evaluation criteria.

STTR: The clear delineation of responsibilities of the SBC and RI for the success of the proposed cooperative R/R&D effort will be evaluated. The offeror must demonstrate the ability to organize for effective conversion of intellectual property into products and services of value to NASA and the commercial marketplace.

Factor 4: Commercial Potential and Feasibility
The proposal will be evaluated for the commercial potential and feasibility of the proposed innovation and associated products and services for NASA mission programs, other government agencies and non-government markets. The offeror's experience and record in technology commercialization, co-funding commitments from private or non-SBIR/non-STTR funding sources, existing and projected commitments for Phase III funding, investment, sales, licensing, and other indicators of commercial potential and feasibility will be considered along with the initial commercialization strategy for the innovation.

Factor 5: Price Reasonableness
The offeror's cost proposal will be evaluated for price reasonableness based on the information provided in the Proposal Budget form. NASA will comply with the FAR and NASA FAR Supplement (NFS) to evaluate the proposed price/cost to be fair and reasonable.

The Contracting Officer shall submit a recommendation for award to the Source Selection Official after completion of evaluation for price reasonableness and determination of responsibility.

Scoring of Factors and Weighting
Factors 1, 2 and 3 will be scored numerically with Factor 1 worth 50 percent and Factors 2 and 3 each worth 25 percent. The sum of the scores for Factors 1, 2 and 3 will comprise the Technical Merit score. The evaluation for Factor 4, Commercial Potential and Feasibility, will be in the form of an adjectival rating (Excellent, Very Good, Good, Fair, Poor). For Phase I proposals, Technical Merit is more important than Commercial Merit. Factors 1 - 4 will be evaluated and used in the selection of proposals for negotiation. Factor 5 will be evaluated and used in the award decision, i.e., NASA will only make award when the price is fair and reasonable.

4.2.3 Prioritization

In prioritizing proposals recommended for negotiations NASA will also consider other factors including recommendations from the centers and mission directorates regarding such things as overall NASA priorities, program balance, and available funding.

4.2.4 Selection

Proposals recommended for negotiations will be forwarded to the Program Management Office for analysis and presented to the Source Selection Official and Mission Directorate Representatives. The Source Selection Official has the final authority for choosing the specific proposals for contract negotiation. Each proposal selected for negotiation will be evaluated for cost/price reasonableness, the terms and conditions of the contract will be negotiated and a responsibility determination made. The Contracting Officer will advise the Source Selection Official on matters pertaining to cost reasonableness, responsibility and known past performance issues.

The list of proposals selected for negotiation will be posted on the NASA SBIR/STTR website (http://sbir.nasa.gov). All firms will receive a formal notification letter. A Contracting Officer will negotiate an appropriate contract to be signed by both parties before work begins.

4.3 I-Corps

For awardees invited to submit an I-Corps proposal pursuant to section 3.3.6.2, NASA will provide a programmatic assessment of firms and their technologies to include:

- Number of previous SBIR/STTR awards received by the firm and the firm's commercialization success rate.
- Potential for commercialization of the selected Phase I research/solution to non-NASA markets (distinct from integration/transition into NASA programs).
- Technical relevance to NASA.

Based on these assessments, STTR offerors will be selected to participate in phone interviews conducted by the NASA SBIR/STTR PMO and the NSF-provided I-Corps instructors. NASA will use these interviews to determine the dynamics of the teams and gauge their level of commitment to meeting requirements for I-Corps to make the final selection. NASA will make the final selections for I-Corps based upon its initial assessments of the I-Corps proposals and the assessments of the phone interviews.

NASA anticipates a total of approximately 42 SBIR/STTR firms will be selected for participation in the I-Corps program for Phase I.

4.4 Phase II Proposals

All Phase II proposals will be evaluated and ranked on a competitive basis. Proposals will be initially screened to determine responsiveness. Proposals determined to be responsive to the administrative requirements of this solicitation and having a reasonable potential of meeting a NASA need, as evidenced by the technical abstract included in the Proposal Summary form, will be technically evaluated by NASA personnel to determine the most promising technical and scientific approaches. Each proposal will be reviewed on its own merit. NASA is under no obligation to fund any proposal or any specific number of proposals in a given subtopic. It also may elect to fund several or none of the proposed approaches to the same subtopic.

4.4.1 Evaluation Process

The Phase II evaluation process is similar to the Phase I process. Each proposal will be reviewed by NASA scientists and engineers and by qualified experts outside of NASA as needed. In addition, the proposals will be reviewed for commercial merit. NASA may use a peer review panel to evaluate commercial merit. Panel membership may include non-NASA personnel with expertise in business development and technology commercialization.

4.4.2 Phase II Evaluation Criteria

NASA intends to select for award those proposals that offer the most advantageous research and development, deliver technological innovation that contributes to NASA's missions, provides societal benefit and grows the US economy. NASA will give primary consideration to the scientific and technical merit and feasibility of the proposal and its benefit to NASA interests. Each proposal will be evaluated and scored on its own merits using the factors described below:

Note: Past performance will not be a separate evaluation factor but will be evaluated under Factors 1 (with respect to performance in Phase I) and 4 (with respect to commercialization past performance, as applicable) below.

Factor 1: Scientific/Technical Merit and Feasibility
The proposed R/R&D effort will be evaluated on its originality, the feasibility of the innovation and potential technical value. In addition, past performance of Phase I will be evaluated to determine the degree to which Phase I objectives were met, and whether the Phase I results indicate a Phase II project is appropriate. The evaluators may review the Phase I final technical report to verify the Phase I results.

Factor 2: Experience, Qualifications and Facilities
The technical capabilities and experience of the PI or project manager, key personnel, staff, consultants and subcontractors, if any, are evaluated for consistency with the research effort and their degree of commitment and availability. The necessary instrumentation or facilities required must show to be adequate and any reliance on external sources, such as government furnished equipment or facilities, addressed (section 3.4.4, Part 8).

Factor 3: Effectiveness of the Proposed Work Plan
The work plan will be reviewed for its comprehensiveness, effective use of available resources, labor distribution and the proposed schedule for meeting the Phase II objectives. The methods planned to achieve each objective or task should be discussed in detail. The proposed path beyond Phase II for further development and infusion into a NASA mission or program will also be reviewed. Please see Factor 5 for price evaluation criteria.

STTR: The clear delineation of responsibilities of the SBC and RI for the success of the proposed cooperative R/R&D effort will be evaluated. The offeror must demonstrate the ability to organize for effective conversion of intellectual property into products and services of value to NASA and the commercial marketplace.

Factor 4: Commercial Potential and Feasibility
The proposal and Commercial Metrics Survey will be evaluated for the commercial potential and feasibility of the proposed innovation and associated products and services. The offeror's experience and record in technology commercialization; current funding commitments from private or non-SBIR funding sources; existing and projected commitments for Phase III funding, investment, sales, licensing, and other indicators of commercial potential and feasibility will be considered along with the commercialization plan for the innovation. Evaluation of the commercialization plan and the overall proposal will include consideration of the following areas:

1. **Commercial Potential and Feasibility of the Innovation:** This includes assessment of (a) the transition of the innovation into a well-defined product or service; (b) a realistic target market niche; (c) a product or service that has strong potential for meeting a well-defined need within the target market and (d) a commitment of necessary financial, physical and/or personnel resources.
2. **Intent and Commitment of the Offeror:** This includes assessing the commercialization of the innovation for (a) importance to the offeror's current business and strategic planning; (b) reliance on (or lack thereof) government markets and (c) adequacy of funding sources necessary to bring technology to identified market.
3. **Capability of the Offeror to Realize Commercialization:** This includes assessment of (a) the offeror's past performance, experience and success in technology commercialization; (b) the likelihood that the offeror will be able to obtain the remaining necessary financial, technical and personnel-related resources; and (c) the current strength and continued financial viability of the offeror.

Factor 5: Price Reasonableness

The offeror's cost proposal will be evaluated for price reasonableness based on the information provided in the Proposal Budget form. NASA will comply with the FAR and NASA FAR Supplement (NFS) to evaluate the proposed price/cost to be fair and reasonable.

The Contracting Officer shall submit a recommendation for award to the Source Selection Official after completion of evaluation for price reasonableness and determination of responsibility.

Scoring of Factors and Weighting

Factors 1, 2, 3 and 4 will be scored numerically with Factor 1 worth 45 percent, Factors 2 and 3 each worth 25 percent and Factor 4 worth five percent. The sum of the scores for Factors 1, 2, 3 and 4 will comprise the Technical Merit score. Factors 1 - 4 will be evaluated and used in the selection of proposals for negotiation. Factor 5 will be evaluated as part of the award decision, i.e., NASA will only make award when the price is fair and reasonable.

4.4.3 Prioritization

In prioritizing proposals recommended for negotiations NASA will also consider other factors including recommendations from the centers and mission directorates regarding such things as overall NASA priorities, program balance, and available funding.

4.4.4 Selection

Proposals recommended for negotiations will be forwarded to the Program Management Office for analysis and presented to the Source Selection Official and Mission Directorate Representatives. The Source Selection Official has the final authority for choosing the specific proposals for contract negotiation. Each proposal selected for negotiation will be evaluated for cost/price reasonableness. After completion of evaluation for cost/price reasonableness and a determination of responsibility the Contracting Officer will submit a recommendation for award to the Source Selection Official.

The list of proposals selected for negotiation will be posted on the NASA SBIR/STTR website (http://sbir.nasa.gov). All firms will receive a formal notification letter. A Contracting Officer will negotiate an appropriate contract to be signed by both parties before work begins.

4.5 Debriefing of Unsuccessful Offerors

After Phase I and Phase II selections for negotiation have been announced, debriefings will be available to the offeror's corporate official or designee via email. Written debriefings will be sent only to the Business Official designated in the proposal. Telephone requests for debriefings will not be accepted. Debriefings are not opportunities to reopen selection decisions. Debriefings will not disclose the identity of the proposal evaluators, proposal scores, the content of, or comparisons with other proposals. The debriefing process for Phase I and Phase II proposals are described below.

4.5.1 Phase I Debriefings

Debriefings will be automatically emailed to the designated Business Official within 60 days of the announcement of selection for negotiation. If you have not received your debriefing by this time, contact the SBIR/STTR Program Support Office at ARC-SBIR-PMO@mail.nasa.gov.

4.5.2 Phase II Debriefings

For Phase II, offerors must send a debriefing request via email to the SBIR/STTR Program Office at ARC-SBIR-PMO@mail.nasa.gov within 60 days after the selection announcement. Late requests will not be honored.

5. Considerations

5.1 Requirement for Contracting

To simplify making contract awards and to reduce processing time, all contractors selected for Phase I and Phase II contracts shall ensure that:

1. All information in your proposal is current, e.g., your address has not changed, the proposed PI is the same, etc. If changes have occurred since submittal of your proposal, notify the Contracting Officer immediately.
2. Your firm is registered with System for Award Management (SAM) (section 2.2).
3. Your firm is in compliance with the VETS-4212 requirement (section 2.3.1). Confirmation of that the report has been submitted to the Department of Labor is current shall be provided to the Contracting Officer within 10 business days of the notification of selection for negotiation.
4. Your firm HAS NOT proposed a Co-Principal Investigator.
5. STTR selectees should provide a copy of their executed Allocation of Rights Agreement to the Contracting Officer within 10 business days of receiving notification of selection for negotiation.
6. Your firm is required to provide timely responses to all communications from the NSSC Contracting Officer.
7. All proposed cost is supported with documentation such as a quote, previous purchase order, published price lists, etc. All letters of commitment are dated and signed by the appropriate person. If a University is proposed as a subcontractor or a RI, the signed letter shall be on the University letterhead from the Office of Sponsored Programs. If an independent consultant is proposed, the signed letter should not be on a University letterhead. If the use of government facility or equipment is proposed, your firm shall submit a signed letter from the government facility stating the availability, cost if any, and authorizing the use of it, and a signed letter from your firm justifying the need to use the facility.

From the time of proposal notification of selection for negotiation, until the award of a contract, all communications shall be submitted electronically to NSSC-SBIR-STTR@nasa.gov.

Note: Costs incurred prior to and in anticipation of award of a contract are entirely the risk of the contractor in the event that a contract is not subsequently awarded. A notification of selection for negotiation is not to be misconstrued as an award notification to commence work.

5.2 Awards

5.2.1 Model Contracts

An example of the Phase I and II contracts can be found in the NASA SBIR/STTR Firm Library: http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html.

Note: Model contracts are subject to change.

5.3 Reporting and Required Deliverables

An IT Security Management Plan is required at the beginning of the contract. Contractors interested in doing business with NASA and/or providing IT services or solutions to NASA should use the list found at the website of the Office of the Chief Information Officer (OCIO) as a reference for information security requirements - https://www.nasa.gov/content/security-requirements-policies. An example of the IT Security Management Plan can be found in the NASA SBIR/STTR Firm Library: http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html. For more information see NASA FAR Supplement clause 1852.204-76

All contracts shall require the delivery of technical reports that present: (1) the work and results accomplished; (2) the scientific, technical and commercial merit and feasibility of the proposed innovation, and project results; (3) its relevance and significance to one or more NASA interests (section 9); and (4) the strategy for development, transition of the proposed innovation, and project results into products and services for NASA mission programs and other potential customers. Deliverables may also include the demonstration of the proposed innovation and/or the delivery of a prototype or test unit, product or service for NASA testing and utilization.

The technical reports and other deliverables are required as described in the contract and are to be provided to NASA. These reports shall document progress made on the project and activities required for completion. Periodic certification for payment will be required as stated in the contract. A final report must be submitted to NASA upon completion of the Phase I or Phase II R/R&D effort in accordance with applicable contract provisions.

A final New Technology Summary Report (NTSR) is due at the end of the contract, and New Technology Report(s) (NTR) are required if technology(s) are developed under the award, prior to submission of the final invoice. For additional information on NTSR and NTR requirements and definitions see section 1.12 and 5.8.

Report deliverables shall be submitted electronically via the EHB. For any reports that require an upload NASA requests the submission in PDF or MS Word format.

Note: To access contract management in the EHB you will be required to have an identity in the NASA Account Management System (NAMS). This is the agency's centralized system for requesting and maintaining accounts for NASA IT systems and applications. The system contains user account information, access requests and account maintenance processes for NASA employees, contractors, and remote users such as educators and foreign users. A basic background check is required for this account. Instructions will be provided during contract negotiations.

5.4 Payment Schedule

All NASA SBIR and STTR contracts are firm-fixed-price contracts. The exact payment terms will be included in the contract.

Although invoices are submitted electronically through the Department of Treasury's Invoice Processing Platform (IPP), as a condition for payment, invoice certifications shall be completed in the EHB for each individual invoice. The certification is pre-set in the EHB and it shall be completed before uploading each invoice in IPP. Upon completion of the certification, a link to IPP is automatically provided in the EHB.

5.5 Profit or Fee

Contracts may include a reasonable profit. The reasonableness of proposed profit is determined by the Contracting Officer during contract negotiations. Reference FAR 15.404-4.

5.6 Rights in Data Developed Under SBIR Funding Agreements

The clause at FAR 52.227-20, Rights in Data—SBIR/STTR Program, governs rights to data used in, or first produced under, any Phase I, Phase II or federally funded SBIR Phase III contract.

Rights in technical data, including software developed under the terms of any funding agreement resulting from applications submitted in response to this solicitation, shall remain with the contractor, except that the government shall have the limited right to use such data for government purposes and shall not release such SBIR/STTR data outside the government without permission of the recipient for a period of not less than 4 years from delivery of the

last deliverable under that agreement (either Phase I, Phase II or federally funded SBIR Phase III). Agencies are released from obligation to protect SBIR data upon expiration of the protection period except that any such data that is also protected and referenced under a subsequent SBIR award must remain protected through the protection period of that subsequent SBIR award. However, effective at the conclusion of the 4-year period, the government shall have unlimited rights in any data delivered under the award.

5.7 Copyrights

The contractor may copyright and publish (consistent with appropriate national security considerations, if any) material developed with NASA support. NASA receives a royalty-free license for the federal government and requires that each publication contain an appropriate acknowledgment and disclaimer statement.

5.8 Invention Reporting, Election of Title, Patent Application Filing and Patents,

Consistent with the SBIR/STTR Program requirements, the Contractor shall complete their electronic submission of the New Technology Report (NTR), for any new subject inventions; and the New Technology Summary Report (NTSR) for the interim and final contract periods, via the SBIR/STTR EHB at http://sbir.nasa.gov under the Handbooks section.

NASA SBIR and STTR contracts will includes FAR 52.227-11 Patent Rights – Ownership by the Contractor and NFS 1852.227-11 Patent Rights – Ownership by the Contractor (APR 2015), which requires SBIR/STTR contractors to disclose all subject inventions to NASA within two (2) months of the inventor's report to the contractor. A subject invention means any invention of the Contractor made in the performance of work under this contract. Once the contractor discloses a subject invention, the contractor has up to 2 years to notify the government whether it elects to retain title to the subject invention. If the contractor elects to retain title, a patent application covering the subject invention must be initiated within the 1 year statutory period. If the contractor fails to do any of these within time specified periods, the government has the right to obtain title.

Small business concerns normally may elect to retain ownership of any subject invention. In such circumstances, the government shall have a nonexclusive, nontransferable, irrevocable, royalty-free license for federal government use. Further, the government reserves the right to require the patent holder to license others in certain circumstances and may require that anyone exclusively licensed to sell the invention in the United States must normally manufacture it domestically. To the extent authorized by 35 U.S.C. 205, the government will not make public any information disclosing a government-supported invention for a minimum 4-year period (that may be extended by subsequent SBIR funding agreements) to allow the contractor a reasonable time to pursue a patent.

5.9 1852.225-70 Export Licenses

The contractor shall comply with all U.S. export control laws including Export Administration Regulations (EAR) and International Traffic in Arms Regulations (ITAR). Offerors are responsible for ensuring that all employees who will work on this contract are eligible under export control laws, EAR and ITAR. Any employee who is not a U.S. citizen or a permanent resident may be restricted from working on this contract if the technology is restricted under export control laws, ITAR, or EAR unless the prior approval of the Department of State or the Department of Commerce is obtained via a technical assistance agreement or an export license. Violations of these regulations can result in criminal or civil penalties.

For additional information on ITAR, please visit the Code of Federal Regulations at https://www.ecfr.gov/cgi-bin/text-idx?tpl=/ecfrbrowse/Title22/22cfr120_main_02.tpl. For additional information on EAR, please visit https://www.bis.doc.gov/index.php/regulations/export-administration-regulations-ear. For additional training, refer to

http://sbir.gsfc.nasa.gov/content/training-resources. For additional assistance contact the NASA SBIR helpdesk at sbir@reisystems.com.

5.10 Government Furnished and Contractor Acquired Property

Title to property furnished by the government will remain with the government. A property loan agreement shall be executed which will include the requirements for return of the property.

Title to property acquired with government funds will be vested with NASA, unless it is determined that transfer of title to the contractor would be more cost effective than recovery of the equipment by NASA.

Note: Use of government facilities is covered in section 5.13 Use of Federal Services, Facilities or Equipment.

5.11 Essentially Equivalent Awards and Prior Work

If an award is made pursuant to a proposal submitted under either SBIR or STTR Solicitations, the firm will be required to certify with every invoice that it has not previously been paid nor is currently being paid for essentially equivalent work by any agency of the federal government. **Failure to report essentially equivalent or duplicate efforts can lead to the termination of contracts and/or civil or criminal penalties.**

5.12 Additional Information

5.12.1 Precedence of Contract over Solicitation

This Program Solicitation reflects current planning. If there is any inconsistency between the information contained herein and the terms of any resulting SBIR/STTR contract, the terms of the contract take precedence over the solicitation.

5.12.2 Evidence of Contractor Responsibility

In addition to the information required to be submitted for Phase II proposals as stated in in section 3.4.12, before award of an SBIR or STTR contract, the government may request the offeror to submit certain organizational, management, personnel and financial information to establish responsibility of the offeror. Contractor responsibility includes all resources required for contractor performance, i.e., financial capability, work force and facilities.

5.13 Use of Federal Services, Facilities or Equipment

Federal Departments and Agencies
Use of SBIR funding for unique federal/non-NASA services, equipment or facilities from a federal department or agency which does not meet the definition of a federal laboratory as defined in the SBA Policy Directive on the SBIR/STIR Program, requires a waiver from the SBA. Proposals requiring waivers must include an explanation of why the waiver is appropriate. NASA will provide the offeror's request, along with an explanation to SBA during the negotiation process. NASA cannot guarantee that a waiver can be obtained from SBA. Specific proposal instructions to request use of Federal Services, Facilities or Equipment are in section 3.3 of the solicitation.

Note: NASA Facilities qualify as Federal Laboratories.

Agreement to Use Any Federal Facility

All offerors selected for award that require the use of any federal facility shall, within twenty (20) business days of notification of selection for negotiations, provide to the NASA Shared Services Center Contracting Officer an agreement by and between the Contractor and the appropriate federal facility, executed by the government official authorized to approve such use. The Agreement must delineate the terms of use, associated costs, facility responsibilities and liabilities. Having a signed agreement for use of federal facilities is a requirement for award.

An executed SBIR/STTR Use Agreement, available in the Firm Library (http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html), is required before a contractor can use NASA services, facilities, or equipment. Offerors should not include an executed SBIR/STTR Use Agreement in the proposals. NASA expects selected offerors to execute the SBIR/STTR Use Agreement during their negotiations with NSSC. The information required in the proposals should facilitate executing the SBIR/STTR Use Agreement.

Contractor Responsibilities for Costs

In accordance with the Federal Acquisition Regulations (FAR) Part 45, it is NASA's policy not to provide services, equipment or facilities (resources) [capital equipment, tooling, test and computer facilities, etc.] for the performance of work under SBIR/STTR contracts. Generally, any contractor will furnish its own resources to perform the proposed work on the contract.

In all cases, the Contractor shall be responsible for any costs associated with services, equipment or facilities provided by NASA or another federal department or agency, and such costs shall result in no increase in the price of this contract.

Note: The Facility Use Agreement has been updated to include additional requirements related to NASA IT Security under Section C. Terms and Conditions (of the Facility Use Agreement).

> 3. If Contractor's use of NASA facilities, equipment, and/or services includes use of or access to NASA Information Technology (IT) resources, the Contractor will at all times remain in compliance with and adhere to all NASA IT security requirements and processes, including those set forth in the Contractor's IT Security Plan. The Contractor's failure to do so may result in NASA's unilateral termination of this Use Agreement.

6. Submission of Proposals

6.1 Submission Requirements

NASA uses electronically supported business processes for the SBIR/STTR Programs. An offeror must have Internet access and an email address. Paper submissions are not accepted.

The EHB for submitting proposals is located at http://sbir.nasa.gov under the Handbooks section. The Proposal Submissions EHB guides the firms through the steps for submitting an SBIR/STTR proposal. All EHB submissions are through a secure connection. Communication between NASA's SBIR/STTR Programs and the firm is primarily through a combination of EHBs and email.

6.2 Submission Process

New SBCs must register in the EHB to begin the submission process. Returning firms can use the same account that they have used for previous submissions. Firms are encouraged to start the proposal process early, to allow for sufficient time to complete the submissions process. It is recommended that the Business Official, or an authorized representative designated by the Business Official, be the first person to register for the SBC. The SBC's Employer Identification Number (EIN)/Taxpayer Identification Number is required during registration.

Note: The designated firm admin, typically the first person to register your firm, is the only individual authorized to update and change the firm level forms.

For successful proposal submission, SBCs shall complete all forms online, upload their technical proposal in an acceptable format, and have the Business Official and Principal Investigator electronically endorse the proposal. Electronic endorsement of the proposal is handled online with no additional software requirements. The term "technical proposal" refers to the part of the submission as described in section 3.3.4 for Phase I proposals or 3.4.4 for Phase II proposals.

STTR: The Research Institution is required to electronically endorse the Research Agreement prior to the SBC endorsement of the completed proposal submission.

6.2.1 What Needs to Be Submitted

The entire proposal package must be submitted via the Submissions EHB located on the NASA SBIR/STTR website.

Note: Other forms of submissions are not acceptable.

The proposal package includes:

1. Proposal Contact Information
2. Proposal Certifications
3. Proposal Summary
4. Proposal Budget
5. Technical Proposal (upload)
6. Research Agreement (STTR only)
7. Briefing Chart
8. NASA Evaluation License Application (only if TAV is being proposed)

9. I-Corps Opt-In (Phase I only) or Capital Commitments Addendum (Phase II only, optional)
10. Firm Level Forms (completed once for all proposals submitted to a single Solicitation)
    a. Firm Certifications
    b. Audit Information
    c. Prior Awards Addendum
    d. Commercial Metrics Survey

6.2.2 Technical Proposal Submissions

NASA converts all technical proposal files to PDF format for evaluation. Therefore, NASA requests that technical proposals be submitted in PDF format.

Note: Embedded animation or video, as well as reference technical papers for "further reading" will not be considered for evaluation.

Virus Check
The offeror is responsible for performing a virus check on each submitted technical proposal. As a standard part of entering the proposal into processing, NASA will scan each submitted electronic technical proposal for viruses.

Note: The detection, by NASA, of a virus on any electronically submitted technical proposal, may cause rejection of the proposal.

6.2.3 Technical Proposal Uploads

Firms will upload their proposals using the Submissions EHB. Directions will be provided to assist users. All transactions via the EHB are encrypted for security. Firms cannot submit security/password protected technical proposal and/or supporting documentation, as reviewers may not be able to open and read the files.

You may upload the technical proposal multiple times, with each new upload replacing the previous version, but only the final uploaded and electronically endorsed version will be considered for review. Before you can submit the proposal package, you must download the entire proposal package and certify that you have reviewed it to ensure that you have uploaded the correct materials.

6.3 Deadline for Phase I Proposal Receipt

All Phase I proposal submissions shall be received no later than *5:00 p.m. EDT on Friday, March 29, 2019* via the NASA SBIR/STTR website (http://sbir.nasa.gov), under the Handbooks section.

The EHB will not allow submissions after this deadline, but firms will have read-only access to the materials they have already submitted.

Note: Offerors are strongly encouraged to start the submission process early in order to allow sufficient time for completing their proposal package.

6.**4 Deadline for Phase II Proposal Receipt**

All Phase II proposal submissions shall be received no later than 5:00 p.m. EST the last day of the Phase I contract original period of performance, via the NASA SBIR/STTR website (http://sbir.nasa.gov) under the Handbooks section.

The EHB will be available for submissions approximately 6 weeks prior to completion date of Phase I contracts. Receipt of Phase II proposals are due on the last day of performance under SBIR/STTR Phase I contracts. The EHB will not be available for submissions after this deadline, but firms will have read-only access to the materials they have already submitted.

6.5 Acknowledgment of Proposal Receipt

The final proposal submission includes successful completion of all firm level forms, Contact Information, Proposal Certifications, Proposal Budget, Proposal Summary, the Technical Proposal upload, the Briefing Chart, the I-Corps Opt-In (if applicable in Phase I), and electronic endorsement by the SBC Official and Principal Investigator. For STTR submissions, it also includes the Research Agreement and endorsement of this agreement by the Research Institution official. NASA will acknowledge receipt of electronically submitted proposals upon endorsement by the SBC Official to the SBC Official's email address as provided on the proposal cover sheet, as well as to the user who created the proposal, if different. If a proposal acknowledgment is not received, the offeror should contact the NASA SBIR/STTR Program Support Office at 301-937-0888 or sbir@reisystems.com.

6.6 Withdrawal of Proposals

Prior to the close of submissions, proposals may be withdrawn via the Proposal Submissions Electronic Handbook hosted on the NASA SBIR/STTR website (http://sbir.nasa.gov) under the Handbooks section. In order to withdraw a proposal after the deadline, the designated SBC Official must send written notification via email to sbir@reisystems.com.

6.7 Service of Protests

Copies of any protests, as defined in section 33.101 of the FAR, shall be served on the Contracting Officer by obtaining written and dated acknowledgement of receipt from the NASA SBIR/STTR Program contact listed below:

Benjamin Benvenutti
NASA Shared Services Center
Building 1111, C Road
Stennis Space Center, MS 39529
benjamin.s.benvenutti@nasa.gov

The copy of any protest shall be received within one calendar day of filing a protest with the GAO.

7. Proposal, Scientific and Technical Information Sources

7.1 NASA Websites

General sources relating to organizational and programmatic information at NASA is available via the following websites:

NASA Budget Documents, Strategic Plans and Performance Reports:
http://www.nasa.gov/about/budget/index.html
NASA Organizational Structure: http://www.nasa.gov/centers/hq/organization/index.html
NASA SBIR/STTR Programs: http://sbir.nasa.gov

Information regarding the mission directorates and the NASA centers can be obtained at the following websites:

Office of the Chief Technologist	
Space Technology Roadmaps	http://www.nasa.gov/offices/oct/home/roadmaps/index.html

NASA Mission Directorates	
Aeronautics Research	http://www.aeronautics.nasa.gov/
Human Exploration and Operations	http://www.nasa.gov/directorates/heo/home/
Science	http://nasascience.nasa.gov
Space Technology	http://www.nasa.gov/directorates/spacetech/home/index.html

NASA Centers	
Armstrong Flight Research Center (AFRC)	http://www.nasa.gov/centers/armstrong/home/index.html
Ames Research Center (ARC)	http://www.nasa.gov/centers/ames/home/index.html
Glenn Research Center (GRC)	http://www.nasa.gov/centers/glenn/home/index.html
Goddard Space Flight Center (GSFC)	http://www.nasa.gov/centers/goddard/home/index.html
Jet Propulsion Laboratory (JPL)	http://www.nasa.gov/centers/jpl/home/index.html
Johnson Space Center (JSC)	http://www.nasa.gov/centers/johnson/home/index.html
Kennedy Space Center (KSC)	http://www.nasa.gov/centers/kennedy/home/index.html
Langley Research Center (LaRC)	http://www.nasa.gov/centers/langley/home/index.html
Marshall Space Flight Center (MSFC)	http://www.nasa.gov/centers/marshall/home/index.html
Stennis Space Center (SSC)	http://www.nasa.gov/centers/stennis/home/index.html
NASA Shared Services Center (NSSC)	https://www.nssc.nasa.gov/

7.2 United States Small Business Administration (SBA)

The Policy Directives for the SBIR/STTR Programs may be obtained from the following source. SBA information can also be obtained at: http://www.sbir.gov.

U.S. Small Business Administration
Office of Technology – Mail Code 6470
409 Third Street, S.W.
Washington, DC 20416
Phone: 202-205-6450

7.3 National Technical Information Service

The National Technical Information Service is an agency of the Department of Commerce and is the federal government's largest central resource for government-funded scientific, technical, engineering and business-related information. For information regarding their various services and fees, call or write:

National Technical Information Service
5285 Port Royal Road
Springfield, VA 22161
Phone: 703-605-6000
URL: http://www.ntis.gov

7.4 Other Sources of Assistance

The U.S. Government invests in a wide variety of resources designed to aid and assist small business owners and their employees. A variety of websites containing these resources and links to additional resources can be found at: http://sbir.nasa.gov/content/additional-sources-assistance.

8. Submission Forms

Note: Previews of all forms and certifications are available via the NASA SBIR/STTR Firm Library, located at: http://sbir.gsfc.nasa.gov/sbir/firm_library/index.html.

SBIR Phase I Check List

For assistance in completing your Phase I proposal, use the following checklist to ensure your submission is complete.

1. The proposal and innovation is submitted for one subtopic only (section 3.1).

2. The entire proposal package is submitted consistently with the requirements outlined in section 3.3.
    a. Proposal Contact Information
    b. Proposal Certifications
    c. Proposal Summary
    d. Proposal Budget
    e. Technical Proposal
    f. Briefing Chart
    g. I-Corps Opt-In
    h. Firm Level Forms (completed once for all proposals submitted to a single Solicitation)
        i. Firm Certifications
        ii. Audit Information
        iii. Prior Awards Addendum
        iv. Commercialization Metrics Survey

3. **The technical proposal shall not exceed a total of 19 8.5 x 11 inch pages and follow the format requirements (section 3.3.2).**

4. The technical proposal contains all ten parts in order (section 3.3.4).

5. Any additional required letters/documentation.
    a. A letter of commitment from the facility manager, if the research or R&D effort requires use of federal facilities (section 3.3.3.4).
    b. Letters of commitment from Subcontractors/Consultants.
    c. If the firm is an eligible joint ventures and limited partnership, a copy or comprehensive summary of the joint venture agreement or partnership agreement is included.
    d. NASA Evaluation License Application if proposing the use of NASA Technology (TAV).
    e. Supporting documentation of budgeted costs.

6. Proposed funding does not exceed $125,000 (section 1.3).

7. Proposed project duration does not exceed 6 months (section 1.3).

8. Proposal package electronically endorsed by the SBC Official and the PI.

9. **Proposals must be received no later than 5:00 p.m. EDT on March 29, 2019 (section 6.3).**

STTR Phase I Check List

For assistance in completing your Phase I proposal, use the following checklist to ensure your submission is complete.

1. The proposal and innovation is submitted for one subtopic only (section 3.1).

2. The entire proposal package is submitted consistently with the requirements outlined in section 3.3.
    a. Proposal Contact Information
    b. Proposal Certifications
    c. Proposal Summary
    d. Proposal Budget
    e. Technical Proposal
    f. Research Agreement (STTR only)
    g. Briefing Chart
    h. I-Corps Opt-In
    i. Firm Level Forms (completed once for all proposals submitted to a single Solicitation)
        i. Firm Certifications
        ii. Audit Information
        iii. Prior Awards Addendum
        iv. Commercialization Metrics Survey

3. **The technical proposal shall not exceed a total of 19 8.5 x 11 inch pages and follow the format requirements (section 3.3.2).**

4. The technical proposal contains all ten parts in order (section 3.3.4).

5. Any additional required letters/documentation.
    a. A letter of commitment from the facility manager, if the research or R&D effort requires use of federal facilities (section 3.3.3.4).
    b. Letters of commitment from Subcontractors/Consultants.
    c. If the firm is an eligible joint ventures and limited partnership, a copy or comprehensive summary of the joint venture agreement or partnership agreement is included.
    d. NASA Evaluation License Application if proposing the use of NASA Technology (TAV).
    e. Supporting documentation of budgeted costs.

6. Proposed funding does not exceed $125,000 (section 1.3).

7. Proposed project duration does not exceed 13 months (section 1.3).

8. Research Agreement electronically endorsed by both the SBC Official and the RI (section 3.3.5, 6.2).

9. Proposal package electronically endorsed by the SBC Official and the PI.

10. **Proposals must be received no later than 5:00 p.m. EDT on March 29, 2019 (section 6.3).**

11. Signed Allocation of Rights Agreement available for Contracting Officer within 10 days of selection.

SBIR Phase II Check List

For assistance in completing your Phase II proposal, use the following checklist to ensure your submission is complete.

1. The proposal and innovation is submitted for one subtopic only (section 3.1).

2. The entire proposal package is submitted consistently with the requirements outlined in section 3.4.
    a. Proposal Contact Information
    b. Proposal Certifications
    c. Proposal Summary
    d. Proposal Budget
    e. Technical Proposal
    f. Briefing Chart
    g. Capital Commitments Addendum (if applicable)
    h. Firm Level Forms (completed once for all proposals submitted to a single Solicitation)
        i. Firm Certifications
        ii. Audit Information
        iii. Prior Awards Addendum
        iv. Commercialization Metrics Survey

3. **The technical proposal shall not exceed a total of 46 8.5 x 11 inch pages and follow the format requirements (section 3.4.2).**

4. The technical proposal contains all ten parts in order (section 3.4.4).

5. Any additional required letters/documentation
    a. A letter of commitment from the facility manager, if the research or R&D effort requires use of federal facilities (section 3.4.4).
    b. Letters of commitment from Subcontractors/Consultants.
    c. Letters in support of Capital Commitments Addendum.
    d. If the firm is an eligible joint ventures and limited partnership, a copy or comprehensive summary of the joint venture agreement or partnership agreement is included.
    e. NASA Evaluation License Application if proposing the use of NASA Technology (TAV).
    f. Supporting documentation of budgeted costs.

6. Proposed funding does not exceed $750,000, excluding the $5,000 Technical and Business Assistance, if requested (section 1.8).

7. Proposed project duration does not exceed 24 months (section 1.3).

8. Proposal package electronically endorsed by the SBC Official and the PI.

9. **Phase II proposal submissions will be due the last day of the Phase I contract (section 6.4).**

STTR Phase II Check List

For assistance in completing your Phase II proposal, use the following checklist to ensure your submission is complete.

1. The proposal and innovation is submitted for one subtopic only (section 3.1).

2. The entire proposal package is submitted consistently with the requirements outlined in section 3.4.
    a. Proposal Contact Information
    b. Proposal Certifications
    c. Proposal Summary
    d. Proposal Budget
    e. Technical Proposal
    f. Research Agreement
    g. Briefing Chart
    h. Capital Commitments Addendum (if applicable)
    i. Firm Level Forms (completed once for all proposals submitted to a single Solicitation)
        i. Firm Certifications
        ii. Audit Information
        iii. Prior Awards Addendum
        iv. Commercialization Metrics Survey

3. **The technical proposal shall not exceed a total of 46 8.5 x 11 inch pages and follow the format requirements (section 3.4.2).**

4. The technical proposal contains all ten parts in order (section 3.4.4).

5. Any additional required letters/documentation
    a. A letter of commitment from the facility manager, if the research or R&D effort requires use of federal facilities (section 3.4.4).
    b. Letters of commitment from Subcontractors/Consultants.
    c. Letter in support of Capital Commitments Addendum.
    d. If the firm is an eligible joint ventures and limited partnership, a copy or comprehensive summary of the joint venture agreement or partnership agreement is included.
    e. NASA Evaluation License Application if proposing the use of NASA Technology (TAV).
    f. Supporting documentation of budgeted costs.

6. Proposed funding does not exceed $750,000 excluding the $5,000 Technical and Business Assistance, if requested (section 1.8).

7. Proposed project duration does not exceed 24 months (section 1.3).

8. Research Agreement electronically endorsed by both the SBC Official and the RI (section 3.4.5, 6.2).

9. Proposal package electronically endorsed by the SBC Official and the PI.

10. **Phase II proposal submissions will be due the last day of the Phase I contract (section 6.4).**

11. Signed Allocation of Rights Agreement, available for the Contracting Officer within 10 days of the selection.

9. Research Topics for SBIR and STTR

Introduction

The SBIR and STTR subtopics are organized into groupings called "Focus Areas". Focus Areas are a way of grouping NASA interests and related technologies with the intent of making it easier for proposers to understand related needs across the agency and thus identify subtopics where their research and development capabilities may be a good match.

Note: The SBIR and STTR Subtopics will appear in one combined listing. The STTR subtopics will begin with a "T" and will be clearly marked so that offerors will know that the additional Research Institution (RI) partnership is required before submitting a proposal.

Subtopic numbering conventions from previous year's solicitations have been maintained for traceability of like-subtopics from previous solicitations. The mapping is as follows:

A – Aeronautics Research Mission Directorate (ARMD)
H – Human Exploration and Operations Mission Directorate (HEOMD)
S – Science Mission Directorate (SMD)
Z – Space Technology Mission Directorate (STMD)
T – Small Business Technology Transfer (STTR)

Note: The former HEOMD Subtopics in In Situ Resource Utilization (previously H1 and H2) and In-Space Manufacturing (previously H7) are now being managed by STMD. In Situ Resource Utilization subtopics will appear as Z12 subtopics and In-Space Manufacturing will appear in Z3.

Proposers should think of the Subtopic Lead Mission Directorates and Lead/Participating Centers as potential customers for their proposals. Multiple Mission Directorates and Centers may have interests across the subtopics within a Focus Area.

Related subtopic pointers are identified when applicable in the subtopic headers to assist proposers with identifying related subtopics that also potentially seek related technologies for different customers or applications. As stated in section 3.1, an offeror shall not submit the same (or substantially equivalent) proposal to more than one subtopic. It is the offeror's responsibility to select which subtopic to propose to.

Moon to Mars Campaign 🌙

NASA is implementing a program for the exploration and utilization of the Moon followed by missions to Mars and other destinations, called the Moon to Mars campaign (see https://www.nasa.gov/feature/nasa-unveils-sustainable-campaign-to-return-to-moon-on-to-mars and https://www.nasa.gov/topics/moon-to-mars). An early element of the exploration campaign is the delivery of payloads to the Moon for scientific study and the advancement of technology capabilities to support sustained lunar surface operations. There are 18 subtopics where proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. These subtopics will be marked by a moon and will include language referring to the Commercial Lunar Payload Services (CLPS) contract. For additional information on the Moon to Mars Campaign please see the Notable Changes section at the front of this solicitation.

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Focus Area 1 In-Space Propulsion Technologies

Lead MD: STMD
Participating MD(s): STTR

NASA is interested in technologies for advanced in-space propulsion systems to reduce travel time, increase payload mass, reduce acquisition costs, reduce operational costs, and enable new science capabilities for exploration and science spacecraft. The future will require demanding propulsive performance and flexibility for more ambitious missions requiring high duty cycles, more challenging environmental conditions, and extended operation. This focus area seeks innovations for NASA propulsion systems in chemical, electric, nuclear thermal and advanced propulsion systems related to human exploration and science missions. Propulsion technologies will focus on a number of mission applications including ascent, descent, orbit transfer, rendezvous, station keeping, and proximity operations.

T2.02 Advanced Technologies for In-Space Electric Propulsion (EP) (STTR)
Lead Center: GRC
Participating Center(s): JPL
Related Subtopic Pointer(s): H10.01, Z8.01, Z10.02

Advanced EP Technologies for Small Spacecraft

NASA seeks new technologies that can be rapidly fielded to drastically decrease mission cost. Small spacecraft (<500 kg) launched as secondary payloads provide such an opportunity; however, sub-kilowatt in-space electric propulsion (EP) systems with high propellant-throughput capability remain immature. Future small spacecraft and constellations of small spacecraft will have high-performance electric propulsion requirements, usually in volume-, mass-, and power-limited envelopes. Furthermore, the cost of electric propulsion systems need to decrease to remain commensurate with the total mission cost of small spacecraft. Toward these ends, proposals must address one of the following specific areas of interest applicable to sub-kilowatt Hall-effect and/or gridded-ion thrusters. Proposers are expected to show an in-depth understanding of the current state-of-the-art (SOA) and quantitatively (not qualitatively) describe improvements over relevant SOA technologies that substantiate investment in the new technology. Proposers must also quantitatively explain the operational benefit of the new technology from the perspective of improving or enabling mission potential. Proposals outside of the scope described below shall not be considered:

- Lower-cost, more compact, heated cathode assemblies, which demonstrate the capability to achieve 10,000 hours of operation and 10,000 cycles with performance and reliability comparable to state-of-the-art flight-heritage cathode assemblies.
- Compact heaterless cathode assemblies capable of reliable ignition below 500 V, and greater than 10,000 hours of operation and 10,000 cycles.
- Innovative high-temperature discharge channel materials and/or designs that permit fabrication of thinner walls, yet still capable of surviving the rigors of launch and repeated thermal cycling.
- Innovative, lower-cost, and reliable xenon flow control systems capable of delivering well-regulated flow rates between 0 and 5 mg/s for Hall-effect or gridded-ion systems. Scale and fault-tolerance are to be consistent with reasonable small spacecraft requirements for a NASA class-D mission.

The Science Mission Directorate (SMD) needs spacecraft with more demanding propulsive performance and flexibility for more ambitious missions requiring high duty cycles, more challenging environmental conditions, and extended operation. Planetary spacecraft need the ability to rendezvous with, orbit, and conduct in-situ exploration of planets, moons, and other small bodies in the solar system. Mission priorities are outlined in the decadal surveys for each of the SMD Divisions (https://science.nasa.gov/about-us/science-strategy/decadal-surveys). Future spacecraft and constellations of spacecraft will have high-precision propulsion requirements, usually in volume-, mass-, and power-limited envelopes.

This subtopic seeks innovations to meet future SMD propulsion requirements in electric propulsion systems related to missions to the moon, Mars, and small bodies (like asteroids, comets, and Near-Earth Objects). Additional electric propulsion technology innovations are also sought to enable low-cost systems for small spacecraft missions. The

roadmap for in space propulsion technologies is covered under the NASA Technology Area- TA-02 In Space Propulsion.

The expected TRL of this project is 4 to 5.

Advanced Diagnostics for Electric Propulsion (EP) Testing Facilities

Advancements in electric propulsion and growing demand for long-life and highly reliable electric propulsion systems necessitates new or improved diagnostic tools for ground test facilities. Diagnostics are generally desirable that increase functionality, improve accuracy, increase reliability, accelerate collection time, require fewer resources to implement, lower cost, are non-intrusive, and/or are compatible with novel propellants such as iodine. Toward these ends, proposers must address one of the following areas of interest. Proposers are expected to show an in-depth understanding of the current state-of-the-art (SOA) and quantitatively (not qualitatively) describe improvements over relevant SOA technologies that substantiate investment in the new technology. Proposals outside of the scope described below shall not be considered:

- Species spectrometers for characterizing plasma with charged species that have overlapping energy distribution, such as that of a charge-exchange plasma generated by a Hall thruster. Magnetically-shielded Hall thrusters tend to have a greater fraction of multiply-charged species than traditional Hall thrusters and the distribution function of the multiply-charged species tend to overlap (Hofer, AIAA-2012-3788; Huang, IEPC-2013-057). There are regions in the plume of a magnetically-shielded Hall thruster where the traditional approaches to measuring fractions of multiply-charged species are inadequate.
- An iodine compatible diagnostics package. Iodine properties make it an interesting alternative to xenon propellant; however, iodine is highly corrosive, and damages most modern plasma diagnostics employed in electric propulsion technology development. An innovative diagnostics package is desired that can be demonstrated immune to iodine both under vacuum and in ambient laboratory conditions where the diagnostic might remain contaminated with small amounts of iodine post-test. Proposers should include a development plan that includes both theoretical and experimental evidence that the probes will remain immune from long-term degradation, providing consistent reliable data.

The Science Mission Directorate (SMD) needs spacecraft with more demanding propulsive performance and flexibility for more ambitious missions requiring high duty cycles, more challenging environmental conditions, and extended operation. Planetary spacecraft need the ability to rendezvous with, orbit, and conduct in-situ exploration of planets, moons, and other small bodies in the solar system. Mission priorities are outlined in the decadal surveys for each of the SMD Divisions (https://science.nasa.gov/about-us/science-strategy/decadal-surveys). Future spacecraft and constellations of spacecraft will have high-precision propulsion requirements, usually in volume-, mass-, and power-limited envelopes. The new/improved diagnostics being solicited will aid in verification and validation of these electric propulsion technologies for their anticipated functional environments.

This subtopic seeks innovations to meet future SMD propulsion requirements in electric propulsion systems related to missions to the moon, Mars, and small bodies (like asteroids, comets, and Near-Earth Objects). The roadmap for in space propulsion technologies is covered under the NASA Technology Area- TA-02 In Space Propulsion.

The expected TRL of this project is 4 to 6.

T2.03 Performance Demonstration of High Payoff Propulsion Technology: Rotating Detonation Engine and Dual Mode Ionic Liquid (STTR)
Lead Center: MSFC
Participating Center(s): GRC, MSFC
Related Subtopic Pointer(s): T2.02, Z8.01

NASA is soliciting proposals for performance demonstrations of lower Technology Readiness Level (TRL) high payoff propulsion technology. The objective is to gain performance data to validate previous or concurrent analytical performance predictions of the technology. Conventional propulsion systems are highly mature with diminishing returns for investments in evolutionary steps of higher performance. Proposals only addressing the following focus areas will be considered:

Rotating Detonation Engines - Rocket Applications

New technologies such as the rotating detonation engine (RDE) offers a step function improvement over state-of-the-art alternatives. However, RDE diagnostics and analytical models are limited for system performance characterization and design optimization. This topic has an objective of anchoring either existing or concurrent RDE model validation efforts. The proposals may include novel diagnostic solutions for system characterization in the challenging environment. This topic seeks to advance the capabilities for RDE thermal design, injector design, and pressure loss optimization. Phase II must include hot-fire testing for analytical model validation activities and/or advanced RDE diagnostics performance demonstration.

Dual Mode Propulsion

The government has spent significant resources to mature and demonstrate the non-toxic propellants (e.g., AF-M315E). In addition to anticipated life cycle cost reductions, these non-toxic propulsion systems have comparable or better performance as state-of-the-art alternatives. Today, many spacecraft carry two propulsion options: high thrust propulsion for high acceleration maneuvers (such as orbit insertion) and high specific impulse (low thrust) for station keeping and less time critical maneuvers. Dual mode operation is conventionally flown in two ways: as either a single propellant system, which typically offers lower performance in lieu of cost and packaging advantages or has independent propellant systems (e.g., hydrazine and xenon) to maximize performance. However, AF-M315E has been shown to have acceptable performance for combustion high thrust systems as well as low thrust variants. Near-term investments are anticipated to field both high thrust and low thrust systems. This solicitation seeks innovative solutions for interfacing with a common propellant tank for dual mode operation and validate integrated system performance. The Phase I proposal must include innovative propellant conditioning solutions with breadboard or higher fidelity hardware and the Phase II deliverables must include flight weight and efficient packaging systems that matches the proposed system architecture.

The expected TRL for this project is 3 to 5.

Z10.01 Cryogenic Fluid Management (SBIR)
Lead Center: GRC
Participating Center(s): JSC, KSC, MSFC
Related Subtopic Pointer(s): Z4.03

The Space Technology Mission Directorate (STMD) strives to provide the technologies that are needed to enable exploration of the solar system, both manned and unmanned systems. Cryogenic Fluid Management (CFM) is a key technology to enable exploration. Whether nuclear thermal propulsion or liquid oxygen/liquid methane is chosen by Human Exploration and Operation Mission Directorate (HEOMD) as the main in-space propulsion element to transport humans, CFM will be required to store propellant for up to five years in various orbital environments. Transfer will also be required, whether to engines or other tanks (e.g., depot/aggregation), to enable the use of cryogenic propellants that have been stored. In conjunction with In-Situ Resource Utilization (ISRU), oxygen will have to be produced, liquefied, and stored, the latter two of which are CFM functions for the surface of Mars. ISRU and CFM liquefaction drastically reduces the amount of mass that must be landed.

This subtopic seeks technologies related to cryogenic propellant (e.g., hydrogen, oxygen, methane) storage and transfer to support NASA's space exploration goals. This includes a wide range of applications, scales, and environments consistent with future NASA missions. Such missions include, but are not limited to a methane upper stage, nuclear thermal propulsion, lander propulsion, aggregation stages, and ISRU in support of the NASA exploration mission objectives. Anticipated outcome of Phase I proposals are expected to deliver proof of the proposed concept with some sort of basic testing or physical demonstration. Proposals shall include plans for a prototype and demonstration in a defined relevant environment (with relevant fluids) at the conclusion of Phase II.

Desired technology concepts are listed below in order of priority:

- Broad area cooling methods for cryogenic thin-walled metallic and/or composite propellant tanks (reduced and/or zero boil-off applications): Design and integration concepts must exhibit low mass, high-heat transfer between cooling fluid and propellant in tank, high heat exchanger efficiency (>90%), and operate in reduced gravity environments (10-6 g worse case). Proposers should consider structural and pressure vessel implications of the proposed concept. If tube-on-tank cooling is proposed, concepts are solicited for reliable, low thermal resistance manufacturing and attachment of cooling tubes to propellant tanks. Target applications include liquid oxygen liquefaction system (16 g/s neon gas, 85K < T < 90K, pressure drop < 0.25 psia, 2.6m diameter, 3m tall tank) and liquid hydrogen nuclear thermal propulsion system (3.5 g/s helium gas, 20K < T < 24K, 7m diameter, 8m tall tank).
- Cryogenic transfer line thermal coatings (ex. nano-structured, micro-structured, vapor deposition) for 0.5'' to 3'' OD tubing that can reduce the chilldown time (amount of time from room temperature to 77K) by at least 30% relative to uncoated standard stainless steel line in low-g. Coated lines should be able to maintain performance (reduction in chilldown time) after multiple (> 15) thermal cycles (room temperature to 77K and back). Proposed coatings should be oxygen compatible. Anticipated maximum allowable working pressure is 500 psia.
- Lightweight all composite spherical tanks for cryogenic propellants. Spherical versus cylindrical tanks have improved thermal storage characteristics (due to reduced surface area to volume ratio), better packaging benefits (when considering engine and plumbing) and have inherently lower stresses due to geometry. While progress has been made on all-composite tanks, there is no state-of-art spherical tank designed for a target diameter range of 4-8ft, a max pressure of 500+ psig, helium permeability less than 1x10-4 sccs/m^2 (at 500 psi), and an operating temperature range of ambient to -320° F (LN_2); with goal of -423° F (LH_2). Proposals shall also include plans for oxygen compatibility and cryogenic LN_2 testing.
- Sub-grid CFD model of the nucleate boiling process for 1-g and low-g to be implemented into commercial industry standard CFD codes. The sub-grid model should capture the nucleation and growth of bubble on a heated wall and estimate the bubble departure frequency to be implemented via Lagrangian-Eulerian or Eulerian-Eulerian approaches for tracking the phases using discrete phase modeling (DPM), volume of fluid (VOF), Level Set, or Population Balance Methods. The boiling sub-grid model should be validated against available experimental data (with a target accuracy of 40%), with emphasis on cryogenic boiling data. The sub-grid model and implementation scheme shall be a contract deliverable.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined; however, at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

The expected TRL for this project is 2 to 4.

References:

- Johnson et al. "Investigation into Cryogenic Tank Insulation Systems for the Mars Surface Environment" 2018 Joint Propulsion Conference Cincinnati, OH, July, 2018. Paper.
- Plachta, D., et al. "Zero Boil-Off System Testing" NASA TP 20150023073.

Z10.02 In-Space Electric Propulsion Component Technologies (SBIR)
Lead Center: GRC
Participating Center(s): JPL
Related Subtopic Pointer(s): T2.02, Z8.01

In-Space Electric Propulsion Component Technologies

Electric Propulsion for space applications has shown tremendous benefit to a variety of NASA, commercial, and DoD missions. The electric propulsion systems currently under development have uncovered challenges and limits to these technologies. This subtopic seeks proposals that explore uses of technologies that will provide superior performance, reduce complexity, increase reliability, and/or lower cost for high specific impulse/low mass electric propulsion systems. Proposers are expected to show an in-depth understanding of the current state-of-art (SOA) and quantitatively (not qualitatively) describe improvements over relevant SOA technologies that substantiate investment in the new technology. Proposers must also quantitatively explain the operational benefits of the new technology from the perspective of improving or enabling mission potential. Proposals outside of the scope described below shall not be considered.

These technologies of interest include:

- Advanced magnetics for Hall/ion thrusters. Specifically:
    - High temperature capable magnetic components (>500° C).
    - 3D printing of magnetic materials.
- Advanced Hall/ion cathode technologies. Experimental demonstration backed by theoretical or computational modeling are preferred. Specifically:
    - Lower cost fabrication techniques for cathode assemblies.
    - Advanced cathode emitter materials
    - Long-life heaters for hollow cathodes made with barium oxide (BaO), lanthanum hexaboride (LaB6) or other materials. In order to achieve reliable cathode ignition, barium oxide cathodes must operate at 1050 - 1200° C while the LaB6 heaters typically must operate at 1500 – 1700° C. Reproducible fabrication processes that minimize unit-to-unit variations in performance and lifetime will be critical for the practical adaptation of a new heater technology.
- Advanced materials for Hall thruster systems. Specific areas of interest include:
    - High emissivity (>0.6) coatings and/or surface treatments suitable for use with high-temperature (300-500° C) electric propulsion components with long operating times (>20 kh).
    - High-voltage (>600 V), high-temperature harnessing capable of long-term (>20 kh) vacuum operation over temperature ranges of -100° C to 400° C.

Low-cost, high-temperature anode gas distributors capable of achieving a high degree of flow uniformity in Hall thruster discharge channels through use of innovative designs and/or fabrication techniques.

The Science Mission Directorate (SMD) needs spacecraft with more demanding propulsive performance and flexibility for more ambitious missions requiring high duty cycles, more challenging environmental conditions, and extended operation. Planetary spacecraft need the ability to rendezvous with, orbit, and conduct in-situ exploration of planets, moons, and other small bodies in the solar system. Mission priorities are outlined in the decadal surveys for each of the SMD Divisions (https://science.nasa.gov/about-us/science-strategy/decadal-surveys). Future spacecraft and constellations of spacecraft will have high-precision propulsion requirements, usually in volume-, mass-, and power limited envelopes.

Additional electric propulsion technology innovations are also sought to enable low-cost systems for Discovery class missions, and low-power, nuclear electric propulsion (NEP) missions. The roadmap for in space propulsion technologies is covered under OCT's TA-02 In-Space Propulsion.

Expected TRL for this project is 4 to 5.

Precision Low-Noise Micropropulsion for Fine Pointing of Astronomical Observatories

Future astronomical observatories are facing two critical challenges:

- Advanced astrophysical and exoplanet science demand increasingly longer observing times with more precise pointing and wavefront error stability requirements.
- The performance and lifetime of reaction wheels is limited.

For the first challenge, solar pressure and torque demand that space-based observatories have active, continuous control of pointing, typically using reaction wheels as the actuator, which have limited pointing performance and create large amounts of mechanical noise. Large and heavy vibration isolation stages are typically used to protect the instruments from the vibrations and jitter induced by reaction wheels. But for many new mission concepts using higher resolution detectors or coronagraphs to block starlight, vibration isolation systems struggle to achieve the necessary wavefront error stability, required to be less than 1 nanometer. New, lower-noise actuators or "active" vibration isolation technology for large deployable structures are an attractive alternative. For the second challenge, reaction wheel failure has limited the lifetime of many astronomical observatory missions. Longer life actuators would be necessary to justify replacement of high-heritage reaction wheels.

Most micropropulsion systems are being developed for high-impulse, low-power applications as a "miniature" equivalent to existing propulsion options with just a few throttle points. However, as fine pointing actuators, mainly pushing back against solar pressure (7 $\mu N / m^2$), precision microthrusters do not need to demonstrate high thrust or impulse. Instead, low thrust noise (< 1 $\mu N/\sqrt{Hz}$) over a continuous throttle range (5-100 µN) is necessary to replace reaction wheels. For any application including precision pointing, increased reliability and lifetime >4 years are also critical. If such propulsion systems could be developed, NASA and commercial space-based observatories would have better pointing performance without need for expensive and heavy structure for vibration isolation or the likely lifetime limitation of using reaction wheels.

Anticipated environments for these devices will be typical for L_1/L_2 or Earth-trailing orbits with similar radiation dose requirements of "deep space" or even GEO-like orbits. Thermal environment varies from mission to mission, but often the thrusters will be placed orthogonal to the plane that is generating the solar pressure, which usually means in the shade or partial shade. As an example, typical operating environment temperatures for ST-7 were 0° C - 50° C with non-operating temperatures stretching to -20° C - 75° C.

This technology is critical to the Physics of the Cosmos and Exoplanet Exploration Programs in the Astrophysics Directorate, both of which have added microthrusters for precision control on their list of high priority technologies. Example missions include Laser Interferometer Space Antenna (LISA) and Habitable Exoplanet Observatory (HabEx), which have both baselined precision microthrusters instead of reaction wheels.

Expected TRL for this project is 4 to 6.

Z10.03 Nuclear Thermal Propulsion (SBIR)
Lead Center: MSFC
Participating Center(s): GRC, SSC
Related Subtopic Pointer(s): H10.01

Reactor and Fuel System

The focus is on highly stable materials for nuclear fuels and non-fuel reactor components (i.e., moderator tie tubes, etc.) that can heat hydrogen to temperatures greater than 2600K without undergoing significant dimensional deformation, cracking, or hydrogen reactions. Current technology hurdles related to ceramic metal fuels center around refractory metal processing and manufacturing (i.e., welding of refractories, refractory metal coatings, etc.). The development of refractory alloys with enhanced/targeted material properties is of key interest (i.e., tungsten or molybdenum with increased ductility, or dispersion strengthen Mo/W alloys). Current technology hurdles with carbide fuels include embedding carbide kernels with coatings in a carbide matrix with potential for total fission product containment and high fuel burn-up. Manufacturing and testing of the insulator and reflector materials is also critical to the success of a Nuclear Thermal Propulsion (NTP reactor.

Technologies being sought include:

- Low Enriched Uranium reactor fuel element designs with high temperature (> 2600K), high power density (>5 MW/L) to optimize hydrogen propellant heating.
- New advanced manufacturing processes to quickly manufacture the fuel with uniform channel coatings and/or claddings that reduce fission product gas release and reactor particulates into the engines exhaust stream.

Fuels focused on Ceramic-metallic (cermet) designs:

- New fuel element geometries which are easy to manufacture and coat, and better performing than the traditional prismatic fuel geometries with small through holes with coatings.
- Best joining and manufacturing processes for thin-walled (0.010”) tungsten, molybdenum, and molybdenum/tungsten alloys.
- Diffusion bonding/other bonding technologies for CERMET materials.
- Machining processes for cooling channel formation in CERMET materials.
- Uranium nitride and uranium dioxide fuel particle production methods and particle coating methods.
- Development of dispersion strengthen molybdenum/tungsten alloys.
- Formation of small diameter (0.100” ID) thin-walled (0.010”) molybdenum, tungsten, and molybdenum/tungsten cylindrical tubes.

Fuels focused on carbide designs:

- Compatibility with high temperature hydrogen.
- High thermal conductivity and other properties (e.g., ductility) needed for high power density operation (~5MW/l).
- Kernel diameters, including coatings for fission product containment, which allow the fuel element to be fabricated with adequate strength for high temperature and high-power density operation.

Insulator design (e.g., of one application is for tie tubes and the other is for interface with the pressure vessel), which has very low thermal conductivity and neutron absorption, withstands high temperatures, compatible with hot hydrogen and radiation environment, and light weight.

Future mission applications for this technology include Human Missions to Mars, Science Missions to Outer Planets, and Planetary Defense. Some technologies may have applications for fission surface power systems.

Desired Deliverables for this technology would include research that could be conducted to demonstrate technical feasibility during Phase I and show a path toward Phase II hardware/software demonstration with delivery of a demonstration unit or software package for NASA testing at the completion of the Phase II contract.

Phase I Deliverables - Feasibility study, including simulations and measurements, proving the proposed approach to develop a given product (TRL 2-3). Verification matrix of measurements to be performed at the end of Phase II, along with specific quantitative pass-fail ranges for each quantity listed.

Phase II Deliverables - Working engineering model of proposed product, along with full report of component and/or breadboard validation measurements, including populated verification matrix from Phase I (TRL 4-5). Opportunities and plans should also be identified and summarized for potential commercialization.

Expected TRL for this project is 2 to 5.

Ground Test Technologies

Included in this area of technology development needs are identification and application of robust materials, advanced instruments and monitoring systems capable of operating in extreme temperature, and pressure and radiation environments. Specific areas of interest include:

- Devices for measurement of radiation, pressure, temperature and strain in a high temperature and radiation environment:
    - Non-intrusive diagnostic technology to monitor engine exhaust for fuel element erosion/failure and release of radioactive particulates.

Future mission applications for this technology include Human Missions to Mars, Science Missions to Outer Planets, and Planetary Defense.

Desired Deliverables for this technology include research that could be conducted to demonstrate technical feasibility during Phase I and show a path towards Phase II hardware/software demonstration with delivery of a demonstration unit or software package for NASA testing at the completion of the Phase II contract.

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Expected TRL for this project is 2 to 5.

Engine System Design

Scope is on a range of modern technologies associated with NTP using solid core nuclear fission reactors. The baseline engines are pump fed with a thrust ~25,000 lbf and a specific impulse goal of 900 seconds (using hydrogen) and are used individually or in clusters for the spacecraft's primary propulsion system. The NTP can have multiple start-ups (>4) with cumulative run time >100 minutes in a single mission, which can last a few years. The Thrust to weight is ~3.5 without the external shield. Specific areas of interest include the following:

- *Subcritical LH_2 Turbopump for NTP Engine* - LH_2 Turbopump design that is capable of operating in a subcritical mode over the full range of turbopump speeds for a NTP Engine. The benefit of a turbopump operating with a subcritical design is that an NTP engine with long transient start-up and shutdown durations can operate over the entire transients without encountering any resonance modes where vibration levels are high. The mass flow rate is less than 28 lbs/sec, pump exit pressure less than 2800 psia and pump inlet pressure 8-30 psia.
- *NTP Engine Instrumentation* - Instrumentation is needed for engine control and health monitoring. Sensors must be designed to withstand a harsh NTP environment such as high temperatures, nuclear radiation composed of neutrons and gamma rays, and high vibration levels, and provide accurate measurements. Non-invasive designs for measuring neutron flux (possibly outside of reactor assembly), chamber temperature, operating pressure, and liquid hydrogen propellant flow rates over a wide range of temperatures are desired. Sensors need to operate for total run times in these harsh environments on the order of a few hours, interspersed over periods of months/years. The radiation environment adjacent to the reactor core assembly may include up to 10^{14} fast (>1MeV) neutrons/cm^2-sec, 10^{15} thermal/epithermal (<1 MeV) neutrons/cm^2-sec, and a gamma ray dose rate up to 10^9 Rad(Si)/hr.

Future mission applications for this technology include Human Missions to Mars, Science Missions to Outer Planets, and Planetary Defense.

Desired Deliverables for this technology include research that could be conducted to demonstrate technical feasibility during Phase I and show a path toward Phase II hardware/software demonstration with delivery of a demonstration unit or software package for NASA testing at the completion of the Phase II contract.

Phase I Deliverables - Feasibility study, including simulations and measurements, proving the proposed approach to develop a given product (TRL 2-3). Verification matrix of measurements to be performed at the end of Phase II, along with specific quantitative pass-fail ranges for each quantity listed.

Phase II Deliverables - Working engineering model of proposed product, along with full report of component and/or breadboard validation measurements, including populated verification matrix from Phase I (TRL 4-5). Opportunities and plans should also be identified and summarized for potential commercialization.

Expected TRL for this project is 2 to 5.

References:

Reactor and Fuel System

- Solid core NTP has been identified as an advanced propulsion concept which could provide the fastest trip times with fewer SLS launches than other propulsion concepts for human missions to Mars over a variety of mission years. NTP had major technical work done between 1955-1973 as part of the Rover and Nuclear Engine for Rocket Vehicle Application (NERVA) programs. A few other NTP programs followed including the Space Nuclear Thermal Propulsion (SNTP) program in the early 1990's. The NTP concept is similar to a liquid chemical propulsion system, except instead of combustion in the thrust chamber, a monopropellant is heated with a fission reactor (heat exchanger) in the thrust chamber and exposes the engine components and surrounding structures to a radiation environment.
- Focus is on a range of modern technologies associated with NTP using solid core nuclear fission reactors and technologies needed to ground test the engine system and components. The engines are pump fed ~25,000 lbf with a specific impulse goal of 900 seconds (using hydrogen) and are used individually or in clusters for the spacecraft's primary propulsion system. The NTP can have multiple start-ups (>4) with cumulative run time >100 minutes in a single mission, which can last a few years. The Rover/NERVA program ground tested a variety of engine sizes, for a variety of burn durations and start-ups with the engine exhaust released to the open air. Current regulations require exhaust filtering of any radioactive noble gases and particulates. The NTP primary test requirements can have multiple start-ups (>8) with the longest single burn time ~50 minutes.

Ground Test Technologies

- Solid core NTP has been identified as an advanced propulsion concept which could provide the fastest trip times with fewer SLS launches than other propulsion concepts for human missions to Mars over a variety of mission years. NTP had major technical work done between 1955-1973 as part of the Rover and Nuclear Engine for Rocket Vehicle Application (NERVA) programs. A few other NTP programs followed including the Space Nuclear Thermal Propulsion (SNTP) program in the early 1990's. The NTP concept is similar to a liquid chemical propulsion system, except instead of combustion in the thrust chamber, a monopropellant is heated with a fission reactor (heat exchanger) in the thrust chamber and exposes the engine components and surrounding structures to a radiation environment.
- Focus is on a range of modern technologies associated with NTP using solid core nuclear fission reactors and technologies needed to ground test the engine system and components. The engines are pump fed ~25,000 lbf with a specific impulse goal of 900 seconds (using hydrogen) and are used individually or in clusters for the spacecraft's primary propulsion system. The NTP can have multiple start-ups (>4) with cumulative run time >100 minutes in a single mission, which can last a few years. The Rover/NERVA program ground tested a variety of engine sizes, for a variety of burn durations and start-ups with the engine exhaust released to the open air. Current regulations require exhaust filtering of any radioactive noble gases and particulates. The NTP primary test requirements can have multiple start-ups (>8) with the longest single burn time ~50 minutes.

Engine System Design

- Solid core NTP has been identified as an advanced propulsion concept which could provide the fastest trip times with fewer SLS launches than other propulsion concepts for human missions to Mars over a variety of mission years. NTP had major technical work done between 1955-1973 as part of the Rover and Nuclear Engine for Rocket Vehicle Application (NERVA) programs. A few other NTP programs followed including the Space Nuclear Thermal Propulsion (SNTP) program in the early 1990's. The NTP concept is similar to a liquid chemical propulsion system, except instead of combustion in the thrust chamber, a monopropellant is heated with a fission reactor (heat exchanger) in the thrust chamber and exposes the engine components and surrounding structures to a radiation environment.

Focus Area 2 Power Energy and Storage

Lead MD: STMD
Participating MD(s): SMD

Power is a ubiquitous technology need across many NASA missions. Within the SBIR Program, power is represented across a broad range of topics in human exploration, space science, space technology and aeronautics. New technologies are needed to generate electrical power and/or store energy for future human and robotic space missions and to enable hybrid electric aircraft that could revolutionize air travel. A key goal is to develop technologies that are multi-use and cross-cutting for a broad range of NASA mission applications. In aeronautics, power technologies are needed to supply large-scale electric power and efficiently distribute the power to aircraft propulsors (see Focus Area 18 – Air Vehicle Technologies). In the space power domain, mission applications include planetary surface power, large-scale spacecraft prime power, small-scale robotic probe power, and smallsat/cubesat power. Applicable technology options include photovoltaic arrays, radioisotope power systems, nuclear fission, thermal energy conversion, motor/generators, fuel cells, batteries or other energy storage devices, power management, transmission, distribution and intelligent control. An overarching objective is to mature technologies from analytical or experimental proof-of-concept (TRL3) to breadboard demonstration in a relevant environment (TRL5). Successful efforts will transition into NASA Projects where the SBIR/STTR deliverables will be incorporated into ground testbeds or flight demonstrations.

S3.01 Power Generation and Conversion (SBIR)
Lead Center: GRC
Participating Center(s): ARC, JPL
Related Subtopic Pointer(s): H5.01, Z1.04

Photovoltaic Energy Conversion

Photovoltaic cell and blanket technologies lead to significant improvements in overall solar array performance by increasing photovoltaic cell efficiency greater than 30%, increasing array mass specific power greater than 300W/ kg, decreased stowed volume, reduced initial and recurring costs, long- term operation in radiation environments, high power arrays, and a wide range of space environmental operating conditions are solicited.

Being sought are proposals that show advances in, but not limited to, the following:

- Photovoltaic cell and blanket technologies capable of low intensity, low-temperature operation applicable to outer planetary (low solar intensity) missions.
- Photovoltaic cell, and blanket technologies that enhance and extend performance in lunar applications including orbital, surface, and transfer.
- Solar arrays to support Extreme Environments Solar Power type missions, including long-lived, radiation tolerant, and cell and blanket technologies applicable to Jupiter missions.
- Lightweight solar array technologies applicable to science missions using solar electric propulsion.

Current missions being studied require solar arrays that provide 1 to 20 kilowatts of power at 1 AU, greater than 300 watts/kilogram specific power, operation in the range of 0.7 to 3 AU, and low stowed volume.

These technologies are relevant to any space science, earth science, planetary surface, or other science mission that requires affordable high-efficiency photovoltaic power production or radioisotope heat sources for orbiters, flyby craft, landers, and rovers. Specific requirements can be found in the references listed below but include many future SMD. Specific requirements for orbiters and flybys to Outer planets include: LILT capability (>38% at 10 AU and <−140° C), radiation tolerance (6e15 1 MeV e-cm^2), high power (>50 kW at 1 AU), low mass (3× lower than SOP), low volume (3× lower than SOP), long life (>15 years), and high reliability.

These technologies are relevant and align to any Space Technology Mission Directorate (STMD) or Human Exploration and Operations Mission Directorate (HEOMD) missions that require affordable, high-efficiency photovoltaic power production. NASA applications for a radioisotope heat source include orbiters, flyby craft, landers, and rovers.

Expected TRL for this project is 3 to 5.

Dynamic Power Conversion

For space applications where solar power is not practical, power convertors are used to convert heat to electrical power with dynamic engines combined with alternators typically providing significantly higher efficiency than current static devices. Being sought are proposals that show advances in, but not limited to, the following:

- Novel Stirling, Brayton or Rankine convertors that can be integrated with one or more 250 watt-thermal General-Purpose Heat Source (GPHS) modules to provide high thermal-to-electric efficiency (>25%), low mass, long life (>10 yrs), and high reliability for planetary spacecraft, landers, and rovers.
- Miniature dynamic power convertors that can be integrated with one or more 1 watt-thermal Radioisotope Heater Units (RHU) to provide long duration electric power for planetary smallsats and distributed instruments.
- Advanced dynamic conversion components including hot-end heat exchangers, cold-end heat exchangers, regenerators/recuperators, alternators, engine controllers, heat pipes, and radiators that improve system performance, reliability, and fault tolerance.

This technology directly aligns with the Science Mission Directorate - Planetary Science Division for space power and energy storage. Investments in more mature technologies through the Radioisotope Power System Program is ongoing. This SBIR subtopic scope provides a lower TRL technology pipeline for advances in this important power capability that improves performance, reliability, and robustness.

Expected TRL for this project is 1 to 4.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

References:

Photovoltaic Energy Conversion

- Solar Power Technologies for Future Planetary Science Missions, found at: https://solarsystem.nasa.gov/resources/548/solar-power-technologies-for-future-planetary-science-missions/
- NASA outlines New Lunar Science, Human Exploration Missions, found at: https://www.nasa.gov/feature/nasa-outlines-new-lunar-science-human-exploration-missions
- NASA Science Missions, found at: https://science.nasa.gov/missions-page?field_division_tid=All&field_phase_tid=3951

Dynamic Power Conversion

- https://rps.nasa.gov/about-rps/overview/
- Oriti, Salvatore, "Dynamic Power Convertor Development for Radioisotope Power Systems at NASA Glenn Research Center," AIAA P&E 2018, AIAA 2018-4498.
- Wilson, Scott D., "NASA Low Power Stirling Convertor for Small Landers, Probes, and Rovers Operating in Darkness," AIAA P&E 2018, AIAA 2018-4499.
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S3.03 Power Electronics and Management, and Energy Storage (SBIR)
Lead Center: GRC
Participating Center(s): GRC, GSFC, JPL
Related Subtopic Pointer(s): S4.04, Z1.03

Power Electronics and Management

NASA's Planetary Science Division is working to implement a balanced portfolio within the available budget and based on a decadal survey that will continue to make exciting scientific discoveries about our solar system. This balanced suite of missions shows the need for low mass/volume power electronics and management systems and components that can operate in extreme environment for future NASA Science Missions.

Advances in electrical power technologies are required for the electrical components and systems of these future spacecraft/platforms to address program size, mass, efficiency, capacity, durability, and reliability requirements. Radioisotope power systems (RPS), Advanced Modular Power Systems (AMPS) and In-Space Electric Propulsion (ISP) are several programs of interest which would directly benefit from advancements in this technology area. These types of programs, including Mars Sample Return using Hall thrusters and power processing units, require advancements in components and control systems beyond the state-of-the-art. Of importance are expected improvements in system robustness, energy density, speed, efficiency, or wide-temperature operation (-125° C to over 450° C) with a number of thermal cycles. Science Mission Directorate (SMD) has a need for intelligent, fault-tolerant Power Management and Distribution (PMAD) technologies to efficiently manage the system power for deep space missions.

Overall technologies of interest include:

- High power density/high efficiency modular power electronics and associated drivers for switching elements.
- Non-traditional approaches to switching devices, such as addition of graphene and carbon nanotubes to material.
- Materials for lightweight, flexible, low voltage (less than 5 volts) power transmission.
- Intelligent power management and fault-tolerant electrical components and PMAD systems.
- Advanced electronic packaging for thermal control and electromagnetic shielding.

The possible programs that could benefit from this technology include AMPS, Solar Electric Propulsion, RPS, and CubeSat/NanoSat Programs.

The expected TRL for this project is 3 to 5.

Energy Storage

Future science missions will require advanced primary and secondary battery systems capable of operating at temperature extremes from -100° C for Titan missions to 400 to 500° C for Venus missions, and a span of -230° C to +120° C for Lunar Quest. Advancements to battery energy storage capabilities that address operation at extreme temperatures combined with high specific energy and energy density (>200 Wh/kg and >200 Wh/l) are of interest in this solicitation.

In addition to batteries, other advanced energy storage/load leveling technologies designed to the above mission requirements, such as mechanical or magnetic energy storage devices, are of interest. These technologies have the potential to minimize the size and mass of future power systems.

Research should be conducted to demonstrate technical feasibility during Phase I and show a path toward a Phase II, and when possible, deliver a demonstration unit for NASA testing at the completion of the Phase II contract. Phase II emphasis should be placed on developing and demonstrating the technology under relevant test conditions. Additionally, a path should be outlined that shows how the technology could be commercialized or further developed into science-worthy systems.

The possible programs that could benefit from this technology include Solar Electric Propulsion, AMPS, RPS, and CubeSat/NanoSat Programs.

The expected TRL for this project is 3 to 5.

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Power Electronics and Management

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Energy Storage

- https://www.nasa.gov/smallsat-institute/smd
- https://science.nasa.gov/
- https://www1.grc.nasa.gov/space/sep/

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

Z1.03 Kilowatt-Class Energy Conversion for Small Fission Reactors (SBIR)
Lead Center: GRC
Participating Center(s): JPL
Related Subtopic Pointer(s): S3.03, Z8.03

NASA is considering the use of kilowatt class Fission Power Systems for surface missions to the moon and Mars. This technology directly aligns with the Space Technology Mission Directorate (STMD) roadmap for space power and energy storage and could be infused into the Kilopower Project to enhance performance or reliability. Current work in fission power systems is focused on the Kilopower project which uses a highly enriched Uranium-Molybdenum reactor core with a Beryllium oxide reflector. Depleted uranium, tungsten, and lithium hydride provide shielding of gamma rays and neutrons to the power conversion system, control electronics, payload, and habitat. Heat is removed from the core at approximately 800° C using sodium heat pipes and delivered to the power conversion system. Waste heat is removed from the power conversion system at approximately 100 to 200° C using water heat pipes coupled to aluminum or composite radiator panels. The Kilopower project targets the 1-10 kW electrical power range with most previous work focused on a demonstration of the 1 kWe design. The current solicitation is focused on innovations that enable the scaling of the 1 kWe design to 10 kWe, with a specific focus on surface power applications. Areas of interest include:

- Robust, efficient, highly reliable, and long-life thermal-to-electric power conversion, controller, and PMAD technology. Power conversion can consist of multiple lower power units which could be combined to create 10 kW of electric power. Stirling, Brayton, and thermoelectric convertors that can be coupled to Kilopower reactors are of interest.
- Reduction in shield mass through increased distance from core with mass effective Power Management and Distribution (PMAD) and transmission or lightweight possibly retractable booms.
- Radiation shield materials selection, design, and fabrication for mixed neutron and gamma environments, with consideration for mass effectiveness, manufacturability, and cost.
- Radiation tolerant electronics designed to withstand an induced radiation environment in addition to the ambient environment in space. Target dose tolerance ranges for fission power system electronics are between 1E11 to 1E13 n/cm^2 total neutron fluence, and between 100 kRad(Si) and 1000kRad(Si) total ionizing gamma dose.

Proposed concepts must identify, compare, and contrast advantages over key metrics pertinent to the technology concept.

The desired deliverables are primarily a prototype hardware to demonstrate concept feasibility. The appropriate research and analysis required to develop the hardware is also desired. The expected TRL for this project is 3 to 5.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

References:

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- Gibson, M.A., et al., "NASA's Kilopower Reactor Development and the Path to Higher Power Missions," 2017 IEEE Aerospace Conference, 4-11 March 2017, Big Sky, MT.
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Z1.04 Long Duration Lunar Energy Storage and Discharge (SBIR)
Lead Center: GRC
Participating Center(s): GRC, JSC
Related Subtopic Pointer(s): S3.01, Z5.05

NASA is seeking innovative solutions for long duration energy storage to support lunar surface operations, including landers, habitats, science platforms, robotic and crewed rovers, and the utilization of in-situ resources. Effective solutions require high-capacity, high-energy density, and long-life energy storage systems with very high reliability. Many of these systems will be required to provide continuous power during a 354-hour lunar night prior to recharge via a solar array during the lunar day. Power levels for these assets range from 500 W to 10 kW. Rovers capable of providing prolonged excursions before returning to the base will also be required. Power requirements for rovers are expected to be in the 1-3 kW range for unpressurized rovers increasing to 7-20 kW for pressurized rovers. It is anticipated that these vehicles will be refueled at the base, possibly directly re-fueled with scavenged propellants, or in-situ produced fuels, or electrically recharged via a solar array or other power technology. Technologies of interest in this solicitation are primary and regenerative fuel cells and rechargeable batteries. Technologies should be lightweight, low cost, and have service lives >5 years to survive multiple crew campaigns. Strong consideration should be given to environmental robustness for surface environments that include day/night thermal cycling, unshielded natural radiation, partial gravity, vacuum, and dust.

Advanced secondary/rechargeable batteries that go beyond lithium-ion and can safely provide >400 Wh/kg at the cell level are of interest for these missions. Secondary batteries that have 4-year shelf life and can provide >1,000 cycles at 70% depth-of-discharge are highly desirable. These secondary batteries are expected to operate safely over a temperature range of -20° C to +70° C with excellent capacity retention, comparable to room temperature operation.

Technological advances are also sought for Primary Fuel Cell (PFC) and Regenerative Fuel Cell (RFC)-based systems and sub-systems that contribute to system simplicity and improved reliability through:

- Innovative, integrated system-level design concepts.
- Passive ancillary components.

An example of these advances at the system level is primary and/or regenerative fuel cell systems that minimize or eliminate reactant re-circulation external to the stacks themselves. Examples at the component level include replacement of pumps and other active, motorized mechanical ancillary components with passive devices that perform the functions of both reactant management and thermal control.

Solutions are sought for PFCs using solid electrolytes in the power classes of 1 to 10 kW. Target specific power for the Lunar applications is >2,000 W/kg with an efficiency of >70% at 1,500 W/kg. PFC nominal current density is >200 mA/cm^2 with peak transients of >750 mA/cm^2. A final operational life of >10,000 hours is desired. Proposers should specify the path to meet this requirement at the system level. Reactant chemistries of interest in this solicitation are H_2/O_2 and CH_4/O_2, as well as other propellants. The ability to operate on scavenged propellants is highly desirable.

RFC systems are also of interest to meet long duration surface power energy storage needs with minimal opportunity for servicing or maintenance. PFC, water electrolyzers, and associated balance-of-plant hardware constitute a RFC system. The most direct approach to achieving mission efficiency, life, and reliability goals is to implement fuel cell, electrolysis, and RFC integrated fluid system functions through passive means and the elimination of as many ancillary and rotating components as possible. The range of energy storage of interest is 36 kW•hr(net) to >350 kW•hr(net) with a system level specific energy of > 600 Wh/kg, Target round trip efficiency is >51% (HHV) at 600 Wh/kg. Discharge power levels are anticipated to be between 100 W and 1 kW for an RFC. The final desired mission

operational lifetime for the RFC is >60 cycles @ 680 hours per cycle with a service/maintenance Interval ≥ 5 years. Proposers should specify the path to meet the life and service requirements at the system level. RFC development should focus exclusively on proton exchange-membrane (PEM) technology utilizing pure hydrogen, oxygen, and water as reactants. Electrolyzer self-pressurization is required to meet round-trip efficiency targets. Any proposal including a full RFC system operation must identify mechanisms to managing high-pressure water quality over the mission duration and de-humidification of gases prior to storage if gases are stored beyond the controlled thermal envelope.

RFC Subsystem Requirements are as follows:

Fuel Cells

- Power Levels: 100 W to 1 kW
- Specific Power: >2,000 W/kg
- Efficiency: >60% at 1,500 W/kg
- Operational Life: >10,000 hours (Specify path to meet this requirement at the system level)
- Nominal Current Density (Peak transient): 150 to 250 mA/cm^2 (>850 mA/cm^2)
- Applicable Chemistries: Solid electrolyte (non-liquid) including polymeric (ionic and anionic) and ceramic
- Reactant Chemistries: pure H_2 and O_2

Electrolyzers

- Production Rates: Generate sufficient reactant to support fuel cell operation with capability of >15% margin
- Efficiency: >70% at 1,500 W/kg
- Operational Pressure: ≥ 2,000 psig (sustained)
- Pressure Configurations: balanced (anode ≈ cathode) Preference given to a design that can also operate in an unbalanced mode with a fully pressurized oxygen cavity and ambient pressure (~15 psia) hydrogen cavity.
- Operational Life: >10,000 hours (Specify path to meet this requirement at the system level)
- Applicable Chemistries: Solid electrolyte (non-liquid) including polymeric (ionic and anionic) capable of meeting the pressure requirement Reactant Feed Configurations: Liquid Anode Feed, Vapor cathode feed
- Feedstock Chemistries: H_2O

The energy storage technologies described in this subtopic have applicability over a broad range of mobile and stationary lunar surface systems. It is believed that Space Technology Mission Directorate (STMD) is the relevant directorate to develop and mature this technology given its wide scope of applicability. It is anticipated that these technologies will be required to enable initial lunar exploration and the establishment of a human presence on the moon. After appropriate development and maturation in STMD, the technology would most likely be transitioned to specific programs in Human Exploration and Operations Mission Directorate (HEOMD) and Science Mission Directorate (SMD) responsible for developing the individual assets for lunar exploration. These assets include landers, unpressurized and pressurized rovers, robotic rovers, and various science platforms.

The desired deliverables would be a prototype of the types of technologies described above and/or a lunar payload package of the technology developed which can meet the size and related limitations of the lunar payload statement in the above subtopic description. The goal is to mature technologies from analytical or experimental proof-of-concept (TRL 3) to breadboard demonstration in a relevant environment (TRL 5). Research should be conducted to demonstrate technical feasibility during Phase I and show a path toward a Phase II hardware demonstration, and when possible, deliver a demonstration unit for functional and environmental testing at the completion of the Phase II contract.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities

are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

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Focus Area 3 Autonomous Systems for Space Exploration

Lead MD: HEOMD
Participating MD(s): SMD, STTR

The exploration of space requires the best of the nation's technical community to provide the technologies that will enable human and robotic exploration beyond Low Earth Orbit (LEO): to establish a lunar presence, to visit asteroids, to extend human reach to Mars, and for increasingly ambitious robotic missions such as a Europa Lander. Autonomous Systems technologies provide the means of migrating mission control from Earth to spacecraft, habitats, and robotic explorers. This is enhancing for missions in the Earth-Lunar neighborhood and enabling for deep space missions. Long light-time delays, for example up to 42 minutes round-trip between Earth and Mars, require time-critical control decisions to be closed on-board autonomously, rather than through round-trip communication to Earth mission control. For robotic explorers this will be done through automation, while for human missions this will be done through astronaut-automation teaming.

Long-term crewed spacecraft and habitats, such as the International Space Station, are so complex that a significant portion of the crew's time is spent keeping it operational even under nominal conditions in low-Earth orbit, while still requiring significant real-time support from Earth. The considerable challenge is to migrate the knowledge and capability embedded in current Earth mission control, with tens to hundreds of human specialists ready to provide instant knowledge, to on-board automation that teams with astronauts to autonomously manage spacecraft and habitats. For outer planet robotic explorers, the opportunity is to autonomously and rapidly respond to dynamic environments in a timely fashion.

Machine learning has made spectacular advances for terrestrial applications, exceeding human capabilities in tasks such as image classification. Machine learning could become an increasingly important aspect of space exploration, from finding novel patterns in the science data transmitted from robotic spacecraft, to the operation of sustainable

habitats. Machine learning and inferencing calls for new computing paradigms; for space, radiation tolerant processors will be enabling.

In order to enable on-board autonomy, both software advances and computing advances need to be addressed.

The autonomous agent subtopic addresses this challenge by soliciting proposals that leverage the growing field of cognitive computing to advance technology for deep-space autonomy.

Fault management is an integral part of space missions. The fault management subtopic spans the lifecycle of fault management for space missions from design through verification and validation to operations. In the past, the predominant operational approach to detected faults has been to safe the spacecraft, and then rely on Earth mission control to determine how to proceed. New mission concepts require future spacecraft to autonomously decide how to recover from detected anomalies and continue the mission. The fault management subtopic solicits proposals that advance fault management technology across architectures, design tools, verification and validation, and operations.

The sustainable habitat subtopic calls for machine learning technology in order to substantially improve diagnostic and prognostic performance for integrated systems health management. This subtopic solicits technology for long-term system health management that goes beyond short-term diagnosis technology to include advances machine learning and other prognostic technologies. Enhancing the capability of astronauts is also critical for future long-duration deep space missions.

The Deep Neural Network accelerator and Neuromorphic computing subtopic addresses extrapolating new terrestrial computing paradigms related to machine learning to the space environment. For machine inferencing and learning computing hardware proposals, metrics related to energy expenditure per operation (e.g., multiply-add) and throughput acceleration in a space environment are especially relevant.

The subtopic on swarms of space vehicles addresses technologies for control and coordination of planetary rovers, flyers, and in-space vehicles in dynamic environments. Co-ordinated swarms can provide a more robust and sensor-rich approach to space missions, allowing simultaneous recording of sensor data from dispersed vehicles and co-ordination especially in challenging environments such as cave exploration.

H6.01 Integrated Systems Health Management for Sustainable Habitats (SBIR)
Lead Center: ARC
Participating Center(s):
Related Subtopic Pointer(s): H6.03, S5.05

Novel Machine Learning Concepts for Automated Space Habitats

Methods and tools are needed that can adapt to novel, but benign configuration changes on space habitat environmental control and life support systems. They should automatically learn how to make the distinction between these nominal conditions that should not be reported and abnormal conditions that need to be reported. Where reports are needed, these are ideally delivered early to provide ample time for the system (or operators) to react to either prevent an impending fault or to prepare to take mitigating action.

Furthermore, proposed tools and techniques should be capable of carrying out adaptation and selection of conditions that need to be reported in an automated way. These tools should have the capability to:

- Recognize acceptable changes.
- Extract relevant features.
- Establish novel threshold conditions upon which to act, either in the parameter space or probabilistically.

Methods based upon machine learning and data mining should aim to reveal latent, unknown conditions while still retaining and improving the ability to provide highly accurate alerts for known issues. However, it is recognized that the cataloguing and selection of threshold parameters to characterize abnormal conditions for known issues is a daunting task, regardless of which space they are represented in. For any given representation, such limit checks are still vulnerable to false positives (incorrectly calling a fault) as well as false negatives (missing the occurrence of a

fault). Both of these types of errors need to be managed and minimized to acceptable levels while also keeping the early warning metric in mind. As such, mechanisms are needed to assure that these techniques will perform as desired relative to these metrics. For the techniques proposed, the performance targets for known faults and failures will be based upon the following specified performance metrics:

- False alarm rate.
- Missed detection rate.
- Detection time (first time prior to the adverse event that the algorithm indicates an impending fault/failure).

Methods should also explore the trade space for Integrated Systems Health Management data and processing needs in order to provide guidance for future habitat sensor and computational resource requirements. Proposals may address specific system health management capabilities required for habitat system elements (life support systems, etc.). In addition, projects may focus on one or more relevant subsystems such as the ones previously described. The Sustainability Base is a green building test-bed whose requirements, as a low-power and low-consumable habitat, are included in those for deep space habitats. Data available includes photo-voltaic array, electrical power, grey water recycling, environmental data (temp, CO_2, etc.) and facility equipment sensors (flowrates, differential pressures, temperatures, etc.). There is also the possibility that data from deep space habitat laboratories and prototypes might become available. Specific technical areas of interest related to integrated systems health management include the following:

- Machine learning and data mining techniques that are capable of learning from operations data to identify statistical anomalies that may represent previously unknown system degradations
- Methods should facilitate the incorporation of human feedback on the operational significance of the statistical anomalies using techniques such as active learning
- Demonstration of advanced predictive capability using machine learning or data mining methods for known system fault or failure modes, within prescribed performance constraints related to detection time and accuracy
- Prognostic techniques able to predict system degradation, leading to system robustness through automated fault mitigation and improved operational effectiveness. Proposals in this area should focus on systems and components commonly found in space habitats or EVA platforms.
- Innovative human-system integration methods that can convey a wealth of health and status information to mission support staff quickly and effectively, especially under off-nominal and emergency conditions.

Deliverables are expected in the Technology Readiness Levels (TRL) range of of 4-6 or higher and ideally include working integrated software framework capable of direct compatibility with existing programmatic tools by the end of Phase II.

References:

- https://www.nasa.gov/feature/nasa-s-lunar-outpost-will-extend-human-presence-in-deep-space
- The solution should be able to interface with software by employing the tools at the links provided below, to enable compatible programmatic interfaces:
    - https://cfs.gsfc.nasa.gov/
    - https://github.com/nasa/trick

H6.03 Spacecraft Autonomous Agent Cognitive Architectures for Human Exploration (SBIR)
Lead Center: JSC
Related Subtopic Pointer(s): H6.01, T11.04, Z5.04

This subtopic solicits intelligent autonomous agent cognitive architectures that are open, modular, and make decisions under uncertainty. It should be feasible for cognitive agents based on these architectures to be certified or licensed for use on deep space missions to act as liaisons that interact both with the mission control operators, the crew, and most if not all of the spacecraft subsystems. With such a cognitive agent that has access to all onboard data and communications, the agent could continually integrate this dynamic information and advise the crew and mission control accordingly by multiple modes of interaction including text, speech, and animated images. This agent could

respond to queries and recommend to the crew courses of action and direct activities that consider all known constraints, the state of the subsystems, available resources, risk analyses, and goal priorities.

Future deep space human missions will place crews at long distances from Earth causing significant communication lag due to the light distance as well as occasional complete loss of communication with Earth. Novel capabilities for crews and ground staff will be required to manage spacecraft operations including spacecraft and systems health, crew health, maintenance, consumable management, payload management, as well as activities such as food production and recycling. Autonomous agents with cognitive architectures could interface directly with the crew as well as with the onboard systems reducing the cognitive loads on the crew as well as perform many of the tasks that would otherwise require scheduling crew time. In addition, this cognitive computing capability is necessary in many circumstances to respond to off-nominal events that overload the crew, particularly when the event limits crew activity, such as high-radiation events or loss of atmospheric pressure events requiring crew safety or sequestration.
In deep space, crews will be required to manage, plan, and execute the mission more autonomously than currently required on the International Space Station (ISS) due to more distant and longer latency ground support provided. NASA will migrate current operations functionality from the flight control room to the spacecraft to be performed by the crew and autonomous agents supervised by the crew, so the crew is not overburdened. Cognitive agents that can effectively communicate with the crew could perform tasks that would otherwise require crew time by providing assistance, operating systems, providing training, performing inspections, and providing crew consulting among other tasks.

Current typical computers agents can easily perform super-human memory recall and computation feats, but at the same time appear to be severely cognitively impaired in that they fail to recognize the values, implications, severity, reasonableness, and likelihood of the assertions they hold and how inferences can be applied. The consequence is that computer agents often fail to recognize what is obvious and important to humans, appear to be easily deceived, and fail to recognize and learn from mistakes. Thus, crew interface to such typical computer agents for the current state of the art can be burdensome.

This subtopic seeks proposals for effective cognitive architectures that can start to provide autonomous computer agents the common-sense humans take for granted amongst ourselves. Likely such agents would maintain some type of prioritized probabilistic belief network that they continually update based on evidence and inference in order to make decisions and respond to queries that take into account the assessed risks of the assertions they believe to be true.

Due to the complexity of such systems and the need for them to be continually updated, the architecture is required to be modular such that modules can dynamically be added, removed, and enhanced. Such a cognitive architecture is consistent with that proposed by Prof. Marvin Minsky in "The Society of Mind", 1988. The cognitive architecture is required to be capable of supporting multiple processes executing on multiple processors to be able to a meet the expected computational loads as well as be robust to processor failure.

An effective cognitive architecture would be capable of integrating a wide variety of artificial intelligence modules depending on mission requirements. The following modules provide capabilities useful for a wide variety of spacecraft cognitive agents:

- Goal manager: enables the simultaneous execution of multiple goals, e.g., keep crew safe, get tasks A and B done
- Planner/scheduler: creates and updates plans and schedules that accomplish tasks
- Smart executive: robustly executes high-level plans on schedule by coordinated commanding of multiple subsystems
- Sensor processing: separating signal from noise in sensor data, extracting and compressing useful information
- Actuator controllers: low-level commanding of subsystems that change the environment, support feed-forward control
- Skill/behavior manager: the coordination of multiple actuator controllers, e.g., manipulation activities
- Internal/external communication manager: coordinates information exchanges with humans and other agents
- Intra-spacecraft path planner/trajectory generator: develops 3D movement plans for humans and machines
- Internal/external resource manager: controls the use of resources such as memory, power, and consumables.

- Image recognition manager: manages extracting information from images
- Image generation manager: dynamically creating images to convey information to humans, e.g., charts, animations
- Declarative knowledge/rule manager: ensures that the system's declarative knowledge is consistent and updated
- Risk manager: assesses the uncertainty and severity of held assertions and the implication of actions or inaction
- Value manager: assess the importance humans place on goals, assertions, activities, etc.
- Symbol manager: create and use symbols created by humans and other agents to effectively convey information
- Script manager: create and update command sequences to reduce computation required to perform tasks
- Explanatory story manager: increase human communication effectiveness through stories and analogies
- Model manager: create and update models of itself, humans, other systems, and the environment
- System health manager: maintains overall system health, performs diagnoses and prognoses
- Crew health manager: monitors crew health, alerts crew to imminent threats, ...
- Communication signal manager: maintains health of communication paths, develops contingencies
- State estimator: maintains a consistent state of the models it manages for itself and its world
- Attention manager: manages its processing power to prevent overloading itself
- Security manager: monitors and prevents threats to itself, humans, and systems it manages
- Internal simulators: simulates plan execution under various conditions prior to actual plan execution

Cognitive architectures capable of being certified for crew support on spacecraft are required to be open to NASA with interfaces open to NASA partners who develop modules that integrate with the agent, in contrast to proprietary black-box agents. A cognitive agent suitable to provide crew support on spacecraft may be suitable for a wide variety of Earth applications, but the converse is not true requiring this NASA investment.

Proposals should emphasize analysis and demonstration of the feasibility of various configurations, capabilities, and limitations of a cognitive architecture suitable for crew support on deep space missions. The software engineering of a cognitive architecture is to be documented and demonstrated by implementing a prototype goal-directed software agent that interacts as an intermediary/liaison between simulated spacecraft systems and humans.

For Phase I, a preliminary cognitive architecture, preliminary feasibility study, and a detailed plan to develop a comprehensive cognitive architecture feasibility study are expected. A preliminary demonstration prototype of the proposed cognitive architecture is highly encouraged.

For Phase II, the Phase I proposed detailed feasibility study plan is executed generating a comprehensive cognitive architecture, comprehensive feasibility study report including design artifacts such as SysML/UML diagrams, a demonstration of an extended prototype of an agent that instantiates the architecture interacting with a spacecraft simulator and humans executing a plausible HEOMD design reference mission beyond cislunar orbit (e.g., Human Exploration of Mars Design Reference Mission: https://www.nasa.gov/pdf/373665main_NASA-SP-2009-566.pdf), and a detailed plan to develop a comprehensive cognitive architecture feasibility study suitable for proposing to organizations interested in funding this flight capability is expected. A Phase II prototype suitable for a compelling flight experiment on the ISS is encouraged.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-5.

References:

IBM (Watson), Apple (Siri), Microsoft (Cortana), and Amazon (Alexa) are just a few of the companies developing intelligent autonomous agents. However, they generally are proprietary and would not meet the requirements for spacecraft software that could potentially put the crew and mission at risk. There is a need to provide cognitive computing for systems like Robonaut.

A survey of cognitive architectures https://arxiv.org/pdf/1610.08602.pdf. Conferences that include cognitive architecture papers include IJCAI, AAAI, as well as the ongoing CogArch series of workshops.

H6.22 Deep Neural Net and Neuromorphic Processors for In-Space Autonomy and Cognition (SBIR)
Lead Center: GRC
Related Subtopic Pointer(s): H9.07, S3.08, Z8.03

Machine Inferencing and Neuromorphic Capabilities

The Deep Neural Net and Neuromorphic Processors for In-Space Autonomy and Cognition subtopic is focused on computing advances for the space environment based on neurological models in contrast to von Neumann architectures. Deep neural net and neuromorphic processors can enable a spacecraft to sense, adapt, act and potentially learn from its experiences and from the unknown environment without needing a ground mission operations team. Neuromorphic processing will enable NASA to meet growing demands for applying artificial intelligence and machine inferencing and learning algorithms on board a spacecraft that is energy efficient. These demands include enabling on-board cognitive systems to improve mission communication and data processing capabilities, provide sensory processing onboard to optimize communication bandwidth and latency, enhance computing performance, and reduce memory requirements. Additionally, deep neural net and neuromorphic processors show promise for minimizing power requirements that traditional computing architectures now struggle to meet in space applications.

The goal of this subtopic is to develop deep neural net and neuromorphic processing hardware, software, algorithms, architectures, simulators, and techniques as an enabling capability for autonomy in the space environment. Additional areas of interest for research and/or technology development include:

- Deep neural net and neuromorphic processing approaches to enhance data processing, computing performance, and memory conservation.
- Spiking neural net algorithms that learn from the environment and improve operations.
- New brain-inspired chips and breakthroughs in machine understanding and intelligence.
- Novel memristor, MRAM, and other radiation tolerant devices that can be incorporated in neuromorphic processors which show promise for space applications.

This subtopic seeks innovations focusing on low size, weight, and power (SWaP) processing suitable for CubeSat operations or direct integration with sensors in the harsh space environment. Focusing on SWaP-constrained platforms opens the potential for applying neuromorphic processors in spacecraft control situations traditionally reserved for power-hungry general-purpose processors. This technology will allow for increased speed, energy efficiency, and higher performance for computing in unknown and uncharacterized space environments.

Phase I will emphasize research aspects for technical feasibility and show a path towards a Phase II proposal. Phase I deliverables include concept of operations of the research topic, simulations and preliminary results. Early development and delivery of prototype hardware/software is encouraged.

Phase II will emphasize hardware and/or software development with delivery of specific hardware and/or software products for NASA targeting demonstration operations on a CubeSat platform. Phase II deliverables include a working prototype of the proposed product and/or software, along with documentation and tools necessary for NASA to use the product and/or modify and use the software. Hardware products should include both layout and simulation. Sample chips – from device level on up – are encouraged. Software products shall include source for government use. Proposed prototypes shall demonstrate a path towards a CubeSat mission. Proposals should include a strategy for tolerance to radiation and other adverse aspects of the space environment.

Background, State of the Art, and References

The current state-of-the-art (SOA) for in-space processing is the High-Performance Spaceflight Computing (HPSC) processor being developed by Boeing for NASA Goddard Space Flight Center (GSFC). The HPSC, called the Chiplet, contains 8 general purpose processing cores in a dual quad-core configuration; initial hardware delivery is expected by December 2020. In a submission to the Space Technology Mission Directorate (STMD) Game Changing Development (GCD) program, the highest computational capability required by current typical space mission is 35-70 GFLOPS (billion floating-point operations per second).

The current SOA does not address the capabilities required for artificial intelligence and machine inferencing and learning applications in the space environment. These applications require significant amounts of multiply and accumulate operations, in addition to a substantial amount of memory to store data and retain intermediate states in a neural network computation. Terrestrially, these operations require general-purpose graphics processing units (GP-GPUs), which are capable of TFLOPS (10^{12}) -- approximately 3 orders of magnitude above the anticipated capabilities of the HPSC.

Neuromorphic processing offers the potential to bridge this gap through novel hardware approaches. Existing research in the area shows neuromorphic processors to be up to 1000 times more energy efficient than GP-GPUs in artificial intelligence applications. Obviously, the true performance depends on the application, but nevertheless neuromorphic processing has demonstrated characteristics that make it well adapted to the power-constrained space environment.

Neuromorphic computing is a technology to tackle the explosion in computing performance and memory requirements to meet growing demands for artificial intelligence and machine learning. While the commercial market for these processors is in its infancy, there is a growing community of small businesses that have been funded by Air Force and Department of Energy grants toward development of neuromorphic capabilities. These companies continue to make great strides in neuromorphic processor technology including new devices such as memristors. This subtopic would put NASA in a position to join its partners in the DoD and DoE to enable a research area that shows tremendous application for space.

The Cognitive Communications Project, through the Human Exploration and Operations Mission Directorate (HEOMD) Space Communications and Navigation (SCaN) Program, is one potential customer of work from this subtopic area. Neuromorphic processors are a key enabler to the cognitive radio and system architecture envisioned by this project. As communications become more complex, cognition and automation will play a larger role to mitigate complexity and reduce operations costs. Machine learning will choose radio configurations, adjust for impairments and failures. Neuromorphic processors will address the power requirements that traditional computing architectures now struggle to meet.

The expected TRL for proposals is 4-6.

References:

- Several reference papers that have been published at the Cognitive Communications for Aerospace Applications (CCAA) workshop are available at: http://ieee-ccaa.com.
- A survey paper on neuromorphic computing and neural networks in hardware: https://arxiv.org/pdf/1705.06963
- References for deep neural network and neuromorphic computing can be found in IEEE, ACM, and conference archives such as NIPS and ICONS (International Conference on Neuromorphic Systems).

S5.05 Fault Management Technologies (SBIR)
Lead Center: JPL
Participating Center(s): ARC, MSFC
Related Subtopic Pointer(s): H6.01, T4.03, Z8.07

Development, Design, and Implementation of Fault Management Technologies

NASA's science program has well over 100 spacecraft in operation, formulation, or development, generating science data accessible to researchers everywhere. As science missions are given increasingly complex goals, often on compressed timetables, and have more pressure to reduce operations costs, system autonomy must increase in response. Fault Management (FM) is one of the key components of system autonomy.

FM consists of operational mitigations of spacecraft failures and is implemented with spacecraft hardware and on-board autonomous software that controls hardware, software, and information redundancy, in concert with ground-based software and operations procedures. Despite a wealth of lessons learned from past missions, spacecraft failures

are still not uncommon, and reuse of FM approaches is very limited, illustrating that advancements are needed in FM Design Tools, FM Visualization Tools, FM Operations Approaches, FM Verification and Validation Tools, and FM Design Architectures.

The specific objectives of this subtopic are to improve FM technologies and approaches, as follows:

- Improve predictability of FM system complexity and estimates of development and operations costs
- Enable cost-effective FM design architectures and operations
- Determine completeness and appropriateness of FM designs and implementations
- Decrease the labor and time required to develop and test FM models and algorithms
- Improve visualization of the full FM design across hardware, software, and operations procedures
- Determine extent of testing required to verify FM, particularly where model-based, and estimate the potential risk resulting from incomplete coverage
- Increase data integrity between multi-discipline tools
- Standardize metrics and calculations across FM, SE, S&MA and operations disciplines
- Increase reliability of FM systems

Specific technology advancements in the areas listed below are needed to improve the capability of fielded FM systems. Guidance for development can be found in the NASA FM Handbook:

- *FM Design Tools* - System modeling and analysis significantly contributes to the quality of FM design, and may prove decisive in trades of new vs. traditional FM approaches. However, the difficulty in translating system design information into system models often impacts modeling and analysis accuracy. Examples of enabling techniques and tools are automated modeling systems, spacecraft modeling libraries, expedited algorithm development, sensor placement analyses, and system model tool integration.
- *FM Visualization Tools* - FM systems incorporate hardware, software, and operations mechanisms. The ability to visualize the full FM system and the contribution of each component to protecting mission functions and assets is critical to assessing the completeness and appropriateness of the FM design to the mission attributes (mission type, risk posture, operations concept, etc.). Fault trees and state transition diagrams are examples of visualization tools that contribute to visualization of the full FM design.
- *FM Operations Approaches* - Typical FM processes attempt to preserve the asset in the event of detected anomalies by safing the vehicle and relying on mission operations to determine how to proceed. However, many new mission concepts require greater autonomy – for example, riding out failures or autonomously restarting system behavior in order to complete science objectives that require timely operations. Future spacecraft must be able to make decisions about how to recover from failures or degradations and continue the mission. FM designs must enable flexible operations that can integrate on-board decision-making with input from mission operations.
- *FM Verification and Validation Tools* - Along with difficulties in system engineering, the challenge of V&V'ing new FM technologies has been a significant barrier to infusion in flight projects. As complexity of spacecraft and systems increases, the testing required to verify and validate FM implementations can become prohibitively resource intensive without new approaches. Automated test case development, false positive/false negative test tools, model verification and validation tools, and test coverage risk assessments are examples of contributing technologies.
- *FM Design Architectures* - FM capabilities may be implemented through numerous system, hardware, and software architecture solutions. The FM architecture trade space includes options such as embedding within the flight control software or deployment as independent onboard software; on-board versus ground-based capabilities; centralized or distributed FM functions; sensor suite implications; integration of multiple FM techniques; innovative software FM architectures implemented on flight processors or on Field Programmable Gate Arrays (FPGAs); and execution in real-time or off-line analysis post-operations. Alternative architecture choices such as model-based approaches could help control FM system complexity and cost and could offer solutions to transparency, verifiability, and completeness challenges.
- *Multi-discipline FM Interoperation* - FM designers, Systems Engineering, Safety and Mission Assurance, and Operations all perform analyses and assessments of system reliability, failure modes and effects, sensor coverage, failure probabilities, anomaly detection and response, contingency operations, etc. These analyses are highly sensitive to inconsistencies and misinterpretations of multi-discipline data, resulting in higher costs

to resolve disconnects in data and analyses, or even reducing mission success due to failure modes that were overlooked. Solutions that address data integrity, identification of metrics, and standardization of data products, techniques and analyses will reduce cost and failures.

Expected outcomes are better estimation and control of FM complexity and development costs, improved FM designs, and accelerated advancement of FM tools and techniques.

FM technologies are applicable to all Science Mission Directorate (SMD) missions, with particular emphasis on medium to large missions as these have much lower tolerance for risk, representing substantial potential benefit. A few examples are provided below, although these may be generalized to a broad class of missions:

- *Europa Exploration (Clipper and Lander)* - Provide on-board capability to detect and correct radiation-induced execution errors. Provide reliable reasoning capability to restart observations after interruptions without requiring ground in-the-loop. Provide MBSE tools to model and analyze FM capabilities in support of design trades, V&V of FM capabilities, and coordinated development with flight software.
- *Mars Exploration (Rovers and Sample Return)* - Provide on-board capability for systems checkout, enabling mobility after environmentally-induced anomalies (e.g., unexpected terrain interaction). Improve reliability of complex activities (e.g., drilling and sample capture, capsule pickup and remote launch).
- *Search for Extrasolar Planets (Observation)* - Provide sufficient system reliability through on-board detection, reasoning, and response to enable long-period, stable observations. Provide on-board or on-ground analysis capabilities to predict system response and optimize observation schedule. Enable reliable operations while out of direct contact (e.g., deliberately occluded from Earth to reduce photon, thermal, and radio frequency background).

It is intended that proposed efforts conduct an initial proof of concept, after which successful efforts would be considered for follow-on funding by SMD missions as risk-reduction and infusion activities. Research should be conducted to demonstrate technical feasibility and NASA relevance during Phase I and show a path toward a Phase II prototype demonstration.

Accordingly, the Final Report should thoroughly document the innovation, its status at the end of the effort, and as much objective evaluation of its strengths and weaknesses as is practical. The report should include a description of the approach, foundational concepts and operating theory, mathematical basis, and requirements for application. Results should include strengths and weaknesses found, measured performance in tests where possible.

Additional deliverables may significantly clarify the value and feasibility of the innovation. These deliverables should be planned to demonstrate retirement of development risk, increasing maturity, and targeted applications of particular interest. While the wide range of innovations precludes a specific list, some possible deliverables are listed below:

- For innovations that are algorithmic in nature, this could include development code or prototype applications, demonstrations of capability, and results of algorithm stress-testing.
- For innovations that are procedural in nature, this may include sample artifacts such as workflows, model prototypes and schema, functional diagrams, example or tutorial applications.
- Where a suitable test problem can be found, documentation of the test problem and a report on test results, illustrating the nature of the innovation in a quantifiable and reproducible way. Test reports should discuss maturation of the technology, implementation difficulties encountered and overcome, results and interpretation.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-4.

References:

- NASA Fault Management Handbook (https://www.nasa.gov/pdf/636372main_NASA-HDBK-1002_Draft.pdf)
- Talks presented at the 2012 FM Workshop:
    - https://www.nasa.gov/offices/oce/documents/2012_fm_workshop.html

  - https://www.nasa.gov/pdf/637595main_day_1-brian_muirhead.pdf
- NASA Technical Memorandum "Introduction to System Health Engineering and Management for Aerospace (ISHEM)" (https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20060003929.pdf)
- Johnson, S. (ed), System Health Management with Aerospace Applications, Wiley, 2011 (https://www.wiley.com/en-us/System+Health+Management%3A+with+Aerospace+Applications-p-9781119998730)

T4.03 Coordination and Control of Swarms of Space Vehicles (STTR)
Lead Center: JPL
Related Subtopic Pointer(s): S4.02, S5.05

Enabling Technologies for Swarm of Space Vehicles

This subtopic is focused on developing and demonstrating technologies that are enabling to cooperative operation of swarms of space vehicles in a dynamic environment. Primary interest is in technologies appropriate for low-cardinality (4-15 vehicle) swarms of small spacecraft, as well as planetary rovers and flyers (e.g., Mars helicopter); Large swarms and other platforms are of interest if well motivated in connection to NASA's strategic plan and needs identified in decadal surveys.

The proposed technology should be motivated by a design reference mission presented in the proposal.

Possible areas of interest include but are not limited to:

- High precision relative localization and time synchronization in orbit and on planet surface.
- Coordinated task planning, operation, and execution.
- Fast, real-time, coordinated motion planning in areas densely crowded by other agents.
- Human-Swarm interaction interfaces for controlling the multi-agent system as an ensemble.
- Distributed fault detection and mitigation due to hardware failures or compromised systems.
- Communication-less coordination by observing and estimating the actions of other agents in the multi-agent system.
- Cooperative manipulation and in-space construction.
- Close proximity operations of spacecraft swarms including sensors required for collision detection and avoidance.

Subtopic technology directly supports NASA Space Technology Roadmap TA-04 regarding Robotics and Autonomous Systems (4.5.4 Multi-Agent Coordination, 4.2.7 Collaborative Mobility, 4.3.5 Collaborative Manipulation) Strategic Space Technology Investment Plan (Core) Robotic and Autonomous Systems: Relative GNC and Supervisory control of an S/C team:

- Multi-robot follow-on to the M2020+Mars Helicopter programs are likely to necessitate close collaboration among flying robots as advance scouts and rovers.
- Pop-Up Flat-Folding Explorer Robots (PUFFERs) are being developed at JPL and promise a low-cost swarm of networked robots that can collaboratively explore lava-tubes and other hard to reach areas on planet surface.
- A convoy of spacecraft is being considered, in which the lead spacecraft triggers detailed measurement of a very dynamic event by the following spacecraft.
- Multiple concepts for distributed space telescopes and distributed synthetic apertures are proposed that rely heavily on coordination and control technologies developed under this subtopic.

Phase I awards will be expected to develop theoretical frameworks, algorithms, software simulation and demonstrate feasibility. The expected Technology Readiness Level (TRL) range at completion of the project is 2-3. Phase II awards will be expected to demonstrate capability on a hardware test bed. The expected Technology Readiness Level (TRL) range at completion of the project is 4-6.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

References:

- D. P. Scharf, F. Y. Hadaegh and S. R. Ploen, "A survey of spacecraft formation flying guidance and control (part 1): guidance," Proceedings of the 2003 American Control Conference, 2003, Denver, CO, USA, 2003, pp. 1733-1739.
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- "Precision Formation Flying," https://scienceandtechnology.jpl.nasa.gov/precision-formation-flying
- "Mars Helicopter to Fly on NASA's Next Red Planet Rover Mission," https://www.nasa.gov/press-release/mars-helicopter-to-fly-on-nasa-s-next-red-planet-rover-mission
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- S. Bandyopadhyay, R. Foust, G. P. Subramanian, S.-J. Chung, and F. Y. Hadaegh, "Review of Formation Flying and Constellation Missions Using Nanosatellites," Journal of Spacecraft and Rockets, vol. 53, no. 3, 2016, pp. 567-578.
- S. Kidder, J. Kankiewicz, and T. Vonder Haar. "The A-Train: How Formation Flying is Transforming Remote Sensing," https://atrain.nasa.gov/publications.php
- T. Huntsberger, A. Trebi-Ollennu, H. Aghazarian, P. Schenker, P. Pirjanian, and H. Nayar. "Distributed Control of Multi-Robot Systems Engaged in Tightly Coupled Tasks," Autonomous Robots 17, 79–92, 2004.
- Space Studies Board, "Achieving Science with CubeSats: Thinking Inside the Box," National Academies of Sciences, Engineering, and Medicine, 2016. http://sites.nationalacademies.org/SSB/CompletedProjects/SSB_160539

Focus Area 4 Robotic Systems for Space Exploration

Lead MD: STMD
Participating MD(s): SMD, STTR

This focus area includes development of robotic systems technologies (hardware and software) to improve the exploration of space. Robots can perform tasks to assist and off-load work from astronauts. Robots may perform this work before, in support of, or after humans. Ground controllers and astronauts will remotely operate robots using a range of control modes, over multiple spatial ranges (shared-space, line of sight, in orbit, and interplanetary) and with a range of time-delay and communications bandwidth. Technology is needed for robotic systems to improve transport of crew, instruments, and payloads on planetary surfaces, on and around small bodies, and in-space. This includes hazard detection, sensing/perception, active suspension, grappling/anchoring, legged locomotion, robot navigation, end-effectors, propulsion, and user interfaces.

In the coming decades, robotic systems will continue to change the way space is explored. Robots will be used in all mission phases: as independent explorers operating in environments too distant or hostile for humans, as precursor systems operating before crewed missions, as crew helpers working alongside and supporting humans, and as caretakers of assets left behind. As humans continue to work and live in space, they will increasingly rely on intelligent and versatile robots to perform mundane activities, freeing human and ground control teams to tend to more challenging tasks that call for human cognition and judgment.

Innovative robot technologies provides a critical capability for space exploration. Multiple forms of mobility, manipulation and human-robot interaction offer great promise in exploring planetary bodies for science investigations and to support human missions. Enhancements and potentially new forms of robotic systems can be realized through advances in component technologies, such as actuation and structures (e.g. 3D printing). Mobility provides a critical capability for space exploration. Multiple forms of mobility offer great promise in exploring planetary bodies for science investigations and to support human missions. Manipulation provides a critical capability for positioning crew members and instruments in space and on planetary bodies, it allows for the handling of tools, interfaces, and materials not specifically designed for robots, and it provides a capability for drilling, extracting, handling and processing samples of multiple forms and scales. This increases the range of beneficial tasks robots can perform and allows for improved efficiency of operations across mission scenarios. Manipulation is important for human missions, human precursor missions, and unmanned science missions. Sampling, sample handling, transport, and distribution to instruments, or instrument placement directly on in-place rock or regolith, is important for robotic missions to locales too distant or dangerous for human exploration.

Future space missions may rely on co-located and distributed teams of humans and robots that have complementary capabilities. Tasks that are considered "dull, dirty, or dangerous" can be transferred to robots, thus relieving human crew members to perform more complex tasks or those requiring real-time modifications due to contingencies. Additionally, due to the limited number of astronauts anticipated to crew planetary exploration missions, as well as their constrained schedules, ground control will need to remotely supervise and assist robots using time-delayed and limited bandwidth communications. Advanced methods of human-robot interaction over time delay will enable more productive robotic exploration of the more distant reaches of the solar system. This includes improved visualization of alternative future states of the robot and the terrain, as well as intuitive means of communicating the intent of the human to the robotic system.

S4.02 Robotic Mobility, Manipulation and Sampling (SBIR)
Lead Center: JPL
Participating Center(s): AFRC, ARC, GRC, GSFC, JSC
Related Subtopic Pointer(s): S4.04, S1.11, S4.06, T4.03, Z5.05

Technologies for robotic mobility, manipulation, and sampling are needed to enable access to sites of interest and acquisition and handling of samples for in-situ analysis or return to Earth from planets and small bodies including Earth's moon, Mars, Venus, comets, asteroids, and planetary moons.

Mobility technologies are needed to enable access to steep and rough terrain for planetary bodies where gravity dominates, such as Earth’s moon and Mars. Wheeled, legged, and aerial solutions are of interest. Technologies to enable surface mobility on small bodies such as through rolling, walking, and hopping are of interest. Ice penetration technologies reaching more than 1 km depth and enabling access to subsurface oceans are desired. Such technologies could include drills, melt-probes, and hybrid approaches. Manipulation technologies are needed to deploy sampling tools to the surface and transfer samples to in-situ instruments and sample storage containers, as well as hermetic sealing of sample chambers. Sample acquisition tools are needed to acquire samples on planetary and small bodies through soft and hard materials, including ice. Minimization of mass and ability to work reliably in the harsh mission environment are important characteristics for the tools. Design for planetary protection and contamination control is important for sample acquisition and handling systems.

Component technologies for low-mass and low-power systems tolerant to the in-situ environment, e.g., temperature, radiation, and dust, are of particular interest. Technical feasibility should be demonstrated during Phase I and a full capability unit of at least Technology Readiness Level (TRL) 4 should be delivered in Phase II. Proposals should show an understanding of relevant science needs and engineering constraints and present a feasible plan to fully develop a technology and infuse it into a NASA program. Specific areas of interest include the following:

- Mobility and sampling systems for planets, small bodies, and moons.
- Near subsurface sampling tools.
- Deep drill systems such as to enable access to subsurface oceans.
- Low mass/power vision systems and processing capabilities that enable fast surface traverse.
- Electro-mechanical connectors enabling tool change-out in dirty environments.
- Tethers and tether play-out and retrieval systems.
- Miniaturized flight motor controllers.
- Sample handling technologies that minimize cross contamination and preserve mechanical integrity of samples.

NASA continues to explore the solar system and future missions will perform in-situ exploration of solar system bodies. These missions could have mobility systems to access locations of scientific interest, manipulators for assembly and deployment of instruments and tools, and sampling systems to acquire and transfer samples. Technologies from this subtopic could be utilized in these future missions.

Proposers should also note a related subtopic exists that is focused solely on lunar robotic missions (see Z5.05, "Enabling Rover Technologies for Lunar Missions", under the Space Technology Mission Directorate). With NASA's present emphasis on lunar exploration, Z5.05 is provided to help develop innovative lunar rover technologies for in-situ resource utilization and for developing more capable and/or lower cost lunar robots.

References:

- https://mars.nasa.gov/programmissions/
- https://solarsystem.nasa.gov/
- https://www.nasa.gov/specials/ocean-worlds/
- https://science.nasa.gov/news-articles/ocean-worlds

T4.01 Information Technologies for Intelligent and Adaptive Space Robotics (STTR)
Lead Center: ARC
Participating Center(s): ARC
Related Subtopic Pointer(s):

Develop Information Technologies to Improve Space Robots

Extensive and pervasive use of advanced space robots can significantly enhance exploration and increase crew efficiency, particularly for missions that are progressively longer, complex, and distant. The performance of these robots is directly linked to the quality and capability of the information technologies used to build and operate them. With few exceptions, however, current information technology used for state-of-the-art robotics is designed only to meet the needs of terrestrial applications and environments.

The objective of this subtopic, therefore, is to encourage the adaptation, maturation, and retargeting of terrestrial information technologies for space robotics. Proposals are specifically sought to address the following technology needs:

- Advanced robot user interfaces that facilitate distributed collaboration, geospatial data visualization, summarization and notification, performance monitoring, or physics-based simulation. The primary objective is to enable more effective and efficient interaction with robots remotely operated with discrete commands or supervisory control. Note: proposals to develop user interfaces for direct teleoperation (manual control) are not solicited and will be considered non-responsive.
- Navigation systems for mobile robot (free-flying and wheeled) operations in man-made (inside the International Space-Station) and unstructured, natural environments (Earth, Moon, Mars). Emphasis on multi-sensor data fusion, obstacle detection, and localization. The primary objective is to radically and significantly increase the performance of mobile robot navigation through new sensors, avionics (including

COTS processors for use in space), perception algorithms and software. Proposals for small size, weight, and power (SWAP) systems are particularly encouraged.
- Robot software systems that support system-level autonomy, instrument/sensor targeting, downlink data triage, and activity planning. The primary objective is to facilitate the creation, extensibility and maintenance of complex robot systems for use in the real-world. Proposals that address autonomy for planetary rovers operating in rough terrain or performing non-traditional tasks (e.g., non-prehensile manipulation) are particularly encouraged.

Information technology for intelligent and adaptive space robotics is highly cross-cutting:

- The technology can be applied to a broad range of unmanned aerial systems (UAS), including both small-scale drones and Predator / Global Hawk type systems. The technology can also be potentially infused into other flight systems that include autonomous capabilities.
- The technology is directly relevant to "caretaker" robots, which are needed to monitor and maintain human spacecraft during dormant/uncrewed periods. The technology can also be used by precursor robots to perform required exploration work prior to the arrival of humans.
- The technology is required for future missions in Earth Science, Heliophysics, and Planetary Science (including the Moon, icy moons and ocean worlds) that require higher performance and autonomy than currently possible. In particular, missions that must operate in dynamic environments, or measure varying phenomena, will require the technology developed by this subtopic.
- The technology is directly applicable to numerous current mid-TRL (Game Changing Development program) and high-TRL (Technology Mission Development program) R&D activities, including Astrobee, In-space Robotic Manufacturing and Assembly, etc.

Proposers should develop technologies that can be demonstrated with or integrated to existing NASA research robots or projects to maximize relevance and infusion potential. Deliverables:

- Identify scenarios, use cases, and requirements.
- Define specifications.
- Develop concepts and prototypes.
- Demonstrate and evaluate prototypes in real-world settings.
- Deliver prototypes (hardware and/or software) to NASA.

The expected Technology Readiness Level (TRL) range at completion of the project is 4-5.

References:

- https://www.nasa.gov/astrobee
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- J. Crusan, et al. 2018. "Deep space gateway concept: Extending human presence into cislunar space", In Proceedings of IEEE Aerospace Conference, Big Sky, MT.
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Z5.04 Technologies for Intra-Vehicular Activity Robotics (SBIR)
Lead Center: ARC
Participating Center(s): JSC
Related Subtopic Pointer(s): H6.03, S4.02, Z5.05

Improve the Capability or Performance of Intra-Vehicular Activity Robots

To support human exploration beyond Earth orbit, NASA is preparing to develop the Gateway, which will be an orbiting facility near the Moon. This facility would serve as a starting point for missions to cis-lunar space and beyond. This facility could enable assembly and servicing of telescopes and deep-space exploration vehicles. This facility could also be used as a platform for astrophysics, Earth observation, heliophysics, and lunar science.

In contrast to the International Space Station (ISS), which is continuously manned, the Gateway is expected to only be intermittently occupied by humans – perhaps only 1 month per year. Consequently, there is a significant need for the Gateway to have autonomous capabilities for performing payload operations and spacecraft caretaking, particularly when astronauts are not present. Intra-Vehicular Activity (IVA) robots can potentially perform a wide variety of tasks including systems inspection, monitoring, diagnostics and repair, logistics and consumables stowage, exploration capability testing, aggregation of robotically returned destination surface samples, and science measurements and ops.

The objective of this subtopic, therefore, is to develop technologies that can improve the capability or performance of IVA robots to perform payload operations and spacecraft caretaking. Proposals are specifically sought to create technologies that can be integrated and tested with the NASA Astrobee or Robonaut 2 robots in the following areas:

- Sensors and perception systems for interior environment monitoring, inspection, modeling and navigation;
- Robotic tools for manipulating logistics and stowage;
- Operational subsystems that enable extended robot operations (power systems, efficient propulsion, etc.), increase robot autonomy (planning, scheduling, and task execution), or improve human-robot teaming (software architecture, remote operations methods, etc.).

The desired deliverables are prototype components or subsystems. Proposals must describe how the technology will make a significant improvement over the current state of the art, rather than just an incremental enhancement, for a specific IVA robot application.

The expected Technology Readiness Level (TRL) range at completion of the project is 4-5.

References:

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Z5.05 Lunar Rover Technologies for In-situ Resource Utilization and Exploration (SBIR)
Lead Center: JSC
Participating Center(s): ARC, GRC, KSC
Related Subtopic Pointer(s): S4.04, S4.02, T4.01, Z12.01, Z1.04, Z12.02

Enabling Rover Technologies for Lunar Missions

The objective of this subtopic is to innovate lunar rover technologies that will enable in-situ resource utilization (ISRU) and exploration missions. In particular, this subtopic will develop ideas, subsystems components, software tools, and prototypes that contribute to more capable and/or lower-cost lunar robots for these missions.

A potential lunar ISRU application is the prospecting, characterization, and collection of volatiles that could be processed to produce oxygen, fuel, etc. Recent remote sensing measurements, modeling, and data from the Lunar Crater Observation and Sensing Satellite (LCROSS) indicates that there may be an abundance of volatiles (e.g., hydrogen) near the lunar poles. However, the distribution of the volatiles at and under the surface is unknown. This subtopic seeks new robotic technologies to support ISRU activities. This does not include new ISRU payload and/or excavation technologies, which are solicited under the "Lunar Resources - Volatiles" and "Extraction of Oxygen from Lunar Regolith" subtopics. Additionally, lunar power technologies are solicited in the sub-topic titled "Long Duration Lunar Energy Storage and Discharge."

The expected environment at the lunar poles involves all the challenges observed during the Apollo mission (thermal extremes, vacuum, radiation, abrasive dust, electrostatic dust) plus the addition of low sun angles, potentially less consolidated regolith, and permanently shadowed regions with temperatures as low as 40K. This subtopic seeks new technology to address these challenges.

Phase I success involves technical feasibility demonstration through analysis, prototyping, proof-of-concept, or testing. Phase II success will advance Technology Readiness Level (TRL) to a level of 4-5. Of specific interest are:

- Mobility architectures, including novel mobility mechanisms and lunar dust tolerant mechanisms
- Cryo-capable actuators capable of operating at extremely cold temperatures (in environments as cold as -230° C). Preferably solutions will not include heaters as they significantly increase the power draw for normal operations during the lunar day. Novel materials capable of maintaining metallurgical properties at cryogenic temperatures will be considered. Also desired are cryo actuators featuring dust tolerances and the ability to operate at high temperatures as well (approaching 150° C).
- Magnetic gearing applications for space. NASA and others are developing relatively low ratio (less than 25:1 per stage) concentric magnetic gearing for aeronautics applications. Space applications demand high speed-reduction ratio (often more than 1000:1) and high specific torque (>50 Nm/kg), operation in environmental temperatures down to -230° C (40K), operation in low-atmosphere or hard vacuum, with high reliability and energy efficiency. Phase I work would include identifying the most suitable magnetic gear topologies to meet these space application needs, defining the technology development challenges including thermal and structural issues, advancing the most critical aspects of the technology, and producing a low-fidelity prototype to prove the feasibility of the concept(s).
- Perception systems and algorithms with a path toward flight for the lunar surface capable of operating in the harsh lighting conditions that might include high dynamic range, shadowed regions, low angle illumination, and opposition effects
- Lunar regolith terramechanical modeling tools and simulations, especially tools that integration with existing commercial and open source robotic analysis and simulation tools
- Rover embedding and entrapment detection and escape approaches including slip monitoring, regolith sensing/modeling, low ground pressure wheels and soft soil tolerant mobility architectures

Example deliverables coming from a successful Phase II within this subtopic, might including some of the following:

- Designs of cryo-capable or dust tolerant mechanisms motor controllers with test data and prototypes
- Prototype rovers or scale versions of prototype rovers showing novel mobility architecture for escaping entrapment in regolith

- Software algorithms including demonstrating slip detection or image processing in harsh lunar lighting conditions
- Software packages either standalone or integrated with commercially available or open-source robotic simulation packages (preferred)
- NASA is also interested in technologies demonstrations that could serve as payloads on commercial landers at the end of Phase II

For all the above, it is desired to have been demonstrated in, or have a clear path to operating in, the lunar environment.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

This SBIR subtopic resides within the Science and Technology Mission Directorate (STMD) as a vehicle for development of technology objectives. It is expected that successful projects would infuse technology into either the STMD Game Changing Development (GCD) or Technology Demonstration Missions (TDM) programs. Technology could also be infused into joint efforts involving STMD's partners (other mission directorates, other government agencies, and the commercial sector). Flights for these technology missions could be supported on small commercial lunar landers (Science Mission Directorate) or possibly mid-size NASA lunar landers (Human Exploration and Operations Mission Directorate).

Potential customers:

- Autonomy and robotics
- Robotic ISRU missions
- Payloads for Commercial Lunar Payload Services landers
- Commercial vendors
- Future prospecting/mining operations

References:

- https://www.sciencedirect.com/science/article/pii/S0032063310003065
- https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/19720013192.pdf
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- https://www.fbo.gov/index?s=opportunity&mode=form&id=46b23a8f2c06da6ac08e1d1d2ae97d35&tab=core&_cview=0
- https://www.hou.usra.edu/meetings/survivethenight2018/, https://www.nasa.gov/feature/nasas-exploration-campaign-back-to-the-moon-and-on-to-mars and https://www.nasa.gov/sites/default/files/thumbnails/image/nasa-exploration-campaign.jpg
- Information on NASA's interest in landers that might host the rovers and rover technology demonstrations can be found at the following:
    - https://www.nasa.gov/feature/nasa-seeks-ideas-to-advance-toward-human-class-lunar-landers
    - https://govtribe.com/project/lunar-surface-transportation-capability-request-for-information-rfi
- See reference about magnetic gearing references below:
    - Tlali, P. M., Wang, R-J., and Gerber, S., “Magnetic gear technologies: A review,” 2014 Intl. Conference on Electrical Machines, p. 544-550, Berlin, Germany, Sept. 2 – 5, 2014.
    - Justin J. Scheidler, Vivake M. Asnani, and Thomas F. Tallerico, “Overview of NASA’s Magnetic Gears Research,” presented at the AIAA / IEEE Electric Aircraft Technology Symposium, Cincinnati, Ohio, July 12 – 13, 2018.

- Vivake M. Asnani, Justin J. Scheidler, and Thomas F. Tallerico, “Magnetic Gearing Research at NASA,” presented at the 74th Annual Forum of the American Helicopter Society, Phoenix, AZ, May 14 – 17, 2018.
- Aaron D. Anderson and Vivake M. Asnani, "Concentric Magnetic Gearing - State of the Art and Empirical Trends", NASA TM, in-press.

Focus Area 5 Communications and Navigation

Lead MD: HEOMD
Participating MD(s): SMD, STTR

NASA seeks proposals to produce high impact developments in communications and navigation technologies to support future space science and exploration missions. Missions are generating ever-increasing data volumes that require increased performance from communications systems while minimizing the impacts to the spacecraft. This requires higher peak data rates from the communications systems, algorithms to increase the overall throughput of the end-to-end communications channel, and lowering the flight system cost, mass, and power per bit transmitted. Effective communications on a non-interference basis are also required in complex RF environments such as inside a launch vehicle fairing. New analysis methods are requested for predicting the RF environment in fairings and spacecraft cavities. Similarly, missions have a need for more precise guidance, navigation, and control to meet their mission objectives. This requires new and more efficient trajectory planning methods, increased onboard autonomous navigation, and increased precision of onboard instrumentation while minimizing cost, mass, and power. This focus area supports development of technologies in optical communications systems, cognitive communications, flight dynamics and navigation, transformational communications technology, electric field prediction methods, and guidance, navigation, and control that will provide a significant improvement over the current state of the art.

H9.01 Long Range Optical Telecommunications (SBIR)
Lead Center: JPL
Participating Center(s): GRC, GSFC
Related Subtopic Pointer(s): H9.05

Free-space optical communications technologies

Free-space Long Range Optical Communications subtopic seeks innovative technologies in free-space optical communications for pushing future data volume returns from space missions in multiple domains: >100 gigabit/s cis-lunar (Earth or lunar orbit to ground), >10 gigabit/s Earth-sun L_1 and L_2, >1 gigabit/s per AU-squared deep space, and >1 Gbit/s planetary lander to orbiter. Additionally, innovative technologies for improving efficiency, reliability, robustness, and longevity of state-of-the-art laser communication systems are also sought.

Proposals are sought in the following specific areas (TRL 2-3 Phase I for maturation to TRL 3-5 in Phase II):

Flight Laser Transceivers

Low-mass, high-effective isotropic radiated power (EIRP) laser transceivers with:

- 30 to 50 cm clear aperture diameter telescopes for laser communications.
- Targeted mass of opto-mechanical assembly per aperture area, less than 100 kg/square-meter
- Cumulative wave-front error and transmission loss not to exceed 2-dB.
- Advanced thermal designs with allowable flight temperatures of the optics and structure, at least -20° C to 50° C operational range
- Stray light design for 3-degrees from edge of sun operations and surviving direct sun-pointing of

Transceivers fitting above characteristics should support robust link acquisition tracking and pointing characteristics, including point-ahead implementation from space, for beacon assisted and/or beaconless architectures:

- Pointing loss allocations not to exceed 1 dB (pointing errors associated loss of irradiance at target less than 20%)
- Receive field-of-view of at least 1 milliradian radius, for beacon assisted operations
- Beaconless pointing subsystems for operations beyond 3 A.U.
- Assume integrated spacecraft micro-vibration angular disturbance of 150 micro-radians (<0.1 Hz to ~500 Hz)
- Low complexity small footprint laser transceivers for bi-directional optical links, >1 -10 Gbit/second, at a nominal link range of 1000-5000 km, for planetary lander/rover to orbiter and/or space-to-space cross links

Flight Laser Transmitters, Receivers and Sensors

High-gigabit/s laser transmitters:

- 1550 nm wavelength
- Space qualifiable laser and optical components
- High rate 10-100 Gb/s for cis-lunar
- 1 Gb/s for deep-space
- Integrated hardware with embedded software/firmware for innovative coding/modulation/interleaving schemes

High peak-to-average power laser transmitters for regular or augmented M-ary PPM modulation with M=4, 8, 16, 32, 64, 128, 256 operating at NIR wavelengths, preferably 1550 nm with average powers from 5 – 50 W:

- Sub-nanosecond pulse
- Low pulse jitter
- Long lifetime and reliability operating in space environment (> 5 and as long as 20 years)
- High modulation and polarization extinction ratio with 1-10 GHz linewidth

Space qualifiable wavelength division multiplexing transmitters and amplifiers with 4 to 20 channels and average output power > 20W per channel; peak-to-average power ratios >200; >10 Gb/s channel modulation capability:

- >20% wall-plug efficiency (DC-to-optical, including support electronics). Describe approach for stated efficiency of space qualifiable lasers. Multi-watt Erbium Doped Fiber Amplifier (EDFA), or alternatives, with high gain bandwidth (> 30nm, 0.5 dB flatness) concepts will be considered.
- Radiation tolerance better than 50 Krad is required, (including resilience to photo-darkening).

Space qualifiable high-speed receivers and low light level sensitive acquisition, tracking, pointing, detectors and detector arrays:

- NIR wavelengths, 1064nm, 1550 nm
- Supporting low irradiance (~ fW/m2 to pW/m2) detection
- Low sub-nanosecond timing jitter and fast rise time
- Novel hybridization of optics and electronic readout schemes with built-in pre-processing capability
- Characteristics compatible with supporting time-of-flight or other means of processing laser communication signals for high precision range and range rate measurements
- Tolerant to space radiation effects, total dose > 50 krad, displacement damage and single event effects

Novel technologies and accessories

Narrow Band Pass Optical Filters:

- Space qualifiable, sub-nanometer to nanometer, noise equivalent bandwidth with ~90% throughput, large spectral range out-of-band blocking (~ 40 dB)
- NIR wavelengths from 1064 – 1550 nm region, with high transmission through Earth's atmosphere

- Reliable tuning over limited range

Novel integrated photonics applications for space with objective of reducing size, weight and power of modulators, improved integration of opto-electronics and efficient coupling to traditional discrete optics
Concepts for offering redundancy to laser transmitters in space:

- Optical fiber routing of high average powers (10's of watts) and high peak powers (1-10 kW)
- Redundancy in actuators and optical components

Ground Assets for Optical Communication

Large aperture receivers for faint optical communication signals from deep space, subsystem technologies:

- Demonstrate innovative subsystem technologies for >10 m diameter deep space ground collector
- Capable of operating to within 3° of solar limb
- Better than 10 microradian spot size (excluding atmospheric seeing contribution)
- Desire demonstration of low-cost primary mirror segment fabrication to meet a cost goal of less than $35K per square meter
- Low-cost techniques for segment alignment and control, including daytime operations.

1550 nm sensitive photon counting detector arrays compatible with large aperture ground collectors:

- Integrated time tagging readout electronics for >5 gigaphotons/s incident rate
- Time resolution <50 ps 1-sigma
- Highest possible single photon detection efficiency, at least 50% at highest incident rate,
- Total detector active area > 0.3 to 1 mm^2
- Integrated dark rate < 3 megacount/s.

Cryogenic optical filters:

- Operate at 40K with sub-nanometer noise equivalent bandwidths
- 1550 nm spectral region, transmission losses < 0.5 dB, clear aperture
- >35 mm, and acceptance angle >40 milliradians with out-of-band rejection of >65 dB from 0.4 to 5 micrometers

Multi-kilowatt laser transmitters for use as ground beacon and uplink laser transmitters:

- Near infrared wavelengths in 1.0 or 1.55 micrometer spectral region
- Capable of modulating with narrow nanosecond with sub-nanosecond rise times
- Low timing jitter and stable operation
- High speed real-time signal processing of serially concatenated pulse position modulation operating at a few bits per photon with user interface outputs

For all technologies lowest cost for small volume production (5 to 20 units) is a driver. Research must convincingly prove technical feasibility (proof-of-concept) during Phase I, ideally with hardware deliverables that can be tested to validate performance claims, with a clear path to demonstrating and delivering functional hardware meeting all objectives and specifications in Phase II.

A number of FSOC related NASA projects are ongoing with launch expected in the 2019-2022 time frame. The Laser Communication Relay Demonstration (LCRD) is an earth-to-geostationary satellite relay demonstration to launch in late 2019. The Illuma -T Project will extend the relay demonstration to include a LEO node on the ISS in 2021. In 2022 the EM-2 Optical to Orion (O2O) demonstration will transmit data from the Orion crewed capsule as it travels to the Moon and back. In 2022 the Deep Space Optical Communications (DSOC) Project technology demonstration will be hosted by the Psyche Mission spacecraft extending FSOC links to astronomical unit distances.

These missions are being funded by NASA's STMD/TDM Program and HEOMD/SCaN Program.

References:

- Don Cornwell, "NASA's optical communications program for 2015 and beyond," SPIE Proceedings, 9354, 2015
- https://www.nasa.gov/mission_pages/tdm/lcrd/index.html
- https://www.nasa.gov/directorates/heo/scan/opticalcommunications/illuma-t
- https://www.nasa.gov/feature/goddard/2017/nasa-laser-communications-to-provide-orion-faster-connections
- https://www.nasa.gov/mission_pages/tdm/dsoc/index.html

H9.03 Flight Dynamics and Navigation Technology (SBIR)
Lead Center: GSFC
Participating Center(s): JSC, MSFC
Related Subtopic Pointer(s): S3.04

Advanced Techniques for Trajectory Optimization

NASA is planning and proposing increasingly ambitious missions such as crewed and robotic missions in cislunar space, multiple small body (comet/asteroid) rendezvous/flyby missions, outer planet moon tours, Lagrange point missions, and small body sample return using low thrust propulsion (including solar sails). Trajectory design for these complex missions can take weeks or months to generate a single reference trajectory. This subtopic seeks new techniques and tools to speed up and improve the trajectory design and optimization process to allow mission designers to more fully explore trade spaces and more quickly respond to changes in the mission. See Reference 1 for NASA Technology Area (TA) roadmaps:
(https://www.nasa.gov/sites/default/files/atoms/files/2015_nasa_technology_roadmaps_ta_5_communication_and_navigation_final.pdf):

- Low-thrust trajectory optimization in a multi-body dynamical environment (TA 5.4.2.1)
- Advanced deep-space trajectory design techniques. (TA 5.4.2.7) and rapid trajectory design near small bodies (TA 5.4.5.1)
- Tools and techniques for orbit/trajectory design for distributed space missions including constellations and formations (TA 11.2.6)
- Tools and techniques for orbit/trajectory design using dynamical systems theory for Earth-Moon and cislunar missions.

Autonomous Onboard Navigation, Guidance and Control

Future NASA missions require precision landing, rendezvous, formation flying, cooperative robotics, proximity operations (e.g., servicing and assembly), non-cooperative object capture, and coordinated platform operations in Earth orbit, cislunar space, libration orbits, and deep space. These missions require a high degree of autonomy. The subtopic seeks advancements in autonomous navigation and maneuvering technologies for applications in Earth orbit, lunar, cislunar, libration, and deep space to reduce dependence on ground-based tracking, orbit determination, and maneuver planning. See Reference 1 for NASA Technical Area (TA) roadmaps:

- Advanced autonomous navigation techniques including devices and systems that support significant advances in independence from Earth supervision while minimizing spacecraft burden by requiring low power and minimal mass and volume (TA 5.4.2.4, TA 5.4.2.6, TA 5.4.2.8).
- Onboard trajectory planning and optimization algorithms, for real-time mission re-sequencing, on-board computation of large divert maneuvers (TA 5.4.2.3, TA 5.4.2.5, TA 5.4.2.6, TA 9.2.6) primitive body/lunar proximity operations and pinpoint landing (TA 5.4.6.1), including the concept of robust onboard trajectory planning and optimization algorithms that account for system uncertainty (i.e., navigation errors, maneuver execution errors, etc.).

- Onboard relative and proximity navigation (TA 5.4.4) multi-platform relative navigation (relative position, velocity and attitude or pose) which support cooperative and collaborative space operations including satellite servicing and in-space assembly.
- Rendezvous targeting (TA 4.6.2.1) Proximity Operations/Capture/ Docking Guidance (TA 4.6.2.2)
- Advanced filtering techniques (TA 5.4.2.4) that address rendezvous and proximity operations as a multi-sensor, multi-target tracking problem; handle non-Gaussian uncertainty; or incorporate multiple-model estimation.
- Vision processing algorithms (TA 5.4.3.2) to extract the maximum amount of information from images used for optical navigation.

Conjunction Assessment Risk Analysis (CARA)

The U.S. Space Surveillance Network currently tracks more than 22,000 objects larger than 10 centimeters and the number of object in orbit is steadily increasing which causes an increasing threat to human spaceflight and robotic missions in the near-Earth environment. The NASA Conjunction Assessment Risk Analysis (CARA) team identifies close approaches (conjunctions) between NASA satellites and other space objects, determines the risk posed by those events, and plans and executes risk mitigation strategies, including collision avoidance maneuvers to protect space assets and humans in Earth orbit. The ability to perform CARA more accurately and rapidly will improve space safety for all near-Earth operations, improve operational support by providing more accurate and longer-term collision predictions and reduce propellant usage for collision avoidance maneuvers. This subtopic seeks innovative technologies to improve the CARA process including (see Reference 1 for NASA Technical Area (TA) roadmaps).:

- Faster and more accurate methods of detecting close approaches and conjunctions (TA 5.7.1) and computing probability of collision (TA 11.3.6).
- Techniques for improving state and covariance characterization and propagation (TA 5.7.2.1, TA 11.3.6), including improved modeling of non-gravitational force effects, Gaussian mixture models, differential algebra, polynomial chaos expansions, etc.
- Techniques for estimation of object characteristics (TA 5.7.2) relevant to accurate orbit propagation such as ballistic coefficient, attitude or attitude profile, mass, configuration, and maneuvers from available radiometric, photometric and/or astrometric data.
- Event evolution prediction methods, models and algorithms with improved ability to predict orbit characteristics for single and ensemble risk assessment, especially using artificial intelligence/machine learning (TA 5.5.3).

Proposals that leverage state-of-the-art software already developed by NASA, or that can optionally integrate with those packages, such as the General Mission Analysis Tool (GMAT), Copernicus, Evolutionary Mission Trajectory Generator (EMTG), Mission Analysis Low-Thrust Optimization (MALTO), Mission Analysis, Operations, and Navigation Toolkit Environment (MONTE), Goddard Enhanced Onboard Navigation System (GEONS), and Optimal Trajectories by Implicit Simulation (OTIS), or other available software tools are encouraged. Proposers who contemplate licensing NASA technologies are highly encouraged to coordinate with the appropriate NASA technology transfer offices prior to submission of their proposals.

Phase I research should be conducted to demonstrate technical feasibility, with preliminary software being delivered for NASA testing, as well as show a plan towards Phase II integration. For proposals that include hardware development, delivery of a prototype under the Phase I contract is preferred, but not necessary. Phase II new technology development efforts shall deliver components at the TRL 5-6 level with mature algorithms and software components complete and preliminary integration and testing in an operational environment.

References:

- NASA Space Technology Roadmaps (2015) : https://www.nasa.gov/offices/oct/home/roadmaps/index.html
- General Mission Analysis Tool (GMAT): http://gmatcentral.org/display/GW/GMAT+Wiki+Home
- Evolutionary Mission Trajectory Generator (EMTG): https://software.nasa.gov/software/GSC-16824-1
- Copernicus: https://www.nasa.gov/centers/johnson/copernicus/index.html
- Optimal Trajectories by Implicit Simulation: http://otis.grc.nasa.gov/

- Mission Analysis Low-Thrust Optimization (MALTO): https://spaceflightsystems.grc.nasa.gov/SSPO/ISPTProg/LTTT/
- Mission Analysis, Operations, and Navigation Toolkit Environment (MONTE): https://montepy.jpl.nasa.gov/
- Goddard Enhanced Onboard Navigation System (GEONS), (https://software.nasa.gov/software/GSC-14687-1), (https://goo.gl/TbVZ7G)
- Navigator GPS Receiver (http://itpo.gsfc.nasa.gov/wp-content/uploads/gsc_14793_1_navigator.pdf)
- NavCube (https://goo.gl/bdobb9)
- NASA Conjunction Assessment Risk Analysis (CARA) Office: https://satellitesafety.gsfc.nasa.gov/cara.html
- NASA Orbital Debris Program Office: https://www.orbitaldebris.jsc.nasa.gov/
- Newman, Lauri, K., "The NASA robotic conjunction assessment process: Overview and operational experiences," Acta Astronautica, Vol. 66, Issues 7-8, Apr-May 2010, pp. 1253-1261, https://www.sciencedirect.com/science/article/pii/S0094576509004913.
- Newman, Lauri K., et al. "Evolution and Implementation of the NASA Robotic Conjunction Assessment Risk Analysis Concept of Operations." (2014). https://ntrs.nasa.gov/search.jsp?R=20150000159
- Newman, Lauri K., and Matthew D. Hejduk. "NASA Conjunction Assessment Organizational Approach and the Associated Determination of Screening Volume Sizes." (2015). https://ntrs.nasa.gov/search.jsp?R=20150011461

H9.05 Transformational Communications Technology (SBIR)
Lead Center: GRC
Related Subtopic Pointer(s): H9.01

Revolutionary Concepts

NASA seeks revolutionary, transformational communications technologies that emphasize not only dramatic reduction in system size, mass, and power but also dramatic implementation and operational cost savings while improving overall communications architecture performance, including security. The proposer is expected to identify new ideas, create novel solutions and execute feasibility demonstrations. For example, there is interest in exploiting the demarcation between quantum and classical communications, specifically quantum coherent transport devices as opposed to ballistic transport devices. Emphasis for this subtopic is on the far-term (≈10yrs.) insofar as mission insertion and commercialization but it is expected that the proposer proves fundamental feasibility via prototyping within the normal scope of the SBIR program. The transformational communications technology development will focus research in the following areas:

- Systems optimized for energy efficiency (information bits per unit energy)
- Advanced materials; smart materials; electronics embedded in structures; functional materials; graphene-based electronics/detectors
- Technologies that address flexible, scalable digital/optical core processing topologies to support both RF and optical communications in a single terminal
- Nanoelectronics and nanomagnetics; quantum logic gates; single electron computing; superconducting devices; technologies to leapfrog Moore’s law.
- Quantum communications, methods for probing quantum phenomenon, methods for exploiting exotic aspects of quantum theory.
- Human/machine and brain-machine interfacing; the convergence of electronic engineering and bio-engineering; neural signal interfacing.
- Integrated photonic circuit quantum memory.

The research should be conducted to demonstrate theoretical and technical feasibility during the Phase I and Phase II development cycles and be able to demonstrate an evolutionary path to insertion within approximately 10 years. Delivery of a prototype of the most critically enabling element of the technology for NASA testing at the completion of the Phase II contract is expected.

Phase I deliverables shall include a final report describing theoretical analysis and prototyping concepts. The technology should have eventual commercialization potential. For Phase II consideration, the final report should include a detailed path towards Phase II prototype hardware.

The expected TRL for this project is 2 to 4.

NASA is seeking cutting-edge technology to keep the U.S. at the forefront of information and communications technology. For example, NASA Space Technology Roadmap TA 05 identifies quantum communications as a critical area. China launched the world's first quantum satellite in 2016 (the Quantum Experiments at Space Scale (QUESS) satellite). A fleet of quantum-enabled craft is likely to follow. Groups from Canada, Japan, Italy and Singapore also have plans for quantum space experiments and this competition represents a new space race.

References:

- NASA Space Technology Roadmaps (2015) : https://www.nasa.gov/offices/oct/home/roadmaps/index.html
- "The Quantum Communications Space Race: A Review of Quantum Key Distribution Initiatives from Around the World," E. Katz, NASA TM 219760 (2018).
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- https://www.nasa.gov/pdf/675092main_SCaN_ADD_Executive_Summary.pdf

H9.07 Cognitive Communication (SBIR)
Lead Center: GRC
Participating Center(s): GSFC, JPL
Related Subtopic Pointer(s): H6.22, Z8.03

Cognitive Capabilities

NASA's Space Communication and Navigation (SCaN) program seeks innovative approaches to increase mission science data return, improve resource efficiencies for both SCaN customers and networks, and ensure resilience in the unpredictable space environment. The Cognitive Communication subtopic focuses on advances in artificial intelligence, machine learning, and signal and data processing including:

- Adaptive, autonomous, and cognitive link technologies to improve mission communication capabilities.
- Networking technologies to move data through and among network nodes in a more efficient and intelligent manner, including on-board processing of data packets.
- System-wide approaches to optimize scheduling of network relay satellites and ground stations to balance utilization and reach maximum data transfer potential.

A cognitive system is envisioned to sense, detect, adapt, and learn from its experiences and environment to optimize the communications capabilities for the user mission satellite or network infrastructure. Goals of this capability are to improve communications capability and efficiency, mitigate channel impairments, and reduce operations complexity and costs through intelligent and autonomous communications and data handling.

The overall goal is to perform research and/or technology development to optimize space communication links, networks, and system-wide resource scheduling. Specific focus areas include:

- Flexible communication platforms, modules, and/or antennas that include novel signal processing technology [e.g., graphics processing units - GPUs - for space applications, neuromorphic approaches, and phased array antennas with integrated processing for interference mitigation].
- Wideband sensing and communications for S-, X-, and Ka-bands, coupled with machine learning algorithms that learn from the environment [e.g., learning channel impairments, spectrum sharing in noisy environments].
- On-board processing technology and decentralized networking techniques to enable data switching, routing, storage, and scheduling on a spacecraft [e.g., routing based on quality of service and data flow-specific requirements such as latency].

- Other innovative, related areas of interest.

This subtopic seeks innovations that address the unique needs of NASA's data communication requirements for the space environment, specifically focusing on low size, weight, power, and cost applications suitable for small satellite or cubesat operations. Proposed systems should highlight advancements to provide the needed communications capability while minimizing on-board resources such as power consumption and thermal dissipation. Proposals should consider how the technology can mature into a successful demonstration using one or several cubesat platforms.

Phase I will emphasize research aspects for technical feasibility, infusion potential into space operations, clear and achievable benefits (e.g., 2x-5x increase in throughput, 25-50% reduction in power, improved quality of service or efficiency, reduction in operations costs), and show a path towards a Phase II proposal. Phase I Deliverables include feasibility and concept of operations of the research topic, simulations and measurements, validation of the proposed approach to develop a given product (TRL 3-4), and a plan for further development of the specific capabilities or products to be performed in Phase II. Early development and delivery of prototype hardware/software is encouraged.

Phase II will emphasize hardware/software development with delivery of specific hardware or software product for NASA targeting demonstration operations on a cubesat platform. Phase II deliverables include a working prototype or engineering model of the proposed product/platform or software, along with documentation of development, capabilities, and measurements (showing specific improvement metrics), documents and tools as necessary for NASA to modify and use the cognitive software capability or hardware component(s). Proposed prototypes shall demonstrate a path towards a flight-capable cubesat platform. Opportunities and plans should also be identified and summarized for potential commercialization or NASA infusion. Software applications and platform/infrastructure deliverables for SDR platforms shall be compliant with the latest NASA standard for software defined radios, the Space Telecommunications Radio System (STRS), NASA-STD-4009 and NASA-HNBK-4009.

Cognitive networks and operations are a key goal of the HEOMD SCaN Program communications plan, including the SCaN Next Generation Architecture. As communications and networks become more complex, cognition and automation will play a larger role to mitigate complexity and reduce operations costs. Machine learning will configure networks, choose radio configurations, adjust for impairments and failures, and monitor short and long-term performance for improvements. SCaN has invested in Phase III CRP SBIR contracts and stands ready for additional investments. STMD recently awarded two Early Career Faculty grants to study topics related to Cognitive Communication including distributed network routing and blockchain-based data processing.

The SCaN Cognitive Communications project intends to fly a multi-cubesat mission in the early 2020s. This mission is intended to demonstrate research results obtained both internally and resulting from prior SBIR awards. Results from this subtopic would be candidates for this initial mission or future demonstrations.

Related Subtopic Areas

The focus of this subtopic is the application of advanced processing power to communication systems, especially for cubesats and small satellites. Development of the requisite processors and low-cost radiation hardening techniques is best suited to the Z8 topic area, particularly Z8.03 (Low Cost Radiation Hardened Integrated Circuit Technology). Development of neuromorphic processors and related enhanced processing capability to enable cognitive algorithms in general spacecraft applications is best suited to the H6 topic area, particularly H6.22 (Neuromorphic Processors for In-Space Autonomy and Cognition).

References:

Several reference papers that have been published through the Cognitive Communications Project include:

- "Implementation of a Space Communications Cognitive Engine" https://ntrs.nasa.gov/search.jsp?R=20180002166
- "Multi-Objective Reinforcement Learning-based Deep Neural Networks for Cognitive Space Communications" https://ntrs.nasa.gov/search.jsp?R=20170009153
- "Assessment of Cognitive Communications Interest Areas for NASA Needs and Benefits" https://ntrs.nasa.gov/search.jsp?R=20170009386

- "Architecture for Cognitive Networking within NASAs Future Space Communications Infrastructure" https://ntrs.nasa.gov/search.jsp?R=20170001295
- "Modulation Classification of Satellite Communication Signals Using Cumulants and Neural Networks" https://ntrs.nasa.gov/search.jsp?R=20170006541
- Results of the Cognitive Communications for Aerospace Applications workshop are available at: http://ieee-ccaa.com/ccaaw-summary/

S3.04 Guidance, Navigation and Control (SBIR)
Lead Center: GSFC
Participating Center(s): JPL, MSFC
Related Subtopic Pointer(s): H9.03

Guidance, Navigation and Control

NASA seeks innovative, groundbreaking, and high impact developments in spacecraft guidance, navigation, and control technologies in support of future science and exploration mission requirements. This subtopic covers mission enabling technologies that have significant Size, Weight and Power, Cost, and Performance (SWaP-CP) improvements over the state-of-the-art COTS in the areas of Spacecraft Attitude Determination and Control Systems, Absolute and Relative Navigation Systems, and Pointing Control Systems, and Radiation-Hardened GN&C Hardware.

Component technology developments are sought for the range of flight sensors, actuators, and associated algorithms and software required to provide these improved capabilities. Technologies that apply to most spacecraft platform sizes will be considered.

Advances in the following areas are sought:

- Spacecraft Attitude Determination and Control Systems: Sensors and actuators that enable <0.1 arcsecond level pointing knowledge and arcsecond level control capabilities for large space telescopes, with improvements in size, weight, and power requirements.
- Absolute and Relative Navigation Systems: Autonomous onboard flight navigation sensors and algorithms incorporating both spaceborne and ground-based absolute and relative measurements. For relative navigation, machine vision technologies apply. Special considerations will be given to relative navigation sensors enabling precision formation flying, astrometric alignment of a formation of vehicles, robotic servicing and sample return capabilities, and other GN&C techniques for enabling the collection of distributed science measurements. In addition, flight sensors and algorithms that support onboard terrain relative navigation are of interest.
- Pointing Control Systems: Mechanisms that enable milli-arcsecond class pointing performance on any spaceborne pointing platforms. Active and passive vibration isolation systems, innovative actuation feedback, or any such technology that can be used to enable other areas within this subtopic applies.
- Radiation-Hardened Hardware: GN&C sensors that could operate in a high radiation environment, such as the Jovian environment.
- Fast light Gyroscopes and Accelerometers: In conventional ring laser gyros, precision increases with cavity size and measurement time. Fast-light media, however, can be used to increase gyro precision without having to increase size or decrease measurement frequency, thereby increasing the time for standalone spacecraft navigation. (The increased precision also opens up new science possibilities such as measurements of fundamental physical constants, improving the sensitivity-bandwidth product for gravity wave detection, and tests of general relativity.) Prototype fast-light gyros are sought that can be implemented in a compact rugged design that is tolerant to variations in temperature and G-conditions, with the ultimate goal of demonstrating decreased angular random walk.

Phase I research should be conducted to demonstrate technical feasibility as well as show a plan towards Phase II integration and component/prototype testing in a relevant environment. Phase II technology development efforts shall deliver component/prototype at the TRL 5-6 level consistent with NASA SBIR/STTR Technology Readiness Level (TRL) Descriptions. Delivery of final documentation, test plans, and test results are required. Delivery of a hardware component/prototype under the Phase II contract is preferred.

Proposals should show an understanding of one or more relevant science or exploration needs and present a feasible plan to fully develop a technology and infuse it into a NASA program.

This subtopic is for all mission enabling Guidance, Navigation, and Control technology in support of SMD missions and future mission concepts. Proposal for the development of hardware, software, and/or algorithm are all welcome. The specific applications could range from cubesat/smallsat, to ISS payload, to any flagship missions.

Relevance to NASA

Science areas: Heliophysics, Earth Science, Astrophysics, and Planetary Missions Capability requirement areas:

- *Spacecraft GN&C Sensors* – optical, RF, inertial, and advanced concepts for onboard sensing of spacecraft attitude and orbit states
- *Spacecraft GN&C Estimation and Control Algorithms* – Innovative concepts for onboard algorithms for attitude/orbit determination and control for single spacecraft, spacecraft rendezvous and docking, and spacecraft formations.

References:

- 2017 NASA Strategic Technology Investment Plan: https://go.usa.gov/xU7sE
- 2015 NASA Technology Roadmaps: https://go.usa.gov/xU7sy

T5.02 Electric field mapping and prediction methods within spacecraft enclosures (STTR)
Lead Center: KSC
Participating Center(s): GSFC, JPL, JSC, MSFC

NASA Launch Services program is responsible for ensuring the safety of NASA payloads on commercial rockets. This includes prediction and mitigation of hazardous electric fields created within the payload enclosure and similar areas of the rocket. NASA and industry have commonly used approximation methods to determine the average fields in enclosures. In the last decade the Launch Services Program has funded studies to support quantification of electromagnetic field characterization in fairing cavities due to internal and external sources. By accurately predicting these fields, acoustic and thermal blanketing can be optimized for RF attenuation and design changes can be quickly evaluated reducing schedule impacts. Cost savings can also be realized by reducing stringent radiated susceptibility requirements and reliability improved by accurately predicting signal transmission/reception environments within enclosures. This methodology can also improve Human exposure safety limits evaluations for manned vehicle enclosures with transmitting systems.

Initially studies focused on computational methods using the recent advances in computing power and the improved efficiency of matrix-based solutions provided by GPU computing. Results indicate solution of an integrated fairing is deterministic, but sensitive to small variation in structures, materials. As of yet, only the empty or sparse cavity can be reliably solved with 3D computational tools even with large computing systems and the use of non-linear basis functions. Results also indicate that computational approximation methods such as physical optics and multilevel fast multipole are not reliable prediction methods within enclosures of this scale because of the underlying assumption sets that are inconsistent with enclosure boundaries. More recently, LSP has concentrated on statistically formulating a compilation of test/computational results to produce a maximum expected environment. Preliminary results are promising in the area of statistical bounding of the desired solution. The researched methodology should offer the following advantages over 3D computational and standard volume-based approximation methods:

- Predict BOTH statistical Mean AND Maximum Expected E field and/or common mode current.
- Consider the over-moded (electrically large conductive cavities) and under-moded (electrically smaller damped enclosures).
- Consider complex materials with multiple joined enclosures.
- Applications of this prediction methodology are far reaching and include shielding effectiveness and prediction of fields within a cavity enclosure due to internal transmitters and operating avionics.

To enable bounded solutions in electromagnetic environment prediction, proposals are solicited to develop technology that does the following:

- Bounds the expected peak electric field environment inside enclosures such as rocket fairings, and spacecraft enclosures. The method should include the technology required, the technique as well as the necessary verification efforts.
- To develop a numerical or statistically based methodology for characterizing shielding effectiveness of enclosures with associated applicable apertures.
- To develop methods for field enhancement/reduction based on thermal/acoustic blanketing and metal/composite components such as avionics and PAF structures.
- Develop preliminary user-friendly modeling software that can be easily customized to support NASA-specific applications.

Phase I Deliverables - Research, identify and evaluate candidate algorithms or concepts for electromagnetic field mapping of typical spacecraft and rocket enclosures. Demonstrate the technical feasibility and show a path towards a computer model development. It should identify improvements over the current state of the art for both time/resource savings and systems development and the feasibility of the approach in a varied-enclosure environment. Lab-level demonstrations are required. Deliverables must include a report documenting findings.

Phase II Deliverables - Emphasis should be placed on developing usable computer model and demonstrating the technology with under and over moded conditions with testing. Deliverables shall include a report outlining the path showing how the technology could be matured and applied to mission-worthy systems, verification test results, computer model with user's and other associated documentation. Deliverable of a functional computer model with associated software is expected at the completion of the Phase II contract.

Relevance to NASA

All NASA payloads, particularly those with hardware sensitive to electric fields will benefit from launch and ascent risk reduction.

References:

Expected Electric Field Prediction methods in Fairing/Aircraft and Spacecraft Enclosures:

- Paul G Bremner, Dawn Trout, Gabriel Vazquaz, Neda Nourshamsi, James C.West, and Charles F. Bunting, "Modal Q Factor and Modal Overlap of Electrically Small Avionics Boxes", Proc. IEEE Intnl. Symp. EMC, Long Beach, August 2018
- D. A. Hill, "Electromagnetic Fields in Cavities. Deterministic and Statistical Theories" John Wiley & Sons, Hoboken, New Jersey 2009
- J. Ladbury, G. Koepke, and D. Camell, "Evaluation of the NASA Langley Research Center Mode-Stirred Chamber Facility," NIST, Technical Note 1508, 1999.
- A. Schaffar and P. N. Gineste, "Application of the power balance methods to E- field calculation in the ARIANE 5 launcher payloads cavities," Presented at International Symposium on EMC, Long Beach, 2011, pp. 284-289.
- D.H. Trout, "Electromagnetic Environment in Payload Fairing Cavities," Dissertation, University of Central Florida, 2012.
- L. Kovalevsky, R.S. Langley, P. Besnier and J. Sol, "Experimental validation of the Statistical Energy Analysis for coupled reverberant rooms", Proc. IEEE Intnl. Symp. EMC, Dresden, August 2015
- Bremner, P.G, Vazquez, G., Trout, D.H and Cristiano, D.J., "Canonical Statistical Model for Maximum Expected Imission of Wire Conductor in an Aperture Enclosure", Proc. IEEE Intnl. Symp. EMC, Ottawa, October 2016
- G.B Tait, C. Hager, M.B. Slocum and M.O. Hatfiled, "On Measuring Shielding Effectiveness of Sparsely Moded Enclosures in a Reverberation Chamber", IEEE Trans. on EMC, Volume:55, Issue: 2, October 2012

Focus Area 6 Life Support and Habitation Systems

Lead MD: HEOMD
Participating MD(s): STTR

The Life Support and Habitation Systems Focus Area seeks key capabilities and technology needs encompassing a diverse set of engineering and scientific disciplines, all which provide technology solutions that enable extended human presence away from Earth, in deep space and on planetary surfaces, such as Moon and Mars. The focus is on those mission systems and elements that directly support astronaut crews, such as Environmental Control and Life Support Systems (ECLSS), Extravehicular Activity (EVA) Systems and Radiation Protection, as well as systems engineering approaches that enable vehicle and system integration.

Environmental Control and Life Support Systems encompass process technologies, equipment and monitoring functions necessary to provide and maintain a livable environment within the pressurized cabin of crewed spacecraft, including environmental monitoring, water recycling, waste management and resource recovery. For future crewed missions beyond low-Earth orbit (LEO) and into the solar system, regular resupply of consumables and emergency or quick-return options will not be feasible. Technologies are of interest that enable long-duration, safe and sustainable deep-space human exploration. Special emphasis is placed on developing technologies that will fill existing gaps, reduce requirements for consumables and other resources, including mass, power, volume and crew time, and which will increase safety and reliability with respect to the state-of-the-art. Because spacecraft may not be tended by crew for long periods, systems must be operable after long periods of dormancy or absence of crew.

As we consider human missions beyond earth, new technologies must be compatible with attributes of the environments we encounter, including partial gravity, atmospheric pressure and composition, space radiation, and presence of planetary dust. Portable Life Support System (PLSS) components that require space vacuum, may not operate in the weak carbon dioxide atmosphere on Mars. For astronauts to walk once again on a distant planetary surface, an effective boot must be incorporated into the design of the exploration space suit's pressure garment. Outside of the protection of the Earth's magnetosphere, radiation in deep space will be a challenge. Electronic systems, including processors for high performance computing and power converters, for avionics within spacecraft cabins and space suits, will need to be radiation hardened or otherwise tolerant to the radiation environment. There is a wealth of commercial off-the-shelf (COTS) hardware that could potentially be used, but only if tested for tolerance to these environments

The current collaborative environment between government, commercial and international sectors will result in the distributed development of human spacecraft elements and systems for human missions of the future, such as Gateway. Their integration may benefit from advances in model based systems engineering approaches.

Please refer to the description and references of each subtopic for further detail to guide development of proposals.

H3.02 Spacecraft Solid Waste Management (SBIR)
Lead Center: JSC
Participating Center(s): ARC, KSC, MSFC

This subtopic has two areas of scope. The primary area of emphasis is water recovery and stabilization of human metabolic waste (feces). The secondary scope seeks robust flow meters for effluent gas measurement from waste processors.

Water Recovery and Stabilization of Human Metabolic Waste (Feces)

Human solid waste (feces) contains ~75% water by mass that is currently not recovered on ISS. Feces are collected and stored in relatively impermeable containers for short term storage (1-3 months) and disposed of in departing logistics vehicles. Quantified, this represents approximately 170 g per crew member per day of recoverable water, which translates to 0.68 kg per day for a crew of 4 and can total as much as 680 kg for a 1,000-day long duration human exploration mission. In addition to water recovery, stabilization of feces is a critical gap for long duration human planetary exploration missions to Moon and Mars. Water removal is a first step in stabilization and has the

potential to decrease odor control technology mass. Technologies are requested to recover water and stabilize feces for use on long duration human exploration missions to Moon and Mars.

Simplified, low temperature, and robust methods for recovery of water from human solid metabolic waste are sought. Low temperature (<110 C) is desired to reduce the release of volatile organic compounds, avoid organic compound oxidation to CO and CO_2 and their subsequent treatment prior to return to the cabin air. The range of technologies can include air drying, vacuum drying, freeze drying and alternative low energy methods. The cost for recovering fecal water, in terms of mass, power, volume and crew time equivalents must not outweigh mass savings obtained by its recovery. Drying and stabilization of feces can reduce odor generation and prevent microbial proliferation if the water activity level is less than 0.6. Technologies must be able to recover >80% of the water content. Captured water should have minimal free gas and be suitable for eventual delivery to a waste water tank. Purification of the water is not requested because it will be processed by downstream treatment systems. However, the chemical constituents of the recovered water must be characterized. Technologies must be able to accommodate a wide range of condensable and non-condensable mass flow rates as the feces are processed and dried. Water recovery should be accommodated directly or with an assumed regenerative heat exchanger to recover energy prior to phase separation (as necessary). Systems must be capable of microgravity and/or planetary surface operation (moon or Mars) for 1 to 18 months at a time, with 11 to 18-month periods of dormancy, and with minimal crew maintenance. Compatibility with existing waste collection hardware is of interest. Planned fecal waste collection (Universal Waste Management System - UWMS) consists of individual defecations and hygiene wipes collected in gas permeable bags. 15-25 individual bags are contained in rigid containers that are changed out every 2-3 days.

It is highly desirable that on-demand manufacturing (i.e., additive manufacturing and post finishing) be considered for consumable or maintenance items. Technologies must consider accumulation of organics and microbial proliferation between normal waste processing cycles and extended dormancy and any change in performance should be characterized. Evolved gases during processing may require treatment and could consider absorbers and or materials (membranes) that prevent the transmission of volatile organic carbon. Thermal and power efficiency must be addressed. It is desirable that rigid UWMS canisters be reusable to reduce logistical resupply mass. Alternatives to the rigid UWMS canister are acceptable if it does not require significant changes of UWMS operations. It is desirable that the processed fecal material and associated wipes and bags occupy less volume than the preprocessed state. Information on UWMS can be found at: Logistics Reduction: Universal Waste Management System (LR-UWMS): https://techport.nasa.gov/view/93128.

Long Life Robust Flow Meters for Effluent Gases from Waste Processors

Currently human space exploration life support waste management systems have need for a robust mass flow measurement method in mixed water vapor and hydrocarbon gas flows that are produced from processing trash and waste streams. Thermal processing of trash and waste will evolve a ‘dirty’ gas flow comprised of large number of complex organic and non-organic gaseous mixtures with a high water vapor content. Mass flow measurement can be used for process control of the heating of the waste or as a non-condensable gas flow control device for a vehicle vacuum system. Current gas flow measurement systems do not operate well in a dirty gaseous stream for the operating conditions required on long duration missions such as to Mars. Gaseous compounds deposit on downstream process tubing and sensors with surface temperature and pressure conditions that are favorable to condensation. Over time, deposits can render sensors inaccurate or inoperative. Such effluents will include but are not limited to time-variant mixture of air, water vapor, gaseous organic and inorganic components, entrained liquids and sticky compounds that appear as precipitates. Information on trash processing and effluent characteristics are provided in the following reference: NextSTEP F: Logistics Reduction in Space by Trash Compaction and Processing System (TCPS): https://www.nasa.gov/nextstep/trash.

The proposed flow sensor technologies will operate in a range of conditions depending on the trash processing application. Sensors should be capable of operation over the range of 6-55 kPa, temperature ranges of 15-180 C, and 0.001-2.0 g/min. The sensor will be exposed to high relative humidity and saturated water vapor conditions, and to volatile organic gases. The sensor will be monitoring flow in fluid passages from 3-20 mm. It is desired that proposed technologies have low pressure drop, contaminant tolerant surfaces, and long-term operation between calibrations. The accuracy of the proposed technology across the range of operation conditions should be defined in the proposal.

For both areas of scope, hardware attributes should include robust design, low volume and compact size, low mass, reduced or zero requirements for crew time, and minimized consumable mass. Phase I Deliverables - Reports demonstrating proof of concept, test data from proof of concept studies, concepts and designs for Phase II. Phase I tasks should answer critical questions focused on reducing development risk prior to entering Phase II. Phase II Deliverables - Delivery of technologically mature components/subsystems that demonstrate performance over the range of expected spacecraft conditions. Prototypes must be full scale unless physical verification in 1-g is not possible. Robustness must be demonstrated with long term operation and with periods of intermittent dormancy. System should incorporate safety margins and design features to provide safe operation upon delivery to a NASA facility.

The expected TRL for these scopes is 2 to 4.

References:

- National Aeronautics and Space Administration, NASA Technology Roadmaps, TA 6: Human Health, Life Support, and Habitation Systems (National Aeronautics and Space Administration, Draft, May 2015, www.nasa.gov/sites/default/files/atoms/files/2015_nasa_technology_roadmaps_ta_6_human_health_life_support_habitation.pdf).
- Miriam Sargusingh, Molly Anderson, Jay Perry, Robyn Gatens, James Broyan, Ariel Macatangay, Walter Schneider and Nikzad Toomarian "Development and Maturation for Exploration: 2017 to 2018 Overview", 48th International Conference on Environmental Systems, Paper ICES-2018-182 https://ttu-ir.tdl.org/ttu-ir/bitstream/handle/2346/74153/ICES_2018_182.pdf
- Logistics Reduction: Universal Waste Management System (LR-UWMS): https://techport.nasa.gov/view/93128
- Advanced Exploration Systems Program, Life Support Systems Project: https://www.nasa.gov/content/life-support-systems
- NextSTEP F: Logistics Reduction in Space by Trash Compaction and Processing System (TCPS): https://www.nasa.gov/nextstep/trash
- Fisher, John W. and Lee, Jeffrey M. "Space Mission Utility and Requirements for a Heat Melt Compactor", 46th International Conference on Environmental Systems, Paper ICES-2016-377 https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20160008947.pdf
- Lee, Jeffrey M., Fisher, John W. and Pace, Gregory "Heat Melt Compactor Development Progress", 47th International Conference on Environmental Systems, Paper ICES-2017-267 https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20170011317.pdf
- John P. Wetzel, Robert J. Surdyk, Joe Klopotic and Krishnaswamy K. Rangan "Heat Melt Compactor Test Unit", 48th International Conference on Environmental Systems, Paper ICES-2017-267 https://ttu-ir.tdl.org/ttu-ir/bitstream/handle/2346/74255/ICES_2018_318.pdf

H3.03 Microbial Monitoring and Control for Spacecraft Cabins (SBIR)
Lead Center: JPL
Participating Center(s): JPL, JSC, KSC, MSFC

This subtopic has two areas of scope. The primary area of emphasis is non-gene based microbial monitoring technologies. The secondary scope is alternative methods and agents for microbial control in potable water systems.

Spacecraft Microbial Monitoring for Long Duration Human Missions

With the advent of molecular methods, emphasis is now being placed on nucleic acids to rapidly detect microorganisms. However, automation for these systems is still in development and the time from sample collection to result output is not instantaneous. Recent advancements in the field of metabolomics have potential to substitute (or augment) current gene-based microbial detection technologies. NASA is soliciting non-gene based microbial detection technologies and systems that target microbial metabolites and which quantify the microbial burden of surfaces, air, and water inside future long-duration deep space habitats.

Airborne Contamination
Future human spacecraft such as Gateway and Mars vehicles may be uncrewed between missions. Crew could be absent from the vehicle for periods that could last up to 1 to 3 years. Before crews can return, these environments must be verified prior to crew return. Novel methods that have the potential to enable remote autonomous microbial monitoring are sought, which do not require manual sample collection, preparation or processing.

Potable Water
A simple integrated, microbial sensor system that enables sample collection, processing, and detection of microbes or microbial activity in a spacecraft potable water supply is sought. A system that is fully-automated and which could be integrated within a spacecraft's water processing system as an in-line detector is preferred. Such a system could be used to monitor microbial burden in the water supply during both uncrewed and crewed operations.

Habitat Surfaces
Future habitats for human habitation of cis-lunar space, such as Gateway, are expected to be crew tended only 1 to 3 months at a time and then left unoccupied for many months between missions. When the crew returns to occupy the habitat they will want to quickly, efficiently, and accurately assess the microbial status of the habitat surfaces. A microbial assessment / monitoring system or hand-held device that requires little to no consumables is sought.

Alternative Sanitation Agents for Potable Water

For water recovery during human exploration missions, NASA ensures compliance with microbial requirements by initially disinfecting the process water and removing organic content that serves as nutrients for microbial growth. In addition, a biocide is added to the potable water as further mitigation against microbial growth. For the Shuttle and International Space Station programs, NASA used iodine as the biocide. However, iodine can create health issues for the crew and thus has to be removed from the potable water prior to crew consumption. This approach is undesirable for future missions and thus NASA is pursuing new sanitation agents and methods for spacecraft potable water.

The use of silver at biocidal concentrations of 0.05 – 0.4 mg/L is under consideration, but dosing and maintenance in potable water systems have not been satisfactorily worked out. Alternative biocides may be available. NASA seeks a biocide that provides effective microbial control at a given concentration, can be reasonably added to the process water, is acceptable for long term storage prior to use, can be consumed by the crew for long duration without undesirable side effects, and is compatible with typical materials used in potable water systems such as Teflon, Viton, 316 L SS, Inconel 718, and Titanium.

Phase I Deliverables
Reports demonstrating proof of concept, test data from proof of concept studies, concepts and designs for Phase II. Phase I tasks should answer critical questions focused on reducing development risk prior to entering Phase II. The expected TRL for these scopes is 2 to 4.

Phase II Deliverables
Delivery of technologically mature components/subsystems that demonstrate performance over the range of expected spacecraft conditions. Prototypes must be full scale. Robustness must be demonstrated with long term operation and with periods of intermittent dormancy. Systems and chemical agents should incorporate safety and design features to provide safe operation upon delivery to a NASA facility.

Hardware attributes should include robust design, low volume and compact size, low mass, reduced or zero requirements for crew time, and minimized consumable mass. For example, typical ISS Express Rack instruments have a volume of 64 L and a mass of 30 kg; a reasonable goal for this subtopic would be 10 L and 10 kg for an autonomous instrument; closer to 1 L for a hand-held device.

References:

- Pierson, D., Botkin, D. J., Bruce, R. J., Castro, V. A., Smith, M. J., Oubre, C. M., Ott, C. M., “Microbial Monitoring of the International Space Station,” in Environmental Monitoring: A Comprehensive Handbook, edited by J. Moldenhauer, DHI Publishing: River Grove, IL., 2012, pp. 1-27.

- A list of targeted abiotic contaminants for environmental monitoring can be found at "Spacecraft Water Exposure Guidelines for Selected Waterborne Contaminants" located at https://www.nasa.gov/feature/exposure-guidelines-smacs-swegs
- Li, Wenyan, Calle, Luz, Hanford, Anthony, Stambaugh, Imelda and Callahan, Michael "Investigation of Silver Biocide as a Disinfection Technology for Spacecraft – An Early Literature Review", 48th International Conference on Environmental Systems, Paper ICES-2018-82 https://ttu-ir.tdl.org/ttu-ir/bitstream/handle/2346/74083/ICES_2018_82.pdf

H4.01 Exploration Portable Life Support System (xPLSS) for Deep Space and Surface Missions (SBIR)
Lead Center: JSC
Participating Center(s): JSC
Related Subtopic Pointer(s): H4.04

The current Extra-Vehicular Mobility Unit (EMU) on International Space Station (ISS) has a limited life span. NASA plans to continue using the current EMU/spacesuit for the life of ISS. Future missions being contemplated will also need a new spacesuit to meet the technology objectives. With the anticipation of a replacement suit for ISS or other future mission, the plan for an Exploration EMU (xEMU) is underway. As part of the xEMU, an Exploration Portable Life Support Subsystem (xPLSS) is currently being developed, integrated, and tested in house at JSC. Technology gaps remain for the xPLSS especially for deep space and surface missions. The first focus area is for small, radiation hardening (Rad-Hard), isolated direct current (DC) to DC (DC/DC) converters with an efficiency of greater than 80 percent (%) which would be helpful in low-earth orbit as well as deep space. The second focus area is for a boost compressor to enable the xPLSS pressure swing adsorption (PSA) carbon dioxide (CO_2) and water (H_2O) removal system to function in a partial atmosphere which exists on Mars. The boost compressor would be helpful in a surface mission to Mars. Based on these xPLSS technology gaps, the two focus areas are specifically detailed below for the SBIR 2019 solicitation cycle:

Small, Rad-Hard, Isolated DC/DC Converters with an Efficiency of >80%

For spacesuit life support systems, there are a number of small point of load applications such as smart instruments, controllers, etc. that require small, low power output, isolated DC/DC converters. With derating and the limited offering available from existing catalog parts, the available efficiency is often much lower than the rated efficiencies advertised for the part of 70-80% as the converter losses become a larger part of the overall output dropping the realized efficiencies below 65%. Develop a converter family with the following attributes:

- Power output families: 500mW, 1W, 2W, 5W:
    - The Electrical, Electronic, and Electromechanical (EEE) and Mechanical Parts Management and Implementation Plan for Space Station requires derating depending on the implementation that can range from 0.5-0.75 permissible of rated power output
- Input voltage: 34VDC max, 28VDC nominal
- Output voltages: Variable set by resistor with 1.5-15 VDC range minimum
- Regulation accuracy < 1%
- Converter efficiencies: >80% after EEE derating
- Total dose: >30 krad (Si)
- Single Event Latch (SEL) immune to 60 MeV-cm^2/mg
- Single Event Upset (SEU) errors less than 10^{-2} events/2000 hrs operating time
- Overcurrent protection, set by a resistor
- Over-temperature protection – shutdown
- Under-voltage lockout
- Soft-start, set by capacitor
- Packages: hermetically sealed flatpacks
- Options: programmable switching frequency, set by a resistor

Boost Compressor to Enable xPLSS Pressure Swing Adsorption (PSA) CO_2/H_2O Removal Function in Partial Atmosphere

The state of the art with respect to continuously regenerable carbon dioxide (CO_2)/water (H_2O) removal functions small enough to deploy in a spacesuit life support system is the amine swingbed using pressure swing adsorption. The current xPLSS system design planned for xEMU Demonstration on ISS will function in 1-2 torr condition. However, the current xPLSS amine swingbed system cannot function in the martian environment. Therefore, develop a boost compressor that can meet the following basic performance goals in a partial environment:

- Outlet Pressure: 0.2-9 torr
- Robust outlet pressure tolerance: System integration via pressure switches and valving can preclude outlet pressures higher than the 9 torr listed above. However, a concept with tolerance of pressures up to 15.2 psia would greatly simplify the integration and potentially produce a more reliable system.
- Inlet Pressure: 5 psia down to 0.1 torr
- Robust inlet pressure: System integration via pressure switches and valving can preclude inlet pressures higher than the 5 psia listed above. However, a concept with tolerance of pressures up to 19.5 psia would greatly simplify the integration and potentially produce a more reliable system.
- Effective pumping Speed > 600 lpm
- Constituents in the flow: oxygen (nearly 100% at initial opening of the bed), CO_2, H_2O, ammonia (NH_3) Motivation from 28VDC (nominal) input BLDC Stepper motor
- Structure born vibration needs to be minimized
- Conduction cooling for waste heat
- Operational life > 5000 hours

Phase I products: By the end of Phase I, it would be beneficial to have a concept design for infusion into the xPLSS. Testing of the concept is desired at this Phase.

Phase II products: By the end of Phase II, a prototype ready for system-level testing in the xPLSS or in a representative loop of the xPLSS is desired.

The xPLSS Subtopic is relevant to the xEMU Project, Human Exploration and Operations Mission Directorate (HEOMD), International Space Station, and Gateway for space suit development. More focused development beyond SBIR could come through Space Technology Mission Directorate (STMD).

The transition from the current Extravehicular Mobility Unit (EMU) to a new Exploration EMU (xEMU) for deep-space missions will necessitate a relevant demonstration of critical life support capabilities including scarring for upgrades to a full xEMU functionality. As the strategy for human space exploration beyond low-Earth Orbit (LEO) progresses, the plan for an Exploration xEMU flight demonstration suit on ISS in the mid-2020s will offer flexibility and adaptability to accommodate several potential outcomes. The progressive development, integration, and testing to prepare for a flight demonstration unit will enable a technical path to the current EMU replacement or a deep-space exploration Extravehicular Activity space suit or both.

The expected TRL for this project is 2 to 4.

H4.04 Exploration Pressure Garment System (xPGS) for deep space and surface missions (SBIR)
Lead Center: JSC
Participating Center(s): JSC
Related Subtopic Pointer(s): H4.01

Surface Space Suit Boot

This subtopic is searching for concepts and technologies to be incorporated into future prototypes of a space suit boot for a space suit that will walk on a planetary surface. Figure 1 (see https://www.nasa.gov/suitup 'EVA Boot SBIR Figures') communicates the rough terrain that astronauts will encounter and will be expected to traverse.

The Phase I effort may focus on a particular challenge of the design in order to mature that concept or technology to progress to a Phase II effort. Test data and detailed design or 3-D models are acceptable results from Phase I. The Phase I final report also must indicate how the Phase I effort allows for successful implementation in the Phase II effort. Expected TRL is from 2 to 5.

The expectation for the Phase II effort is that a functional boot prototype will be created, tested at pressurized conditions, and delivered. As stated above, the results from Phase I are expected to provide confidence that the technology or design can be carried into a Phase II resulting in a boot prototype delivery. A functional boot prototype from Phase II would need to meet the following requirements:

- Carry pressure load of suit pressurized to 8 psi:
    - Indicates that a load path must be incorporated into the design and proven to carry the pressure load
    - Indicates that the entire structure can be pressurized to 16 psi without failing
- Provide mobility at the ankle:
    - No less than 40° of flexion (toes toward shin; 0° position to be used as the reference is foot flat on the floor with the leg 90° to the foot), and 20° extension (pointed toes)
- Provide boot sole flexibility when pressurized equivalent to that of an Air Force jump boot, sturdy hiking boot, or work boot:
    - This requirement has not yet been quantified, but subjectively the boot sole must function in a similar manner to these soles.
- Indicate ability to meet boot cycle life:
    - The vendor shall not be required to test the boot to the anticipated cycle life of 800,000 steps, unless that is the primary requirement being addressed by the proposal.
    - All Phase II boot designs delivered shall be delivered with test and/or analysis data that provides evidence/rationale for being able to meet the cycle life requirement over a reasonable development effort (1-3 years).
- Provide information to assure the design will function in an environment with surface temperatures of the Lunar Surface: -180 to 210° F
- While surface boots will need to operate in a dust environment, dust mitigation methods are currently being developed under separate efforts and is not the focus of this topic. However, a discussion of why the design/technology/materials are expected to perform in a dust environment shall be expected in the proposal.

Space suit walking boot continues to be a challenge. Integrating the high performance of hiking boots into a pressurized garment creates challenges of its own and are beyond those of a typical boot. While all hiking boots need to provide good fit and to be durable for thousands to hundreds of thousands of walking steps over rough terrain, meeting these requirements for a boot that is inflated to 8 pounds per square inch of pressure introduces new aspects to the challenge. Each of these challenges that can be addressed are described below:

Boot-to-foot integration

The boot is integral with the knee of the pressure garment in order to close the air retention layer function for the leg, the diameter of the ankle opening is matched to that of the knee which creates a challenge for good fit at the ankle. Additionally, the integration of the boot to knee of the suit means that the boot is donned when it is connected to the leg. Therefore, these two drivers make donning a space suit boot similar to stepping into galoshes. However, when walking, the foot must be well integrated with the boot in order to avoid injury to the foot and maintain walking stability and control. Mitigating slipping of the heel in the boot has been the focus of several efforts. Below is a brief summary of boot-to-foot integration concepts that have been investigated previously:

- Boot indexing concepts:
    - Straps:
        - Strap over the arch of the foot has been helpful:
            - See Figure 2 (see https://www.nasa.gov/suitup 'EVA Boot SBIR Figures')
        - Thickness, location, and tenacity of closure method are all critical design factors
    - Magnets:
        - Did not work well:

            - Attached to Liquid Cooling Garment (= Footed longjohns worn under the suit) bootie
        - Slippage; magnets not powerful enough resulted in heel lift:
            - Comfort can also be an issue due to a hard object under the heel
        - Could be worth looking at again with an overboot concept and/or electromagnets
    - Heel clip (overboot):
        - Developed for the AX-5 prototype space suit, however test data has not been located, if it exists:
            - See Figure 3 (see https://www.nasa.gov/suitup 'EVA Boot SBIR Figures')
        - Could be worth looking at again
    - Air bladders:
        - Reebok Pumps-style – Implementation of repeated attempts has been problematic:
            - Location of bladders
            - Shape of bladders
            - Pressurization system complexity and reliability
            - Achieving acceptable pressure
    - Boa lacing:
        - Z-2 added heel lace in addition to dorsum lace, which was helpful, but did not completely resolve heel slippage issue:
            - See Figure 4 (see https://www.nasa.gov/suitup 'EVA Boot SBIR Figures')
    - Other ideas:
        - Vacuum pack forming to grab ankle using ambient vacuum
        - Power Strap (ski boots)

Boot design concepts that, in addition to performing its function of pressure retention/providing pressure to the foot, also allows for ease of don/doff and provides excellent and comfortable foot-to-boot integration are sought.

Boot material durability

Looking forward to long-duration exploration missions, the materials of the boot will be exposed to rough terrain over hundreds of thousands of walking cycles. Boot sole and boot upper materials are being sought for durability and flexibility in the rough environment and extreme temperatures of the space environment. In addition to being durable, the boot sole material also needs to serve as a functional boot sole when incorporated in a pressure garment, which means that the sole cannot allow the pressurization to modify its shape. For example, past materials that have been investigated have allowed the boot sole to become convex or rocker-shaped indicating that the boot sole material was too flexible. Similarly, boot sole materials that are too stiff so as not to allow natural gait have also been rejected.

The Phase I effort proposals shall be expected to address one or more aspects of these challenges in an innovative way.

This project could enable sustained Lunar surface EVAs as part of a human lunar program. More focused development beyond SBIR could come through Space Technology Mission Directorate (STMD).

References

- Ross, Amy, Joseph Kosmo, Nikolay Moiseyev, Anatoly Stoklitsky. "Comparative Space Suit Boot Test." 32nd ICES. SAE. July 2002, San Antonio, TX: SAE, 2002.
- Ross, Amy, Richard Rhodes, Shane McFarland. “NASA’s Advanced Extra-vehicular Activity2 Space Suit Pressure Garment 2018 Status and Development Plan.” 48th International Conference on Environmental Systems (ICES). July 2018, Albuquerque, New Mexico: ICES-2018-273.
- Ross, Amy, Richard Rhodes, David Graziosi, Bobby Jones, Ryan Lee, Bazle Haque, John W. Gillespie, Jr. “Z-2 Prototype Space Suit Development.” 44th ICES. July 2014, Tucson, AZ: NTRS JSC-CN-31290.
- Ross, Amy. ‘Z-1 Prototype Space Suit Testing Summary.” 43rd International Conference on Environmental Systems (ICES). American Institute of Aeronautics and Astronautics (AIAA). July 2013, Vail, CO: AIAA, 2013: NTRS JSC-CN-28415.

H6.04 Model Based Systems Engineering for Distributed Development (SBIR)
Lead Center: ARC
Participating Center(s):
Related Subtopic Pointer(s): S5.05, T11.03, Z8.03, Z8.07

As NASA looks to develop a cis-lunar infrastructure, starting with components like the Gateway, there will be considerable interest in partnerships with a wide variety of communities. Building from the success of the international partnerships for ISS, space agencies from multiple governments are looking for roles on the Gateway. The rapidly growing commercial space industry is also likely to seek roles in supporting this infrastructure. All of these potential partners will have their own design capabilities, their own development processes, and internal constituencies to support. Enabling disparate systems built in different locations by different owners to all work cohesively together will require a significant upgrade to NASA's core systems engineering toolset.

Model Based Systems Engineering holds considerable promise for facilitating this type of distributed development process, but we need to significantly improve and expand the engineering support infrastructure to enable the systems we will need for lunar exploration. Methodologies that support integration amongst tools and exchange of information between multidisciplinary artifacts are important development opportunities. The definition of interface standards and tools that enable inspection of distributed models across domains are very important. Tools or systems that allow models to be shared across development environments and trace the resulting systems back to contributions from multiple partners are also of high interest. SysML related tools are relevant to this subtopic, but need to address distributed development, multi-disciplinary system development, and the engineering of interfaces between subsystems built by different communities from requirements through testing, verification, and validation.

Model Based Systems Engineering for distributed development is relevant to all Human Exploration Operations Mission Directorate (HEOMD) missions, and of timely interest for Gateway development. Over the next 3 to 5 years, there will be considerable opportunity for small business contributions to be matured and integrated into the engineering support infrastructure as Gateway evolves from concept to development program.

During Phase I, research should be conducted to demonstrate methodologies and tools that support distributed multi-disciplinary development efforts, their technical feasibility, and NASA relevance.

Phase I proposals should clearly indicate how the research will go beyond state of the art engineering practices. Prototypes are strongly encouraged and could take several forms such as augmentations/plugins to existing SySML tools. Phase II deliverables should include at a minimum demonstration of a prototype tool or methodology on a small system(s) that is representative or analog of a portion of lunar infrastructure, and documentation with source for NASA to explore use of the tool. The expected TRL for this project is 5 to 7.

References:

References documenting current State of Practice within NASA - proposals shall address technology advances beyond state of practice:

- NASA Office of Safety and Mission Assurance: Model-Based Mission Assurance https://sma.nasa.gov/sma-disciplines/model-based-mission-assurance
- NASA Engineering and Safety Academy: Systems Engineering https://nescacademy.nasa.gov/category/3/sub/17
- NASA/SP-2016-6105/SUPPL/Vol 2: Expanded Guidance for NASA Systems Engineering. Volume 2: Crosscutting Topics, Special Topics, and Appendices. Section 8.2 Model Based Systems Engineering https://ntrs.nasa.gov/search.jsp?R=20170007239

General References for Model-Based System Engineering for Distributed Development, and relevant NASA Missions:

- Research Challenges in Modeling & Simulation for Engineering Complex Systems http://trainingsystems.org/publications/Research-Challenges-in-Modeling-and-Simulation-for-Engineering-Complex-Systems.pdf
- https://ieeexplore.ieee.org/document/6899129/
- https://www.incose.org/docs/default-source/space-systems-working-group/2015-gsfc-se-seminar---incose-sswg-cubesat-model-status.pdf?sfvrsn=4d3087c6_0
- https://www.sae.org/publications/technical-papers/content/2011-01-2664/
- https://www.nasa.gov/feature/deep-space-gateway-to-open-opportunities-for-distant-destinations
- https://www.nasa.gov/topics/moon-to-mars/lunar-outpost

Papers where MBSE was implemented as a pathfinder on a NASA project:

- Modeling to Mars: a NASA Model Based Systems Engineering Pathfinder Effort https://ntrs.nasa.gov/search.jsp?R=20170009110
- Using A Model-Based Systems Engineering Approach for Exploration Medical System Development https://ntrs.nasa.gov/search.jsp?R=20170008864
- Using Model-Based Systems Engineering to Provide Artifacts for NASA Project Life-Cycle and Technical Reviews https://ntrs.nasa.gov/search.jsp?R=20170008864

Forward-looking documents describing challenges and opportunities for using MBSE at NASA:

- NASA/TM-2017-219633, M-1435: Digital Model-Based Engineering: Expectations, Prerequisites, and Challenges of Infusion https://ntrs.nasa.gov/search.jsp?R=20170006995
- Research Challenges in Modeling & Simulation for Engineering Complex Systems http://trainingsystems.org/publications/Research-Challenges-in-Modeling-and-Simulation-for-Engineering-Complex-Systems.pdf

T6.05 Testing of COTS Systems in Space Radiation Environments (STTR)
Lead Center: LaRC
Participating Center(s): LaRC
Related Subtopic Pointer(s): Z8.03, Z1.03, Z6.01

The use of COTS (Commercial Off-The-Shelf) parts in space for electronics is a potential significant enabler for many capabilities during a mission. This subtopic is seeking a better understanding of the feasibility of COTS electronics for High Performance Computing (HPC) in space environments which are already heavily shielded. It seeks strategies based on a complete system analysis of HPC COTS that include, but not limited only to, failure modes to mitigate radiation induced impacts to potential HPC systems in those highly shielded space environments.

As background, spacecraft experience exposure to damaging radiation and that amount of exposure from various sources, (e.g., sun and galactic cosmic radiation sources) increases notably as the spacecraft ventures further away from the Earth's magnetic field, since the magnetic field offers some level of protection. As spacecraft, and their electronic systems, proceed again to the moon and further into deep space, considerable work has and continues to be done to evaluate and determine how to appropriately protect the astronauts and to shield or otherwise protect various spacecraft, habitats, and their electronic systems, depending upon the needs of the missions.

Many of the most protective physical shielding approaches known result in infrastructure which is too heavy for what is considered acceptable for many missions' intended launch and spaceflight conditions. Therefore, typically lighter infrastructure shielding is presently being used when and where possible. Spacecraft faring deeper into space for fly-by missions (e.g., New Horizons), orbiters (e.g., Mars Orbiter), or landers (e.g., Mars Rover) are examples of such relatively lightly shielded systems. The lighter shielding sacrifices some radiation protection and therefore results in some limitations in what their electronic systems, especially High-Performance Computing (HPC), could do (e.g., more on-board processing which could reduce by orders of magnitude the volume of data needed to be transmitted back and forth to Earth; and increase actual data collection rates for the mission at hand). There are already ongoing projects to upgrade the current radiation workhorse CPU (RAD750) by an order of magnitude, but this is not a COTS item and is expensive to manufacture and to buy. For critical systems that must be operational continuously and which

may also have more lightly shielded systems, there is no other option at this time. This subtopic does not seek work of that nature.

Unlike the lightly shielded space environments discussed above, space environments which are highly shielded from radiation, such as is inherently the case for the interiors of manned missions and for habitats where humans live and work, high level radiation hardened systems like the relatively expensive RAD750 may not be as necessary even in deeper space beyond most of the present day LEO (Low Earth Orbit) situations. Instead, a less expensive COTS solution for the HPC system may be acceptable for a number of non-critical tasks that are not harmed by power interruptions, hardware failures, radiation upsets, etc. in those environments over what may have been thought likely. In order to assess the feasibility of a COTS solution for those types of highly shielded space environments, this subtopic is seeking proposals.

Successful Small Business Concern/Research Institution teams would be able to do space radiation modeling and a complete analysis of the COTS related HPC systems (e.g., modelling for an appropriate space relevant environment; destructive testing and analysis; and testing in an appropriate space relevant environment (e.g., in particle beams)). Further, since all parts in these HPC systems cannot be tested, an understanding of what parts are susceptible to radiation damage (e.g., Solid State Drives - SSDs) is crucial so as to create the list of potential test candidates.

Phase I Proposers are expected to develop a plan or strategy that explains and details how they would approach solving the problem that helps NASA mitigate radiation induced failures in the HPC system/components, identify COTS equipment that are likely candidates based on environmentally relevant testing, as well as modeling of interior environment and data analysis of similarly known/used approaches like the Orion vehicle testing (EM-1 when released). They should highlight the innovation in the suggested approach and explain why it would be a better solution over what may presently be used. Additionally, they should also indicate how the proposed strategies could be used commercially if developed. Phase I concept studies are expected raise the TRL to at least a 3/4 when completed. Phase II proposals would use that innovative approach to refine any and conduct further relevant interior environmental modeling and conduct the space radiation relevant testing and analysis on the selected COTS HPC parts/systems which could lead toward creating prototypes of the potential commercial items that come from the analysis. The deliverables from a successful Phase II is expected to raise the TRL to 5/6. Phase III would commercialize those items.

Relevance to NASA

The results from a project addressing this subtopic could be relevant to any NASA mission or project (e.g., as originating out of the Human Exploration and Operations Mission Directorate - HEOMD and the Space Technology Mission Directorate - STMD) and any commercial space activity that intends to send humans beyond LEO - Low Earth Orbit with a HPC system for reasons relevant to the mission.

References:

There are many references on each individual aspect of the work involved but very few references on the entire process wanted. For a tool that can model the radiation environment inside a spacecraft:
OLTARIS: On-line Tool for the Assessment of Radiation in Space, NASA/TP-2010-216722, July 2010. R.C.Singleterry, S.R.Blattnig, M.S.Clowdsley, G.D.Qualls, C.A.Sandridge, L.C.Simonsen, J.W.Norbury, T.C.Slaba, S.A.Walker, F.F.Badavi, J.L.Spangler, A.R.Aumann, E.N.Zapp, R.D.Rutledge, K.T.Lee, R.B.Norman.
A reference to help understand the radiation testing of powered COTS parts, see:
Correlation of Neutron Dosimetry Using a Silicon Equivalent Proportional Counter Microdosimeter and SRAM SEU Cross Sections for Eight Energy Spectra, IEEE Transaction on Nuclear Science, Vol.~50, No.~6, pp.~2363-2366, Decmeber 2003. B.Gersey, R.Wilkins, H.Huff, R.C.Dwivedi, B.Takala, J.O'Donnell, S.A.Wender, R.C.Singleterry.

T6.06 Spacecraft Water Sustainability through Nanotechnology (STTR)
Lead Center: JSC
Participating Center(s): ARC, JSC, KSC, MSFC

Nanotechnology Innovations for Spacecraft Water Management Applications

Water recovery from wastewater sources is key to long duration human exploration missions. Without substantial water recovery, life support system launch weights are prohibitively large. Regenerative systems are utilized on the ISS to recycle water from humidity condensate and urine, but the Urine Processor and Water Processor Assemblies contain rotary systems and produce brines (Distillation Assembly), utilize non-regenerable consumables (Multi-Filtration Beds) and operate at high temperature and pressures (Catalytic Reactor). To stabilize urine and protect components from biofouling and precipitation, a toxic pretreatment formula is added to collected urine. Simple measurements of water composition are made during flight, including conductivity, total organic carbon and iodine concentration. For determination of ionic or organic species in water and wastewater, samples must be returned to earth.

This subtopic solicits improvements to reduce complexity, decrease consumable mass, improve safety and reliability, and to achieve a higher degree of autonomy are of interest. In the past decade, technology developers have used nanotechnology to improve capabilities of catalytic oxidation, microbial control, surface fouling, disinfection, water quality monitoring, nano-photonic heating and distillation, selective and reversible removal of trace contaminants, and transport and delivery of treatment systems using nano-carriers. This solicitation deliberately requests for “technology building blocks” that demonstrate new nanotechnology capability which can favorably impact the NASA water recovery application. Because of the interconnected nature of water recovery systems, it is hard to insert new technology into an existing system. When key subsystem technologies are developed and demonstrated, new system level approaches can be implemented.

This solicitation targets three key aspects of water management for human spacecraft. These areas of scope are aligned with the three specific thrusts described within the white paper of the Nanotechnology Signature Initiative (NSI) "Water Sustainability through Nanotechnology". Please see references for additional information, including water quality requirements and guidelines.

Water Recovery from Wastewater: Increasing Water Availability Using Nanotechnology

- NASA is seeking nanotechnology based technologies capable of processing up to 10 liters/day urine, with >95% water recovery, system energy use <300 Watts, and contaminant levels in distillate less than 1.5 mg/l for organics, and less than 0.3 mg/l for ammonia.
- Technologies for water recovery from mixed streams of an exploration wastewater (containing hygiene, clothes wash, etc.) are also of interest.
- Water reuse systems must be capable of operating for 6 months at a time with dormancy periods of up to 2 years in between operations.
- For potential future bioregenerative life support applications involving growth of crop plants for production of food, process water may include agricultural waste waters, and there may be interest in separation of sodium chloride, nitrogen, potassium, phosphorous and other nutrients from waste water for reuse in plant growth systems.

Stabilization of Water and Water Recovery System Hardware - Improving the Efficiency of Water Delivery and Use with Nanotechnology

Biological growth on condensing heat exchanger surfaces and in plumbing lines and tanks (for both potable water and wastewater) is a significant concern in water systems for future manned missions:

- NASA is seeking methods to maintain concentrations of biocidal silver (0.05 – 0.4 mg/L) in potable water including surface treatments that may limit silver loss.
- Biofilm growth can obstruct flow paths in operational wastewater collection and processing systems, especially in tanks where stagnant conditions lead to consistent growth. A greater concern is missions beyond

ISS that include dormant periods when the spacecraft is not tended by crew, during which biofilm growth would be even more significant. NASA is currently considering the concept of flushing the wastewater plumbing with potable water to reduce microbial and organic content before dormancy, though additional controls are required to insure biofilm growth does not impact operations once the crew returns to the vehicle. NASA seeks robust design solutions that mitigate (but not necessarily eliminate) biofilm growth in plumbing and tanks during nominal operations and to prepare a system for dormancy. Design solutions must be viable for implementation with minimal crew time (automated concepts are much preferred) and must be compatible with materials typically used in water plumbing (for example viton, Teflon, 316L SS, Inconel 718). Treatments that also reduce scale and solids build up are of interest.

- Alternative pretreatment methods are of interest for urine and wastewater, to inhibit microbial growth and to prevent precipitation of calcium salts and production/evolution of ammonia. Nanotechnology solutions may allow for the elimination of use of pretreatment chemicals classified as toxicity level 2 or higher.

Enabling Next-Generation Water Monitoring Systems with Nanotechnology

- Multi-species analyte measurement capability is of interest that would be competitive to standard water monitoring instruments such as ion-chromatography, inductively coupled plasma spectroscopy, and high-performance liquid chromatography. Components that enable the miniaturization of these monitoring systems, such as microfluidics and small-scale detectors, will be considered.
- NASA is seeking nano-sensors that measure pH, ionic silver (Continuous in-line measurement of ionic silver (range 10 to 1000 ppb), conductivity, TOC (minimum detection level 50 ppb) with >3-year service life and >50% size reduction compared to current SOA.
- Applications exist for monitoring species within regenerated potable water and/or wastewater (potential waste streams: urine, humidity condensate, Sabatier product water, waste hygiene, and waste laundry water).

While NASA is looking for innovative solutions to any aspect of water management as described above, several focused areas are of particular interest. Innovations that target improvements to delivery and maintenance of silver for use as a biocide in potable water, surface treatments and methods that suppress biofilm growth and support system dormancy, multi-analyte species monitoring capability and/or energy efficient distillation, are especially welcome. Expected TRL is from 2 to 4.

References:

- NASA is a collaborating agency with the NTSC Committee on Technology Subcommittee on Nanoscale Science, Engineering and Technology's Nanotechnology Signature Initiative (NSI): "Water Sustainability through Nanotechnology" (Water NSI). For a white paper on the NSI, see https://www.nano.gov/node/1580
- A high-level overview of NASA's spacecraft water management was presented at a webinar sponsored by the Water NSI: "Water Sustainability through Nanotechnology: A Federal Perspective, Oct. 19, 2016" https://www.nano.gov/publicwebinars
- A general overview of the state of the art of spacecraft water monitoring and technology needs was presented at a webinar sponsored by the Water NSI: "Water Sustainability through Nanotechnology: Enabling Next-Generation Water Monitoring Systems, Jan. 18, 2017" located at https://www.nano.gov/publicwebinars
- For a list of targeted contaminants and constituents for water monitoring, see "Spacecraft Water Exposure Guidelines for Selected Waterborne Contaminants" located at https://www.nasa.gov/feature/exposure-guidelines-smacs-swegs
- Technical papers on a wide variety of Environmental Control and Life Support System (ECLSS) topics are available at https://www.ices.space/conference-proceedings/

Focus Area 7 Human Research and Health Maintenance

Lead MD: HEOMD

NASA’s Human Research Program (HRP) investigates and mitigates the highest risks to astronaut health and performance for exploration missions. The goal of the HRP is to enable space exploration beyond low earth orbit by reducing the risks to human health and performance through a focused program of basic, applied and operational research leading to the development and delivery of:

- Human health, performance, and habitability standards.
- Countermeasures and other risk mitigation solutions.
- Advanced habitability and medical support technologies.

HRP has developed an Integrated Research Plan (IRP) to describe the requirements and notional approach to understanding and reducing the human health and performance risks. The IRP describes the Program's research activities that are intended to address the needs of human space exploration and serve HRP customers. The Human Research Roadmap (http://humanresearchroadmap.nasa.gov) is a web-based version of the IRP that allows users to search HRP risks, gaps, and tasks.

The HRP is organized into several research Elements:

- Human Health Countermeasures.
- Human Factors and Behavioral Performance.
- Exploration Medical Capability.
- Space Radiation.

Each of the HRP Elements address a subset of the risks. A fifth Element, ISS Medical Projects (ISSMP), is responsible for the implementation of the research on various space and ground analog platforms. HRP subtopics are aligned with the Elements and solicit technologies identified in their respective research plans.

H12.01 Radioprotectors and Mitigators of Space Radiation-induced Health Risks (SBIR)
Lead Center: JSC

Space radiation is a significant obstacle to sending humans on long duration missions beyond low earth orbit. NASA is concerned with the health risks to astronauts following exposures to galactic cosmic rays (GCR), the high-energy particles found outside Earth's atmosphere. Astronaut health risks from space radiation exposure are categorized into cancer, late and early central nervous systems (CNS) effects, and degenerative risks, which include cardiovascular diseases (CVD) and premature aging.

This subtopic is for development of biological countermeasures that can target common pathways (e.g., inflammation) across aging, cancer, cardiovascular disease, and neurodegeneration in order to minimize or prevent adverse health effects from space radiation. Drugs that target senolytic agents for anti-aging are the emphasis of this solicitation. The proposed project should focus on repurposing of technology and compounds for NASA applications. Expected TRL for this project is 5 to 8.

In Phase I of the project, the company should test radioprotectors or mitigators using protons or other charged particles at doses simulating exposure to space radiation. This testing can be done with cell models at the location of choice. Deliverables for the Phase I will be data generated from this exposure with the radioprotector selected. After contract award, due to the nature of this research, the contractor should immediately coordinate with their technical monitor for any special considerations for testing. In Phase II of the project, we would expect the company to expand testing radioprotectors or mitigators with combinations of different particles and energies that simulate the space radiation environment.

This subtopic seeks technology development that benefits the Space Radiation Element of the NASA Human Research Program (HRP). Biomedical countermeasures are needed for all of the space radiation risks. Anti-aging drugs are relevant to cancer, degenerative tissue damage and CNS damage.

References:

The following references discuss the different health effects NASA has identified in regard to space radiation exposure:

- Evidence report on central nervous systems effects
  - https://humanresearchroadmap.nasa.gov/evidence/reports/CNS.pdf.
- Evidence report on degenerative tissue effects
  - https://humanresearchroadmap.nasa.gov/evidence/reports/Degen.pdf.
- Evidence report on carcinogenesis - https://humanresearchroadmap.nasa.gov/evidence/reports/Cancer.pdf.

H12.05 Reduced Oxygen Usage for Medical Events (SBIR)
Lead Center: JSC
Participating Center(s): GRC

Human exploration missions beyond low earth orbit (LEO) require a variety of medical interventions to address planned and un-planned operations. One intervention involves the delivery of medical grade oxygen, specifically during Advanced Life Support (ALS) protocols. NASA currently uses a pneumatic, portable ventilator where rate and volume can be independently controlled and oxygen is supplied via pressurized tanks on the Space Station. Computational models show that, when operating the device, the addition of oxygen into the close vehicle environment via enriched exhalation and/or blow-by quickly violates NASA Flight Rules to NOT exceed greater than 30% oxygen concentration. The Flight Rule was put in place to minimize the likelihood of a fire on NASA vehicles. Specifically, within 20-30 minutes on the International Space Station, a localized high percentage oxygen bubble forms around the patient and within 12 hours the entire cabin exceeds NASA Flight Rules regarding oxygen concentration. These limitations significantly impair NASA's ability to respond to ALS events and only worsen as vehicle volumes become smaller for the Orion Program, Commercial Crew Program, and future Exploration Programs (like Gateway).

NASA requires new technologies that will enable the delivery of medical grade oxygen while reducing/eliminating elevated oxygen concentration levels in the cabin atmosphere. Specifically, NASA seeks technologies/methods to reduce enriched oxygen exhalation and/or reduce oxygen blow-by. Examples of technology developments can include, but are not limited to, improved oxygen delivery (e.g., mask) design, improved ventilator modes, and/or shaped ventilator output (e.g., oxygen leading with air following).

For the above technology, research should, at a minimum, be conducted to analyze technical feasibility during Phase I and show a path toward Phase II demonstration and/or prototype hardware/process.

This technology would reduce the mass/volume/power required to deliver medical oxygen to a sick or injured astronaut and simultaneously reduce the spaceflight cabin fire hazard risk. It supports NASA's Human Research Program Exploration Medical Capabilities, the ISS Health Maintenance System, and the Commercial Crew Program.

References:

- https://www.nasa.gov/hrp/elements/exmc
- https://ntrl.ntis.gov/NTRL/dashboard/searchResults/titleDetail/N20110022379.xhtml
- https://www.nasa.gov/exploration/commercial/crew/index.html
- https://www.nasa.gov/directorates/heo/index.html

H12.06 Continuous Crew Health Monitoring (SBIR)
Lead Center: JSC
Participating Center(s): GRC

Human exploration missions beyond low earth orbit (LEO) require physiologic monitoring of the crew. Currently, NASA employs a wide variety of commercial off the shelf (COTS) crew-worn biosensors and devices that provide minutes to hours of high quality physiologic information. All these devices require mass, volume, power, and crew time to operate, each of which will be in short supply during missions beyond LEO. Additionally, existing technologies typically do not provide continuous physiologic monitoring and instead require either electrode replacement, battery replacement or some other constraint that limits the operation of the technology. The exploration vehicle, however, will already provide a variety of technologies that could potentially be used to extrapolate human physiologic data in a continuous manner that does not require additional mass, volume, power, and/or crew time to operate. Examples of technology embedded within the vehicle include, but are not limited to, high quality video and

audio, wireless networks, radio frequency identification, and other electromagnetic (EM) sources/detectors.

NASA requires new technologies that will exploit vehicle infrastructure to continuously monitor the crew's physiologic parameters without crew intervention. Ideally, these solutions should not require additional mass, volume, power, and/or crew time and should leverage an existing capability already being provided by the vehicle. However, NASA is amenable to incorporating novel and innovative technologies that could be added to the vehicle or the crew. Examples of technology developments can include, but are not limited to, heart and respiration rate detection via HD video, temperature detection via infrared camera, or stress detection via voice analysis.

Phase I Deliverable - Conceptual prototype of a monitoring device/algorithm and final report detailing the conceptual prototype and hardware/software development plans.

Phase II Deliverable - Completed monitoring device/algorithm, and final report on the development, testing, and validation of the tool.

The expected TRL for this project is 2 to 4.

This technology would reduce the mass/volume/power required to execute physiological monitoring and supports NASA's Human Research Program Exploration Medical Capabilities, the ISS Health Maintenance System, and the Commercial Crew Program.

References:

- https://www.nasa.gov/hrp/elements/exmc
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- https://www.nasa.gov/exploration/commercial/crew/index.html
- https://www.nasa.gov/directorates/heo/index.html

Focus Area 8 In-Situ Resource Utilization

Lead MD: STMD

In-Situ Resource Utilization (ISRU) involves any hardware or operation that harnesses and utilizes 'in-situ' resources (natural and discarded) to create products and services for robotic and human exploration. ISRU encompasses a broad range of systems, and is typically divided into six focus areas: Resource Assessment, Resource Acquisition, Resource Processing/Consumable Production, In Situ Manufacturing, In Situ Construction, and In-Situ Energy. ISRU products and services can be used to reduce Earth launch mass or lander mass by not bringing everything from Earth, reduce risks to the crew and/or mission by reducing logistics, increasing shielding, and providing increased self-sufficiency, or reduce costs by needing less launch vehicles to complete the mission and/or through the reuse of hardware and lander/space transportation vehicles. Since ISRU can be performed wherever resources may exist, ISRU technologies and systems may need to operate in a variety of environments and gravities, and may need to consider a wide variety of potential resource physical and mineral characteristics. This year's solicitation will focus on critical technologies needed in the areas of Resource Acquisition and Consumable Production for the Moon and Mars.

Z12.01 Extraction of Oxygen from Lunar Regolith (SBIR)
Lead Center: JSC
Participating Center(s): GRC, JPL, KSC, MSFC
Related Subtopic Pointer(s): Z12.02, Z5.05

NASA has a strong interest in technologies that enable In-situ Resource Utilization (ISRU), where commodities such as propellant and breathing air are made from lunar materials to enable exploration beyond low earth orbit. Several categories of technologies related to the extraction of oxygen from lunar regolith are sought in the following subtopic. These include solar concentrator technologies, molten oxide electrolysis, and beneficiation/size sorting.

Solar Concentrator Technologies for Oxygen Extraction and In-Situ Construction

Solar concentrators have been used to successfully demonstrate multiple ISRU technologies including hydrogen and carbothermal reduction, sintering of surfaces pads, and production of blocks for construction. Terrestrial state of the art solar concentrators are heavy, not designed for easy packaging/shipping and assembly/installation, and can be maintained and cleaned on a periodic basis to maintain performance. For in-situ resource utilization (ISRU) space applications, NASA is interested in solar concentrators that are able to be packaged into small volumes, are light weight, easily deployed and set up, can autonomously track the sun, and can perform self-cleaning operations to remove accumulated dust. Materials, components, and systems that would be necessary for the proposed technology must be able to operate on the lunar surface: up to 110 C (230 F) during sunlit periods and survive temperatures down to -170 C (-274 F) during periods of darkness. Systems must also be able to operate for at least one year with a goal of 5 years. Each of the following specific areas of technology interest may be developed as a standalone technology, but proposals that address multiple areas are encouraged.

Lightweight Mirrors/Lenses - Proposals must clearly state the estimated W/kg for the proposed technology. Phase I efforts, if prototyped, can be demonstrated at any scale, but must be scalable up to 26 kW of reflected solar energy assuming an incoming solar flux of 1000 W/m^2. Phase II deliverables include prototype(s) that must be deployed and supported in Earth 1-g (without wind loads) but should include design recommendations for mass reductions for lunar gravity (1/6-g) deployment. Proposals should address the following attributes: high reflectivity, low coefficient of thermal expansion, strength, mass, reliability and cost.

Dust Repellent Mirrors/Lenses - Dust particles that cling to the surface of a mirror or lens will degrade the performance of a solar concentrator. Proposals must demonstrate a scalable means to remove or repel dust from mirrors and/or lenses without the use of consumables.

Efficient transmission of energy for oxygen/metal extraction - While the solar concentrator will need to move to track the sun, reactors requiring direct thermal energy for oxygen extraction will be in a fixed position and orientation. Concentrated sunlight must be able to be directed to a single or multiple spots to effectively heat or melt the regolith. Options, such as adjustable mirrors and fiber optics, must be included in the proposed development effort as well as the expected transition losses from collection to point delivery. For carbothermal reduction, surface temperatures of >1600° C are required.

Sintering end effector - Concepts must produce a focal point temperature of 1050° C for the purpose of sintering lunar regolith with a fiber optic interface efficiency of greater than 90%.

Molten Oxide Electrolysis

This particular method of oxygen extraction has the potential to provide relatively high yields of oxygen per mass of regolith. Proposals must specify the expected wear and replacement rate of Anodes/Cathodes. Proposals must also specify the expected loss and replacement of any additives such as flux or ionic liquids. Phase I demonstrations may be any scale. Phase II demonstrations should be scalable up to 1.6 kg/hr oxygen. Multiple units are acceptable if required but need to be specified. Specify which metals will be extracted during oxygen removal and how the metals will be separated and captured.

Beneficiation/Size Sorting

Mineral beneficiation and size sorting systems can greatly improve the effectiveness of oxygen extraction techniques such as hydrogen reduction. Proposals should demonstrate a means to remove particles larger than 1 mm and increase the concentration of minerals such as FeO, Fe_2O_3 and $FeTiO_3$. Phase I demonstrations can be at any scale, but Phase II demonstrations should be scalable up to 80 kg/hr of bulk regolith at the inlet of the device.

Relevance to NASA

Each of these technologies are considered key for ISRU processing. There is currently an ISRU project being funded by AES/STMD, and the last time NASA was focused on lunar ISRU, solar concentrators were used for multiple applications, and both molten oxide electrolysis and beneficiation of minerals was being demonstrated at a small scale.

References:

Solar Concentrator Technologies for Oxygen Extraction and In-Situ Construction

- Gordon, P. E., Colozza, A. J., Hepp, A. F., Heller, R. S., Gustafson, R., Stern, T., & Nakamura, T. (2011). Thermal energy for lunar in-situ resource utilization: technical challenges and technology opportunities.
- Nakamura, T., & Smith, B. (2011, January). Solar thermal system for lunar ISRU applications: development and field operation at Mauna Kea, HI. In 49th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition (p. 433).
- Gustafson, R., White, B., Fidler, M., & Muscatello, A. (2010). Demonstrating the solar carbothermal reduction of lunar regolith to produce oxygen. In 48th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition (p. 1163).

Molten Oxide Electrolysis

- Sibille, L., Sadoway, D. R., Sirk, A., Tripathy, P., Melendez, O., Standish, E. & Poizeau, S. (2009). Production of Oxygen from Lunar Regolith using Molten Oxide Electrolysis.
- Vai, A., Yurko, J., Wang, D. H., & Sadoway, D. (2010). Molten oxide electrolysis for lunar oxygen generation using in-situ resources. Minerals, Metals and Materials Society/AIME, 420 Commonwealth Dr., P. O. Box 430 Warrendale PA 15086 USA. [np]. 14-18 Feb.
- Sibille, L., & Dominguez, J. (2012, January). Joule-heated molten regolith electrolysis reactor concepts for oxygen and metals production on the moon and mars. In 50th AIAA Aerospace Sciences Meeting including the New Horizons Forum and Aerospace Exposition (p. 639).
- Sibille, L., Sadoway, D., Sirk, A., Tripathy, P., Melendez, O., Standish, E. & Poizeau, S. (2009). Recent advances in scale-up development of molten regolith electrolysis for oxygen production in support of a lunar base. In 47th AIAA Aerospace Sciences Meeting including The New Horizons Forum and Aerospace Exposition (p. 659).

Beneficiation/Size Sorting

- Trigwell, S., Captain, J., Weis, K., & Quinn, J. (2012). Electrostatic Beneficiation of Lunar Regolith: Applications in In-Situ Resource Utilization. Journal of Aerospace Engineering, 26(1), 30-36.

Z12.02 Payloads for Lunar Resources: Volatiles (SBIR)
Lead Center: KSC
Participating Center(s): GRC, JPL, JSC, LaRC
Related Subtopic Pointer(s): Z12.01, Z5.05

Whereas the Moon was once thought to be dry, more recent discoveries indicate that there are a variety of resources that exist on the Moon in an embedded or frozen state in the regolith. When acquired and exposed to higher temperatures and vacuum, these resources will change state into the vapor phase and are known as volatiles. Examples are polar water ice and other polar volatiles, or hydrogen and helium-3 embedded in the regolith grains by the sun.

Lunar volatiles are a meaningful first focus area for a space exploration strategy because:

- Use of local space resources, including lunar volatiles; for propellant production, life support, radiation shielding, growing plants, industrial processes, etc. will improve the sustainability of human space exploration.
- Technologies and methods for accessing lunar volatiles are relevant to potential future Mars resource utilization.
- Volatiles are of great interest to the science community and provide clues to help understand the solar wind, comets, and the history of the inner solar system.

NASA is interested in this proposal solicitation for small payloads up to 15 kg in mass which are needed to characterize and map lunar volatile resources, which will enable their inclusion in a future lunar ISRU strategy, as listed in selective NASA Strategic Knowledge Gaps (SKG) below. These payloads may be delivered to the surface of the Moon on a small commercial lunar lander and could be stationary on the lander, mobile on a mobility device, or it may itself be mobile and/or deployable. Impactors and other devices that are used or released in lunar orbit are not within the scope of this solicitation.

All proposals need to identify the state-of-the-art of applicable technologies and processes and Technology Readiness Level (TRL) expected at the end of Phase I, with a credible development plan. The Phase I proposal shall also indicate the type of lunar surface assets, interfaces and commodities that are required to carry and support the payload. By the end of Phase I, feasibility of the proposed payload technology should be established with a notional payload packaging concept and evidence that the payload is feasible. If a Phase II is awarded, then further development of the payload technologies and payload packaging shall be required, including a payload prototype delivered to NASA at the end of the two-year project with a goal of achieving TRL 6.

Due to the fact that frozen lunar volatiles primarily exist in, or near, permanently shadowed regions (PSR), if the prototype hardware proposed will need to operate under lunar vacuum conditions in PSR, it will either need to be designed to operate and be tested at extremely low temperatures (down to 40 K) or include estimates on thermal management and power to operate under these temperatures. Other proposals for finding and characterizing frozen buried volatiles near PSR's are also in scope, as well as mining hydrogen and helium volatiles embedded in the regolith. Methods to collect the volatiles without significant loss to sublimation are of high interest. Proposals must include plans for the design and test of critical or high-risk attributes associated with the proposed technology that enable its eventual use as flight hardware. At the end of Phase II, successful payload designs will be considered for funding applied to a commercial lunar lander flight in a potential Phase III award.

Proposals will be evaluated on the basis of feasibility, mass, power, volume, and complexity. All proposals shall identify the SKG(s) from the list below that will be met. Payloads with a proposed mass of greater than 15 kg will not be considered in this subtopic.

The following information is only provided so that proposers understand the context and purpose of the small payloads being solicited for a robotic lunar landing mission.

Recent data from NASA's Lunar CRater Observation and Sensing Satellite (LCROSS), and Lunar Reconnaissance Orbiter (LRO) missions indicate that as much as 20% of the material kicked up by the LCROSS impact was volatiles, including water, methane, ammonia, hydrogen gas, carbon dioxide and carbon monoxide. The instruments also discovered relatively large amounts of light metals such as sodium, mercury and possibly even silver.

The following criteria are relevant to this SBIR solicitation, as reported by the Lunar Exploration Analysis Group (LEAG):

Significant uncertainties remain regarding to the distribution of volatiles at the 10 to 100 m resolution scales accessible to near term orbital missions. Data and models are clear that volatiles are distributed unevenly at this scale and mission success scenarios should accommodate this likelihood. We also found that a range of new orbital missions and science support activities could reduce this risk by improving both the empirical data upon which site selections are based, and the scientific understanding of polar volatile evolution. Regarding landed experiments, there are several key measurements-- such as compositional variation and soil geotechnical and thermal properties--within the capabilities of small near-term missions that would greatly improve the understanding of polar volatiles; obtaining any of the needed quantities would benefit subsequent missions.

There are sufficient data to support near-term landing site selections – Enhanced hydrogen is widespread across the polar regions and is sometimes concentrated in permanently shadowed regions (PSRs). Data show that average annual surface temperatures below 110K are also widespread, including both PSRs and areas sometimes illuminated. This characteristic allows preservation of shallow buried ice for geologic time. LCROSS demonstrated hydrogen and water do occur at shallow depths at the LCROSS target site PSR. However, arguments derived from lunar surface processes suggest volatiles will be distributed irregularly and high-water abundance observed by LCROSS was not consistent with the regional H abundance indicating sampling of a local concentration.

The expected patchy nature of hydrogen distributions constitutes significant risk to missions requiring detection and sampling of hydrogen. Higher resolution definitive hydrogen data would reduce this risk.

LEAG Volatiles Specific Action Team (SAT) Landed Measurements Finding #1

Small near-term missions can provide critical data to resolve important unknowns regarding polar volatile science and resource utilization:

- Lateral and vertical distribution of volatiles
- Chemical phases that contain volatile elements
- Geotechnical and thermal properties of polar soils
- Mobility of volatiles and associated timescale(s)
- Landed experiments obtaining any of the important quantities are of great science and exploration value.

LEAG Volatiles Specific Action Team (SAT) Landed Measurements Finding #2

Early characterization of the variation in volatile abundance at ISRU and scientifically relevant spatial scales would greatly benefit all future missions:

- Current understanding of the spatial variation of volatile abundance at the scale of landers and small rovers is a major uncertainty. This ignorance is a strong inhibitor for the use of static landers
- Several studies suggest that near surface volatiles will be very unevenly distributed due to the impact process and other mechanisms
- A small rover traversing several hundred meters could characterize the variation in volatiles at this scale with simple instrumentation. A rover traverse of several hundred meters to several kilometers is required. The minimum distance for ground truthing is 20 km. Minimum distance to confirm if there are volatiles present is likely to be ~1 km.
- This would provide ground-truth for orbital volatile measurements by beginning to close the gap in scales.

LEAG Volatiles Specific Action Team (SAT) Landed Measurements Finding #3

The physical and chemical forms of abundant volatile elements are critical to understanding the resource and its origins:

- Early measurements should include unambiguous determination of the chemical phase of volatiles present to a depth of one or more meters
- Measurements should not be restricted to the detection of water, but include other volatile species
- Profiling is desirable, but a bulk analysis would be of very high value.
- It is necessary to measure the isotopic composition of volatile elements. Both with respect to fundamental volatile science and with respect to assessing quantitatively potential landing-induced contamination of the surface materials.

LEAG Volatiles Specific Action Team (SAT) Landed Measurements Finding #4

Successful exploitation of in-situ resources requires knowledge of the physical (geotechnical) and thermal properties of polar regolith in addition to the volatile abundance:

- The utility of a resource is highly dependent on the cost of extraction that is in turn dependent on the physical and chemical state of the volatile and its refractory matrix
- The ISRU community should develop specific measurement objectives for geotechnical and temperature dependent properties
- Thermal analysis of polar soils such as differential scanning calorimetry would greatly enhance the ability to develop ISRU regolith processing strategies, even in a volatile poor polar target

- Thermal analysis can also be made sensitive to volatiles found in the LCROSS plume that could cause significant concerns for contamination and degradation of ISRU hardware including H_2S, Hg, and Na.
- Physical and thermal properties of polar regolith should be measured. The potential effect of some volatile compounds such as Hg and Na on instrument degradation should be quantified.

LEAG Volatiles Specific Action Team (SAT) Landed Measurements Finding #8

In addition to ISRU goals, landed experiments should include measurements of current volatile flux to aid understanding volatile transport mechanism:

- Apollo surface experiments revealed a dynamic exosphere and produced a lengthy list of potential volatile atmospheric species
- Measurements might include:
- Pressure
- Atmospheric species
- Flux directions
- Measurements at PSR contacts to measure the volatile flux into cold traps

The relevant lunar Strategic Knowledge Gaps (SKG's) for this subtopic are listed below:

I-C. Regolith 2: Quality/ quantity/distribution/form of H species and other volatiles in mare and highlands regolith (requires robotic precursor missions).
Robotic in-situ measurements of volatiles and organics on the lunar surface and eventual sample return of "pristine" samples. Enables prospecting for lunar resources and ISRU. Feeds forward to Near Earth Asteroids (NEA)-Mars. Relevant to the Planetary Science Decadal survey.
I-D-1. Composition/quantity/distribution/form of water/H species and other volatiles associated with lunar cold traps. Required "ground truth" in-situ measurement within permanently shadowed lunar craters or other sites identified using LRO data. Technology development required for operating in extreme environments. Enables prospecting of lunar resources and ISRU. Relevant to Planetary Science Decadal survey.
I-D-3 Subsection c: Geotechnical characteristics of cold traps
Landed missions to understand regolith densities with depth, cohesiveness, grain sizes, slopes, blockiness, association and effects of entrained volatiles.
I-D-7 Subsection g: Concentration of water and other volatiles species with depth 1-2 m scales
Polar cold traps are likely less than ~2 Ga, so only the upper 2-3 m of regolith are likely to be volatile-rich.
I-D-9 Subsection I: mineralogical, elemental, molecular, isotopic make up of volatiles
Water and other exotic volatile species are present; must know species and concentrations.
I-D-10 Subsection j: Physical nature of volatile species (e.g., pure concentrations, inter-granular, globular)
Range of occurrences of volatiles; pure deposits (radar), mixtures of ice/dirt (LCROSS), H_2-rich soils (neutron).
I-E. Composition/volume/distribution/form of pyroclastic/dark mantle deposits and characteristics of associated volatiles.
Required robotic exploration of deposits and sample return. Enables prospecting for lunar resources and ISRU. Relevant to Planetary Science Decadal survey.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

References:

- https://www.nasa.gov/exploration/library/skg.html
- https://www.hou.usra.edu/meetings/leag2016/pdf/5025.pdf
- https://www.lpi.usra.edu/leag/GAP_SAT_03_09_12.pdf
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- https://www.globalspaceexploration.org/wordpress/wp-content/uploads/IAC64/IAC-13.A3.1.4%20ISECG%20Assessment%20of%20the%20Strategic%20Knowledge%20Gaps%20for%20Exploration.pdf
- https://www.nasa.gov/feature/nasa-expands-plans-for-moon-exploration-more-missions-more-science

Focus Area 9 Sensors, Detectors and Instruments

Lead MD: SMD

NASA's Science Mission Directorate (SMD) (http://nasascience.nasa.gov/) encompasses research in the areas of Astrophysics, Earth Science, Heliophysics and Planetary Science. The National Academy of Science has provided NASA with recently updated Decadal surveys that are useful to identify technologies that are of interest to the above science divisions. Those documents are available at the following locations:

- Astrophysics - http://sites.nationalacademies.org/bpa/BPA_049810 (link is external).
- Planetary - http://sites.nationalacademies.org/ssb/completedprojects/ssb_065878 (link is external).
- Earth Science - http://science.nasa.gov/earth-science/decadal-surveys/.
- Heliophysics the 2014 technology roadmap can be downloaded here: http://science.nasa.gov/heliophysics/.

A major objective of SMD instrument development programs is to implement science measurement capabilities with smaller or more affordable spacecraft so development programs can meet multiple mission needs and therefore make the best use of limited resources. The rapid development of small, low-cost remote sensing and in-situ instruments is essential to achieving this objective. For Earth Science needs, in particular, the subtopics reflect a focus on instrument development for airborne and Unmanned Aerial Vehicle (UAV) platforms. Astrophysics has a critical need for sensitive detector arrays with imaging, spectroscopy, and polarimetric capabilities, which can be demonstrated on ground, airborne, balloon, or suborbital rocket instruments. Heliophysics, which focuses on measurements of the sun and its interaction with the Earth and the other planets in the solar system, needs a significant reduction in the size, mass, power, and cost for instruments to fly on smaller spacecraft. Planetary Science has a critical need for miniaturized instruments with in-situ sensors that can be deployed on surface landers, rovers, and airborne platforms. For the 2019 program year, we are restructuring the Sensors, Detectors and Instruments Topic, adding new, rotating out, and retiring some of the subtopics. Please read each subtopic of interest carefully. We continue to emphasize Ocean Worlds and solicit development of in-situ instrument technologies and components to advance the maturity of science instruments focused on the detection of evidence of life, especially extant of life, in the Ocean Worlds. The microwave technologies subtopic was split last year into two subtopics one focused on active microwave remote sensing and the second on passive systems such as radiometers and microwave spectrometers. A key objective of this SBIR topic is to develop and demonstrate instrument component and subsystem technologies that reduce the risk, cost, size, and development time of SMD observing instruments and to enable new measurements. Proposals are sought for development of components, subsystems and systems that can be used in planned missions or a current technology program. Research should be conducted to demonstrate feasibility during Phase I and show a path towards a Phase II prototype demonstration. The following subtopics are concomitant with these objectives and are organized by technology.

S1.01 Lidar Remote Sensing Technologies (SBIR)
Lead Center: LaRC
Participating Center(s): GSFC, JPL
Related Subtopic Pointer(s): T8.02

NASA recognizes the potential of lidar technology in meeting many of its science objectives by providing new capabilities or offering enhancements over current measurements of atmospheric and topographic parameters from ground, airborne, and space-based platforms. To meet NASA's requirements for remote sensing from space, advances are needed in state-of-the-art lidar technology with an emphasis on compactness, efficiency, reliability, lifetime, and high performance. Innovative lidar subsystem and component technologies that directly address the measurement of atmospheric constituents and surface topography of the Earth, Mars, the Moon, and other planetary bodies will be considered under this subtopic. Compact, high-efficiency lidar instruments for deployment on unconventional platforms, such as balloon, small sat, and CubeSat are also considered and encouraged.

Proposals must show relevance to the development of lidar instruments that can be used for NASA science-focused measurements or to support current technology programs. Meeting science needs leads to four primary instrument types:

- *Backscatter* - Measures beam reflection from aerosols to retrieve the opacity of a gas.
- *Ranging* - Measures the return beam's time-of-flight to retrieve distance.
- *Doppler* - Measures wavelength changes in the return beam to retrieve relative velocity.
- *Differential absorption* - Measures attenuation of two different return beams (one centered on a spectral line of interest) to retrieve concentration of a trace gas.

The proposed subtopic addresses many missions programs, and project identified by the Science Mission Directorate including:

- *Aerosols* - missions ongoing and planned include ACE (Aerosols/Clouds/Ecosystems), PACE (Plankton, Aerosol, Cloud, ocean Ecosystems), and MESCAL (Monitoring the Evolving State of Clouds and Aerosols).
- *Greenhouse Gases* - missions planned include sensing of carbon dioxide and methane. The ASCENDS (Active Sensing of CO_2 Emissions over Nights, Days, and Seasons) mission was recommended by the Decadal Survey.
- *Ice Elevation* - missions ongoing and planned include ICESat (Ice, Cloud, and land Elevation Satellite), as well as aircraft-based projects such as IceBridge.
- *Terrestrial Ecosystem Structure* - missions ongoing and planned include GEDI (Global Ecosystems Dynamics Investigation). Ocean sensing applications are also of interest to NASA.
- *Atmospheric Winds* - missions planned include 3D-Winds, as recommended by the Decadal Survey. Lidar wind measurements in the Mars atmosphere are also under study in the MARLI (Mars Lidar for Global Climate Measurements from Orbit) program.
- *Planetary Topography* - altimetry similar to Earth applications is being planned for planetary bodies such as Titan and Europa.
- *Automated Landing, Hazard Avoidance, and Docking* - technology development is called for under programs and missions such as ALHAT (Autonomous Landing and Hazard Avoidance Technology), COBALT (COoperative Blending of Autonomous Landing Technologies), and Kodiak.

Phase I research should demonstrate technical feasibility and show a path toward a Phase II prototype unit. Phase II prototypes should be capable of laboratory demonstration and preferably suitable for operation in the field from a ground-based station, an aircraft platform, or any science platform amply defended by the proposer.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-6.

References:

- NASA missions are aligned with the National Research Council's decadal surveys, with the latest survey published in 2018 under the title "Thriving on Our Changing Planet: A Decadal Strategy for Earth Observation from Space" (http://sites.nationalacademies.org/DEPS/esas2017/index.htm).
- NASA lidar applications and technology needs for Earth Science are also summarized in the report "NASA ESTO Lidar Technologies Investment Strategy: 2016 Decadal Update" (https://ntrs.nasa.gov/search.jsp?R=20180002566).
- Conference proceeding on NASA lidar interests in earth science, exploration, and aeronautics can be found at the Technical Interchange Meeting on Active Optical Systems (https://www.nasa.gov/nesc/tim-active-optical-systems).

S1.02 Technologies for Active Microwave Remote Sensing (SBIR)
Lead Center: JPL
Participating Center(s): GSFC
Related Subtopic Pointer(s): T8.02

1 Watt G-band (167-175 GHz) Solid State Power Amplifier for Remote Sensing Radars

Development of 1 Watt G-band (167-175 GHz) Solid State Power Amplifier for Remote Sensing Radars. Future Cloud, water and precipitation missions require higher frequency electronics, with small form factors and high power added efficiencies (PAE). Solid state amplifiers that meet high efficiency (>20 % PAE) and have small form factors would be suitable for SmallSats, enabling single satellite missions, such as RainCube, and would enable future swarm techniques.

Relevance to NASA

Cloud, water and precipitation measurements Increase capability of measurements to smaller particles, and enabling much more compact instruments.

The desired deliverables are design and simulation of potential amplifiers meeting the 1 Watt G-band (167-175 GHz) with 20% PAE. The expected Technology Readiness Level (TRL) range at completion of the project is 2-4.

Ultra-Wide Band (UWB) Non-Contact Ground Penetrating Radar (GPR) Antenna

Development of UWB (ultra-wideband) non-contact GPR (ground penetrating radar) antenna for terrestrial and planetary mobility (aka rover or drone) platforms. Antenna designed to be mounted under rovers and other autonomous vehicles. Planar, or other low-profile antenna desired for easy accommodation onto the underside of a drone or rover. Frequency of operation 120 MHz - 2 GHz, linearly polarized, 3 dB beamwidth > 90°, 50 Ohm input, optimized to couple into ice/regolith (er = 1.7 to 3.1) at a standoff distance of 10-20 cm.

Relevance to NASA

Future Earth and planetary science small payload missions.

The desired deliverables are mechanical drawing of antenna, with electromagnetic analysis (such as HFSS) of the antenna performance.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-4.

GPS (Global Positioning System) Denied Timing Synchronization

Development of solutions to GPS-denied multi-static radar timing synchronization. This would enable multi-platform instruments to share timing, which is enabling for GPS denied environments, which could be for planetary science or GPS hostile locations on Earth (such as subsurface). Desire to wirelessly distribute a synchronized PPS and/or 10 MHz clock in a GPS-denied environment between multiple radar units with <0.5 ns accuracy. Perform in specification

at distances of up to 5 km. Synchronization hardware should be low mass (<1 kg), low power (<1 W), small size (<5x5x10 cm). Should have a path to flight qualification to be used for lunar and planetary science.

Relevance to NASA

Future Earth and planetary science small payload missions.

The desired deliverables are design and analysis of potential solutions, for which realizable hardware exists or is plausibly able to be developed with current technology.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-4.

V Band Switch (65-70 GHz)

Currently funded RTD to build technology for developing a pressure sensing absorption radar at V-band is in need for a wideband switch operating over 65-70 GHz range. This technology if developed will allow airborne demonstration of first ever remote measurement of surface pressure that helps better predict path and strength of hurricanes.

Relevance to NASA

Surface Pressure Sensing Absorption Radar using V-band.

The desired deliverables are:

- V-band SPDT switch
- Frequency: 65-70 GHz
- Insertion Loss< 0.5dB
- Isolation>35dB
- Should be able to handle 2W of input power
- Compact, light weight

References:

- 1 Watt G-band (167-175 GHz) Solid State Power Amplifier for Remote Sensing Radars:
    - https://www.jpl.nasa.gov/cubesat/missions/raincube.php
    - https://www.nap.edu/read/11952/chapter/9
    - https://www.nasa.gov/mission_pages/GPM/overview/index.html

S1.03 Technologies for Passive Microwave Remote Sensing (SBIR)
Lead Center: GSFC
Participating Center(s): JPL
Related Subtopic Pointer(s): T8.02, Z8.03

NASA employs passive microwave and millimeter-wave instruments for a wide range of remote sensing applications from measurements of the Earth's surface and atmosphere to cosmic background emission. Proposals are sought for the development of innovative technology to support future science and exploration missions MHz to THz sensors. Technology innovations should either enhance measurement capabilities (e.g., improve spatial, temporal, or spectral resolution, or improve calibration accuracy) or ease implementation in spaceborne missions (e.g., reduce size, weight, or power, improve reliability, or lower cost). Specific technology innovations of interest are listed below, however other concepts will be entertained.

Ultra-Compact Radiometer

An ultra-compact radiometer of either a switching or pseudo-correlation architecture with internal calibration sources is needed. Designs with operating frequencies at the conventional passive microwave bands of 36.6 GHz (priority),

18.65 GHz, and 23.8 GHz enabling dual-polarization inputs. Interfaces include waveguide input, control, and digital data output. Ideal design features enable subsystems of multiple (10's of) integrated units to be efficiently realized.

This technology, in conjunction with deployable antenna technology, would enable traditional Earth land and ocean radiometry with significantly reduced instrument size, making it suitable for CubeSat or SmallSat platforms.

The expected Technology Readiness Level (TRL) range at completion of the project is 4-5.

Compact, scalable, 3D routing of LO, IF and DC signals for focal plane arrays at room and cryogenic temperatures

Compact, scalable, 3D routing of LO, IF and DC signals for focal plane arrays at room and cryogenic temperatures. A single routing block should perform the following functions: Accept 32 IF inputs, 16 LO inputs and 160 DC inputs, on one side of the routing block. Input interfaces to IF, LO and DC should facilitate blind-mating (e.g., push-on connectors). At the output, all IF signals should be concentrated into no more than 4 connectors (using e.g., multi-core coaxial connectors). The 16 LO input connections should be internally combined into a single connector at the output. All DC signals should be concentrated into no more than 4 connectors at the output. All output signals connectors should be on the opposite side of the routing block to the inputs. The LO should be able to route signals up to 60 GHz and the IF up to 12 GHz with max. 8dB loss at LO and package of 4”x 4”x 4”. This routing block should be scalable by forming close-packing arrays of such blocks to arbitrary sizes.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-4.

Photonic Integrated Circuits for Microwave Remote Sensing Systems

Photonic Integrated Circuits are an emerging technology for passive microwave remote sensing. NASA is looking for photonic integrated circuits for utilization in processing microwave signals in spectrometers, beam forming arrays, correlation arrays and other active or passive microwave instruments. Small businesses are encouraged to identify, propose, and utilize designs where PIC technology would be most beneficial for a microwave remote sensing instrument subsystem.

PICs may enable significantly increased bandwidth of Earth viewing, astrophysics, and planetary science missions. In particular, this may allow for increased bandwidth or resolution receivers, with applications such as hyperspectral radiometry.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-5.

Low power RFI mitigating receiver back ends for broad band microwave radiometers

NASA requires a low power, low mass, low volume, and low data rate RFI mitigating receiver back-end that can be incorporated into existing and future radiometer designs. The system should be able to channelize up to 1 GHz with 16 sub bands and be able to identify RFI contamination using tools such as kurtosis.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-5.

Miniature W-band Diplexer

As NASA seeks to develop broadband and array microwave radiometer technology, there is a need for miniaturized diplexers to separate W-band signals from Ka-band and lower frequency signals. Specifically, a diplexer unit that separates and passes the frequency bands allocated to and traditionally used for passive sensing is needed. A successful design has features enabling integration into subsystems including other supporting elements such as broadband antenna array elements and MMIC LNA's.

The expected Technology Readiness Level (TRL) range at completion of the project is 4-5.

Low power, compact lasers for THZ time domain and frequency domain spectroscopy

NASA is developing a compact broadband THz spectrometer based on asynchronous optical sampling time domain spectroscopy (TDS). Erbium femtosecond lasers with low volume, low mass and low power are required. The lasers are to use 1550 nm erbium technology with pulse width < 100 fs and repetition rate of 80-100 MHz. The lasers should operate with single mode-lock state, high stability and low amplitude and phase noise. The fiber coupled output power should be > 100 mW.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-3.

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S1.04 Sensor and Detector Technologies for Visible, IR, Far-IR, and Submillimeter (SBIR)
Lead Center: JPL
Participating Center(s): ARC, GSFC, JPL, LaRC
Related Subtopic Pointer(s): S1.01, S2.04, S2.01

Sensor and Detector Technologies for Visible, IR, Far-IR, and Submillimeter

NASA is seeking new technologies or improvements to existing technologies to meet the detector needs of future missions, as described in the most recent decadal surveys:

- *Earth science* - (http://www.nap.edu/catalog/11820.html)
- *Planetary science* - (http://www.nap.edu/catalog/10432.html)
- *Astronomy and astrophysics* - (http://www.nap.edu/books/0309070317/html/)

Sensor and detector technologies operating in the visible range are not being solicited this year.

Low-power and low-cost digital readout integrated circuits (DROICs):

- In pixel digital readout integrated circuit (DROIC) for high dynamic range infrared imaging and spectral imaging (10-60 Hz operation) focal plane arrays to circumvent the limitations in charge well capacity, by using in-pixel digital counters that can provide orders of magnitude larger effective well depth, thereby affording longer integration times. Longer integration times provide improved signal-to-noise ratio and/or higher operating temperature, which reduces cooler capacity, resulting in savings in size, weight, power and cost.
- High speed (> 1 KHz full frame) shallow well (LSB between 32 – 128 electrons), integrate-while-read mode, with global shutter, 2-color bias-switchable focal planes with DROIC are required for high speed applications such as Fourier transform spectrometers. One color must be responsive to the solar spectrum and the other color must be responsive to thermal emission spectrum, with linear e-APD in both colors.

Low Size, Weight, and Power (SWaP) novel spectrometers:

- Compact low size, weight, and power (SWaP) novel spectrometers for space applications. This could include the conventional high-performance spectrometers based on dispersive elements, Fourier transform spectrometers, tilted grating concepts, etc. Furthermore, an integrated optics based low SWaP spectrometer also applicable to CubeSat and SmallSat applications.

MKID/TES Readout:

- Compact, low power, ASICs for readout of Kinetic Inductance Detector (KID) arrays each with a low operating power and capable of operation at both room temperature and cryogenic temperatures to perform one of the following functions: 8192 point FFT processor with 5 bits of depth using a polyphase oversampling or a Hanning window. Input format would be SERDES (2-4Gsamples/sec) and output format USB2.0 or similar and Power <=2W. >10bit ADC at >1GHz sampling rate with >2000 bands, ~5kHz bandwidth, power <0.3W. Of particular interest are SQUID based systems with a first stage operating at sub-Kelvin temperatures and compatible with 32X40 detector array format.
- Low power, low noise, cryogenic multiplexed readout for large format two-dimensional bolometer arrays with 1000 or more pixels, operating at 65-350 mK. We require a superconducting readout capable of reading two Transition Edge Sensors (TESs) per pixel within a 1 mm-square spacing. The wafer-scale readout of interest will be capable of being indium-bump bonded directly to two dimensional arrays of membrane bolometers, after the application of indium-bumps possibly at another facility. We require row and column readout with very low crosstalk, low read noise, and low detector Noise Equivalent Power degradation.

Lidar Detectors:

- Single photon (Geiger-mode) avalanche photodiode detector array technology for high-speed, imaging or non-imaging lidar applications. Detector array should be 32x32 or larger, demonstrating scalability to 256x256 or larger to cover $2x10^{12}$ photon/s dynamic range, with crosstalk and after pulsing probability < 2%, photon detection probability > 50% @ 532nm, and dark count rate < 10Hz per pixel at non-cryogenic temperatures, and radiation tolerance for 5 year low earth orbit mission. Detector should be compatible with hybridization techniques allowing connection to readout integrated circuit. Future missions and applications include the Aerosols Lidar Mission called for by the 2017 Decadal Survey for Earth Science, planetary surface mapping, vegetation, and trace gas lidar.
- Space qualify a commercial 2k x 2k polarization camera for a solar coronograph for low Earth orbit and Earth-Sun Lagrange point environments.

IR and Far-IR/Submillimeter-wave Detector Technologies:

- Tunable IR Detector: Development of an un-cooled broadband photon detector with average QE>50% over the spectral range from 3um to 50um. The Detectivity D* must be greater than 5x109. The detector may have electrically tunable spectral range.

Novel Materials and Devices: New or improved technologies leading to measurement of trace atmospheric species (e.g., CO, CH_4, N_2O) or broadband energy balance in the IR and far-IR from geostationary and low-Earth orbital platforms. Of particular interest are new direct detectors or heterodyne detectors technologies made using high temperature superconducting films (YBCO, MgB_2) or engineered semiconductor materials, especially 2Dimensional Electron Gas (2DEG) and Quantum Wells (QW). Candidate missions are thermal imaging, LANDSAT Thermal InfraRed Sensor (TIRS), Climate Absolute Radiance and Refractivity Observatory (CLARREO), BOReal Ecosystem Atmosphere Study (BOREAS), Methane Trace Gas Sounder or other infrared earth observing missions.

Array Receivers: Development of a robust wafer level packaging/integration technology that will allow high-frequency capable interconnects and allow two dissimilar substrates (i.e., Silicon and GaAs) to be aligned and mechanically 'welded' together. Specially develop ball grid and/or Through Silicon Via (TSV) technology that can support submillimeter-wave (frequency above 300 GHz) arrays.

Receiver Components: Local Oscillators capable of spectral coverage 2-5 THz; Output power up to >2 mW; Frequency agility with > 1GHz near chosen THz frequency; Continuous phase-locking ability over the THz tunable range with <100 kHz line width. Both solid-state (low parasitic Schottky diodes) as well as Quantum Cascade Lasers (for f>2 THz) will be needed. Components and devices such as mixers, isolators, and orthomode transducers, working in the THz range, that enable future heterodyne array receivers are also desired. GaN based power amplifiers at frequencies above 100 GHz and with PAE> 25% are also needed. ASIC based SoC solutions are needed for heterodyne receiver backends. ASICs capable of binning >6GHz intermediate frequency bandwidth into 0.1-0.5 MHz channels

with low power dissipation <0.5W would be needed for array receivers. Low-power Low Noise Amplifiers (LNA) with 15-20 dB Gain and <5 Kelvin Noise over the 4-8 GHz bandwidth must be demonstrated while operating linearly and biasing at 200uW or less. The P1dB and OIP3 data should be collected at different biases to recommend gain and gain stages at temperatures from 4 Kelvin to 300K. An intermediate set point of particular interest is 20 Kelvin.

Relevance to NASA

- Future short-wave, mid-wave, and long-wave infrared Earth science and planetary science missions all require detectors that are sensitive, broadband, and require low-power for operation.
- Future Astrophysics instruments require cryogenic detectors that are super sensitive, broadband, and provide imaging capability (multi-pixel).
- Aerosol spaceborne lidar as identified by 2017 decadal survey. Reduces uncertainty about climate forcing in aerosol-cloud interactions and ocean ecosystem carbon dioxide uptake. Additional applications in planetary surface mapping, vegetation, and trace gas lidar.
- Earth Radiation Budget measurement per 2007 decadal survey Clouds and Earth's Radiant Energy System (CERES) Tier-1 designation. To maintain the continuous radiation budget measurement for climate modeling and better understand radiative forcings.
- Astrophysical missions such as Origins Space Telescope (OST) will need IR and Far-IR detector and related technologies.
- LANDSAT Thermal InfraRed Sensor (TIRS), Climate Absolute Radiance and Refractivity Observatory (CLARREO), BOReal Ecosystem Atmosphere Study (BOREAS), Methane Trace Gas Sounder or other infrared earth observing missions

Current Science missions utilizing two-dimensional, large-format cryogenic readout circuits:

- HAWC + (High Resolution Airborne Wideband Camera Upgrade) for SOFIA (Stratospheric Observatory for Infrared Astronomy)
- PIPER (Primordial Inflation Polarization Experiment), Balloon-borne

The expected Technology Readiness Level (TRL) range at completion of the project is 2-4.

Two-Dimensional Cryogenic Readout for Far IR Bolometers

Low power, low noise, cryogenic multiplexed readout for large format two-dimensional bolometer arrays with 1000 or more pixels, operating at 65-350 mK. We require a superconducting readout capable of reading two Transition Edge Sensors (TESs) per pixel within a 1 mm-square spacing. The wafer-scale readout of interest will be capable of being indium-bump bonded directly to two dimensional arrays of membrane bolometers, after the application of indium-bumps possibly at another facility. We require row and column readout with very low crosstalk, low read noise, and low detector Noise Equivalent Power degradation.

Current Science missions utilizing two-dimensional, large-format cryogenic readout circuits:

- HAWC + (High Resolution Airborne Wideband Camera Upgrade) for SOFIA (Stratospheric Observatory for Infrared Astronomy)
- PIPER (Primordial Inflation Polarization Experiment), Balloon-borne

Future missions requiring two-TES per pixel readout with two-dimensional cryogenic circuits:

- PIPER Dual Polarization Upgrade
- PICO (Probe of Inflation and Cosmic Origins, a Probe-class Cosmic Microwave Background mission concept

The expected Technology Readiness Level (TRL) range at completion of the project is 4-5.

Sub-milliWatt amplifiers enabling multiplexed readout systems (MRS)

Sub-milliWatt amplifiers enabling multiplexed readout systems (MRS) in the 4-8 GHz bandwidth are needed to maintain the thermal stability of Focal Plane Array and Origins Space Telescope(OST) instruments Origins Survey Spectrometer (OSS) microwave kinetic inductance detectors (MKIDs) and Far-infrared Imager and Polarimeter (FIP) and Lynx Telescope X-ray Microcalorimeter using microwave SQUID multiplexers.

Another bandwidth 0.5-8.5 GHz, would also be useful for Heterodyne Receiver for OST (HERO). Other NASA systems in the Space Geodesy Project (SGP) would be interested in bandwidths up to 2-14 GHz. All these systems include a comb generator coupled in periodically to calibrate out system drifts. A 30 dB coupler is being baselined.

Regardless of bandwidth or thermal dissipation requirements (both OST and Lynx instruments have tight self-heating requirements), the linearity of these amplifiers over the bandwidth is critical. With 200 microWatt power dissipation up to optimal biasing, we seek devices' P1dB and OIP3 data characterized at these low biases and packaging that provides matching circuits and calibration coupling at set temperatures from 4 Kelvin (~200uW biases) up to 300 Kelvin (nominal biases). We need to trade off Gain Flatness and Gain stages, with Noise Temperature that's achievable without upsetting the thermal stability and isolation of the overall telescope.

The dual objectives of controlling self-heating and optimizing linearity and noise temperature maintenance, trading off gain and gain stages is not unique (e.g., SGP), but NASA's OST and Lynx missions drive the state-of-the-art technology to new levels that other NASA programs and industry can benefit from.

15-20 dB Gain and <5 Kelvin Noise over the 4-8 GHz bandwidth must be demonstrated while operating linearly, biasing at 200uW (e.g., Vd=0.09V, Id=2.2mA) or less. The P1dB and OIP3 data should be collected at different biases to recommend gain and gain stages at temperatures from 4 Kelvin to 300K. An intermediate set point of particular interest is 20 Kelvin.

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The expected Technology Readiness Level (TRL) range at completion of the project is 3-4.

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- A Time Domain SQUID Multiplexing System for Large Format TES Arrays": PDF download link: http://ws680.nist.gov/publication/get_pdf.cfm?pub_id=30767

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S1.05 Detector Technologies for UV, X-Ray, Gamma-Ray Instruments (SBIR)
Lead Center: JPL
Participating Center(s): GSFC, MSFC

Detectors

This subtopic covers detector requirements for a broad range of wavelengths from UV through to gamma ray for applications in Astrophysics, Earth Science, Heliophysics, and Planetary Science. Requirements across the board are for greater numbers of readout pixels, lower power, faster readout rates, greater quantum efficiency, single photon counting, and enhanced energy resolution.

The proposed efforts must be directly linked to a requirement for a NASA mission. These include Explorers, Discovery, Cosmic Origins, Physics of the Cosmos, Solar-Terrestrial Probes, Vision Missions, and Earth Science Decadal Survey missions. Proposals should reference current NASA missions and mission concepts where relevant. Specific technology areas are:

- Solid-state single photon counting radiation tolerant detectors in CCD or CMOS architecture for astrophysics, heliophysics, and planetary missions.
- Large area array, low noise, high efficiency CMOS, potentially in 3D stacked technology for the very large focal plane arrays of large aperture telescopes as well for heliophysics and planetary science measurements.
- Significant improvement in wide band gap semiconductor materials, such as AlGaN, ZnMgO and SiC, individual detectors, and detector arrays for operation at room temperature for astrophysics missions and planetary science composition measurements.
- Highly integrated, low noise (< 300 electrons rms with interconnects), low power (< 100 uW/channel) mixed signal ASIC readout electronics as well as charge amplifier ASIC readouts with tunable capacitive inputs to

match detector pixel capacitance. See needs of National Research Council's Earth Science Decadal Survey (NRC, 2007).

- Visible-blind SiC Avalanche Photodiodes (APDs) for EUV photon counting are required. The APDs must show a linear mode gain >10E6 at a breakdown reverse voltage between 80 and 100V. The APD's must demonstrate detection capability of better than 6 photons/pixel/s down to 135nm wavelength. See needs of National Research Council's Earth Science Decadal Survey (NRC, 2007): Tropospheric ozone.
- Visible-blind UV and EUV detectors with small pixels, large format, photon-counting sensitivity and detectivity, low voltage and power requirements.
- Large area (3 m^2) photon counting near-UV detectors with 3 mm pixels and able to count at 10 MHz. Array with high active area fraction (>85%), 0.5 megapixels and readout less than 1 mW/channel. Imaging from low-Earth orbit of air fluorescence will require the development of high sensitivity and efficiency detection of 300-400 nm UV photons to measure signals at the few photon (single photo-electron) level. A secondary goal minimizes the sensitivity to photons with a wavelength greater than 400 nm. High electronic gain (10E4 to 10E6), low noise, fast time response (<10 ns), minimal dead time (<5% dead time at 10 ns response time), high segmentation with low dead area (<20% nominal, <5% goal), and the ability to tailor pixel size to match that dictated by the imaging optics. Optical designs under consideration dictate a pixel size ranging from approximately 2 x 2 mm^2 to 10 x 10 mm^2. Focal plane mass must be minimized (2g/cm^2 goal). Individual pixel readout is required. The entire focal plane detector can be formed from smaller, individual sub-arrays.
- Neutral density filter for hard x-rays (> 1 keV) to provide attenuation by a factor of 10 to 1000 or more. The filter must provide broad attenuation across a broad energy range (from 1 keV to ~100 keV or more) with a flat attenuation profile of better than 20%.
- Solar X-ray detectors with small independent pixels (< 250 ¼m) and fast read-out (>10,000 count/s/pixel) over an energy range from < 5 keV to 300 keV.
- Supporting technologies that would help enable X-ray Surveyor mission that requires the development of X-ray microcalorimeter arrays with much larger field of view, ~105-106 pixels, of pitch ~ 25-100 um, and ways to read out the signals. For example, modular superconducting magnetic shielding is sought that can be extended to enclose a full-scale focal plane array. All joints between segments of the shielding enclosure must also be superconducting.
- Improved long-wavelength blocking filters are needed for large-area, x-ray microcalorimeters. Filters with supporting grids are sought that, in addition to increasing filter strength, also enhance EMI shielding (1 - 10 GHz) and thermal uniformity for decontamination heating. X-ray transmission of greater than 80% at 600 eV per filter is sought, with infrared transmissions less than 0.01% and ultraviolet transmission of less than 5% per filter. Means of producing filter diameters as large as 10 cm should be considered.

NASA flagship missions under study are LUVOIR, HabEx, Lynx, New Frontier-IO:

- http://luvoir.stsci.edu
- https://www.jpl.nasa.gov/habex/
- https://wwwastro.msfc.nasa.gov/lynx/
- https://science.nasa.gov/astrophysics/
- https://explorers.gsfc.nasa.gov

The desired deliverables are results of tests and analysis of designs and/or prototype hardware. The expected Technology Readiness Level (TRL) range at completion of the project is 3-5.

References:

- https://cor.gsfc.nasa.gov
- https://planetarymissions.nasa.gov
- https://ehpd.gsfc.nasa.gov

S1.06 Particles and Fields Sensors & Instrument Enabling Technologies (SBIR)
Lead Center: GSFC
Participating Center(s): GSFC
Related Subtopic Pointer(s): Z12.02

While the size distribution of matter in space that ranges from large-scale (planets – moons – asteroids – dust) objects is quite well characterized down to micron-sized dust particles, below that there is a significant, largely unobserved gap down to single ions/electrons/ENAs. To cover the observational gap between 10-6m and 10-10m in particle size that includes nano-dust and molecules in space, new technology investment is needed. Advanced sensors for the detection of elementary particles (atoms, molecules and their ions) and electric and magnetic fields in space and associated instrument technologies are often critical for enabling transformational science from the study of the sun's outer corona, to the solar wind, to the trapped radiation in Earth's and other planetary magnetic fields, and to the atmospheric composition of the planets and their moons. Improvements in particles and fields sensors and associated instrument technologies enable further scientific advancement for upcoming NASA missions such as CubeSats, Explorers, STP, LWS, and planetary exploration missions. Technology developments that result in a reduction in size, mass, power, and cost will enable these missions to proceed. Of interest are advanced magnetometers, electric field booms, ion/atom/molecule detectors, dust particle detectors, and associated support electronics and materials.

Low energy particle instruments often require significant high voltage power supplies up to 20KV. Linear control of high voltage with optical isolation is highly desirable in space plasma instrument. General specifications 3.3 to5V control, 10KV to 20KV high voltage, low leakage current, up to 25KV isolation voltage, Fast slew rate >200V/us; temperature insensitivity on the range -35° C to +55° C, radiation hardness >1~200Keads.

Subtopic is relevant to NASA Explorer missions, Decadal survey missions MIDEX, GDC, DYNAMICS, DRIVE Initiative, DISCOVERY, New Frontiers; CubeSat and SmallSat missions; and Sub-orbitals.

The desired deliverables of a Phase II are prototype and hardware. A prototype component that can be tested in engineering model plasma instrument. The expected Technology Readiness Level (TRL) range at completion of the project is 5-7.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

S1.07 In-Situ Instruments/Technologies for Lunar and Planetary Science (SBIR)
Lead Center: JPL
Participating Center(s): ARC, GRC, GSFC, MSFC
Related Subtopic Pointer(s): Z12.02

This subtopic solicits development of advanced instrument technologies and components suitable for deployment on in-situ planetary and lunar missions. These technologies must be capable of withstanding operation in space and planetary environments, including the expected pressures, radiation levels, launch and impact stresses, and range of survival and operational temperatures. Technologies that reduce mass, power, volume, and data rates for instruments and instrument components without loss of scientific capability are of particular importance. In addition, technologies that can increase instrument resolution and sensitivity or achieve new & innovative scientific measurements are solicited. For example, missions, see http://science.hq.nasa.gov/missions. For details of the specific requirements see the National Research Council's, Vision and Voyages for Planetary Science in the Decade 2013-2022 (http://solarsystem.nasa.gov/2013decadal/). Technologies that support NASA˘s New Frontiers and Discovery missions to various planetary bodies are of top priority.

In-situ technologies are being sought to achieve much higher resolution and sensitivity with significant improvements over existing capabilities. In-situ technologies amenable to Cubesats and Smallsats are also being solicited. Atmospheric probe sensors and technologies that can provide significant improvements over previous missions are also sought. Specifically, this subtopic solicits instrument development that provides significant advances in the following areas, broken out by planetary body:

- *Mars* - Sub-systems relevant to current in-situ instrument needs (e.g., lasers and other light sources from UV to microwave, X-ray and ion sources, detectors, mixers, mass analyzers, etc.) or electronics technologies (e.g., FPGA and ASIC implementations, advanced array readouts, miniature high voltage power supplies). Technologies that support high precision in-situ measurements of elemental, mineralogical, and organic composition of planetary materials are sought. Conceptually simple, low risk technologies for in-situ sample extraction and/or manipulation including fluid and gas storage, pumping, and chemical labeling to support analytical instrumentation. Seismometers, mass analyzers, technologies for heat flow probes, and atmospheric trace gas detectors. Improved robustness and g-force survivability for instrument components, especially for geophysical network sensors, seismometers, and advanced detectors (iCCDs, PMT arrays, etc.). Instruments geared towards rock/sample interrogation prior to sample return are desired.
- *Venus* - Sensors, mechanisms, and environmental chamber technologies for operation in Venus's high temperature, high-pressure environment with its unique atmospheric composition. Approaches that can enable precision measurements of surface mineralogy and elemental composition and precision measurements of trace species, noble gases and isotopes in the atmosphere are particularly desired.
- *Small Bodies* - Technologies that can enable sampling from asteroids and from depth in a comet nucleus, improved in-situ analysis of comets. Imagers and spectrometers that provide high performance in low light environments. Dust environment measurements & particle analysis, small body resource identification, and/or quantification of potential small body resources (e.g., oxygen, water and other volatiles, hydrated minerals, carbon compounds, fuels, metals, etc.). Advancements geared towards instruments that enable elemental or mineralogy analysis (such as high-sensitivity X-ray and UV-fluorescence spectrometers, UV/fluorescence systems, electron probes including collimated e-beam sources for micro-analyzers, mass spectrometry, gas chromatography and tunable diode laser sensors, calorimetry, imaging spectroscopy, and LIBS) are sought.
- *Saturn, Uranus and Neptune* - Components, sample acquisition, and instrument systems that can enhance mission science return and withstand the low-temperatures/high-pressures of the atmospheric probes during entry.
- *The Moon* – For lunar science, solicited are advancements in the areas of compact, light-weight, low power instruments geared towards in- situ lunar surface measurements, geophysical measurements, lunar atmosphere and dust environment measurements & regolith particle analysis. Specifically, advancements geared towards instruments that enable elemental or mineralogy analysis (such as high-sensitivity X-ray and Raman spectrometers, UV/fluorescence systems, scanning electron microscopy with chemical analysis capability, mass spectrometry, gas chromatography and tunable diode laser sensors, calorimetry, laser-Raman spectroscopy, imaging spectroscopy, and LIBS) are sought. These developments should be geared towards sample interrogation, prior to possible sample return. Systems and subsystems for seismometers and heat flow sensors capable of long-term continuous operation over multiple lunar day/night cycles with improved sensitivity at lower mass and reduced power consumption are sought. Also, of interest are portable surface ground penetrating radars to characterize the thickness of the lunar regolith, as well as low mass, thermally stable hollow cubes and retro-reflector array assemblies for lunar surface laser ranging. Of secondary importance are instruments that measure the micrometeoroid and lunar secondary ejecta environment, plasma environment, surface electric field, secondary radiation at the lunar surface, and dust concentrations and its diurnal dynamics. Further, lunar regolith particle analysis techniques are desired (e.g., optical interrogation or software development that would automate integration of suites of multiple back scatter electron images acquired at different operating conditions, as well as permit integration of other data such as cathodoluminescence and energy-dispersive x-ray analysis.). This topic seeks advancement of concepts and components to develop a Lunar Geophysical Network as envisioned in the Vision and Voyages for Planetary Science in the Decade 2013 - 2022. Understanding the distribution and origin of both shallow and deep moonquakes will provide insights into the current dynamics of the lunar interior and its interplay with external phenomena (e.g., tidal interactions with Earth). The network is envisioned to be comprised of multiple free-standing seismic stations which would operate over many years in even the most extreme lunar

temperature environments. Technologies are sought to advance all aspects of the network including sensor emplacement, power, and communications in addition to seismic, heat flow, magnetic field and electromagnetic sounding sensors.

Proposers are strongly encouraged to relate their proposed development to:

- NASA's future planetary exploration goals
- Existing flight instrument capability, to provide a comparison metric for assessing proposed improvements

Proposed instrument architectures should be as simple, reliable, and low risk as possible while enabling compelling science. Novel instrument concepts are encouraged particularly if they enable a new class of scientific discovery. Technology developments relevant to multiple environments and platforms are also desired.

Proposers should show an understanding of relevant space science needs and present a feasible plan to fully develop a technology and infuse it into a NASA program.

In-situ instruments and technologies are essential bases to achieve SMD's planetary science goals summarized in Decadal Study (National Research Council's, Vision and Voyages for Planetary Science in the Decade 2013-2022. In-situ instruments and technologies play indispensable role for NASA's New Frontiers and Discovery missions to various planetary bodies.

NASA SMD has two excellent programs to bring this subtopic technologies to higher level: PICASSO and MatISSE. The Planetary Instrument Concepts for the Advancement of Solar System Observations (PICASSO) Program invests in low-TRL technologies and funds instrument feasibility studies, concept formation, proof-of-concept instruments, and advanced component technology. The Maturation of Instruments for Solar System Exploration (MatISSE) Program invests in mid-TRL technologies and enables timely and efficient infusion of technology into planetary science missions. The PICASSO and MatISSE are in addition to Phase III opportunities.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-5.

References:

- For example, missions see http://science.hq.nasa.gov/missions
- National Research Council's, Vision and Voyages for Planetary Science in the Decade 2013-2022 (http://solarsystem.nasa.gov/2013decadal/)

S1.08 Suborbital Instruments and Sensor Systems for Earth Science Measurements (SBIR)
Lead Center: LaRC
Participating Center(s): ARC, GSFC, JPL, LaRC

In-situ sensors & sensor systems targeting trace gas measurements

Earth science measurements from space are considerably enhanced by observations from generally far-less costly suborbital instruments and sensor systems. These instruments and sensors support NASA's ESD science, calibration/validation and environmental monitoring activities by providing ancillary data for satellite calibration and validation; algorithm development/refinement; and finer-scale process studies. Accordingly, instrument and sensor systems are sought that include air quality, greenhouse gases, flux measurements, advancement of methods for assessing air mass photochemical age or for differentiating emissions sources (for example, real-time, fast response

isotopic carbon measurements) and atmospheric composition. In-situ sensor systems (airborne, land and water-based) can comprise stand-alone instrument and data packages; instrument systems. This subtopic solicits instrument systems configured for ground-based/mobile surface deployments, as well as for integration on NASA's Airborne Science aircraft fleet or commercial providers, UAS, or balloons. An important goal is to create sustainable measurement capabilities to support NASA's Earth science objectives – most notably support of its Earth Venture programs especially validation and verification of LEO and GEO AQ/AC satellites through involvement with NASA's intensive targeted field campaigns and or its ground-based networks. Instrument prototypes as a deliverable in Phase II proposals and/or field demonstrations are encouraged.

Complete instrument systems are generally desired, including features such as remote/unattended operation and data acquisition, and minimum size, weight, and power consumption. All proposals must summarize the current state of the art, and demonstrate how the proposed sensor or sensor system represents a significant improvement over the state of the art.

Desired passive sensors/instruments, in-situ/airborne sensors or mated platform/sensors include:

- Small, turn-key trace gas measurement sensors with 1-10 Hz time response that are suitable for autonomous aircraft and/or UAV deployment and capable of detecting:
    - NOx, NOy, CH_2O, O_3, benzene, toluene at < 5 % uncertainty
    - CO, CH_4, OCS and N_2O at < 1% uncertainty
    - CO_2 at < 0.05% uncertainty,
- Where these uncertainties apply to measurements made on airborne platforms under flight conditions (variable ambient pressure and temperature)
- Real-time, 0.1-1 Hz gas-phase radioisotopic (especially radiocarbon) measurements suitable for distinguishing emissions sources and for deployment on aircraft or UAVs
- Bulk or film retroreflector subsystems that advance NASA open path trace gas measurements (similar to the widely used NASA LaRC Diode Laser Hygrometer). Operational at wavelengths of 2-5 um and/or 8-12 um bands with low return light cone divergence (<2°).
- Low-volume (<0.1 L) multi-pass cell spectrometer subsystems that advance NASA extractive trace gas measurements. Operational at wavelengths 2-5 um or greater with pathlengths of 50+ meters.
- Aircraft static air temperature sensor measurement to better than 0.1° C accuracy under upper troposphere / lower stratosphere conditions.
- Miniaturized passive sensor systems that observe both trace gases *and* aerosols at a similar price point but beyond the capabilities of both the Pandora spectrometer (http://sciglob.com) and Cimel sun photometer (https://www.cimel.fr) systems are sought. System should be stand-alone, user-friendly, autonomous/remotely operated instruments actively tracking the Sun and Moon (with a pointing precision of at least if not better than 0.1°) and capable of making sky/surface observations on the scale of tens of seconds from both stable (e.g., roofs/towers) and mobile platforms (e.g., ship and or vehicle) while having integrated real-time preliminary data processing for trace gases and aerosols. Systems must be capable of providing high-resolution UV-VIS-NIR solar/lunar/clear sky spectra that can be used to determine atmospheric abundance of O_3, NO_2, HCHO, SO_2, BrO, HONO, CHOCHO, H_2O_v and aerosols. TG observations require a S/N ratio of better than 2500:1 whereas aerosol observations require an accuracy of at least 3%. Proposed systems must maintain an absolute calibration while deployed.

Desired ocean color sensors/instruments include:

- In-situ instruments to measure in-situ and lab-based absorption, backscatter and beam attenuation in the ocean, extending the current commercial capability beyond what is available today (410-750 nm) and obtaining measurements that extend into the UV and Near-IR regions of the spectrum.
- In-situ instruments to measure ocean Volume Scattering Function (VSF) and backscatter, extending the current capacity of few specific wavelengths to a hyperspectral capability extending from the UV to NIR with high angular (<10 °) resolution.
- Instrumentation with improved methods and measurement platform for upwelling radiances just below the water surface (Lu(0-)), extending spectrally from the UV to NIR.

- Instrumentation for in-situ measurements of polarization IOPs (Mueller Matrix: S11, S12 and S22) spanning from UV<->NIR, with high angular resolution (<=10 º) of scattering components.

The S1.08 subtopic is and remains highly relevant to NASA SMD and Earth Science research programs, in particular, the Earth Science Atmospheric Composition and Climate focus areas. In-situ sensors and, specifically trace gas sensors, inform directed Airborne Science field campaigns led by these programs and provide important validation of airborne and ground-based remote sensors (e.g., GCAS, 4STAR, AERONET, and Pandoras) as well as the current and next generation of satellite-based sensors (e.g., OCO, TEMPO). The solicited measurements are highly relevant to past and future NASA airborne campaigns (e.g., FIREX-AQ, CAMP2EX, KORUS-AQ, DISCOVER-AQ). Given the on-going and continuing need for such airborne science missions, it is expected that the sensors and sensor systems developed under this subtopic would directly benefit these missions and those expected in the coming decade.

Other programs relevant to NASA are ESD Tropospheric Composition Program and ESD Radiation Sciences Program.

Instruments developed for this subtopic would provide synergistic trace gas and aerosol observations that would contribute to the validation and or verification of the following satellites (both U.S. and international):

- Active Satellites:
    - AURA NASA LEO.
    - MetOp-A EUMETSAT LEO.
    - S-NPP NASA LEO.
    - MetOp-B EUMETSAT LEO.
    - DSCOVR NASA L1.
    - Sentinel 3A EUMETSAT LEO.
    - Sentinel 5P ESA LEO.
    - GaoFen-5 CSA LEO.
    - NOAA-20 NOAA LEO.
    - Sentinel 3B EUMETSAT LEO.
- To be launched:
    - GEO-KOMPSAT 2 NIER GEO.
    - TEMPO NASA GEO.
    - Sentinel 4 EUMETSAT GEO.
    - Sentinel 5 EUMETSAT LEO.
    - MAIA NASA LEO.

The need horizon of the subtopic sensors and sensors systems is BOTH near (<5 years) and mid-term (5-10 years). The expected Technology Readiness Level (TRL) range at completion of the project is 4-7.

References:

Relevant current and past field campaign websites include:

- KORUS-AQ: https://espo.nasa.gov/korus-aq
- DISCOVER-AQ: https://discover-aq.larc.nasa.gov/
- CAMP2Ex: https://espo.nasa.gov/camp2ex/content/CAMP2Ex
- FIREX-AQ: https://espo.nasa.gov/firex-aq/content/FIREX-AQ, https://www.esrl.noaa.gov/csd/projects/firex/science.html
- AToM: https://espo.nasa.gov/home/atom

S1.09 Cryogenic Systems for Sensors and Detectors (SBIR)
Lead Center: GSFC
Participating Center(s): JPL
Related Subtopic Pointer(s): T13.01

Cryogenic systems provide the necessary environment for low temperature detectors and sensors, as well as for telescopes and instrument optics on infrared observatories. As such, technological improvements to cryogenic systems further advance the mission goals of NASA through enabling performance (and ultimately science gathering) capabilities of flight detectors and sensors. There are five areas in which NASA is seeking to expand state of the art capabilities:

Low Temperature/High Efficiency Cryocoolers

NASA seeks improvements to multistage low temperature spaceflight cryocoolers. Coolers are sought with the lowest temperature stage typically in the range of 4 to 10 K, with cooling power at the coldest stage larger than currently available, and high efficiency. The desired cooling power is application specific, but two examples are 0.3 Watts at 10 K and 0.2 Watts at 4 K. In applications where the device is coupled to an advanced magnetic cooler, it needs to tolerate large swings in heat load on a time scale of the order of minutes to tens of minutes. Devices that produce extremely low vibration, particularly at frequencies below a few hundred Hz, are of special interest. System or component level improvements that improve efficiency and reduce complexity and cost are desirable.

Coolers in this class are of interest for space telescopes and instruments for infrared astronomy, as well as for instruments using low temperature detectors, particularly those using advanced sub-Kelvin detectors. Examples of future missions that require this technology include two of the large missions under study for the 2020 Astrophysics Decadal Survey:

- Origins Space Telescope.
- LYNX (microcalorimeter instrument).

Low temperature cryocoolers are listed as a "Technology Gap" in the latest (2017) Cosmic Origins Program Annual Technology Report.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-5.

Miniaturized/Efficient Cryocooler Systems

NASA seeks miniature, highly efficient cryocoolers for instruments on Earth and planetary missions. A range of cooling capabilities sought. Two examples include 0.2 Watt at 30 K with heat rejection at 300 K, and 0.3 W at 35K with heat rejection of 150 K. For both examples, an input power of ≤ 5 Watt and a total mass of ≤ 400 grams is desired. The ability to fit within the volume and power limitations of a SMALLSAT platform would be highly advantageous. Components, such as low-cost cryocooler electronics that are sufficiently rad hard for lunar or planetary missions, are also sought.

NASA is moving toward the use of small, low cost satellites to achieve many of its Earth science, and some of its planetary science goals. The development of cryocoolers that fit within the size and power constraints of these platforms will greatly expand their capability, for example, by enabling the use of infrared detectors. In planetary science, progress on cryogenic coolers will enable the use of far- to mid-infrared sensors with orders of magnitude improvement in sensitivity for outer planetary missions. These will allow thermal mapping of outer planets and their moons.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-4.

Sub-Kelvin Cooling Systems

Future NASA missions require sub-Kelvin coolers for extremely low temperature detectors. Systems are sought that will provide continuous cooling with high cooling power (> 5microWatts at 50 mK), low operating temperature (< 35

mK), and higher heat rejection temperature (preferably > 10 K), while maintaining high thermodynamic efficiency and low system mass. Improvements in components for adiabatic demagnetization refrigerators are also sought. Specific components include:

- Compact, lightweight, low current superconducting magnets capable of producing a field of at least 4 Tesla while operating at a temperature of at least 10 K, and preferably above 15 K. Desirable properties include:
    - A high engineering current density, preferably > 300 Amp/mm
    - A field/current ratio of > 0.5 Tesla/Amp, and preferably > 0.8 Tesla/Amp
    - Low hysteresis heating
    - Mass < 2.5 kg
- Lightweight Active/Passive magnetic shielding (for use with 4 Tesla magnets) with low hysteresis and eddy current losses, and low remanence
- Heat switches for operation at < 10 K with on/off conductance ratio > 30,000, actuation time of < 10 s, and an off conductance of < 50 microWatt/K. Materials are also sought for gas-gap heat switch shells: these are tubes with extremely low thermal conductance below 1 K; they must be impermeable to helium gas, have high strength, including stability against buckling, and have an inner diameter > 20 mm
- High cooling power density magnetocaloric materials, especially single crystals with volume > 20 cc. Examples of desired single crystals include GdF3, GdLiF4, and Gd elpasolite
- 10 mK- 300 mK high-resolution thermometry
- Suspensions with the strength and stiffness of Kevlar, but lower thermal conductance from 4 K to 0.050 K

Advanced superconducting detectors, such as Transition Edge Sensors (TESs) and Microwave Kinetic Inductance Detectors (MKIDs), operate at extremely low temperatures. Large arrays of such detectors will require advanced subKelvin coolers with large cooling power. These detectors offer orders of magnitude improvement in sensitivity, and thus are slated for a number of future astrophysics missions. Examples of future missions that advanced subKelvin coolers include two of the large missions under study for the 2020 Astrophysics Decadal Survey: Origins Space Telescope and LYNX (microcalorimeter instrument). Other future missions include Probe of Inflation and Cosmic Origins. SubKelvin coolers are listed as a "Technology Gap" in the latest (2017) Cosmic Origins Program Annual Technology Report.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-4.

Rad-hard Cryogenic Accelerometers

NASA seeks accelerometers that can operate at 150 K, withstand a 0.01 Tesla magnetic field and are radiation hard to 2-5 megarads.

Cryocoolers are needed for for the operation of high sensitivity infrared detectors that are planned for missions to the outer planets and their moons. Most cryocooler components are easily made rad-hard. However, accelerometers, which are required for vibration cancellation, are currently not available that can operate in extreme conditions, especially in the high radiation environments around Jupiter's moons.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-4.

Ultra-lightweight Dewars

NASA seeks extremely lightweight thermal isolation systems for scientific instruments. An important example is a large cylindrical, open top dewar to enable large, cold balloon telescopes. Such a dewar would be launched warm, and so would not need to function at ambient pressure, but at altitude, under ~4 millibar external pressure, it would need to contain cold helium vapor. In operation, heat flux through the walls should be less than 0.5 Watts per square meter. The ability to rapidly pump and hold a vacuum at altitude is necessary. Initial demonstration units of greater than 1 meter diameter and height are desired, but the technology must be scalable to 3 – 4 meters with a mass that is a small fraction of the net lift capability of a scientific balloon (~2000 kg).

The potential for ground-based infrared astronomy is extremely limited. Even in airborne observatories, such as SOFIA, observations are limited by the brightness of the atmosphere and the warm telescope itself. However, high altitude scientific balloons are above enough of the atmosphere, that with a telescope large enough and cold enough, background-limited observations are possible. The ARCADE project demonstrated that at high altitudes, it is possible to cool instruments in helium vapor. Development of ultra-lightweight dewars that could be scaled up to large size, yet still be liftable by a balloon would enable ground-breaking observational capability.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-4.

References:

Low temperature/high efficiency cryocoolers

- For more information on the Origins Space Telescope, see https://asd.gsfc.nasa.gov/firs/

Miniaturized/Efficient Cryocooler Systems

- An example of cubesat mission using cryocoolers is given at: https://www.jpl.nasa.gov/cubesat/missions/ciras.php

Sub-Kelvin cooling systems

- For a description of the state-of-the-art sub-Kelvin cooler in the Hitomi mission, see: Shirron, et al. "Thermodynamic performance of the 3-stage ADR for the Astro-H Soft-X-ray Spectrometer instrument," Cryogenics 74 (2016) 24–30, and references therein.
- For articles describing magnetic sub-Kelvin coolers and their components, see the July 2014 special issue of Cryogenics: Cryogenics 62 (2014) 129–220.

Rad-hard cryogenic accelerometers

- I.M. McKinley, M.A. Mok, D.L. Johnson, and J.I. Rodriguez, 2018. Characterization Testing of Lockheed Martin Micro1-2 Cryocoolers Optimized for 220 K Environment, International Cryocooler Conference, Burlington, VT, USA. June 18-21, 2018. Cryocoolers 20.
- M.A. Mok, I.M. McKinley, and J.I. Rodriguez, 2018. Low Temperature Characterization of Mechanical Isolators for Cryocoolers, International Cryocooler Conference, Burlington, VT, USA. June 18-21, 2018. Cryocoolers 20.
- D. Glaister, E. Marquardt and R. Taylor, "Ball Low Vibration Cryocooler Assemblies," preesented at the ICC20, June 2018, Burlington, VT.
- http://iopscience.iop.org/article/10.1088/1757-899X/278/1/012005

Ultra-lightweight dewars

- For a description of a state-of-the art balloon cryostat, see Singal, et al. "The ARCADE 2 instrument," The Astrophysical Journal, 730:138 (12pp), 2011 April 1

S1.10 Atomic Interferometry (SBIR)
Lead Center: GSFC
Participating Center(s): GSFC, JPL
Related Subtopic Pointer(s): T8.02

Recent developments of laser control and manipulation of atoms have led to new types of precision inertial force and gravity sensors based on atom interferometry. Atom interferometers exploit the quantum mechanical wave nature of atomic particles and quantum gases for sensitive interferometric measurements. Ground-based laboratory experiments and instruments have already demonstrated beyond the state-of-the-art performances of accelerometer, gyroscope, and gravity measurements. The microgravity environment in space provides opportunities for further drastic

improvements in sensitivity and precision. Such inertial sensors will have great potential to provide new capabilities for NASA Earth and planetary gravity measurements, for spacecraft inertial navigation and guidance, and for gravitational wave detection and test of properties of gravity in space.

Currently the most mature development of atom interferometers as measurement instruments are those based on light pulsed atom interferometers with freefall cold atoms. There remain a number of technical challenges to infuse this technology in space applications. Some of the identified key challenges are (but not limited to):

- Compact high flux ultra-cold atom sources for free space atom interferometers (Example: >1e+06 total useful free-space atoms, <1 nK, Rb, K, Cs, Yb, Sr, and Hg. Performance and species can be defined by offerors). Other related innovative methods and components for cold atom sources are of great interest, such as a highly compact and regulatable atomic vapor cell.
- Ultra-high vacuum technologies that allow completely sealed, non-magnetic enclosures with high quality optical access and the base pressure maintained <1e-09 torr. Consideration should be given to the inclusion of cold atom sources of interest.
- Beyond the state-of-the-art photonic components at wavelengths for atomic species of interest, particularly at NIR and visible: efficient acousto-optic modulators (low RF power ~200 mW, low thermal distortion, ~80% or greater diffraction efficiency); efficient electro-optic modulators (low bias drift, residual AM, and return loss, fiber-coupled preferred), miniature optical isolators (~30 dB isolation or greater, ~ -2 dB loss or less), robust high-speed high-extinction shutters (switching time < 1 ms, extinction > 60 dB are highly desired).
- Flight qualifiable lasers or laser systems of narrow linewidth, high tunability, and/or higher power for clock and cooling transitions of atomic species of interest. Cooling and trapping lasers: 10 kHz linewidth and ~1 W or greater total optical power. Compact clock lasers: 5e-15 Hz/$\sqrt{\tau}$ near 1 s (wavelengths for Yb+, Yb, Sr clock transitions are of special interest).
- Analysis and simulation tool of a cold atom system in trapped and freefall states relevant to atom interferometer and clock measurements in space.

All proposed system performances can be defined by offerors with sufficient justification. Subsystem technology development proposals should clearly state the relevance, define requirements, relevant atomic species and working laser wavelengths, and indicate its path to a space-borne instrument.

Currently, no technology exists that can compete with the potential sensitivity, (potential) compactness, and robustness of Atom Optical-based gravity and time measurement devices. Earth science, planetary science, and astrophysics all benefit from unprecedented improvements in gravity and time measurement. Specific roadmap items supporting science instrumentation include, but are not limited to:

- TA-7.1.1: Destination Reconnaissance, Prospecting, and Mapping (gravimetry)
- TA-8.1.2: Electronics (reliable control electronics for laser systems)
- TA-8.1.3: Optical Components (reliable laser systems)
- TA-8.1.4: Microwave, Millimeter, and Submillimeter-Waves (ultra-low noise microwave output when coupled w/ optical frequency comb)
- TA-8.1.5: Lasers (reliable laser system w/ long lifetime)

The desired deliverables are prototype hardware/software, documented evidence of delivered TRL (test report, data, etc.), summary analysis, and supporting documentation. The expected Technology Readiness Level (TRL) range at completion of the project is 3-5.

References:

- 2017 NASA Strategic Technology Investment Plan: https://go.usa.gov/xU7sE
- 2015 NASA Technology Roadmaps: https://go.usa.gov/xU7sy

S1.11 In-Situ Instruments/Technologies for Ocean Worlds Life Detection (SBIR)
Lead Center: JPL
Participating Center(s): ARC, GRC, GSFC
Related Subtopic Pointer(s): S4.04, S4.02

This subtopic solicits development of in-situ instrument technologies and components to advance the maturity of science instruments focused on the detection of evidence of life, especially extant of life, in the Ocean Worlds (e.g., Europa, Enceladus, Titan, Ganymede, Callisto, Ceres, etc.). These technologies must be capable of withstanding operation in space and planetary environments, including the expected pressures, radiation levels, launch and impact stresses, and range of survival and operational temperatures. Technologies that reduce mass, power, volume, and data rates for instruments and instrument components without loss of scientific capability are of particular importance. In addition, technologies that can increase instrument resolution and sensitivity or achieve new & innovative scientific measurements are solicited.

Specifically, this subtopic solicits instrument technologies and components that provide significant advances in the following areas, broken out by planetary body:

- *Europa, Enceladus, Titan and other Ocean Worlds in general* - Technologies and components relevant to life detection instruments (e.g., microfluidic analyzer, MEMS chromatography/mass spectrometers, laser-ablation mass spectrometer, fluorescence microscopic imager, Raman spectrometer, tunable laser system, liquid chromatography/mass spectrometer, X-ray fluorescence, digital holographic microscope-fluoresce microscope, Antibody microarray biosensor, nanocantilever biodetector etc.) Technologies for high radiation environments, e.g., radiation mitigation strategies, radiation tolerant detectors, and readout electronic components, which enable orbiting instruments to be both radiation-hard and undergo the planetary protection requirements of sterilization (or equivalent).
- *Europa* - Life detection approaches optimized for evaluating and analyzing the composition of ice matrices with unknown pH and salt content. Instruments capable of detecting and identifying organic molecules (in particular biomolecules), salts and/or minerals important to understanding the present conditions of Europa s ocean are sought (such as high-resolution gas chromatograph or laser desorption mass spectrometers, dust detectors, organic analysis instruments with chiral discrimination, etc.). These developments should be geared towards analyzing and handling very small sample sizes (mg to mg) and/or low column densities/abundances. Also, of interest are imagers and spectrometers that provide high performance in low-light environments (visible and NIR imaging spectrometers, thermal imagers, etc.), as well as instruments capable of providing improving our understanding Europa s habitability by characterizing the ice, ocean, and deeper interior and monitoring ongoing geological activity such as plumes, ice fractures, and fluid motion (e.g., seismometers, magnetometers). Improvements to instruments capable of gravity (or other) measurements that might constrain properties such as ocean and ice shell thickness will also be considered.
- *Enceladus* - Life detection approaches optimized for analyzing plume particles, as well as for determining the chemical state of Enceladus icy surface materials (particularly near plume sites). Instruments capable of detecting and identifying organic molecules (in particular biomolecules), salts and/or minerals important to understand the present conditions of the Enceladus ocean are sought (such as high-resolution gas chromatograph or laser desorption mass spectrometers, dust detectors, organic analysis instruments with chiral discrimination, etc.). These developments should be geared towards analyzing and handling very small sample sizes (mg to mg) and/or low column densities/abundances. Also, of interest are imagers and spectrometers that provide high performance in low-light environments (visible and NIR imaging spectrometers, thermal imagers, etc.), as well as instruments capable of monitoring the bulk chemical composition and physical characteristics of the plume (density, velocity, variation with time, etc.). Improvements to instruments capable of gravity (or other) measurements that might constrain properties such as ocean and ice shell thickness will also be considered.
- *Titan* - Life detection approaches optimized for searching for biosignatures and biologically relevant compounds in Titan s lakes, including the presence of diagnostic trace organic species, and also for analyzing Titan s complex aerosols and surface materials. Mechanical and electrical components and subsystems that work in cryogenic (95K) environments; sample extraction from liquid methane/ethane, sampling from organic 'dunes' at 95K and robust sample preparation and handling mechanisms that feed into mass analyzers are sought. Balloon instruments, such as IR spectrometers, imagers, meteorological instruments, radar sounders, solid, liquid, air sampling mechanisms for mass analyzers, and aerosol detectors are also solicited.

Low mass and power sensors, mechanisms and concepts for converting terrestrial instruments such as turbidimeters and echo sounders for lake measurements, weather stations, surface (lake and solid) properties packages, etc. to cryogenic environments (95K).
- Other Ocean Worlds targets may include Ganymede, Callisto, Ceres, etc.

In-situ instruments and technologies are essential bases to achieve SMD's planetary science goals summarized in Decadal Study (National Research Council's, Vision and Voyages for Planetary Science in the Decade 2013-2022. In-situ instruments and technologies play indispensable role for NASA's New Frontiers and Discovery missions to various planetary bodies.

NASA SMD has two excellent programs to bring this subtopic technologies to higher level: PICASSO and MatISSE. The Planetary Instrument Concepts for the Advancement of Solar System Observations (PICASSO) Program invests in low-TRL technologies and funds instrument feasibility studies, concept formation, proof-of-concept instruments, and advanced component technology. The Maturation of Instruments for Solar System Exploration (MatISSE) Program invests in mid-TRL technologies and enables timely and efficient infusion of technology into planetary science missions. The PICASSO and MatISSE are in addition to Phase III opportunities.

Proposers are strongly encouraged to relate their proposed development to:

- NASA's future Ocean Worlds exploration goals
- Existing flight instrument capability, to provide a comparison metric for assessing proposed improvements

Proposed instrument architectures should be as simple, reliable, and low risk as possible while enabling compelling science. Novel instrument concepts are encouraged particularly if they enable a new class of scientific discovery. Technology developments relevant to multiple environments and platforms are also desired.

Proposers should show an understanding of relevant space science needs and present a feasible plan to fully develop a technology and infuse it into a NASA program. The expected Technology Readiness Level (TRL) range at completion of the project is 3-5.

References

- For synergistic NASA technology solicitation, see ROSES 2016/C.20 Concepts for Ocean worlds Life Detection Technology (COLDTECH) call: https://nspires.nasaprs.com/external/solicitations/summary.do?method=init&solId={5C43865B-0C93-6ECA-BCD2-A3783CB1AAC8}&path=closedPast
- For another synergistic NASA technology solicitation, see ROSES 2018/C.23 for Instrument Concepts for Europa Exploration 2 call: https://nspires.nasaprs.com/external/solicitations/summary!init.do?solId=%7b17B73E96-6B65-FE78-5B63-84C804831035%7d&path=open

S4.06 Sample Collection for Life Detection in Outer Solar System Ocean World Plumes (SBIR)
Lead Center: JPL
Participating Center(s): ARC, GSFC, JPL
Related Subtopic Pointer(s): S1.11

This subtopic solicits development of technologies for sample collection from plumes in the Ocean Worlds Exploration Program (e.g., Europa, Enceladus, Titan, Ganymede, Callisto, Ceres, etc.). This sample collection system would be used as the front-end system in conjunction with in-situ instruments developed under subtopic S1.11. This fly-through sampling subtopic is distinct from S4.02, which solicits sample collection technologies from surface platforms. These technologies must be capable of withstanding operation in space and planetary environments, including the expected pressures, radiation levels, launch and impact stresses, and range of survival and operational temperatures. Technologies that allow collection during high speed (>1 km/sec) velocity passes through a plume are of interest as are technologies that can maximize total sample mass collected while passing through tenuous plumes. Technologies that reduce mass, power, volume, and data rates without loss of scientific capability are of particular

importance. This technology would enable high-priority sampling and potential sample return from the plumes of Enceladus with a fly-by mission. This would be a substantial cost savings over a landed mission.

The icy moons of the outer Solar System are of astrobiological interest. The most dramatic target for sampling from a plume is for Enceladus. Enceladus is a small icy moon of Saturn, with a radius of only 252km. Cassini data have revealed about a dozen or so jets of fine icy particles emerging from the south polar region of Enceladus. The jets have also been shown to contain organic compounds, and the south-polar region is warmed by heat flow coming from below.

As a target for future missions, Enceladus rates high because fresh samples of interest are jetting into space ready for collection. Indeed, Enceladus has been added to the current call for New Frontiers missions with a focus on habitability and life detection. Particles from Enceladus also form the E-ring around Saturn. The particles in the E-ring are known to contain organics and are thus also an important target for sample collection and analysis. Recent data have indicated a possible plume at Europa that may also be carrying ocean water from that world into space. In addition to plumes, there are other energetic processes that can spray material from the surface of these low-gravity worlds into space where they could also be collected in-flight and analyzed.

Collecting samples for a variety of science purposes is required. These include samples that allow for determination of the chemical and physical properties of the source ocean, samples for detailed characterization of the organics present in the gas and particle phases, and samples for analysis for biomarkers indicative of life. Thus, these Ocean Worlds of the outer Solar System offer the opportunity for a conceptually new approach to life detection focusing on in-flight sample collection of material freshly injected into space. Technologies of particular interest include sample collection systems and subsystems capable of:

- Capture, containment, and/or transfer of gas, liquid, ice, and/or mineral phases from plumes to sample processing and/or instrument interfaces.
- Technologies for characterization of collected sample parameters including mass, volume, total dissolved solids in liquid samples, and insoluble solids.
- Sample collection and sample capture for in-situ imaging.
- Systems capable of high-velocity sample collection with minimal sample alteration to allow for habitability and life detection analyses.
- Microfluidic sample collection systems that enable sample concentration and other manipulations.
- Plume material collection technologies that minimize risk of terrestrial contamination, including organic chemical and microbial contaminates.

Proposers are strongly encouraged to relate their proposed development to NASA's future Ocean Worlds exploration goals. Proposed instrument architectures should be as simple, reliable, and low risk as possible while enabling compelling science. Novel instrument concepts are encouraged particularly if they enable a new class of scientific discovery. Technology developments relevant to multiple environments and platforms are also desired.

Proposers should show an understanding of relevant space science needs and present a feasible plan to fully develop a technology and infuse it into a NASA program. The desired deliverables are well-conceived and analyzed designs, prototypes, and test data. The expected Technology Readiness Level (TRL) range at completion of the project is 2-5.

References:

- For synergistic NASA technology solicitation, see ROSES 2016/C.20 Concepts for Ocean worlds Life Detection Technology (COLDTECH) call: https://nspires.nasaprs.com/external/solicitations/summary.do?method=init&solId={5C43865B-0C93-6ECA-BCD2-A3783CB1AAC8}&path=init.
- For the NASA Roadmap for Ocean World Exploration see: http://www.lpi.usra.edu/opag/ROW.

T8.02 Photonic Integrated Circuits (STTR)
Lead Center: GSFC
Participating Center(s): GRC, JSC
Related Subtopic Pointer(s): H9.01, H9.03, H9.07, H6.01, H6.22, S1.11, S1.01, S1.04, S1.02, S1.03, S1.07, T8.02, Z5.05, Z7.01, Z6.01

Integrated photonics generally is the integration of multiple lithographically defined photonic and electronic components and devices (e.g., lasers, detectors, waveguides/passive structures, modulators, electronic control and optical interconnects) on a single platform with nanometer-scale feature sizes. The development of photonic integrated circuits permits size, weight, power and cost reductions for spacecraft microprocessors, communication buses, processor buses, advanced data processing, free space communications and integrated optic science instrument optical systems, subsystems and components, which is particularly critical for small spacecraft platforms. This subtopic solicits methods, technology and systems for development and incorporation of active and passive circuit elements for integrated photonic circuits for:

- *Integrated photonic sensors (physical, chemical and/or biological) circuits* - NASA application examples include but are not limited to: Lab-on-a-chip systems for landers, astronaut health monitoring, front-end and back-end for remote sensing instruments including trace gas lidars, large telescope spectrometers for exoplanets using photonic lanterns and narrow band filters. On-chip generation and detection of light of appropriate wavelength may not be practical, requiring compact hybrid packaging for providing broadband optical input-output and also, as a means to provide coupling of light between the sensor-chip waveguides and samples, unique optical components (e.g., plasmonic waveguides, microfluidic channel) may be beneficial. Examples: Terahertz spectrometer, optical spectrometer, gyroscope, magnetometer, urine/breath/blood analysis.
- *Integrated photonic circuits for analog RF applications* - NASA applications include new methods due to size, weight and power improvements, passive and active microwave signal processing, radio astronomy, and Terahertz spectroscopy. As an example, integrated photonic circuits having very low insertion loss (e.g., ~1dB) and high spur free dynamic range for analog and RF signal processing and transmission which incorporate, for example, monolithic high-Q waveguide microresonators or Fabry-Perot filters with multi-GHz RF pass bands. These components should be suitable for designing chip-scale tunable opto-electronic RF oscillator and high precision optical clock modules. Examples: Ka, W, V band radar/receivers.
- *Integrated photonic circuits for very high-speed computing and free space communications* - advanced computing engines that approach TeraFLOP per second computing power for spacecraft in a fully integrated combined photonic and electronic package. Free space communications downlink modems at the > 1 Terabit per second level for Near-Earth (Low-Earth Orbit to ground) and > 100 Mbls for > 1 AU distances. Examples: transmitters, receivers, microprocessors.

This subtopic also investigates new science that may be enabled by quantum mechanical technologies in space implemented in a photonic integrated circuit e.g.:

- Space-based atomic and optical clocks.
- Atomic inertial sensors.
- Nitrogen-vacancy diamond (or other) magnetometers.
- Atomic vapor magnetometers.
- Additional quantum sensors that provide an advantage (e.g., sensitivity, SWaP, cost, operating temperature) over present-day sensors.

The expected Technology Readiness Level (TRL) range at completion of this project is 2 to 4.

There are multiple Mission Directorates within NASA for which this technology is relevant:

- *Human Exploration & Operations Mission Directorate (HEOMD)* - astronaut health monitoring
- *Science Mission Directorate (SMD)* - Earth, planetary and astrophysics compact science instrument (e.g., optical and terahertz spectrometers, magnetometers on a chip)

- *Space Technology Mission Directorate (STMD)* - game changing technology for small spacecraft communication and navigation (optical communication, laser ranging, gyroscopes)
- *STTR* - Exponentially increasing interest and programs at universities and start-ups in integrated photonics.

References:

NASA Space Technology Area Roadmaps - 6.2.2, 13.1.3, 13.3.7, all sensors, 6.4.1, 7.1.3, 10.4.1, 13.1.3, 13.4.3, 14.3

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- Integrated photonics in the 21st century By: Thylen, Lars; Wosinski, Lech PHOTONICS RESEARCH Volume: 2 Issue: 2 Pages: 75-81 Published: APR 2014
- Photonic Integrated Circuits for Communication Systems By: Chovan, Jozef; Uherek, Frantisek RADIOENGINEERING Volume: 27 Issue: 2 Pages: 357-363 Published: JUN 2018
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T8.04 Metamaterials and Metasurfaces Technology for Remote Sensing Applications (STTR)
Lead Center: GSFC
Participating Center(s): GSFC

Metamaterials are manmade (synthesized) composite materials whose electromagnetic, acoustic, optical, etc. properties are determined by their constitutive structural materials and their configurations. Metamaterials can be precisely tailored to manipulate electromagnetic waves, including visible light, microwaves, and other parts of the spectrum, in ways that no natural materials can (Kock 1946, 1948; Cotton 2003; Alici et al. 2007). The development of metamaterials continues to redefine the boundaries of materials science. In the field of electromagnetic research and beyond, these materials offer excellent design flexibility with their customized properties and their tunability under external stimuli. The vast possibilities for metamaterial technology to apply to remote sensing applications could apply to various areas across SMD, including Earth, lunar, and planetary science.

Topics of potential interest to explore for NASA's applications are listed below:

- Antenna beam shaping with metamaterials (at optical as well as microwave wavelengths).
- Reconfigurable metamaterial filters covering microwave to optical frequency bands
- Development of microwave and millimeter-wave metamaterials: radar scanning systems, flat panel antennas, novel magnetic materials and high-performance absorbing and shielding materials for electromagnetic compatibility or interference (EMC/EMI). Single feed horn antennas to cover multiple frequencies, including 10, 18, 36, and 89 GHz (Caloz et al. 2001; Caloz and Itoh 2002).
- Development of fabrication processes for metamaterials with nanoparticles
- Tunable, reconfigurable metamaterials using liquid crystal medium (Applications: IR and Optical spectrometers).
- Development of artificial ferrites and artificial dielectrics using metamaterial concepts to design electrically small, lightweight, and efficient RF components.

- Use of Gradient Indexed Metamaterial (GIM) for on-chip routing of light and THz frequency signals. Design and prototype development of broadband (covering 10 GHz) THz components such as transmission line bends, power splitters, filters, and photonics.

Phase I should provide a comprehensive feasibility study to address an applicable area of interest within the field of metamaterial technology. Phase II Deliverables may include prototypes and demonstration of performance. Expected TRL is from 1 to 3.

Relevance to NASA

Metamaterial technology has the biggest potential to impact the future of spaceborne instrumentation by reducing size, weight, and power (SWaP) as well as the overall cost of future space missions. Due to the nature of metamaterials, there are a multitude of possible applications for this technology. For example, applications of metamaterials for remote sensing include tunability, complex filtering, light channeling/trapping, superbeaming, and determination of optical angular momentum modes via metamaterials. For additional information regarding SMD technology needs, please review https://science.nasa.gov/about-us/science-strategy/decadal-surveys.

References:

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Focus Area 10 Advanced Telescope Technologies

Lead MD: SMD

The NASA Science Missions Directorate seeks technology for cost-effective high-performance advanced space telescopes for astrophysics and Earth science. Astrophysics applications require large aperture light-weight highly reflecting mirrors, deployable large structures and innovative metrology, control of unwanted radiation for high-contrast optics, precision formation flying for synthetic aperture telescopes, and cryogenic optics to enable far infrared telescopes. A few of the new astrophysics telescopes and their subsystems will require operation at cryogenic temperatures as cold a 4 K. This topic will consider technologies necessary to enable future telescopes and observatories collecting electromagnetic bands, ranging from UV to millimeter waves, and also include gravity waves. The subtopics will consider all technologies associated with the collection and combination of observable signals. Earth science requires modest apertures in the 2 to 4 meter size category that are cost effective. New technologies in innovative mirror materials, such as silicon, silicon carbide and nanolaminates, innovative structures, including nanotechnology, and wavefront sensing and control are needed to build telescopes for Earth science.

S2.01 Proximity Glare Suppression for Astronomical Direct Detection of Exoplanets (SBIR)
Lead Center: JPL
Participating Center(s): GSFC
Related Subtopic Pointer(s): S1.04

Control of Scattered Starlight with Coronagraphs and Starshades

This subtopic addresses the unique problem of imaging and spectroscopic characterization of faint astrophysical objects that are located within the obscuring glare of much brighter stellar sources. Examples include planetary systems beyond our own, the detailed inner structure of galaxies with very bright nuclei, binary star formation, and stellar evolution. Contrast ratios of one million to ten billion over an angular spatial scale of 0.05-1.5 arcsec are typical of these objects. Achieving a very low background requires control of both scattered and diffracted light. The failure to control either amplitude or phase fluctuations in the optical train severely reduces the effectiveness of starlight cancellation schemes.

This innovative research focuses on advances in coronagraphic instruments, starlight cancellation instruments, and potential occulting technologies that operate at visible and near infrared wavelengths. The ultimate application of these instruments is to operate in space as part of a future observatory mission concepts such as the Habitable Exoplanet Observatory (HabEx) and the Large UV Optical Infrared Surveyor (LUVOIR). Measurement techniques include imaging, photometry, spectroscopy, and polarimetry. There is interest in component development and innovative instrument design, as well as in the fabrication of subsystem devices to include, but not limited to, the following areas:

Starlight Suppression Technologies:

- Hybrid metal/dielectric, and polarization apodization masks for diffraction control of phase and amplitude for coronagraph scaled starshade experiments.
- Low-scatter, low-reflectivity, sharp, flexible edges for control of solar scatter in starshades.
- Systems to measure spatial optical density, phase inhomogeneity, scattering, spectral dispersion, thermal variations, and to otherwise estimate the accuracy of high-dynamic range apodizing masks.
- Methods to distinguish the coherent and incoherent scatter in a broad band speckle field.

Wavefront Measurement and Control Technologies:

- Small stroke, high precision, deformable mirrors and associated driving electronics scalable to 10,000 or more actuators (both to further the state-of-the-art towards flight-like hardware and to explore novel

concepts). Multiple deformable mirror technologies in various phases of development and processes are encouraged to ultimately improve the state-of-the-art in deformable mirror technology. Process improvements are needed to improve repeatability, yield, and performance precision of current devices.
- Multiplexers with ultra-low power dissipation for electrical connection to deformable mirrors.
- Low-order wavefront sensors for measuring wavefront instabilities to enable real-time control and post-processing of aberrations.
- Thermally and mechanically insensitive optical benches and systems.

Optical Coating and Measurement Technologies:

- Instruments capable of measuring polarization cross-talk and birefringence to parts per million.
- Polarization-insensitive coatings for large optics.
- Methods to measure the spectral reflectivity and polarization uniformity across large optics.
- Methods to apply carbon nanotube coatings on the surfaces of the coronagraphs for broadband suppression from visible to NIR.

Other:

- Artificial star and planet point sources, with 1e10 dynamic range and uniform illumination of an f/25 optical system, working in the visible and near infrared.

These technologies are directly applicable to the Wide Field Infrared Survey Telescope (WFIRST) CGI, the HabEx, and LUVOIR concept studies.

The expected TRL for this project is 3 to 5.

References:

- See the International Society for Optics and Photonics (SPIE) conference papers and articles published in Journal of Astronomical Telescopes and Instrumentation on high contrast coronagraphy, segmented coronagraph design and analysis, and starshades.
- https://wfirst.gsfc.nasa.gov/

S2.02 Precision Deployable Optical Structures and Metrology (SBIR)
Lead Center: JPL
Related Subtopic Pointer(s): H9.01

Assembled Deployable Optical Metering Structures and Instruments

Planned future NASA Missions in astrophysics, such as the Wide-Field Infrared Survey Telescope (WFIRST) and the New Worlds Technology Development Program (coronagraph, external occulter, and interferometer technologies) will push the state of the art in current optomechanical technologies. Mission concepts for New Worlds science would require 10 - 30 m class, cost-effective telescope observatories that are diffraction limited at wavelengths from the visible to the far IR, and operate at temperatures from 4 - 300 K. In addition, ground based telescopes, such as the Cerro Chajnantor Atacama Telescope (CCAT), require similar technology development.

The desired areal density is 1 - 10 kg/m^2 with a packaging efficiency of 3- 10 deployed/stowed diameter. Static and dynamic wavefront error tolerances to thermal and dynamic perturbations may be achieved through passive means (e.g., via a high stiffness system, passive thermal control, jitter isolation, or damping) or through active opto-mechanical control. Large deployable multi-layer structures in support of sunshades for passive thermal control and 20m to 50m class planet finding external occulters are also relevant technologies. Potential architecture implementations must package into an existing launch volume, deploy, and be self-aligning to the micron level. The target space environment is expected to be the Earth-Sun L_2.

This subtopic solicits proposals to develop enabling, cost effective component and subsystem technology for assembling large aperture telescopes with low cost. Research areas of interest include:

- Precision deployable modules for assembly of optical telescopes (e.g., innovative active or passive deployable primary or secondary support structures).
- Hybrid Deployable/Assembled Architectures, packaging and deployment designs for large sunshields and external occulters.
- Innovative concepts for assembling fully integrated modules without multiple external connections for power, heat transfer, or communications, such as:
    - Mechanical connections providing micro-dynamic stability suitable for robotic assembly.
    - Data and power concepts between assemble modules which minimize complexity and mass.
    - Thermal heat transfer concepts between assembled modules which minimize complexity and mass.
- Innovative testing and verification methodologies.

NASA APD's 30-year roadmap calls out several technical needs:

- Under Optics deployment and co-phasing "an 8-16 m telescope will require a segmented approach and advanced options for optics deployment such as robotic assembly."
- Under New Technology Mirrors, On-orbit Fabrication and Assembly Technologies "The key to bigger and better space telescopes may rely, instead, on assembly and testing telescopes on-orbit."
- In 6.5 Technology Summary Optics deployment and assembly is listed for the FIR Surveyor, Large UV Optical Infrared Surveyor, and the X-ray surveyor in the Formative Era, as well as the Cosmic Dawn Mapper and ExoEarth Mapper in the Visionary Era.

The goal for this effort is to mature technologies that can be used to fabricate 16 m class or greater, lightweight, ambient, or cryogenic flight qualified observatory systems. Proposals to fabricate demonstration components and subsystems with direct scalability to flight systems through validated models will be given preference. The target launch volume and expected disturbances, along with the estimate of system performance, should be included in the discussion. Proposals with system solutions for large sunshields and external occulters will also be accepted. A successful proposal shows a path toward a Phase II delivery of demonstration hardware scalable to 5 meter diameter for ground test characterization.

Proposals should show an understanding of one or more relevant science needs, as well as present a feasible plan to fully develop the relevant subsystem technologies and to transition into future NASA program(s).

The expected technology readiness level (TRL) or TRL range at completion of the project is from 3-5.

A successful Phase II would include a demonstration of assembly and disassembly of a stable, stiff structural connection which transfers significant heat as well as data/power. Such a component would be supported by analysis of an observatory optomechanical architecture suitable for future observatories.

References:

Assembled Deployable Optical Metering Structures and Instruments

- https://exoplanets.nasa.gov/exep/technology/in-space-assembly/iSAT_study/ (contains many links to useful recent studies that are ongoing)

S2.03 Advanced Optical Systems and Fabrication/Testing/Control Technologies for EUV/Optical and IR Telescope (SBIR)
Lead Center: MSFC
Participating Center(s): GRC, GSFC, JPL, LaRC
Related Subtopic Pointer(s): H9.01

Optical Components and Systems for Large Telescope Missions

To accomplish NASA's high-priority science requires low-cost, ultra-stable, large-aperture, normal incidence mirror systems with low mass-to-collecting area ratios. Where a mirror system is defined as the mirror substrate, supporting structure, and associated actuation and thermal management systems. After performance, the most important metric for an advanced optical system is affordability or areal cost (cost per square meter of collecting aperture). Current normal incidence space mirrors cost $4 million to $6 million per square meter of optical surface area. This research effort seeks to improve the performance of advanced precision optical components while reducing their cost by 5 to 50 times, to between $100K/m^2 to $1M/m^2.

Specific metrics are defined for each wavelength application region:

Aperture Diameter for all wavelengths, except Far-IR:

- Monolithic: 1 to 8 meters
- Segmented: 3 to 20 meters

For UV/Optical:

- Areal Cost < $500K/m^2
- Wavefront Figure < 5 nm RMS (via passive design or active deformation control)
- Wavefront Stability < 10 pm/10 min
- First Mode Frequency 60 to 500 Hz
- Actuator Resolution < 1 nm RMS
- Optical Path-length Stability < 1 pm/10,000 seconds for precision metrology
- Areal density < 15 kg/m^2 (< 35 kg/m^2 with backplane)
- Operating Temperature Range of 250 to 300K

For Far-IR:

- Aperture diameter 1 to 4 m (monolithic), or 5 to 10 m (segmented)
- Telescope diffraction-limited at <30 microns at operating temperature 4 K
- Cryo-Deformation < 100 nm RMS
- Areal cost < $500K/m^2
- Production rate > 2 m^2 per month
- Areal density < 15 kg/m^2 (< 40 kg/m^2 with backplane)
- Thermal conductivity at 4 K > 2 W/m*K
- Survivability at temperatures ranging from 315 K to 4 K

For EUV:

- Surface Slope < 0.1 micro-radian

Also needed is ability to fully characterize surface errors and predict optical performance.

Proposals must show an understanding of one or more relevant science needs and present a feasible plan to develop the proposed technology for infusion into a NASA program: sub-orbital rocket or balloon; competed Small Explorers (SMEX) or Medium-Class Explorers (MIDEX); or, Decadal class mission. Successful proposals will demonstrate an ability to manufacture, test, and control ultra-low-cost optical systems that can meet science performance requirements

and mission requirements (including processing and infrastructure issues). Material behavior, process control, active and/or passive optical performance, and mounting/deploying issues should be resolved and demonstrated.

An ideal Phase I deliverable would be a precision optical system of at least 0.25 meters; or a relevant sub-component of a system; or a prototype demonstration of a fabrication, test or control technology leading to a successful Phase II delivery; or a reviewed preliminary design and manufacturing plan which demonstrates feasibility. While detailed analysis will be conducted in Phase II, the preliminary design should address how optical, mechanical (static and dynamic), and thermal designs and performance analysis will be done to show compliance with all requirements. Past experience or technology demonstrations which support the design and manufacturing plans will be given appropriate weight in the evaluation.

An ideal Phase II project would further advance the technology to produce a flight-qualifiable optical system greater than 0.5 meters or relevant sub-component (with a TRL in the 4 to 5 range); or a working fabrication, test or control system. Phase I and Phase II mirror system or component deliverables would be accompanied by all necessary documentation, including the optical performance assessment and all data on processing and properties of its substrate materials. A successful mission-oriented Phase II would have a credible plan to deliver for the allocated budget a fully assembled and tested telescope assembly which can be integrated into the potential mission; and, demonstrate an understanding of how the engineering specifications of their system meets the performance requirements and operational constraints of the mission (including mechanical and thermal stability analysis).

S2.03 primary supports potential Astrophysics Division missions. S2.03 has made optical systems in the past for potential balloon experiments. Future potential Decadal missions include Laser Interferometer Space Antenna (LISA), Habitable Exoplanet Observatory (HabEx), Large UV/Optical/Near-IR Surveyor (LUVOIR), and the Origins Space Telescope (OST).

Phase I deliverable should be a precision optical system of at least 0.25 meters; a relevant sub-component; or a prototype demonstration of a fabrication, test or control technology; or a reviewed preliminary design and manufacturing plan which demonstrates feasibility. The preliminary design should address how optical, mechanical (static/dynamic) and thermal designs and performance analysis will be done. Past experience which supports the design and manufacturing plans will be given appropriate weight. Phase II project would further advance the technology to produce a flight-qualifiable optical system greater than 0.5 meters or relevant sub-component (with a TRL in the 4 to 5 range); or a working fabrication, test or control system. Deliverables should be accompanied by all necessary documentation, including optical performance assessment and all data on processing and properties of its substrate materials. Phase II should have a credible plan to deliver for the allocated budget a fully assembled and tested telescope assembly which can be integrated into the potential mission; and demonstrate an understanding of how the engineering specifications of their system meets the performance requirements and operational constraints of the mission.

Expected TRL for this project is 3 to 5.

Balloon Planetary Telescope

Astronomy from a stratospheric balloon platform offers numerous advantages for planetary science. At typical balloon cruise altitudes (100,000 to 130,000 ft.), 99%+ of the atmospheric is below the balloon and the attenuation due to the remaining atmosphere is small, especially in the near ultraviolet band and in the infrared bands near 2.7 and 4.25 µm. The lack of atmosphere nearly eliminates scintillation and allows the resolution potential of relatively large optics to be realized, and the small amount of atmosphere reduces scattered light and allows observations of brighter objects even during daylight hours.

For additional discussion of the advantages of observations from stratosphere platforms, refer to “Planetary Balloon-Based Science Platform Evaluation and Program Implementation - Final Report,” Dankanich et.al. (NASA/TM-2016-218870, available from https://ntrs.nasa.gov/)

To perform Planetary Science requires a 1-meter class telescope 500 nm diffraction limited performance or Primary Mirror System that can maintain < 10 nm rms surface figure error for elevation angles ranging from 0 to 60° over a temperature range from 220K to 280K.

Phase I will produce a preliminary design and report including initial design requirements such as wave-front error budget, mass allocation budget, structural stiffness requirements, etc., trade studies performed and analysis that compares the design to the expected performance over the specified operating range. Development challenges shall be identified during Phase I including trade studies and challenges to be addressed during Phase II with subsystem proof of concept demonstration hardware. If Phase II can only produce a sub-scale component, then it should also produce a detailed final design, including final requirements (wave-front error budget, mass allocation, etc.) and performance assessment over the specified operating range.

Additional information about Scientific Balloons can be found at https://www.csbf.nasa.gov/docs.html.

Telescope Specifications:

- Diameter > 1 meter
- System Focal Length 14 meter (nominal)
- Diffraction Limit < 500 nm
- Mass < 300 kg
- Shock 10G without damage
- Elevation 0 to 60°
- Temperature 220 to 280 K

Primary Mirror Assembly Specifications:

- Diameter > 1 meter
- Radius of Curvature 3 meters (nominal)
- Surface Figure Error < 10 nm rms
- Mass < 150 kg
- Shock 10G without damage
- Elevation 0 to 60°
- Temperature 220 to 280 K

The relevance to NASA can be found in "Vision and Voyages for Planetary Science in the Decade 2013-2022", page 22, last paragraph of NASA Telescope Facilities within the Summary Section:

- Balloon- and rocket-borne telescopes offer a cost-effective means of studying planetary bodies at wavelengths inaccessible from the ground.6 Because of their modest costs and development times, they also provide training opportunities for would-be developers of future spacecraft instruments. Although NASA's Science Mission Directorate regularly flies balloon missions into the stratosphere, there are few funding opportunities to take advantage of this resource for planetary science, because typical planetary grants are too small to support these missions. A funding line to promote further use of these suborbital observing platforms for planetary observations would complement and reduce the load on the already oversubscribed planetary astronomy program.

And page 203, 5th paragraph, section titled Earth and Space-Based Telescopes:

- Significant planetary work can be done from balloon-based missions flying higher than 45,000 ft. This altitude provides access to electromagnetic radiation that would otherwise be absorbed by Earth's atmosphere and permits high-spatial-resolution imaging unaffected by atmospheric turbulence. These facilities offer a combination of cost, flexibility, risk tolerance, and support for innovative solutions that is ideal for the pursuit of certain scientific opportunities, the development of new instrumentation, and infrastructure support. Given the rarity of giant-planet missions, these types of observing platforms (high-altitude telescopes on balloons and sounding rockets) can be used to fill an important data gap.154,155,156

Potential Advocates include: Planetary Scientists at Goddard Space Flight Center (GSFC), APL, and Southwest

Research Institute, etc., the NASA Balloon Workshop

Potential Projects: Gondola for High Altitude Planetary Science (GHAPS)

If Phase II can only produce a sub-scale component, then it should also produce a detailed final design, including final requirements (wave-front error budget, mass allocation, etc) and performance assessment over the specified operating range.

Expected TRL for this project is 3 to 5.

Large UV/Optical (LUVOIR) and Habitable Exoplanet (HabEx) Missions

Potential UV/Optical missions require 4 to 16 meter monolithic or segmented primary mirrors with < 5 nm RMS surface figures. Active or passive alignment and control is required to achieve system level diffraction limited performance at wavelengths less than 500 nm (< 40 nm RMS wavefront error, WFE). Additionally, potential Exoplanet mission, using an internal coronagraph, requires total telescope wavefront stability on order of 10 pico-meters RMS per 10 minutes. This stability specification places severe constraints on the dynamic mechanical and thermal performance of 4 meter and larger telescope. To meet this requirement requires active thermal control systems, ultra-stable mirror support structures, and vibration compensation.

Mirror areal density depends upon available launch vehicle capacities to Sun-Earth L_2 (i.e., 15 kg/m^2 for a 5 m fairing EELV vs. 150 kg/m^2 for a 10 m fairing SLS). Regarding areal cost, a good goal is to keep the total cost of the primary mirror at or below $100M. Thus, an 8-m class mirror (with 50 m^2 of collecting area) should have an areal cost of less than $2M/m^2. And, a 16-m class mirror (with 200 m^2 of collecting area) should have an areal cost of less than $0.5M/m^2.

Key technologies to enable such a mirror include new and improved:

- Mirror substrate materials and/or architectural designs
- Processes to rapidly fabricate and test UVO quality mirrors
- Mirror support structures that are ultra-stable at the desired scale
- Mirror support structures with low-mass that can survive launch at the desired scale
- Mechanisms and sensors to align segmented mirrors to < 1 nm RMS precisions
- Thermal control (< 1 mK) to reduce wavefront stability to < 10 pm RMS per 10 min
- Dynamic isolation (> 140 dB) to reduce wavefront stability to < 10 pm RMS per 10 min

Also needed is ability to fully characterize surface errors and predict optical performance via integrated opto-mechanical modeling.

Potential solutions for substrate material/architecture include but are not limited to: ultra-uniform low CTE glasses, silicon carbide, nanolaminates or carbon-fiber reinforced polymer. Potential solutions for mirror support structure material/architecture include, but are not limited to: additive manufacturing, nature inspired architectures, nano-particle composites, carbon fiber, graphite composite, ceramic or SiC materials, etc. Potential solutions for new fabrication processes include, but are not limited to, additive manufacture, direct precision machining, rapid optical fabrication, roller embossing at optical tolerances, slumping or replication technologies to manufacture 1 to 2-meter (or larger) precision quality components. Potential solutions for achieving the 10 pico-meter wavefront stability include, but are not limited to: metrology, passive, and active control for optical alignment and mirror phasing; active vibration isolation; metrology, passive, and active thermal control.

S2.03 primary supports potential Astrophysics Division missions. S2.03 has made optical systems in the past for potential balloon experiments. Future potential Decadal missions include LISA, HabEx, LUVOIR and OST.

Phase I deliverable should be a precision optical system of at least 0.25 meters; a relevant sub-component; or a prototype demonstration of a fabrication, test or control technology; or a reviewed preliminary design and manufacturing plan which demonstrates feasibility. The preliminary design should address how optical, mechanical

(static/dynamic) and thermal designs and performance analysis will be done. Past experience which supports the design and manufacturing plans will be given appropriate weight. Phase II project would further advance the technology to produce a flight-qualifiable optical system greater than 0.5 meters or relevant sub-component (with a TRL in the 4 to 5 range); or a working fabrication, test or control system. Deliverables should be accompanied by all necessary documentation, including optical performance assessment and all data on processing and properties of its substrate materials. Phase II should have a credible plan to deliver for the allocated budget a fully assembled and tested telescope assembly which can be integrated into the potential mission; and demonstrate an understanding of how the engineering specifications of their system meets the performance requirements and operational constraints of the mission.

Expected TRL for this project is 2 to 4.

NIR LIDAR Beam Expander Telescope

Potential airborne coherent LIDAR missions need compact 15-cm diameter 20X magnification beam expander telescopes. Potential space based coherent LIDAR missions need at least 50-cm 65X magnification beam expander telescopes. Candidate coherent LIDAR systems (operating with a pulsed 2-micrometer laser) have a narrow, almost diffraction limited field of view, close to 0.8 lambda/D half angle. Aberrations, especially spherical aberration, in the optical telescope can decrease the signal. Additionally, the telescope beam expander should maintain the laser beam's circular polarization. The incumbent telescope technology is a Dahl-Kirkham beam expander. Technology advance is needed to make the beam expander more compact with less mass while retaining optical performance, and to demonstrate the larger diameter.

Science Mission Directorate (SMD) desires both an airborne coherent-detection wind-profiling lidar systems and a space-based wind measurement. The space mission has been recommended to SMD by both the 2007 and 2017 earth science Decadal Surveys. SMD has incorporated the wind lidar mission in its planning and has named it "3-D Winds". SMD recently held the Earth Venture Suborbital competition for 5-years of airborne science campaigns. The existing coherent wind lidar at Langley, Doppler Aerosol WiNd lidar (DAWN), was included in three proposals which are under review. Furthermore, SMD is baselining DAWN for a second Convective Processes Experiment (CPEX)-type airborne science campaign, and for providing cal/val assistance to the ESA AEOLUS space mission. DAWN flies on the DC-8 and it is highly desired to fit DAWN on other NASA and NOAA aircraft. DAWN needs to lower its mass for several of the aircraft, and a low-mass telescope retaining the required performance is needed. Additionally, an electronic remote control of telescope focus is needed to adapt to aircraft cruise altitude and weather conditions during science flights.

A detailed design or a small prototype or a full-sized beam expander.

Expected TRL for this project is 3 to 4.

Fabrication, Test and Control of Advanced Optical Systems

Future UV/Optical/NIR telescopes require mirror systems that are very precise and ultra-stable.

Regarding precision, this subtopic encourages proposals to develop technology which makes a significant advance in the ability to fabricate and test an optical system.

Regarding stability, to achieve high-contrast imaging for exoplanet science using a coronagraph instrument, systems must maintain wavefront stability to < 10 pm RMS over intervals of ~10 minutes during critical observations. The ~10-minute time period of this stability is driven by current wavefront sensing and control techniques that rely on stellar photons from the target object to generate estimates of the system wavefront. This subtopic aims to develop new technologies and techniques for wavefront sensing, metrology, and verification and validation of optical system wavefront stability.

Current methods of wavefront sensing include image-based techniques such as phase retrieval, focal-plane contrast techniques such as electric field conjugation and speckle nulling, and low-order and out-of-band wavefront sensing

that use non-science light rejected by the coronagraph to estimate drifts in the system wavefront during observations. These techniques are limited by the low stellar photon rates of the dim objects being observed (~5 - 11 Vmag), leading to 10s of minutes between wavefront control updates.

New methods may include: new techniques of using out-of-band light to improve sensing speed and spatial frequency content, new control laws incorporating feedback and feedforward for more optimal control, new algorithms for estimating absolute and relative wavefront changes, and the use of artificial guide stars for improved sensing signal to noise ratio and speed.

Current methods of metrology include edge sensors (capacitive, inductive, or optical) for maintaining segment co-phasing, and laser distance interferometers for absolute measurement of system rigid body alignment. Development of these techniques to improve sensitivity, speed, and component reliability is desired. Low power, high-reliability electronics are also needed.

Finally, metrology techniques for system verification and validation at the picometer level during integration and test (I&T) are needed. High speed spatial and speckle interferometers are currently capable of measuring single-digit picometer displacements and deformations on small components in controlled environments. Extension of these techniques to large-scale optics and structures in typical I&T environments is needed.

These technologies are enabling for coronagraph-equipped space telescopes, segmented space telescopes, and others that utilize actively controlled optics. The LUVOIR and HabEx mission concepts currently under study provide good examples.

Phase I deliverable should be a prototype demonstration of a fabrication, test or control technology; or a reviewed preliminary design and manufacturing plan which demonstrates feasibility. The preliminary design should address how optical, mechanical (static and dynamic) and thermal designs and performance analysis will be done to show compliance with all requirements. Past experience or technology demonstrations which support the design and manufacturing plans will be given appropriate weight. Phase II project would further advance the technology to produce a flight-qualifiable optical system greater than 0.5 meters or relevant sub-component (with a TRL in the 4 to 5 range); or a working fabrication, test or control system. Deliverables should be accompanied by all necessary documentation, including optical performance assessment and all data on processing and properties of its substrate materials. Phase II should have a credible plan to deliver for the allocated budget a fully assembled and tested telescope assembly which can be integrated into the potential mission; and demonstrate an understanding of how the engineering specifications of their system meets the performance requirements and operational constraints of the mission.

Expected TRL for this project is 2 to 4.

Optical Components and Systems for Potential Infrared/Far-IR Missions

The Far-IR Surveyor Mission described in NASA's Astrophysics Roadmap, "Enduring Quests, Daring Visions":

In the context of subtopic S2.03, the challenge is to take advantage of relaxed tolerances stemming from a requirement for long wavelength (30 micron) diffraction-limited performance in the fully-integrated optical telescope assembly to minimize the total mission cost through innovative design and material choices and novel approaches to fabrication, integration, and performance verification.

The Far-IR Surveyor is a cryogenic far-infrared mission, which could be either a large single-aperture telescope or an interferometer. There are many common and a few divergent optical system requirements between the two architectures.

Common requirements:

- Telescope operating temperature ~4 K
- Telescope diffraction-limited at 30 microns at the operating temperature
- Mirror survivability at temperatures ranging from 315 K to 4 K
- Mirror substrate thermal conductivity at 4 K > 2 W/m*K

- Zero or low CTE mismatch between mirror substrate and backplane

Divergent requirements:

- Large single-aperture telescope
- Segmented primary mirror, circular or hexagonal
- Primary mirror diameter 5 to 10 m
- Possible 3 dof (tip, tilt and piston) control of mirror segments on orbit
- Interferometer:
- Monolithic primary mirrors
- Afocal, off-axis telescope design
- Primary mirror diameter 1 to 4 m

Success metrics:

- Areal cost < \$500K/m^2
- Areal density < 15 kg/m^2 (< 40 kg/m^2 with backplane)
- Production rate > 2 m^2 per month
- Short time span for optical system integration and test

The technology is relevant to the Far-IR Surveyor mission described in NASA's Astrophysics Roadmap and prioritized in NASA's Program Annual Technology Reports for Cosmic Origins and Physics of the Cosmos. A future NASA far-infrared astrophysics mission will answer compelling questions, such as: How common are life-bearing planets? How do the conditions for habitability develop during the process of planet formation? And how did the universe evolve in response to its changing ingredients (build-up of heavy elements and dust over time)? To answer these questions, NASA will need telescopes and interferometers that reach fundamental sensitivity limits imposed by astrophysical background photon noise. Only telescopes cooled to a cryogenic temperature can provide such sensitivity.

Novel approaches to fabrication and test developed for a far-infrared astrophysics mission may be applicable to far-infrared optical systems employed in other divisions of the NASA SMD, or to optical systems designed to operate at wavelengths shorter than the far-infrared.

Mirrors or optical systems that demonstrably advance TRL to address the overall challenge described under Scope Description while meeting requirements for a single-aperture or interferometric version of the notional Far-IR Surveyor mission.

Expected TRL for this project is 3 to 5.

Ultra-Stable Telescopes and Telescope Structures

Multiple potential balloon and space missions to perform Astrophysics, Exoplanet and Planetary science investigations require a complete optical telescope system with 0.5 meter or larger of collecting aperture. 1-m class balloon-borne telescopes have flown successfully, however, the cost for design and construction of such telescopes can exceed \$6M, and the weight of these telescopes limits the scientific payload and duration of the balloon mission. A 4X reduction in cost and mass would enable missions which today are not feasible. Space-based gravitational wave observatories (LISA) need a 0.5-meter class ultra-stable telescope with an optical path length stability of a picometer over periods of roughly one hour at temperatures near 300K in the presence of large applied static thermal gradients, but a stable thermal environment with expected thermal fluctuations of only ~ 10 microK/√Hz. The telescope will be operated in simultaneous transmit and receive mode, so an unobstructed design is required to achieve extremely low coherent backscatter light performance.

LISA Mission: Space-based gravitational wave observatories require precision displacement measurements between widely spaced proof masses. Displacements of ~ 10 pm over 1,000 seconds between masses spaced at 2.5 million km are required. Telescope systems must contribute at most ~ 1/10th of this displacement budget, or ~ 1 pm over 1,000 seconds.

Prototype unobscured telescope with the required scale size (0.3 m primary, ~ 700 mm length) that can demonstrate the required dimensional stability at room temperature. Very low coherent backscatter.

Expected TRL for this project is 3 to 5.

References:

Optical Components and Systems for Large Telescope Missions

- The Habitable Exoplanet Imager (HabEx) and Large UVOIR (LUVOIR) space telescope studies are developing concepts for UVOIR space telescopes for exoEarth discovery and characterization, exoplanet science, general astrophysics and solar system astronomy. The HabEx Interim Report is available at: https://www.jpl.nasa.gov/habex/pdf/interim_report.pdf. The LUVOIR Interim Report is available at: https://asd.gsfc.nasa.gov/luvoir/.
- The Origins Space Telescope (OST) is a single-aperture telescope concept for the Far-Infrared Surveyor mission described in the NASA Astrophysics Roadmap, "Enduring Quests, Daring Visions" (https://smd-prod.s3.amazonaws.com/science-pink/s3fs-public/atoms/files/secure-Astrophysics_Roadmap_2013_0.pdf).
- The OST mission is described on the website https://origins.ipac.caltech.edu
- The Space Infrared Interferometric Telescope (SPIRIT) and its optical system requirements are described on the website https://asd.gsfc.nasa.gov/cosmology/spirit/
- LISA mission description: https://lisa.nasa.gov/

Balloon Planetary Telescope

- For additional discussion of the advantages of observations from stratosphere platforms, refer to “Planetary Balloon-Based Science Platform Evaluation and Program Implementation - Final Report,” Dankanich et.al. (NASA/TM-2016-218870, available from https://ntrs.nasa.gov/)

Large UV/Optical (LUVOIR) and Habitable Exoplanet (HabEx) Missions

- The Habitable Exoplanet Imager (HabEx) and Large UVOIR (LUVOIR) space telescope studies are developing concepts for UVOIR space telescopes for exoEarth discovery and characterization, exoplanet science, general astrophysics and solar system astronomy. The HabEx Interim Report is available at: https://www.jpl.nasa.gov/habex/pdf/interim_report.pdf. The LUVOIR Interim Report is available at: https://asd.gsfc.nasa.gov/luvoir/.
- The Origins Space Telescope (OST) is a single-aperture telescope concept for the Far-Infrared Surveyor mission described in the NASA Astrophysics Roadmap, "Enduring Quests, Daring Visions" (https://smd-prod.s3.amazonaws.com/science-pink/s3fs-public/atoms/files/secure-Astrophysics_Roadmap_2013_0.pdf).
- The OST mission is described on the website https://origins.ipac.caltech.edu
- The Space Infrared Interferometric Telescope (SPIRIT) and its optical system requirements are described on the website https://asd.gsfc.nasa.gov/cosmology/spirit/

NIR LIDAR Beam Expander Telescope

- NRC Decadal Surveys at: http://sites.nationalacademies.org/DEPS/ESAS2017/index.htm
- https://smd-prod.s3.amazonaws.com/science-pink/s3fs-public/atoms/files/Weather_Focus_Area_Workshop_Report_2015_0.pdf
- A. K. DuVivier, J. J. Cassano, S. Greco and G. D. Emmitt, 2017, “A Case Study of Observed and Modeled Barrier Flow in the Denmark Strait in May 2015” Monthly Weather Review 145, 2385 – 2404 (2017). See also Supplemental Material
- M. J. Kavaya, J. Y. Beyon, G. J. Koch, M. Petros, P. J. Petzar, U. N. Singh, B. C. Trieu, and J. Yu, “The Doppler Aerosol Wind Lidar (DAWN) Airborne, Wind-Profiling, Coherent-Detection Lidar System: Overview, Flight Results, and Plans,” J. of Atmospheric and Oceanic Technology 34 (4), 826-842 (2014)

- Scott A. Braun, Ramesh Kakar, Edward Zipser, Gerald Heymsfield, Cerese Albers, Shannon Brown, Stephen L. Durden, Stephen Guimond, Jeffery Halverson, Andrew Heymsfield, Syed Ismail, Bjorn Lambrigtsen, Timothy Miller, Simone Tanelli, Janel Thomas, and Jon Zawislak, "NASA's Genesis and Rapid Intensification Processes (GRIP) Field Experiment," Bull. Amer. Meteor. Soc. (BAMS) 94(3), 345-363 (2013)

Fabrication, Test and Control of Advanced Optical Systems

- The HabEx Interim Report is available at: https://www.jpl.nasa.gov/habex/pdf/interim_report.pdf. The LUVOIR Interim Report is available at: https://asd.gsfc.nasa.gov/luvoir/resources/docs/LUVOIR_Interim_Report_Final.pdf.

Optical Components and Systems for Potential Infrared/Far-IR Missions

- The Far-Infrared Surveyor is described in NASA's Astrophysics Roadmap, "Enduring Quests, Daring Visions," which can be downloaded from https://smd-prod.s3.amazonaws.com/science-pink/s3fs-public/atoms/files/secure-Astrophysics_Roadmap_2013_0.pdf
- Program Annual Technology Reports (PATR) can be downloaded from the NASA PCOS/COR Technology Development website at https://apd440.gsfc.nasa.gov/technology/

Ultra-Stable Telescopes and Telescope Structures

- LISA mission description: https://lisa.nasa.gov/
- Sanjuan, et al. Note: Silicon carbide telescope dimensional stability for space-based gravitational wave detectors, Rev.Sci. Instrum. 83, 116107 (2012) URL: http://link.aip.org/link/?RSI/83/116107
- DOI: 10.1063/1.4767247

S2.04 X-Ray Mirror Systems Technology, Coating Technology for X-Ray-UV-OIR, and Free-Form Optics (SBIR)
Lead Center: GSFC
Participating Center(s): JPL, MSFC
Related Subtopic Pointer(s): S1.04

X-Ray Mirror Systems Technology, Coating Technology for X-Ray-UV-OIR, and Free-Form Optics

The National Academy Astro2010 Decadal Report identifies studies of optical components and ability to manufacture, coat, and perform metrology needed to enable future X-Ray observatory missions such as Next Generation of X-Ray Observatories (NGXO).

The Astrophysics Decadal specifically calls for optical coating technology investment for future UV, Optical, Exoplanet, and IR missions while Heliophysics 2009 Roadmap identifies the coating technology for space missions to enhance rejection of undesirable spectral lines, improve space/solar-flux durability of EUV optical coatings, and coating deposition to increase the maximum spatial resolution.

Future optical systems for NASAs low-cost missions, CubeSat and other small-scale payloads, are moving away from traditional spherical optics to non-rotationally symmetric surfaces with anticipated benefits of free-form optics such as fast wide-field and distortion-free cameras.

This subtopic solicits proposals in the following three focus areas:

- X-Ray manufacturing, coating, testing, and assembling complete mirror systems in addition to maturing the current technology.
- Coating technology including Carbon Nanotubes (CNT) for wide range of wavelengths from X-Ray to IR (X-Ray, EUV, LUV, VUV, Visible, and IR).

- Free-form Optics design, fabrication, and metrology for CubeSat, SmallSat and various coronagraphic instruments.

S2.04 supports variety of Astrophysics Division missions. The technologies in this subtopic encompasses fields of X-Ray, coating technologies ranging from UV to IR, and Free-form optics in preparation for Decadal missions such as the Habitable Exoplanet Observatory (HabEx), Large UV Optical Infrared Surveyor (LUVOIR), and Origins Space Telescope (OST).

Optical components, systems, and stray light suppression for X-ray missions: The 2010 National Academy Decadal Report specifically identifies optical components and the ability to manufacture and perform precise metrology on them needed to enable several different future missions (NGXO). The NRC NASA Technology Roadmap Assessment ranked advanced mirror technology for new x-ray telescopes as the #1 Object C technology requiring NASA investment.

Free-form Optics: NASA missions with alternative low-cost science and small size payload are increasing. However, the traditional interferometric testing as a means of metrology are unsuited to free-form optical surfaces due to changing curvature and lack of symmetry. Metrology techniques for large fields of view and fast F/#s in small size instruments is highly desirable specifically if they could enable cost-effective manufacturing of these surfaces. (CubeSat, SmallSat, NanoSat, various coronagraphic instruments)

Coating for X-ray, EUV, LUV, UV, Visible, and IR telescopes: Astrophysics Decadal specifically calls for optical coating technology investment for: Future UV/Optical and Exoplanet missions (THEIA or ATLAST). Heliophysics 2009 Roadmap identifies optical coating technology investments for: Origins of Near-Earth Plasma (ONEP); Ion-Neutral Coupling in the Atmosphere (INCA); Dynamic Geospace Coupling (DGC); Fine-scale Advanced Coronal Transition-Region Spectrograph (FACTS); Reconnection and Micro-scale (RAM); & Solar-C Nulling polarimetry/coronagraph for exoplanet imaging and characterization, dust and debris disks, extra-galactic studies and relativistic and non-relativistic jet studies (VNC).

Typical Phase I deliverables, based on sub-elements of S2.04, include:

- X-ray optical mirror system: Analysis, reports, and prototype
- Coating: Analysis, reports, software, demonstration of the concept and prototype
- Freeform Optics: Analysis, design, software and hardware prototype of optical components

Expected TRL for this project is 3 to 6.

X-Ray Mirror Systems Technology

NASA large X-Ray observatory requires low-cost, ultra-stable, light-weight mirrors with high-reflectance optical coatings and effective stray light suppression. The current state-of-art of mirror fabrication technology for X-Ray missions is very expensive and time consuming. Additionally, a number of improvements such as 10 arc-second angular resolutions and 1 to 5 m^2 collecting area are needed for this technology. Likewise, the stray-light suppression system is bulky and ineffective for wide-field of view telescopes.

In this area, we are looking to address the multiple technologies including: improvements to manufacturing (machining, rapid optical fabrication, slumping or replication technologies), improved metrology, performance prediction and testing techniques, active control of mirror shapes, new structures for holding and actively aligning of mirrors in a telescope assembly to enable X-Ray observatories while lowering the cost per square meter of collecting aperture and effective design of stray-light suppression in preparation for the Decadal Survey of 2020. Additionally, we need epoxies to bond mirrors that are made of silicon. The epoxies should absorb IR radiation with wavelengths between 1.5 um and 6 um that traverses silicon with little or no absorption, and therefore can be cured quickly with a beam of IR radiation. Currently, X-Ray space mirrors cost $4 million to $6 million per square meter of optical surface area. This research effort seeks a cost reduction for precision optical components by 5 to 50 times, to less than $1M to $100 K/$m^2$.

The 2010 National Academy Decadal Report specifically identifies optical components and the ability to manufacture and perform precise metrology on them needed to enable several different future missions (NGXO).

The NRC NASA Technology Roadmap Assessment ranked advanced mirror technology for new x-ray telescopes as the #1 Object C technology requiring NASA investment.

Typical Phase I deliverables, based on sub-elements of S2.04, include:

- X-ray optical mirror system: Demonstration, analysis, reports, software and hardware prototype

Expected TRL for this project is 3 to 6.

Coating Technology for X-Ray-UV-OIR

The optical coating technology is a mission-enabling feature that enhances the optical performance and science return of a mission. Lowering the areal cost of coating determines if a proposed mission could be funded in the current cost environment. The most common forms of coating used on precision optics are anti-reflective (AR) coating and high reflective coating.

The current coating technology of optical components needed to support the 2020 Astrophysics Decadal process. Historically, it takes 10 years to mature mirror technology from TRL-3 to 6. To achieve these objectives requires sustained systematic investment.

The telescope optical coating needs to meet low temperature operation requirement. It's desirable to achieve 35 K in future.

A number of future NASA missions require suppression of scattered light. For instance, the precision optical cube utilized in a beam-splitter application forms a knife-edge that is positioned within the optical system to split a single beam into two halves. The scattered light from the knife-edge could be suppressed by CNT coating. Similarly, the scattered light for gravitational-wave application and lasercom system where the simultaneous transmit/receive operation is required, could be achieved by highly absorbing coating such as CNT. Ideally, the application of CNT coating needs to achieve:

- Broadband (visible plus Near IR), reflectivity of 0.1% or less
- Resist bleaching of significant albedo changes over a mission life of at least 10 years
- Withstand launch conditions such vibe, acoustics, etc.
- Tolerate both high continuous wave (CW) and pulsed power and power densities without damage. ~10 W for CE and ~ 0.1 GW/cm^2 density, and 1 kW/nanosecond pulses
- Adhere to the multi-layer dielectric or protected metal coating including Ion Beam Sputtering (IBS) coating

Coating for X-ray, EUV, LUV, UV, Visible, and IR telescopes: Astrophysics Decadal specifically calls for optical coating technology investment for: Future UV/Optical and Exoplanet missions. Heliophysics 2009 Roadmap identifies optical coating technology investments for: Origins of Near-Earth Plasma (ONEP); Ion-Neutral Coupling in the Atmosphere (INCA); Dynamic Geospace Coupling (DGC); Fine-scale Advanced Coronal Transition-Region Spectrograph (FACTS); Reconnection and Micro-scale (RAM); & Solar-C.

Laser Interferometer Space Antenna (LISA) requires low scatter HR coatings and low reflectivity coatings for scatter suppression near 1064 nm. Polarization-independent performance is important.

Nulling polarimetry/coronagraph for Exoplanets imaging and characterization, dust and debris disks, extra-galactic studies and relativistic and non-relativistic jet studies (VNC).

Desired deliverables for this include analysis, reports, software, demonstration of the concept and prototype.

Expected TRL for this project is 3 to 6.

Free-form Optics

Future NASA science missions demand wider fields of view in a smaller package. These missions could benefit greatly by free-form optics as they provide non-rotationally symmetric optics which allow for better packaging while maintaining desired image quality. Currently, the design and fabrication of freeform surfaces is costly. Even though various techniques are being investigated to create complex optical surfaces, small-size missions highly desire efficient small packages with lower cost that increase the field of view and expand operational temperature range of un-obscured systems. In addition to the free-form fabrication, the metrology of free-form optical components is difficult and challenging due to the large departure from planar or spherical shapes accommodated by conventional interferometric testing. New methods such as multibeam low-coherence optical probe and slope sensitive optical probe are highly desirable.

Specific metrics are:

- Design: Innovative reflective optical designs with large fields of view (> 5°) and fast F/#s
- Fabrication: 10 cm diameter optical surfaces (mirrors) with free form optical prescriptions with surface figure tolerances are 1-2 nm rms, and roughness < 5 Angstroms. Larger mirrors are also desired for flagship missions for UV and coronagraphy applications, with $10cm^{-1}$ diameter surfaces having figure tolerances <5nm RMS, and roughness <1 Angstroms RMS
- Metrology: Accurate metrology of 'freeform' optical components with large spherical departures (>1 mm), independent of requiring prescription specific null lenses or holograms.

NASA missions with alternative low-cost science and small size payload are increasing. However, the traditional interferometric testing as a means of metrology are unsuited to free-form optical surfaces due to changing curvature and lack of symmetry. Metrology techniques for large fields of view and fast F/#s in small size instruments is highly desirable specifically if they could enable cost-effective manufacturing of these surfaces. (CubeSat, SmallSat, and NanoSat).

Desired deliverables for this include demonstration, analysis, design, software, and hardware prototype of optical components.

Expected TRL for this project is 3 to 6.

References:

X-Ray Mirror Systems Technology, Coating Technology for X-Ray-UV-OIR, and Free-Form Optics

- The Habitable Exoplanet Observatory (HabEx) is a concept for a mission to directly image planetary systems around Sun-like stars. HabEx will be sensitive to all types of planets; however its main goal is, for the first time, to directly image Earth-like exoplanets, and characterize their atmospheric content. By measuring the spectra of these planets, HabEx will search for signatures of habitability such as water and be sensitive to gases in the atmosphere possibility indicative of biological activity, such as oxygen or ozone. The HabEx study interim report is available at: https://www.jpl.nasa.gov/habex/pdf/interim_report.pdf
- The Large UV/Optical/IR Surveyor (LUVOIR) is a concept for a highly capable, multi-wavelength space observatory with ambitious science goals. This mission would enable great leaps forward in a broad range of science, from the epoch of re-ionization, through galaxy formation and evolution, star and planet formation, to solar system remote sensing. LUVOIR also has the major goal of characterizing a wide range of exoplanets, including those that might be habitable - or even inhabited. The LUVOIR Interim Report is available at: https://asd.gsfc.nasa.gov/luvoir/.
- The Origins Space Telescope (OST) is the mission concept for the Far-IR Surveyor study. NASA's Astrophysics Roadmap, Enduring Quests, Daring Visions, recognized the need for an Origins Space Telescope mission with enhanced measurement capabilities relative to those of the Herschel Space Observatory, such as a three order of magnitude gain in sensitivity, angular resolution sufficient to overcome spatial confusion in deep cosmic surveys or to resolve protoplanetary disks, and new spectroscopic capability.

The community report is available at: http://science.nasa.gov/media/medialibrary/2013/12/20/secure-Astrophysics_Roadmap_2013.pdf

X-Ray Mirror Systems Technology

- NASA High Energy Astrophysics (HEA) mission concepts including X-Ray missions and studies are available at https://heasarc.gsfc.nasa.gov/docs/heasarc/missions/concepts.html.

Coating Technology for X-Ray-UV-OIR

- Laser Interferometer Space Antenna (LISA) is a space-based gravitational wave observatory building on the success of LISA Pathfinder and LIGO. Led by ESA, the new LISA mission (based on the 2017 L3 competition) is a collaboration of ESA and NASA.
- More information could be found at https://lisa.nasa.gov

Free-form Optics

- A presentation on application of Freeform Optics at NASA is available at https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20170010419.pdf

Focus Area 11 Spacecraft and Platform Subsystems

Lead MD: SMD
Participating MD(s): STMD, STTR

The Science Mission Directorate will carry out the scientific exploration of our Earth, the planets, moons, comets, and asteroids of our solar system and the universe beyond. SMD's future direction will be moving away from exploratory missions (orbiters and flybys) into more detailed/specific exploration missions that are at or near the surface (landers, rovers, and sample returns) or at more optimal observation points in space. These future destinations will require new vantage points or would need to integrate or distribute capabilities across multiple assets. Future destinations will also be more challenging to get to, have more extreme environmental conditions and challenges once the spacecraft gets there, and may be a challenge to get a spacecraft or data back from. A major objective of the NASA science spacecraft and platform subsystems development efforts are to enable science measurement capabilities using smaller and lower cost spacecraft to meet multiple mission requirements thus making the best use of our limited resources. To accomplish this objective, NASA is seeking innovations to significantly improve spacecraft and platform subsystem capabilities while reducing the mass and cost that would in turn enable increased scientific return for future NASA missions. A spacecraft bus is made up of many subsystems like: propulsion; thermal control; power and power distribution; attitude control; telemetry command and control; transmitters/antenna; computers/on-board processing/software; and structural elements. High performance space computing technologies are also included in this focus area. Science platforms of interest could include unmanned aerial vehicles, sounding rockets, or balloons that carry scientific instruments/payloads, to planetary ascent vehicles or Earth return vehicles that bring samples back to Earth for analysis. This topic area addresses the future needs in many of these sub-system areas, as well as their application to specific spacecraft and platform needs. For planetary missions, planetary protection requirements vary by planetary destination, and additional backward contamination requirements apply to hardware with the potential to return to Earth (e.g., as part of a sample return mission). Technologies intended for use at/around Mars, Europa (Jupiter), and Enceladus (Saturn) must be developed so as to ensure compliance with relevant planetary protection requirements. Constraints could include surface cleaning with alcohol or water, and/or sterilization treatments such as dry heat (approved specification in NPR 8020.12; exposure of hours at 115° C or higher, non-functioning); penetrating radiation (requirements not yet established); or vapor-phase hydrogen peroxide (specification pending). The following references discuss some of NASA's science mission and technology needs:

- The Astrophysics Roadmap: http://nasascience.nasa.gov/about-us/science-strategy.
- Astrophysics Decadal Survey - "New Worlds, New Horizons: in Astronomy and Astrophysics": http://www.nap.edu/catalog.php?record_id=12951 (link is external).
- The Earth Science Decadal Survey: http://books.nap.edu/catalog.php?record_id=11820 (link is external).

- The Heliophysics roadmap: "The Solar and Space Physics of a New Era: Recommended Roadmap for Science and Technology 20092030": http://hpde.gsfc.nasa.gov/2009_Roadmap.pdf.
- The 2011 Planetary Science Decadal Survey was released March 2011. This decadal survey is considering technology needs: https://solarsystem.nasa.gov/2013decadal/.

S3.05 Terrestrial Balloons and Planetary Aerial Vehicles (SBIR)
Lead Center: GSFC
Participating Center(s): GSFC, JPL

Satellite Communications for Terrestrial Balloons

NASA's Scientific Balloons provide practical and cost-effective platforms for conducting discovery science, development and testing for future space instruments, as well as training opportunities for future scientists and engineers. Balloons can reach altitudes above 36 kilometers, with suspended masses up to 3600 kilograms, and can stay afloat for several weeks. Currently, the Balloon Program is on the verge of introducing an advanced balloon system that will enable 100-day missions at mid-latitudes and thus resemble the performance of a small spacecraft at a fraction of the cost. In support of this development, NASA is seeking cost efficient innovative technologies that can provide high bitrates satellite communications for supporting current and future science needs during long duration missions.

Improved and innovative downlink bitrates using satellite relay communications from balloon payloads are needed. Long duration balloon flights currently utilize satellite communication systems to relay science and operations data from the balloon to ground based control centers. The current maximum downlink bit rate is 150 kilobits per second operating continuously during the balloon flight. Future requirements are for bit rates of 1 megabit per second or more. Improvements in bit rate performance, reduction in size and mass of existing systems, or reductions in cost of high bit rate systems are needed. TDRSS and Iridium satellite communications are currently used for balloon payload applications. A commercial S-band TDRSS transceiver and mechanically steered 18 dBi gain antenna provide 150 kbps continuous downlink. TDRSS K-band transceivers are available but are currently cost prohibitive. Open Port Iridium service is also currently being used.

The expected Technology Readiness Level (TRL) range at completion of the project is 1-3.

Planetary Aerial Vehicles for Titan

Innovations in materials, structures, and systems concepts have enabled aerial vehicles to play an expanding role in NASA's future Solar System Exploration Program. Aerial vehicles are expected to carry scientific payloads at Titan that will perform in-situ investigations of its atmosphere, surface and interior. Titan features extreme environments that significantly impact the design of aerial vehicles.

NASA is interested in conducting long term monitoring of the Titan atmosphere and planetary surface using aerial vehicles at altitudes ranging from the surface up to 20 km. Concepts for Lighter-than-Air (e.g., balloons, airships) and Heavier-than-Air (e.g., fixed wing, rotary wing) vehicles are encouraged. The aerial platforms should be capable of operation in Titan's atmosphere and interaction with the surface is strongly desired. Surface interaction may involve sample collection from surfaces that may contain frozen water ice, organic dunes or hydrocarbon lakes. Concepts that do not have surface interaction and focus on continuous flight are acceptable for consideration. The proposal may assume that a radioisotope thermoelectric generator could be part of the system architecture for providing basic power to the vehicle. The proposal should describe how the vehicle concept would be deployed into the atmosphere or from the surface and operated for its mission. Concepts for any of the following capabilities of aerial vehicle are encouraged:

- Technology demonstration with science payload less than 5 kg.
- Pathfinder mission with science payload less than 30 kg.
- Flagship mission with science payload up to 60 kg.

Small companies can play a major role in planetary aerial vehicles. We expect that a small company with innovative technologies may put together a mission concept that would later be desirable for NASA/JPL to pick up as a mission proposal partner for New Frontiers or Discovery. In our call we state that we are looking for several mission classes

from Technology Demonstration (like the Mars helicopter) to a Flagship mission. It is expected that a Phase I effort will consist of a system-level design and a proof-of-concept experiment on one or more key components.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-3.

References:

- Mars Helicopter Website: https://www.jpl.nasa.gov/news/news.php?feature=7121
- Satellite Communications for Balloons: https://sites.wff.nasa.gov/code820/

S3.08 Command, Data Handling, and Electronics (SBIR)
Lead Center: GSFC
Participating Center(s): JPL, LaRC
Related Subtopic Pointer(s): H9.07, H6.22, S5.06, T8.02, Z8.03, Z6.01

NASA's space-based observatories, fly-by spacecraft, orbiters, landers, and robotic and sample return missions require robust command and control capabilities. Advances in technologies relevant to command and data handling and instrument electronics are sought to support NASA's goals and several missions and projects under development.

The 2019 subtopic goals are to develop platforms for the implementation of miniaturized highly integrated avionics and instrument electronics that:

- Are consistent with the performance requirements for NASA science missions.
- Minimize required mass/volume/power as well as development cost/schedule resources.
- Can operate reliably in the expected thermal and radiation environments.

Successful proposal concepts should significantly advance the state-of-the-art. Furthermore, proposals developing hardware should indicate an understanding of the intended operating environment, including temperature and radiation. It should be noted that environmental requirements can vary significantly from mission to mission. For example, some low earth orbit missions have a Total Ionizing Dose (TID) radiation requirement of less than 10 krad(Si), while some planetary missions can have requirements well in excess of 1 Mrad(Si).

Specific technologies sought by this subtopic include:

- *Fault tolerant Implementation System-on-a-Chip (SOC) Architectures* – Technologies are sought that implement fault tolerant SOC architectures, while leveraging emerging industry standard processor instruction set architectures (ISAs) and on-chip busses. Of particular interest is the RISC-V processor ISA. Offerors should identify coding language of IP cores, use of architecture-specific modules which would limit the ability to embed code into differing chipsets, options for scaling fault tolerance, code size and features versus power and speed. Offerors should identify operating system/toolchain support. Fault tolerant SOC architectures are relevant to increasing science return for missions across all Science Mission Directorate (SMD) divisions. However, the benefits are most significant for miniaturized instruments and subsystems that must operate in harsh environments. These missions include interplanetary cubesats and smallsats, outer planet instruments, and heliophysics missions to harsh radiation environments. For these missions, the inherent fault tolerance would provide an additional level of protection on top of the radiation tolerance of the FPGA or ASIC on which the SOC is implemented. Additionally, for missions with large communication delays, the inherent fault tolerance can limit the need for ground intervention.
- *Radiation Tolerant Onboard Wireless Networks* – Technologies are sought to enable onboard wireless networks that can operate reliably in space environments. Potential applications of interest include monitoring of passive wireless sensor nodes for housekeeping, point-to-point links to communicate to instruments on booms and rotating assemblies, as well as the full implementation of a spacecraft onboard network via wireless. Offerors should identify the concept of operations for the proposed onboard network, and also describe the proposed methodology for ensuring the wireless sensor nodes (transceiver and antenna) will operate reliably in the space environment (especially radiation). Offerors should identify network type (point to point, mesh), frequencies, bandwidth, and power dissipation. Onboard wireless networks can have

relevance across all SMD divisions. However, the most immediate benefits can be for earth science with rotating instrument assemblies. For these applications, wireless networks can significantly simplify communicating high rate data from instruments such as radiometers. Additionally, heliophysics and astrophysics missions using instruments or telescopes on deployable booms could benefit by reducing the amount of wiring that must be integrated into those boom assemblies.

- *System-In-Package Integrated Assemblies* – Technologies are sought enabling highly integrated System-In-Package (SIP) assemblies integrating multiple die from different processes and foundries, enabling implementation of miniaturized, highly-reliable embedded processing, sensor readout, or motor/actuator control modules. The offeror should propose both the SIP technology to be developed, as well as a proof of concept application (relevant to spaceflight subsystems or instruments) that demonstrates the technology. The offeror should address key technical issues in the SIP implementation including thermal management, reliability, and signal integrity. Of particular interest is SIP utilizing 2.5D technology where existing die are integrated using a silicon interposer. SIP has relevance to missions across all SMD divisions where onboard resources are at a minimum. Specifically, SIP can reduce board level functions to the size of a small module, which would be especially relevant to instruments and subsystems on cubesats and outer planet missions.

The expected Technology Readiness Level (TRL) range at completion of the project is 3 to 5.

References:

- For descriptions of radiation effects in electronics, the proposer may visit http://radhome.gsfc.nasa.gov/radhome/overview.htm.

S4.03 Spacecraft Technology for Sample Return Missions (SBIR)
Lead Center: JPL
Participating Center(s): GRC, GSFC
Related Subtopic Pointer(s): T2.02, T2.03, Z8.01, Z10.02, Z7.04, Z8.06

Sample Return Missions that require landing on an extraterrestrial body are the most mass critical missions in NASA's portfolio. The feasibility of scientific missions depends to a very large extent on the mass criticality dictated by the orbital mechanics of the mission design. The least mass critical mission is a single fly-by (e.g., New Horizons), followed by an orbiter or multiple fly-by (e.g., Juno), followed by a lander or rover (e.g., Mars Science Lab), and followed by a sample return (e.g., Mars Sample Return). The mass ratio of the orbit-injected spacecraft mass to the science payload (or return sample) mass varies by several orders of magnitude over these missions. Thus a one-kilogram sample returned from Mars requires three launches of the most powerful launch vehicles available. Therefore, early investments in technologies that could significantly reduce the mass requirements and improve the propulsion efficiency of spacecraft for sample return missions have particularly high payoff potential.

NASA plans to perform sample return missions from a variety of scientifically important targets including Mars, small bodies such as asteroids and comets, and outer planet moons. These types of targets present a variety of spacecraft technology challenges. Some targets, such as Mars and some moons, have relatively large gravity wells and will require ascent chemical propulsion. Propellant possibilities include those that are transported from Earth or propellants that can be generated using local resources. Other targets are small bodies with very complex geography and very little gravity, which present difficult navigational and maneuvering challenges. In addition, the spacecraft will be subject to extreme environmental conditions including low temperatures (-270° C), dust, and ice particles. Reducing the mass associated with these complex design issues (e.g., thermal and power subsystems) is of similar importance.

Technology innovations should either enhance vehicle capabilities (e.g., increase performance, decrease risk, and improve environmental operational margins) or facilitate sample return mission implementation (e.g., reduce size, mass, power, cost). Current and future NASA projects that could use this technology include Mars Sample Return (MSR) and Comet Nucleus Sample Return (CNSR). The drastic mass reductions and propulsion efficiency improvements sought in this subtopic could enable these projects, or significantly enhance their feasibility, as, for example, by reducing the number of launches or the size of the launch vehicles required. An ideal Phase II deliverable would be a successful demonstration of an appropriate-TRL (expected TRL range at completion of this project is 4 to 6) performance test, such as at representative scale and environment, along with all the supporting analyses, design, and hardware specifications.

References:

Mass-Efficient Sample Return Technologies - Vision and Voyages for Planetary Science in the Decade 2013-2022:

- https://www.nap.edu/catalog/13117/vision-and-voyages-for-planetary-science-in-the-decade-2013-2022

MSR Mission:

- http://mars.jpl.nasa.gov:80/missions/samplereturns.html
- http://www.pioneerastro.com/Team/RZubrin/A_Comparison_of_Methods_for_the_Mars_Sample_Return_Mission.pdf

CNSR Mission:

- https://ntrs.nasa.gov/search.jsp?R=20180002990

S4.04 Extreme Environments Technology (SBIR)
Lead Center: JPL
Participating Center(s): GRC, GSFC, LaRC
Related Subtopic Pointer(s): S1.11, S4.02, S4.06, Z8.03, Z12.02, Z5.05

This subtopic addresses NASA's need to develop technologies for producing space systems that can operate without environmental protection housings in the extreme environments of NASA missions. Key performance parameters of interest are survivability and operation under the following conditions:

- Very low temperature environments (Example: temperatures on the surface of Moon as low as -180° C).
- Combination of low temperature and radiation environments (Example: surface conditions at Europa of -180° C with very high radiation).
- Very high temperature, high pressure and chemically corrosive environments (Example: Venus surface conditions, which include very high pressure of 93 bar and extreme temperatures of 485° C).

NASA is interested in expanding its ability to explore the deep atmospheres and surfaces of the Moon, planets, asteroids, and comets through the use of long-lived (days or weeks) balloons and landers. Survivability in extreme high temperatures and high pressures is also required for deep atmospheric probes to the giant planets. Proposals are sought for technologies that are suitable for remote sensing applications at cryogenic temperatures, and in-situ atmospheric and surface explorations in the high temperature, high pressure environment at the Venusian surface (485° C, 93 bar), or in low-temperature environments such as those of Titan (-180° C), Europa (-220° C), Ganymede (-200° C), Mars, the Moon, asteroids, comets and other small bodies. Also, Europa-Jupiter missions may have a mission life of 10 years and the radiation environment is estimated at 2.9 Mega-rad total ionizing dose (TID) behind 0.1 inch thick aluminum. Proposals are sought for technologies that enable NASA's long duration missions to extreme wide-temperature and cosmic radiation environments. High reliability, ease of maintenance, low volume, low mass, and low out-gassing characteristics are highly desirable. Special interest lies in development of the following technologies that are suitable for the environments discussed above:

- Wide temperature range precision mechanisms i.e., beam steering, scanner, linear and tilting multi-axis mechanisms.
- Radiation-tolerant/radiation-hardened low-power, low-noise, mixed-signal mechanism control electronics for precision actuators and sensors.
- Wide temperature range feedback sensors with sub-arc-second/nanometer precision.
- Long life, long stroke, low power, and high torque/force actuators with sub-arc-second/nanometer precision.
- Long life bearings/tribological surfaces/lubricants.
- High temperature energy storage systems. High-temperature actuators and gear boxes for robotic arms and other mechanisms.

- Long life high temperature electronics (including components, circuits and tools) and high temperature electronic packaging.
- Low-power and wide-operating-temperature radiation-tolerant/radiation-hardened RF electronics.
- Radiation-tolerant/radiation-hardened low-power/ultra-low power, wide-operating-temperature, low-noise mixed-signal electronics for space-borne systems such as guidance and navigation avionics and instruments.
- Radiation-tolerant/radiation-hardened power electronics.
- Radiation-tolerant/radiation-hardened electronics packaging (including, shielding, passives, connectors, wiring harness and materials used in advanced electronics assembly).

Research should be conducted to demonstrate technical feasibility during Phase I and show a path toward a Phase II hardware demonstration, and when possible, deliver a demonstration unit for functional and environmental testing at the completion of the Phase II contract.

There is a high relevance to NASA's Science Mission Directorate (SMD). As mentioned above, low temperature survivability is required for surface missions to Titan, Europa, Ganymede, small bodies and comets. Mars diurnal temperatures range from -120° C to +20° C. For the Europa Clipper baseline concept, with a mission life of 10 years, the radiation environment is estimated at 2.9 Mega-rad total ionizing dose (TID) behind 100 mil thick aluminum. Lunar equatorial region temperatures swing from -180° C to +130° C during the lunar day/night cycle, and shadowed lunar pole temperatures can drop to -230° C. Advanced technologies for high temperature systems (electronics, electro-mechanical and mechanical) and pressure vessels are needed to ensure NASA can meet its long duration (days instead of hours) life target for its missions in high temperature and high pressure environments.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

The expected Technology Readiness Level (TRL) range at completion of this project is 3 to 5.

References:

- Proceedings of the Extreme Environment Sessions of the IEEE Aerospace Conference. https://www.aeroconf.org/
- Proceedings of the meetings of the Venus Exploration Analysis Group (VEXAG). https://www.lpi.usra.edu/vexag/
- Proceedings of the meetings of the Outer Planet Assessment Group (OPAG). https://www.lpi.usra.edu/opag/

Z6.01 High Performance Space Computing Technology (SBIR)
Lead Center: JPL
Participating Center(s): GSFC
Related Subtopic Pointer(s): H9.07, H6.22, T8.02, T6.05, Z8.03

The NASA state-of-the-art in space computing utilizes 20-year-old technology and is inadequate for future missions. In conjunction with the US Air Force, NASA is investing in the development of the High Performance Space Computing (HPSC) Chiplet, a radiation-hardened multi-core processor that will improve space computing capabilities by two orders of magnitude. While these efforts will provide an underlying platform, they do not provide the full range of advanced computing capabilities and programming support that developers will require to support missions currently in the planning stage for the mid-2020s and beyond. Topics of interest include:

- *Fault Tolerant, Real Time Linux* - a flight qualifiable version of Linux for the HPSC Chiplet, capable of supporting parallel and heterogeneous processing for autonomy, robotics and science codes is desired. Initial design of a verifiably reliable, fault tolerant, real time Linux kernel is desired. A successful development will potentially result in an eventual Phase 3 award, or alternate funding, to develop a complete, qualified, operating system.
- *HPSC Chiplet Hypervisor* - a bare metal hypervisor capable of supporting symmetric and asymmetric multi-processing, as well as high levels of fault tolerance is desired.
- *Network Switches/Routers* - rad hard, low power switches and routers that support system level fault tolerance and testability are required for sRIO (3.1, 4.0, and above).
- *Neuromorphic computing and Machine Learning* - general purpose neural networks and other machine learning accelerators for robotic vision, system health management and similar applications are needed to meet performance power requirements in future autonomous robotic systems. Initial design of this ASIC and a validated FPGA implementation of critical portions of the design is desired. A successful development will potentially result in an eventual Phase 3 award, or alternate funding, to implement the final chiplet.
- *Graphics Processing* - low power, high performance GPU capability to support crewed vehicle displays, including virtual and augmented reality hardware is desired. An initial GPU chiplet design with validated FPGA implementation of critical portions of the design is desired. A successful development will potentially result in an eventual Phase 3 award, or alternate funding, to implement the final chiplet.

An HPSC ecosystem is of interest to all major programs in Human Exploration & Operations Mission Directorate (HEOMD) and Science Mission Directorate (SMD). Immediate infusion targets include Mars Fetch Rover, WFIRST/Chronograph, Gateway, SPLICE/Lunar Lander. Desired deliverables with regards to hardware elements include a preliminary detailed design ready for fabrication and productization.

The expected Technology Readiness Level (TRL) range at completion of this project is 4 to 6.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

References:

- https://www.nasa.gov/press-release/goddard/2017/nasa-selects-high-performance-spaceflight-computing-hpsc-processor-contractor

Focus Area 12 Entry, Descent, and Landing Systems

Lead MD: STMD
Participating MD(s): HEOMD

The SBIR focus area of Entry, Descent and Landing (EDL) includes the suite of technologies for atmospheric entry as well as descent and landing on both atmospheric and non-atmospheric bodies. EDL mission segments are used in both robotic planetary science missions and human exploration missions beyond Low Earth Orbit, and some technologies have application to commercial space capabilities.

Robust, efficient, and predictable EDL systems fulfill the critical function of delivering payloads to planetary surfaces through challenging environments, within mass and cost constraints. Future NASA missions will require new technologies to break through historical constraints on delivered mass, or to go to entirely new planets and moons. Even where heritage systems exist, no two planetary missions are exactly “build-to-print,” so there are frequently

issues of environmental uncertainty, risk posture, and resource constraints that can be dramatically improved with investments in EDL technologies. New capabilities and improved knowledge are both important facets of this focus area.

Because this topic covers a wide area of interests, subtopics are chosen to enhance and or fill gaps in the existing technology development programs. Future subtopics will support one or more of four broad capability areas, which represent NASA's goals with respect to planetary Entry, Descent and Landing:

- High Mass to Mars Surface
- Precision Landing and Hazard Avoidance
- Planetary Probes and Earth Return Vehicles
- EDL Data Return and Model Improvement

A cross-cutting set of disciplines and technologies will help mature these four capability areas, to enable more efficient, reliable exploration missions. These more specific topics and subtopics may include, but are not limited to:

- Thermal Protection System materials, modeling, and instrumentation
- Deployable and inflatable decelerators (hypersonic and supersonic)
- Guidance, Navigation, and Control sensors and algorithms
- Aerodynamics and Aerothermodynamics advances, including modeling and testing
- Precision Landing and Hazard Avoidance sensors
- Multifunctional materials and structures

This year the Entry, Descent and Landing focus area is seeking innovative technology for:

- Deployable Decelerator Technologies
- EDL Sensors, including those embedded in thermal protection systems and those used for proximity operations and landing
- Hot Structure Technology for Atmospheric Entry Vehicles
- Lander Systems Technology

The specific needs and metrics of each of these specific technology developments are described in the subtopic descriptions.

H5.02 Hot Structure Technology for Aerospace Vehicles (SBIR)
Lead Center: LaRC
Participating Center(s): AFRC, JSC, MSFC
Related Subtopic Pointer(s): A1.10, Z7.03

This subtopic encompasses the development of reusable, hot structure technology for structural components exposed to extreme aerodynamic heating environments on aerospace vehicles. A hot structure system is a multifunctional structure that can reduce or eliminate the need for a separate thermal protection system (TPS). The potential advantages of using a hot structure system in place of a TPS with underlying cool structure are: reduced mass, increased mission capability such as reusability, improved aerodynamics, improved structural efficiency, and increased ability to inspect the structure. Hot structure is an enabling technology for reusability between missions or mission phases, such as aerocapture followed by entry, and has been used in prior NASA programs (HyperX and X-37) on control surfaces and leading edges, as well as Department of Defense programs.

This subtopic seeks to develop innovative low-cost, damage tolerant, reusable and lightweight hot structure technology applicable to aerospace vehicles exposed to extreme temperatures between 1000° C to 2200° C. The aerospace vehicle applications are unique in requiring the hot structure to carry primary structural vehicle loads and to be reusable after exposure to extreme temperatures during atmospheric entry. The material systems of interest for use in developing the hot structure technology include: advanced carbon-carbons (C-C), ceramic matrix composites (CMC), or advanced high temperature metals. Potential applications of the hot structure technology include: primary load-carrying

aeroshell structure, control surfaces, and propulsion system components (such as hot gas valves and passively-cooled nozzle extensions).

Proposals should introduce novel approaches to address the current need for improvements in operating temperature capability, toughness/durability, and material system strength properties. Focus areas should address one or more of the following:

- Improvements in manufacturing process and/or material design to achieve repeatable and uniform material properties, that should be scalable to actual vehicle components - specifically, property data obtained from flat-panel test coupons should represent the properties of flight articles.
- Material/structural architectures and multifunctional systems providing significant improvements of interlaminar mechanical properties while maintaining in-plane and thermal properties compared to state-of-the-art C-C or CMC. Examples include: incorporating through the thickness stitching or 3D woven preforms.
- Functionally graded manufacturing approaches to optimize oxidation protection, damage tolerance, and structural efficiency, in an integrated hot structure concept, to extend performance for multiple cycles up to 2200° C.

For this subtopic, research, testing, and analysis should be conducted to demonstrate technical feasibility during Phase I and show a path towards Phase II hardware demonstration. Phase I feasibility studies should also address cost and risk associated with the hot structures technology. At completion of Phase I, project deliverables should include: coupon specimens of components adequate for thermal/mechanical and/or arc-jet testing and a final report that is acceptable for publication as a NASA Technical Memorandum. Emphasis should be on the delivery of a manufacturing demonstration unit for NASA testing at the completion of the Phase II contract. In addition, Phase II studies should address vehicle integration. Opportunities and plans should also be identified and summarized for potential commercialization.

Hot structures technology is relevant to Human Exploration & Operations Mission Directorate (HEOMD) where the technology can be infused in spacecraft and launch vehicles to provide either improved performance or to enable advanced missions with reusability, increased damage tolerance and durability to withstand long-term space exploration, and to allow for delivery of larger payloads to space destinations. The Advanced Exploration Systems program would be ideal for further funding a prototype hot structure system and technology demonstration. The Commercial Space Transportation program also has interest in this technology for their flight vehicles.

Additionally, Exploration Systems Development programs that could use this technology include the Space Launch System (SLS) for propulsion applications. Potential NASA users of this technology exist for a variety of propulsion systems, including the following:

- Upper stage engine systems, such as those for the Space Launch System.
- In-space propulsion systems.
- Lunar/Mars lander descent/ascent propulsion systems.
- Nuclear thermal rocket propulsion systems.
- Solid motor systems, including those for primary propulsion, hot gas valve applications, and small separation/attitude-control systems.
- Propulsion systems for the Commercial Space industry which is supporting NASA efforts.

Also, the Air Force is interested in such technology for its Evolved Expendable Launch Vehicle (EELV), ballistic missile, and hypersonic vehicle programs. Other non-NASA users include Navy, Army, the Missile Defense Agency (MDA), and the Defense Advanced Research Projects Agency (DARPA). The subject technology can be both enhancing to systems already in use or under development, as well as enabling for applications that may not be feasible without further advancements in high temperature composite technology.

The expected Technology Readiness Level (TRL) range at completion of this project is 1 to 4.

References:

Hot Structures Technology for Aerospace Vehicles

- Glass, D. "Ceramic matrix composite (CMC) thermal protection systems (TPS) and hot structures for hypersonic vehicles." 15th AIAA International Space Planes and Hypersonic Systems and Technologies Conference. 2008.
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- "Carbon-Carbon Nozzle Extension Development in Support of In-Space and Upper-Stage Liquid Rocket Engines" paper; Paul R. Gradl and Peter G. Valentine; 53rd AIAA/SAE/ASEE Joint Propulsion Conference, Atlanta, GA; AIAA-2017-5064; July 2017; https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20170008949.pdf
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Note: The above references are open literature references. Other references exist regarding this technology, but they are all International Traffic in Arms Regulations (ITAR) restricted. Numerous online references exist for the subject technology and projects/applications noted, both foreign and domestic.

Z7.01 Entry Descent & Landing Sensors for Environment Characterization, Vehicle Performance, and Guidance, Navigation and Control (SBIR)
Lead Center: ARC
Participating Center(s): JSC, LaRC
Related Subtopic Pointer(s): Z7.03

NASA manned and robotic missions to the surface of planetary or airless bodies require Entry, Descent, and Landing (EDL). For many of these missions, EDL represents one of the riskiest phases of the mission. Despite the criticality of the EDL phase, NASA has historically gathered limited engineering data from such missions, and use of the data for real-time Guidance, Navigation and Control (GN&C) during EDL for precise landing (aside from Earth) has also been limited. Recent notable exceptions are the Orion Exploration Flight Test 1 (EFT-1) flight test, Mars Science Laboratory (MSL) Entry, Descent and Landing Instrumentation (MEDLI) sensor suite, and the planned sensor capabilities for Mars 2020 (MEDLI2 and map-relative navigation). NASA requires EDL sensors to:

- Understand the in-situ entry environment.
- Characterize the performance of entry vehicles.
- Make autonomous and real-time onboard GN&C decisions to ensure a precise landing.

This subtopic describes three related technology areas where innovative sensor technologies would enable or enhance future NASA EDL missions. Proposers may submit solutions to any of these following subtopic areas:

- High Accuracy, Light Weight, Low Power Fiber Optic Sensing System for EDL Instrumentation Systems.
- Miniaturized Spectrometers for Vacuum Ultraviolet & Mid-wave Infrared In-Situ Radiation Measurements during Atmospheric Entry.
- Novel Sensing Technologies for EDL GN&C and Small-Body Proximity Operations.

NASA seeks innovative sensor technologies to enable and characterize EDL operations on missions to planetary and airless bodies. This subtopic describes three related technology areas where innovative sensor technologies would enable or enhance future NASA EDL missions. Candidate solutions are sought that can be made compatible with the

environmental conditions of deep spaceflight, the rigors of landing on planetary bodies both with and without atmospheres.

High Accuracy, Light Weight, Low Power Fiber Optic Sensing System for EDL Instrumentation Systems

Current NASA state-of-the-art EDL sensing systems are very expensive to design and incorporate on planetary missions. Commercial fiber optic systems offer an alternative that could result in a lower overall cost and weight, while actually increasing the number of measurements. Fiber optic systems are also immune to Electro-Magnetic Interference (EMI) which reduces design and qualification efforts. This would be highly beneficial to future planetary missions requiring thermal protection system (TPS).

The upcoming Mars 2020 mission will fly the Mars EDL Instrumentation 2 (MEDLI2) sensor suite consisting of a total 24 thermocouples, 8 pressure transducers, two heat flux sensors, and a radiometer embedded in the TPS. This set of instrumentation will directly inform the large performance uncertainties that contribute to the design and validation of a Mars entry system. A better understanding of the entry environment and TPS performance could lead to reduced design margins enabling a greater payload mass fraction and smaller landing ellipses. Fiber optic sensing systems can offer benefits over traditional sensing system like MEDLI and MEDLI2 and can be used for both rigid and flexible TPS. Fiber optic sensing benefits include but are not limited to: sensor immunity to EMI, the ability to have thousands of measurements per fiber using Fiber Bragg Grating (FBG), multiple types of measurements per fiber (i.e., temperature, strain, and pressure), and resistance to metallic corrosion.

To be considered against NASA state-of-the-art TPS sensing systems for future flight missions, fiber optic systems must be competitive in sensing capability (measurement type, accuracy, quantity), and sensor support electronics (SSE) mass, size and power. Therefore, NASA is looking for a fiber optic system that can meet the following requirements:

Sensing Requirements:

- TPS Temperature: Measurement Range: -200 to 1250° C (up to 2000° C preferred), Accuracy: +/- 5° C desired
- Surface Pressure: Measurement Range: 0-15 psi, Accuracy: +/-1%

Sensor Support Electronics Requirements (including enclosure):

- Weight: 12 lbs or less
- Size: 240 cubic inches or smaller
- Power: 15W or less
- Measurement Resolution: 14-bit or higher
- Acquisition Rate per Measurement: 16Hz or higher
- Compatibility with other sensors types (e.g.) Heat Flux, Strain, Radiometer, TPS recession

Miniaturized Spectrometers for Vacuum Ultraviolet & Mid-wave Infrared In-Situ Radiation Measurements during Atmospheric Entry

The current state-of-the-art for flight radiation measurements includes radiometers and spectrometers. Radiometers can measure heating integrated over a wide wavelength range (e.g., MEDLI2 Radiometer), or over narrow-wavelength bands (COMARS+ ICOTOM at 2900 nm and 4500 nm). Spectrometers gather spectrally resolved signals and have been developed for Orion EM-2 (combined Ocean Optics STS units with range of 190-1100 nm). A spectrometer provides the gold standard for improving predictive models and improving future entry vehicle designs.

For NASA missions through CO_2 atmospheres (Venus and Mars), a majority of the radiative heating occurs in the Midwave Infrared Range (MWIR: 1500 nm - 6000 nm) [Brandis, AIAA 2015-3111]. Similarly, for entries to Earth, the radiation is dominated by the Vacuum Ultraviolet (VUV) range (100 - 190 nm) [Cruden, AIAA 2009-4240]. Both of these ranges are outside of those detectable by available miniaturized spectrometers. While laboratory scale spectrometers and detectors are available to measure these spectral ranges, there are no versions of these spectrometers

which would be suitable for integration into a flight vehicle due to lack of miniaturization. This subtopic calls for miniaturization of VUV and MWIR spectrometers to extend the current state of the art for flight diagnostics.

Advancements in either VUV or MWIR measurements are sought, preferably for sensors with:

- Self-contained with a maximum dimension of ~10 cm or less
- No active liquid cooling
- Simple interfaces compatible with spacecraft electronics, such as RS232, RS422, or Spacewire
- Survival to military spec temperature ranges [-55 to 125° C]
- Power usage of order 5W or less

Novel Sensing Technologies for EDL GN&C and Small-Body Proximity Operations

NASA seeks innovative sensor technologies to enhance success for EDL operations on missions to other planetary bodies (including the Moon, Mars, Venus, Titan, and Europa). Sensor technologies are also desired to enhance proximity operations (including sampling and landing) on small bodies such as asteroids and comets.

Sensing technologies are desired that determine any number of the following:

- Terrain relative translational state (altimetry/3-axis velocimetry)
- Spacecraft absolute state in planetary/small-body frame (either attitude, translation, or both)
- Terrain characterization (e.g., 3D point cloud) for hazard detection, absolute and/or relative state estimation, landing/sampling site selection, and/or body shape characterization
- Wind-relative vehicle state and environment during atmospheric entry (e.g., velocity, density, surface pressure, temperature)

Successful candidate sensor technologies can address this call by:

- Extending the dynamic range over which such measurements are collected (e.g., providing a single surface topology sensor that works over a large altitude range such as 1m to >10km, and high attitude rates such as greater than 45°/sec)
- Improving the state-of-the-art in measurement accuracy/precision/resolution for the above sensor needs
- Substantially reducing the amount of external processing needed by the host vehicle to calculate the measurements
- Significantly reducing the impact of incorporating such sensors on the spacecraft in terms of Size, Weight, and Power (SWaP), spacecraft accommodation complexity, and/or cost
- Providing sensors that are robust to environmental dust/sand/illumination effects
- Mitigation technologies for dust/particle contamination of optical surfaces such as sensor optics, with possible extensibility to solar panels and thermal surfaces for lunar, asteroid, and comet missions
- Sensing for wind-relative vehicle velocity, local atmospheric density, and vehicle aerodynamics (e.g., surface pressures and temperatures)

NASA is also looking for high-fidelity real-time simulation and stimulation of passive and active optical sensors for computer vision at update rates greater than 2Hz to be used for signal injection in terrestrial spacecraft system test beds. These solutions are to be focused on improving system-level performance Verification and Validation during spacecraft assembly and test.

EDL instrumentation directly informs and addresses the large performance uncertainties that drive the design, validation and in-flight performance of planetary entry systems. Improved understanding of entry environments and TPS performance could lead to reduced design margins enabling a greater payload mass-fraction and smaller landing ellipses. Improved real-time measurement knowledge during entry could also minimize the landing dispersions for placing advanced payloads onto the surface of atmospheric and airless bodies.

NASA Science Mission Directorate (SMD) missions are frequently proposed for high speed Earth return (NF4, Discovery, and Mars Sample Return) and Venus and Mars entry. Capsules used for these missions must withstand

both convective and radiative aeroheating, and NASA now requires EDL instrumentation for these missions. Current radiative measurement techniques (radiometers) provide only an integrated heating over limited wavelength range; past interpretation of such flight data [Johnston, JSR 2015] indicate the need for spectrally resolved measurements from spectrometers. For Earth and Venus, the radiative component may be the dominant source of heating, and emission comes from the VUV that NASA currently has no capability to measure. For Mars and Venus, the aftbody radiation is dominated by MWIR. Again, NASA does not have a method to measure MWIR radiation in flight; the current radiometers integrate across several band systems. Miniaturized spectrometers that can measure in VUV and MWIR would have immediate application to SMD planetary missions. Such spectrometers may also inform what ablation species are emitted from the heatshield and backshell during entry.

The expected Technology Readiness Level (TRL) range at completion of this project is 3 to 5.

Z7.03 Deployable Aerodynamic Decelerator and Weave Diagnostic Technology (SBIR)
Lead Center: LaRC
Participating Center(s): ARC, LaRC
Related Subtopic Pointer(s): A1.10, H5.02, Z7.01, Z2.01, Z8.06

NASA is advancing deployable aerodynamic decelerators to enhance and enable robotic and human space missions. The benefit of deployable decelerators is that the entry vehicle structure and thermal protection system (TPS) is not constrained by the launch vehicle shroud. It has the flexibility to more efficiently use the available shroud volume and can be accommodated within a smaller volume for Earth departure than a traditional rigid heat shield. For Mars, this technology enables delivery of very large (20 metric tons or more) usable payloads, which may be needed to support human exploration. The technology also allows for reduced cost access to space by enabling the recovery of launch vehicle assets. The specialized equipment used to weave 3D woven preforms is based on standard textile equipment that is substantially modified to allow hundreds of layers to be interwoven together. As these complex woven structures are scaled up, it is critical to understand the dynamics of the 3D weaving equipment/hardware and how interactions between different components affect the unit cell of the woven structure and ultimately the material properties. This subtopic area solicits innovative technology solutions applicable to 3D woven TPS and deployable entry concepts. Specific technology development areas include:

- Advancements in textile manufacturing technologies that can be used to simplify production, reduce the mass, or reduce the stowed volume of mechanically deployed structures, inflatable structures, or their flexible TMS are of interest. Thermal protection concepts can also lead to improvements in thermal management efficiency of radiant and conductive heat transport at elevated temperatures (exceeding 1200° C). Concepts can be either passive or active dissipation approaches. For smaller scale inflatable systems, less than 1.5 meters in diameter, thin-ply or thin-film manufacturing approaches that can be used to reduce the minimum design gauge are of particular interest for inflatable structures. Focus of Phase I development can be subscale manufacturing demonstrations that demonstrate proof of concept and lead to Phase II manufacturing scale-up for applications related to Mars entry, Earth return, launch asset recovery, or the emergent small satellite community.
- Concepts designed to augment the drag or provide guidance control for any class of entry vehicle are of interest. Concepts can be either deployable or rigid design systems that are suitable to deployable vehicle designs, including methods that modulate vehicle symmetry or adjust lift for active flight control to improve landing accuracy. Designs that decrease the ballistic coefficient by a factor of two to three times are to be considered. Of particular interest are concepts that can be used to modulate the life or drag of a vehicle for enhanced control. Phase I proofs of concept and preliminary design efforts that will lead to, or can be integrated into, flight demonstration prototypes in a Phase II effort are of interest.
- High temperature capable structural elements to support mechanically deployable decelerators that surpass the performance capability of metallic ribs, joints, and struts are of interest to NASA. High speed entry at Venus or return from cislunar space will require advanced hot structures to enable these future missions. Significant mass savings can be achieved with the utilization of lightweight composite materials that utilize continuous fiber or 3D woven fiber preforms. The composite systems should maintain structural integrity at operating temperatures from 900-1400° C. This subtopic seeks innovative manufacturing approaches that significantly improve in-plane and through the thickness material properties over laminated composite structures. The goal is to achieve at least 50% mass savings over conventional metallic structural elements made from aluminum, titanium, and steel. Anticipated systems would include composite elements such as

flanges, tubes, ribs and struts comprised of 3D woven net shape preforms. Design, analysis, and manufacturing demonstration would establish feasibility in Phase I towards providing test coupons and a scaled-up manufacturing demonstration unit in Phase II.

- Until now, off the shelf 3D weaving equipment has typically been over-designed for the small, relatively thin materials woven commercially. However, the high fiber volume 3D woven TPS that is proposed for future NASA missions will exceed the capability of commercial machines and will require the development of new equipment. Predicting the forces required to interlace these complex 3D woven structures is crucial to a successful build. Furthermore, aspects of the weave change constantly in conventional weaving and adjustments are constantly being made. The material that will be supplied from future equipment must be uniform with predictable material properties and must also be weavable in a predictable time frame with minimum defects. Clearly, many variables dynamically interact with each other during a weaving cycle to create a given unit cell. In this subtopic NASA would like to understand the variables controlling a typical 3D woven unit cell and develop real time measurement systems to ensure that high quality material is consistently produced; without this diagnostics data, it is difficult to make the correct adjustments to ensure consistent material is produced. These data will also feed into computational models of the materials necessary for system performance. Anticipated diagnostic systems could include (but are not limited to) instrumentation to track the load and position of individual beams, beat up bars, take up and other parameters of interest to allow fast reaction time to correct any detrimental changes in the woven product during manufacturing. Phase I awards would perform an assessment of potential diagnostic techniques, and Phase II is expected to produce a prototype and/or actual production instrumentation installed on a weaving machine demonstrating increased control capabilities.

NASA needs advanced deployable aerodynamic decelerators and advanced weave diagnostic technology to enhance and enable robotic and human space missions. Applications include Mars, Venus, Titan, as well as payload return to Earth from orbit and beyond. NASA's Space Technology Mission Directorate (STMD), Human Exploration and Operations Mission Directorate (HEOMD), and Science Mission Directorate (SMD) can all benefit from this technology for various exploration missions.

The expected Technology Readiness Level (TRL) range at completion of this project is 1 to 4.

References:

- Hughes, S. J., et al, “Hypersonic Inflatable Aerodynamic Decelerator (HIAD) Technology Development Overview,” AIAA Paper 2011-2524
- Bose, D. M, et al, “The Hypersonic Inflatable Aerodynamic Decelerator (HIAD) Mission Applications Study,” AIAA Paper 2013-1389
- Hollis, B. R., “Boundary-Layer Transition and Surface Heating Measurements on a Hypersonic Inflatable Aerodynamic Decelerator with Simulated Flexible TPS,” AIAA Paper 2017-3122
- Cassell, A., et al, “ADEPT, A Mechanically Deployable Re-Entry Vehicle System, Enabling Interplanetary CubeSat and Small Satellite Missions,” SSC18-XII-08, 32nd Annual AIAA/USU Conference on Small Satellites

Z7.04 Lander Systems Technologies (SBIR)
Lead Center: MSFC
Participating Center(s): GRC, JSC, LaRC
Related Subtopic Pointer(s): S4.03

Plume/Surface Interaction Analysis & Ground Testing

As NASA and commercial entities prepare to land robotic and crewed vehicles on the Moon, it will be important to understand the terminal descent environments to which both the landing vehicle and the surrounding area on the lunar surface will be subjected. The ability to model and predict the extent to which regolith is transported in the vicinity of the lander vehicle will be critical to setting requirements on lander configurations, instrument placement and protection, and landing stability, among other characteristics. Understanding this phenomenon will also influence landing precision requirements on vehicles and assets that are located in close proximity to increase surface operations

efficiency. The characteristics and behavior of airborne particles during descent is important for designing descent sensor systems that will be effective. Furthermore, although the physics of the atmosphere and the characteristics of the regolith are different for the Moon, the capability to model plume/surface interactions on the Moon will feed forward to Mars, where it is critical for human exploration.

NASA is looking to increase analysis capability that can be applied to predict the plume/surface interaction and nature and behavior of the ejecta, for landing missions both inside and outside NASA. Currently, flight data are collected from early planetary landing, and those data are fed into developmental tools, for validation purposes. The validation data set, as well as the expertise, grows as a result of each mission, and is shared across and applied to all other missions. We gain an understanding of how various parameters, including different types of surfaces, lead to different cratering effects and plume behaviors. The information helps NASA and industry make lander design and operations decisions. Ground testing ("unit tests") is used early in the development of the capability, to provide data for tool validation.

The current post-landing analysis of planetary landers (on Mars) is performed in a cursory manner with only partially empirically-validated tools, because there has been no dedicated fundamental research investment in this area. Flight test data does not exist, in the environments of interest. The community needs ground test and flight test data, together with comprehensive computational fluid dynamics (CFD) tools and methods, to devise validated models for different conditions that can be applicable to a variety of landing missions. A consistent toolset is important for assessing risk and could be utilized by the commercial sector as well as NASA.
Specifically, NASA is seeking:

- Computational methods and analyses that can be applied to the problem of predicting plume effects at the moon with extension to Mars.
- Low-cost ground testing methods, facilities, and/or diagnostics to produce computational model validation data.

For item 1, Phase I efforts should produce a model that predicts either crater size and shape or ejecta field, with a validation plan executed in Phase II (this validation could be performed against NASA-supplied or open source data). For item 2, Phase I efforts should prototype or show proof-of-concept in meeting proposed ground test objectives, and Phase II should implement the methods and produce an operational test bed and/or diagnostic method.

High Temperature, Lightweight Nozzle Extensions

Upper stage and in-space liquid rocket engines are optimized for performance through the use of high area ratio nozzles to fully expand combustion gases to low exit pressures, increasing exhaust velocities. Due to the large size of such nozzles, and the related engine performance requirements, carbon-carbon (C-C), carbon-silicon carbide (C-SiC), and carbon matrix composite (CMC) nozzle extensions are being considered to reduce weight impacts. NASA and industry partners are working towards advancing the domestic supply chain for these composite nozzle extensions. New and emerging carbon matrix-based material systems may also enable nozzle extension designs that offer further increased performance at even lower masses. As such, NASA is seeking the following:

- High temperature capable (such as carbon-carbon (C/C), carbon-silicon carbide (C/SiC), Carbon Matrix Composite (CMC), or other materials) nozzle extension design and material capabilities, for use on Liquid Oxygen/Liquid Methane or Liquid Hydrogen engines. Material systems should be capable of withstanding temperatures greater than 3000° F (> 1925 K), with a target of ~4000° F (~2500 K), for durations of 1500 to 2000 seconds with limited erosion characteristics (<2% weight loss). Joint designs and manufacturing scalability should consider applicability to any number of commercially available and/or in-development in-space class engines, although a preliminary target engine size for consideration would be approximately a 25,000 lb (~100-kN) class engine. Nozzle geometries of interest may be traditional bell/conical nozzle shapes or may include altitude compensating shapes, such as aerospike nozzles.
- Specific technologies to address joining CMC, C/C and C/SiC nozzle extension or chamber components to metal engine components, where significant Coefficient of Thermal Expansion (CTE) mismatch may occur. Solutions should describe how they address CTE mismatch, appropriate hot-gas sealing, and high temperature application (>1500 K) while minimizing joint weight relative to state-of-the-art solutions.

Proposers should be prepared to deliver a proof-of-concept prototype with documented results validating predicted performance at the conclusion of Phase I, and a brassboard prototype and demonstration in a defined relevant environment at the conclusion of Phase II.

The expected Technology Readiness Level (TRL) range at completion of the project is 3 to 6. Relevant current and future lander architectures for this technology include: FLEX-1 and subsequent missions, commercial robotic lunar landers, and planetary mission landers.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

References:

- https://www.nasa.gov/content/lander-technologies

Focus Area 13 Information Technologies for Science Data

Lead MD: SMD

NASA Missions and Programs create a wealth of science data and information that are essential to understanding our earth, our solar system and the universe. Advancements in information technology will allow many people within and beyond the Agency to more effectively analyze and apply these data and information to create knowledge. For example, modeling and simulation are being used more pervasively throughout NASA, for both engineering and science pursuits, than ever before. These tools allow high fidelity simulations of systems in environments that are difficult or impossible to create on Earth, allow removal of humans from experiments in dangerous situations, provide visualizations of datasets that are extremely large and complicated, and aid in the design of systems and missions. In many of these situations, assimilation of real data into a highly sophisticated physics model is needed. Information technology is also being used to allow better access to science data, more effective and robust tools for analyzing and manipulating data, and better methods for collaboration between scientists or other interested parties. The desired end result is to see that NASA data and science information are used to generate the maximum possible impact to the nation: to advance scientific knowledge and technological capabilities, to inspire and motivate the nation's students and teachers, and to engage and educate the public.

S5.01 Technologies for Large-Scale Numerical Simulation (SBIR)
Lead Center: ARC
Participating Center(s): GSFC

Exascale Computing

NASA scientists and engineers are increasingly turning to large-scale numerical simulation on supercomputers to advance understanding of complex Earth and astrophysical systems, and to conduct high-fidelity aerospace engineering analyses. The goal of this subtopic is to increase the mission impact of NASA's investments in supercomputing systems and associated operations and services. Specific objectives are to:

- Decrease the barriers to entry for prospective supercomputing users.
- Minimize the supercomputer user's total time-to-solution (e.g., time to discover, understand, predict, or design).

- Increase the achievable scale and complexity of computational analysis, data ingest, and data communications.
- Reduce the cost of providing a given level of supercomputing performance for NASA applications.
- Enhance the efficiency and effectiveness of NASA's supercomputing operations and services.

Expected outcomes are to improve the productivity of NASA's supercomputing users, broaden NASA's supercomputing user base, accelerate advancement of NASA science and engineering, and benefit the supercomputing community through dissemination of operational best practices.

The approach of this subtopic is to seek novel software and hardware technologies that provide notable benefits to NASA's supercomputing users and facilities, and to infuse these technologies into NASA supercomputing operations. Successful technology development efforts under this subtopic would be considered for follow-on funding by, and infusion into, NASA's high-end computing (HEC) projects - the High End Computing Capability project at Ames Research Center (ARC) and the Scientific Computing project at Goddard Space Flight Center (GSFC). To assure maximum relevance to NASA, funded SBIR contracts under this subtopic should engage in direct interactions with one or both HEC projects, and with key HEC users where appropriate. Research should be conducted to demonstrate technical feasibility and NASA relevance during Phase I and show a path toward a Phase II prototype demonstration.

Offerors should demonstrate awareness of the state-of-the-art of their proposed technology and should leverage existing commercial capabilities and research efforts where appropriate. Open source software and open standards are strongly preferred. Note that the NASA supercomputing environment is characterized by:

- HEC systems operating behind a firewall to meet strict IT security requirements.
- Communication-intensive applications.
- Massive computations requiring high concurrency.
- Complex computational workflows and immense datasets.
- The need to support hundreds of complex application codes - many of which are frequently updated by the user/developer.

Projects need not benefit all NASA HEC users or application codes, but demonstrating applicability to an important NASA discipline, or even a key NASA application code, could provide significant value. For instance, a GPU accelerated (or multi-core) planetary accretion code such as Lagrangian Integrator for Planetary Accretion and Dynamics (LIPAD) could be one possible project.

The three main technology areas of this subtopic are aligned with three objectives of the National Strategic Computing Initiative (NSCI), announced by the White House in July 2015. The overarching goal of NSCI is to coordinate and accelerate U.S. activities in HEC, including hardware, software, and workforce development, so that the U.S. remains the world leader in HEC technology and application. NSCI charges every agency that is a significant user of HEC to make a significant contribution to this goal. This SBIR subtopic is an important part of NASA's contribution to NSCI. See https://www.nitrd.gov/nsci/index.aspx for more information about NSCI. The three main elements of this subtopic are:

- Many NASA science applications demand much faster supercomputers. This area seeks technologies to accelerate the development of an efficient and practical exascale computing system (10^{18}operations per second). Innovative file systems that leverage node memory and a new exascale operating system geared toward NASA applications are two possible technologies for this element. At the same time, this area calls for technology to support co-design (i.e., concurrent design) of NASA applications and exascale supercomputers, enabling application scaling to billion-fold parallelism while dramatically increasing memory access efficiency. This supports NSCI Objective 1. (Accelerating delivery of a capable exascale computing system that integrates hardware and software capability to deliver approximately 100 times the performance of current 10 petaflop systems across a range of applications representing government needs.)
- Data analytics is becoming a bigger part of the supercomputing workload, as computed and measured data expand dramatically, and the need grows to rapidly utilize and understand that data. This area calls for technologies that support convergence of computing systems optimized for modeling & simulation and those optimized for data analytics (e.g., data assimilation, data compression, image analysis, machine learning,

visualization, and data mining). In-situ data analytics that can run in-memory side-by-side with the model run is another possible technology for this element. This supports NSCI Objective 2. (Increasing coherence between the technology base used for modeling and simulation and that used for data analytic computing.)

- Presently it is difficult to integrate cyberinfrastructure elements (supercomputing system, data stores, distributed teams, instruments, mobile devices, etc.) into an efficient and productive science environment. This area seeks technologies to make elements of the supercomputing ecosystem much more accessible and composable, while maintaining security. This supports NSCI Objective 4. (Increasing the capacity and capability of an enduring national HPC ecosystem by employing a holistic approach that addresses relevant factors such as networking technology, workflow, downward scaling, foundational algorithms and software, accessibility, and workforce development.)

Expected TRL for this project is 5 to 7.

References:

- https://www.nas.nasa.gov/hecc/about/hecc_project.html (NASA High-End Computing Capability)
- https://www.nitrd.gov/nsci/index.aspx (The National Strategic Computing Initiative)

S5.02 Commercial Geospatial Analysis Platforms for Earth Science Applications (SBIR)
Lead Center: JPL
Participating Center(s): JPL, MSFC
Related Subtopic Pointer(s): H6.03, S5.03, T11.04

The mission of the NASA Applied Sciences Program within the Earth Science Division is to transfer the results of earth science research for use by commercial firms, other government agencies, and non-profit organizations as part of decision support systems. As a research organization, NASA works via collaboration with the private sector understand and deliver earth science application tools and data sets in formats useful to end users.

NASA Earth Science Division and the NASA Applied Science Program within the Science Mission Directorate (SMD) are continually looking to increase utilization and extend the benefit of NASA Earth Science data and research. Relevant goals and challenges include:

- Increasing the utilization of NASA Earth Science data in organizations' policy, business, and management decisions through the commercialization and operationalization of applied research and information products;
- Accelerating the transfer of NASA Earth Science data and science data systems into the cloud, enabling the re-formatting and reprocessing required for more geospatial analysis organizations, particularly commercial companies, to use and apply NASA data more easily;
- Accelerating NASA's leveraging of driving commercial technologies (e.g., cloud-enabled global-scale analyses, modern data science tools, advanced algorithms and visualization) to increase the utilization of NASA tools and services for a broad suite of non-research users;
- Accelerating NASA's utilization of open-source tools and NASA software tools hosted within open source communities;
- Enabling focused commercial activities on specific challenging NASA problems related to future SMD missions, such as extracting insights from the hyperspectral data cube or resolving characteristics of features using multi-modalities of data from multiple sources (e.g., fusing radar, radiometer, and GNSS-derived data) to both support future missions (e.g., hyperspectral) and ongoing missions;
- Aligning NASA SBIR investments more closely with DoD and Intelligence Community-funded programs such as the NGA Commercial GEOINTActivity (CGA) and DARPA STO's Geospatial Cloud Analytics (GCA) program, as well as help accelerate NASA's own EOSDIS Cloud Evolution.

The NASA Earth (http://science.nasa.gov/earth-science/) and Applied Science (http://appliedsciences.nasa.gov/) programs seeks to increase the utilization and extend the benefit of Earth Science data and research to better meet societal needs. The objective of this subtopic is to provide commercial geospatial analytics firms with improved access

to and translation of NASA data and applied research to support advancement of commercial geospatial analytics capabilities, specifically applied to hyperspectral measurements.

Hyperspectral measurements have been demonstrated by NASA through the Hyperion instrument on EO-1 as well as other developments like the Hyperspectral Thermal Emission Spectrometer (HyTES) and the Airborne Visible / Infrared Imaging Spectrometer (AVIRIS) to have significant value to both Decadal-class science requirements such as surface biology and geology (SBG) as well as non-NASA applications including, disaster response, agricultural and food security, water resource management, ecological forecasting, land surface modeling, air quality and health. Hyperspectral imagery in the visible and shortwave infrared, multi- or thermal IR is identified as a designated observable in the latest Earth Science Decadal Survey demonstrating a long-term need by NASA for geospatial analysis platforms specifically able to address the hyperspectral data cube problem to accelerate access and use.

Hyperspectral image processing is a challenging problem due to the several hundred continuous spectral channels produced for a given multi-spatial dimensional scene, particularly for global-scale, low latency processing of continuously imaging missions. Maintaining research-grade science data systems to perform orthorectification and radiometric calibration for a single sensor is challenging and will become more so as the number and diversity of hyperspectral measurements from air and space grows. Providing the results in formats and at locations, in particular in commercial cloud environments, that lower the barriers to utilization by various communities including commercial companies, but with proper provisions for access to other communities in compliance with NASA's open data standards will become increasingly important. Innovative methods to process the data at low-cost and at high sampling rate are currently in development across industry and academia, as well as within NASA. Following these data processing steps, the ability to extract insights without humans in the loop presents another "big data" problem. Likewise, innovative methods to extract insights from hyperspectral data at low cost and at high sampling rates are currently in development across industry and academia using traditional image processing techniques as well as machine learning and other modern methods

This subtopic seeks proposals to develop data science tools that efficiently process NASA hyperspectral data, provision the data broadly to the commercial and applied research community, particularly in commercial cloud environments, and autonomously extract new insights aimed at driving information product development for the commercial sector and supporting increased utilization of NASA data by non-NASA users. Licensing of NASA-developed tools such as the Hyperspectral Image Interpretation and Holistic Analysis Tools (HiiHAT) (https://hyperspectral.jpl.nasa.gov/) is encouraged and is available open source (https://sourceforge.net/projects/hiihat). Note that licensing of NASA Caltech/JPL software for government use under an SBIR subcontract, typically at no cost, can be an effective approach to evaluating new intellectual property before committing to commercial licensing. Transfer and reformatting of applicable NASA hyperspectral data such as from the Hyperion instrument flown on the EO-1 mission (https://earthexplorer.usgs.gov/) and the ongoing Airborne Visible / Infrared Imaging Spectrometer (AVIRIS) (https://aviris.jpl.nasa.gov/alt_locator/) mission into commercial cloud environments to enable low-cost, high rate processing at global scales is highly encouraged.

Leveraging of existing NASA Earth Science data tools and service are encouraged. Key components include:

- Common Metadata Repository (CMR), a spatial and temporal metadata registry and order broker for all NASA Earth science data (https://cmr.earthdata.nasa.gov/search/);
- Earthdata Search Client (ESDC) (https://search.earthdata.nasa.gov/search), a search tool for CMR and other repositories;
- Global Imagery Browse Services (GIBS) image repository (https://earthdata.nasa.gov/gibs).

Use of open source tools and developments within open source communities is highly encouraged. A sample of NASA open source resources include:

- Apache Science Data Analytics Platform (sdap.apache.org) – A suite of GIS-based data analytics services integrated as a platform for the cloud;
- Open Climate Workbench (climate.apache.org) – used for climate model evaluation;
- Apache OODT (oodt.apache.org) – framework for science data processing and management systems;

- NASA GIBS (github.com/nasa-gibs) – the core for NASA's Global Image Browse Services (GIBS) and many NASA's current browser-based GIS solutions, including the PO.DAAC SOTO and Mars Trek, Moon Trek, Water Trek, Vesta Trek, etc.;
- HySDS (github.com/hysds/hysds) – Hybrid cloud-based data processing framework;
- Pomegranate (pomegranate.jpl.nasa.gov) – webservice for data access and geospatial subsetting;
- NASA Common Mapping Client (https://github.com/nasa/common-mapping-client) – browser-based GIS visualization framework.

To promote interoperability and the use of NASA data services, the proposals should consider metadata and service interface standards that are already part of the NASA data infrastructure, including standards such as Open Search, ISO-19115-2, FGDC, and Open Geospatial Consortium (OGC).

Use of commercial cloud environments is encouraged. NASA and other government agencies are increasingly leveraging commercial cloud vendors for secure, maintainable, cost-effective, and versatile computing infrastructure. Proposals should consider developing cloud-agnostic architecture to take advantage of commercially-provisioned solutions (e.g., serverless solutions, analytic and machine learning services, GPUs, etc.). Proposals that employ robust automated testing infrastructure and continuous integration tools to ensure maintainable, modern software through development cycles are encouraged.

Proposers must describe their commercial information product end use to demonstrate commercial potential and feasibility. Work under this subtopic should be performed consistent with NASA's NASA Earth Science Data Policy (https://science.nasa.gov/earth-science/earth-science-data/data-information-policy/).

Desired deliverables would be to describe the deployment of:

- NASA data in the cloud.
- Contractor-developed software and tools in the cloud; potentially.
- Case studies of application of tools to commercial space information products/insight generation; and potentially.
- Description of process and lessons learned during the development.

Expected TRL for this project is 3 to 6.

Subset of missions that would benefit from this subtopic:

- The Hyperion instrument flown on the EO-1 mission;
- The Hyperspectral Thermal Emission Spectrometer (HyTES) mission;
- The Airborne Visible / Infrared Imaging Spectrometer (AVIRIS) mission;
- The selected Mapping Imaging Spectrometer for Europa (MISE) planned for the Europa Clipper mission;
- The planned Hyperspectral Infrared Imager (HyspIRI) mission, a potential realization of the decadal-class science measurement need.

This subtopic was developed in collaboration between the Applied Science System Engineering group and the Computer Science For Data Intensive Applications group, both within the Instrument Software and Science Data Systems section at JPL and represent the two key groups behind the strategy to engage in a more focused way with innovative small businesses to advanced Earth science data science and applications objectives.

References:

- SMD Earth Science http://science.nasa.gov/earth-science/
- SMD Applied Sciences http://appliedsciences.nasa.gov/
- NASA Earth Data Portal https://earthdata.nasa.gov/
- AVIRIS Mission https://aviris.jpl.nasa.gov/
- AVIRIS Data Portal https://aviris.jpl.nasa.gov/alt_locator/

- EO1 Extended Mission and Data Portal https://eo1.usgs.gov/
- Europa Clipper Mission https://www.jpl.nasa.gov/missions/europa-clipper/
- HyspIRI Mission https://hyspiri.jpl.nasa.gov/
- Hyperspectral Thermal Emission Spectrometer (HyTES) https://hytes.jpl.nasa.gov/
- HiiHAT Hyperspectral Analysis https://hyperspectral.jpl.nasa.gov/ https://sourceforge.net/projects/hiihat.
- USGS Data Portal https://earthexplorer.usgs.gov/
- DAPRA Geospatial Cloud Analytics https://www.darpa.mil/news-events/2017-10-11
- NGA Commercial GEOINT Activity https://www.nga.mil/Partners/Pages/Commercial-GEOINT-Activity.aspx
- JPL Tech Transfer https://ott.jpl.nasa.gov/
- NASA Software Catalog https://software.nasa.gov/
- NASA Earth Science Data Policy https://science.nasa.gov/earth-science/earth-science-data/data-information-policy/
- Apache Science Data Analytics Platform https://sdap.apache.org
- Open Climate Workbench https://climate.apache.org
- NASA Global Imagery Browse Service https://github.com/nasa-gibs
- NASA Common Mapping Client https://github.com/nasa/common-mapping-client
- HySDS https://github.com/hysds/hysds
- Pomegranate https://pomegranate.jpl.nasa.gov

S5.03 Bridging the Gap of Applying Machine Learning to Earth Science (SBIR)
Lead Center: GSFC
Participating Center(s): GSFC
Related Subtopic Pointer(s): S5.01, S5.02, S5.06, T11.04

NASA researchers have begun exploring the application of Machine Learning (ML) to accelerate science and open up new understandings. While there are many problems that can be addressed with ML, the adoption of these techniques and technologies are slow due to the large learning curve associated with the application of this technology and the applicability of commercial tools to specific problems of interest for NASA.

This subtopic area seeks to close those gaps and accelerate the use of ML for NASA Earth Science applications. Proposals MUST be in alignment with existing and/or future NASA programs and address or extend a specific need or question for those programs. In addition, proposals must demonstrate Earth science relevant results.

Innovative proposals using ML are being sought to solve unique problems across the following Earth science challenges:

- Improvements in data assimilation (land, atmosphere, and/or ocean);
- Creation of trained model components for use in Earth system simulations;
- Application of ML models to observation and/or model data, including classification, segmentation, downscaling, transfer learning, and other techniques;
- Combining disparate data sets to lower the uncertainty of observed quantities or to derive new observations.

Research proposed to this subtopic should demonstrate technical feasibility during Phase I, and show a path toward a Phase II prototype demonstration, with significant communication with missions and programs to later plan a potential Phase III infusion. It is highly desirable that the proposed projects lead to solutions that will be infused into NASA programs and projects.

Tools and products developed under this subtopic may be developed for broad public dissemination or used within a narrow scientific community. These tools can be plug-ins or enhancements to existing software, on-line data/computing services, or new stand-alone applications or web services, provided that they promote interoperability and use standard protocols, file formats, and Application Programming Interfaces (APIs).
The desired outcome would be for the algorithms and capabilities developed during the SBIR work would be used and infused in NASA science projects and potentially used to develop new missions.

Expected TRL for this project is 4 to 6.

Relevance to NASA

- Global Modeling and Assimilation Office Assimilation (GMAO) - augment Earth system modeling or data assimilation.
- Carbon Cycle Ecosystems Office (CCOE) - wide variety of applications given the diversity of data sets from sparse in-situ to global satellite measurements.
- EOSDIS (DAACs) - harnessing the potential for new discoveries across the wide array of observation data.
- Earth Science Technology Office (ESTO/AIST) - new technology and services to exploit NASA and non-NASA data.

References:

- https://gmao.gsfc.nasa.gov/
- https://www.giss.nasa.gov/
- https://earthdata.nasa.gov/
- https://www.nccs.nasa.gov/

S5.04 Integrated Science Mission Modeling (SBIR)
Lead Center: JPL
Participating Center(s): GSFC
Related Subtopic Pointer(s): T11.04

Innovative System Modeling Methods and Tools

Several concept/feasibility studies for potential large (flagship) Astrophysics missions are in progress: Large UV Optical Infrared Surveyor (LUVOIR), Origins Space Telescope (OST), Habitable Exoplanet Observatory (HabEx), and Lynx. Following the 2020 Astrophysics decadal rankings, one of these will likely proceed to early Phase A where the infusion of new and advanced systems modeling tools and methods would be a potential game-changer in terms of rapidly navigating architecture trades, requirements development and flow-down, and design optimization.

A variety of Planetary missions require significant modeling and simulation across a variety of possible trade spaces. The portions of this topic area focused on breadth and variable fidelity will support them.

NASA seeks innovative systems modeling methods and tools addressing the following needs:
Define, design, develop, and execute future science missions, by developing and utilizing advanced methods and tools that empower more comprehensive, broader, and deeper system and subsystem modeling, while enabling these models to be developed earlier in the lifecycle. The capabilities should also allow for easier integration of disparate model types and be compatible with current agile design processes. They should enable disciplined system analysis for the design of future missions, including modeling of decision support for those missions and integrated models of technical and programmatic aspects of future missions. Similarly, they should enable evaluation of technology alternatives and impacts, science valuation methods, and programmatic and/or architectural trades.

Proposers are encouraged to address more than one of these areas with an approach that emphasizes integration with others on the list. Specific areas of interest are listed below:

- Conceptual phase models and tools that allow design teams to easily develop, populate, and visualize very broad, multidimensional trade spaces; methods for characterizing and selecting optimum candidates from those trade spaces, particularly at the architectural level. There is specific interest in models and tools that facilitate comprehensive comparison of architectural variants of systems.
- Capabilities for rapid generation of models of function or behavior of complex systems, at either the system or the subsystem level. Such models should be capable of eliciting robust estimates of system performance given appropriate environments and activity timelines, and should be tailored:

  - To support design efforts at the conceptual and preliminary design phases, while being compatible with transition to later phases.
  - To operate within highly distributed, collaborative design environments, where models and/or infrastructure that support/encourage designers are geographically separated (including Open Innovation environments). This includes considerations associated with near-real-time (concurrent?) collaboration processes and associated model integration and configuration management practices.
  - To be capable of execution at variable levels of fidelity. Ideally, models should have the ability to quickly adjust fidelity to match the requirements of the simulation (e.g., from broad-and-shallow to in-depth and back again).
  - To provide cutting-edge methodologies for quantifying, characterizing, tracing, and managing uncertainty in computational and real-world systems.
- Target models (e.g., phenomenological or geophysical models) that represent planetary surfaces, interiors, atmospheres, etc. and associated tools and methods that allow for integration into system design/process models for simulation of instrument responses. These models may be algorithmic or numeric but should be useful to designers wishing to optimize systems remote sensing of those planets.

At the completion of Phase II, a working prototype that is suitable for demonstrations with "real" data to make a compelling case for NASA usage is desired.

Expected TRL for this project is 3 to 5.

References:

- LUVOIR: https://asd.gsfc.nasa.gov/luvoir/
- OST: https://asd.gsfc.nasa.gov/firs/
- HabEx: https://www.jpl.nasa.gov/habex/
- Lynx: https://wwwastro.msfc.nasa.gov/lynx/
- https://mars.nasa.gov/programmissions/
- https://www.jpl.nasa.gov/missions/

Note that this sub-topic area addresses a broad potential range of science mission-oriented modeling tools and methods. While we are interested in the integration of these tools into broader model-based engineering frameworks, proposals with MBSE/SysML as the primary focus are encouraged to propose to H6.04 ("Model Based Systems Engineering for Distributed Development" a new FY19 sub-topic area) instead.

S5.06 Space Weather R2O/O2R Technology Development (SBIR)
Lead Center: GSFC
Participating Center(s): GSFC, MSFC
Related Subtopic Pointer(s): A1.05, H6.01, S4.04, S5.01, S1.06, T6.05

Space weather has the potential to disrupt telecommunications, aircraft and satellite systems, electric power subsystems, and position, navigation, and timing services. Given the importance of these systems to our national well-being, NASA's Heliophysics Division invests in activities to improve the understanding of these phenomena and to enable new monitoring, prediction, and mitigation strategies.

The national direction for this work has been codified by the presidential executive order of October 13, 2016 that coordinates agency efforts to prepare the nation for space weather events. Each agency under this order has specific responsibilities to the Space Weather Operations, Research and Mitigation (SWORM) activity as outlined in the National Space Weather Action Plan (NSWAP). NASA's role under NSWAP is to provide increased understanding of the fundamental physics of the Sun-Earth system through space-based observations and modeling, the development of new space-based space weather technologies and missions and monitoring of space weather for NASA's space missions. This includes research that advances operational space weather needs.

This SBIR subtopic enables NASA to demonstrate progress against NASA Goal 1.4: Understand the Sun and its interactions with Earth and the solar system, including space weather. Specifically, this subtopic provides a means for NASA's Science Mission Directorate (SMD) to meet its obligations under the presidential executive order of October 13, 2016 that coordinates agency efforts to prepare the nation for space weather events. The Heliophysics Living with a Star Program has established a path forward to meet the NASA's obligations to the research portion of these mandates. Further involvement by the emerging Heliophysics space weather commercial community has the potential to significantly advance the space weather applications portion of the mandate. Additionally, space explorers are not protected by the Earth's atmosphere and are exposed to space radiation such as galactic cosmic rays and solar energetic particles. A robust space weather program and the associated forecasting capabilities are essential for NASA's future exploration success.

This SBIR subtopic solicits new, enabling space weather technologies as part of NASA's response to these national objectives. While this subtopic will consider all concepts demonstrably related to NASA's Research-to-Operations/Operations-to-Research (R2O/O2R) responsibilities outlined in the NSWAP, four areas have been identified for priority development:

- NASA supports the Community Coordinated Modeling Center (CCMC), located at Goddard Space Flight Center (GSFC), as a centralized government-run facility that hosts, maintains, and validates heliophysics models, some of which will become suitable for use by the space weather operations and forecasting community. Innovations solicited include the preparation and validation of existing science models that may be suitable for transition to operational use. Areas of special interest include, but are not limited to:
    - Specifications and/or forecasts of the energetic particle and plasma conditions encountered by spacecraft within Earth's magnetosphere, as well as products that directly benefit end-users such as spacecraft operators;
    - Approaches that potentially lead to a 2-3 day forecasting of atmospheric drag effects on satellites and improvement in the quantification of orbital uncertainties in LEO altitude ranges (up to ~2000 km)
    - Longer-range (2-3 days) forecasting of SPEs (Solar Particle Events) and an improved all-clear SPE forecasting capability.
- The Heliophysics System Observatory (HSO) data archives include a vast array of spacecraft observations suitable for the development of space weather benchmarks, which are the set of characteristics against which space weather events are measured. Baseline benchmarks have been established (https://www.whitehouse.gov/wp-content/uploads/2018/06/Space-Weather-Phase-1-Benchmarks-Report.pdf). Innovations to produce and/or further refine these benchmarks are solicited, as are concepts for future creative approaches utilizing new data types or models that could become available.
- A particular challenge is to combine the sparse, vastly distributed data sources available with realistic models of the near-Earth space environment. Data assimilation innovations are solicited that enable tools and protocols for the operational space weather community. Priority will be given to proposals that:
    - Develop data assimilation space weather applications or technologies desired by established operational organizations;
    - Integrate data from assets that typically do not share similar time series, utilize different measurement techniques (e.g., imaging vs in-situ particles and fields), or are distributed throughout the heliosphere;
    - Provide new data-assimilation operational forecasting tools that can be straightforwardly validated by the CCMC or another equally robust validation methodology; and/or,
    - Integrate underutilized resources (e.g., space-based radio occultation for ionospheric specification or USGS ground conductivity measurements related to geomagnetically induced currents).
- Heliophysics science relies on a wide variety of instrumentation for its research and often makes its data available in near-real-time for space weather forecasting purposes. Concepts are solicited for instrumentation concepts, flight architectures, and reporting systems suitable for data assimilation into space weather monitoring and forecasting systems. This includes the miniaturization of existing systems and/or technologies deployable as an array of CubeSats. In order to be considered for investment, SBIR technologies should demonstrate comparable, or better, precision and accuracy when compared to the current state-of-the art. Further, SBIR instrument designs should avoid duplicating current NASA research spacecraft arrays or

detector systems including those currently in formulation (e.g., SDO, Van Allen Probes, MMS, IMAP, GDC, Medici, Explorer concepts, etc.).

Proposals must demonstrate an understanding of the current state-of-the-art, describe how the proposed innovation is superior, and provide a feasible plan to develop the technology and infuse into a specific NSWAP activity.

Space weather is a broad umbrella encompassing science, engineering, applications and operations. The ultimate goal of this SBIR is to generate products or services ("deliverables") that enable end-user action. The deliverables can be applied, for example, to provide space weather hazard assessments, real-time situational awareness, or to plan protective mitigation actions. Deliverables can be in the form of new data, new techniques, new instrumentation, or predictive models that are prepared/validated for transition into operations.

Expected TRL for this project is 3 to 8.

References:

- Pursuant to Executive Order 13744, the Space Weather Operations, Research, and Mitigation (SWORM) Subcommittee was chartered to coordinate Federal Government departments and agencies to meet the activities defined in the National Space Weather Action Plan (NSWAP). See https://www.sworm.gov/.
- Executive Order 13744 -- Coordinating Efforts to Prepare the Nation for Space Weather Events: https://www.federalregister.gov/documents/2016/10/18/2016-25290/coordinating-efforts-to-prepare-the-nation-for-space-weather-events
- National Space Weather Action Plan (NSWAP): https://www.sworm.gov/publications/2015/SWAP%20final__20151028.pdf
- Space Weather Phase I Benchmarks: https://www.sworm.gov/publications/2018/Space-Weather-Phase-1-Benchmarks-Report.pdf

Focus Area 14 In-Space and Advanced Manufacturing

Lead MD: STMD
Participating MD(s): STTR

NASA is seeking technological innovations that will accelerate development and adoption of advanced manufacturing technologies supporting a wide range of NASA Missions. NASA has an immediate need for more affordable and more capable materials and processes across its unique missions, systems, and platforms. Cutting-edge manufacturing technologies offer the ability to dramatically increase performance and reduce the cost of NASA's programs. This topic is focused on technologies for both the ground-based advancements and in-space manufacturing capabilities required for sustainable, long-duration space missions to destinations such as Mars. The terrestrial subtopic areas concentration is on research and development of advanced metallic materials and processes and additive manufacturing technologies for their potential to increase the capability and affordability of engines, vehicles, space systems, instruments and science payloads by offering significant improvements over traditional manufacturing methods. Technologies should facilitate innovative physical manufacturing processes combined with the digital twin modeling and simulation approach that integrates modern design and manufacturing. The in-space manufacturing subtopic areas which focus on the ability to manufacture parts in space rather than launch them from Earth represents a fundamental paradigm shift in the orbital supply chain model for human spaceflight. In-space manufacturing capabilities will decrease overall launch mass, while increasing crew safety and mission success by providing on-demand manufacturing capability to address known and unknown operational scenarios. In addition, advances in lighter-weight metals processing (on ground and in-space) will enable the delivery of higher-mass payloads to Mars and beyond. In order to achieve necessary reliabilities, in-situ process assessment and feedback control is urgently needed. Research should be conducted to demonstrate technical feasibility and prototype hardware development during Phase I and show a path toward Phase II hardware and software demonstration and delivering an engineering development unit for NASA testing at the completion of the Phase II that could be turned into a proof-of-concept system for flight demonstration.

T12.05 In-situ Curing of Thermoset Resin Mixtures (STTR)
Lead Center: JSC
Participating Center(s): LaRC

NASA has a need to significantly improve the manufacturing processes of Thermal Protection Systems (TPS) used in human rated spacecraft with the intention of reducing cost and improving quality and system performance. The fabrication and installation of current TPS are labor intensive, cost prohibitive, and result in many seams between the segments. Future human missions to Mars will require the landing of large-mass payloads on the surface, and these large entry vehicles will require large areas of TPS to protect the structure. In order to reduce the cost and complexity of these vehicles, new TPS materials and compatible additive manufacturing techniques are being developed such that the thermoset-resin based materials can be deposited, bonded and cured on spacecraft structures. Typically, thermoset resin mixtures require thermal cycles at elevated temperatures to be cured and commonly that is done in ovens or autoclaves. Technologies are sought that cure thermoset resin mixtures deposited on the flight structure without placing the structure into large ovens. Instead, the material would be cured in-situ on the structure shortly after deposition.

This subtopic seeks to develop a cost effective and modular method of curing TPS materials on Earth that could be incorporated into additive manufacturing processes. The design concept and process should be able to support curing/setting of high-temperature thermoset resin based materials deposited on composite structures. The goal deliverable for Phase II would be to demonstrate a prototype of the system.

Both Human Exploration and Operations Mission Directorate (HEO) and Science Mission Directorate (SMD) would benefit from this technology. All missions that include a spacecraft that enters a planetary atmosphere require TPS to protect the structure from the high-heating associated with hypersonic flight. Improved performance and lower cost heat shields benefit the development and operation of these spacecraft. Human missions to the moon and Mars would benefit from this technology. Commercial Space programs would also benefit from TPS materials and manufacturing processes developed by NASA.

It is desired that the Phase II deliverable be the engineering design and working prototype of the system. If the solution involves the development of a new self-curing material that meets TPS requirements, a sample or proof of concept will be required.

Expected TRL for this project is 2 to 3.

References:

- https://techcollaboration.center/workshops/advanced-manufacturing-carbon-materials-workshop/ (See presentation on Technical Challenges with 3D Printing Heat Shields)

Z3.01 Advanced Metallic Materials and Processes Innovation (SBIR)
Lead Center: MSFC
Participating Center(s): JPL, LaRC
Related Subtopic Pointer(s): Z3.02

Solid State Joining

This subtopic addresses specific NASA needs in the broad area of metals and metals processes with the focus for this solicitation on solid state welding and processing of specialty material: bulk metallic glasses.

Topic areas for solid state welding revolve around joining metallic materials preferably using solid state welding processes such as friction stir, thermal stir, and ultrasonic stir welding. Higher melting point materials of interest include the nickel based super-alloys such as Inconel 718, Inconel 625, titanium alloys such as Ti-6Al-4V, GRCop, and Mondaloy. Lower melting point materials of interest include Aluminum alloys such as 2195 and 2219. The technology needs for solid state welding should be focused on process improvement, structural efficiency, quality, and reliability for propulsion and propulsion-related components and hardware. This year, NASA is also looking for a mobile friction stir prototype unit for in-space applications

Note: For 2019 solicitation, additive manufacturing has been deleted, unless it specifically addresses needs in specialty metals.

For the 2019 solicitation, the solid state joining focus is:

- Development of diagnostics to accurately measure temperature and forces during welding with the goal of temperature feedback control and accurate force measurement during self-reacting friction-stir welding (FSW) for improved force control.
- Development of new technologies to overcome tool wear issues for friction-stir welding, specifically to have enhanced life with a single pass weld of 1200 inches of 0.625 inches thick Aluminum 2000 series.
- Friction-stir processing of high melting point materials such as Ni-base alloys, titanium alloys, and ferrous alloys.
- Development of a mobile friction-stir welding prototype technologies for in-space manufacturing that addresses:
    - Welding machine operation (tolerances, forge forces, travel speed, spindle speed) in microgravity.
    - Thermal changes in microgravity where convective heat transport no longer occurs.

Potential benefits include Earth-based Flight Software (FSW), such as Space Launch System, Orion, and Commercial Crew Program. Increasing the quality, shortening the development time for FSW processing parameters, and increasing the life of the FSW tool will decrease costs and shorten schedule time for developing and manufacturing of any large scale Aerospace hardware like fuel and oxidizer tanks and crew modules. In addition, increasing the use of lightweight aluminum tanks on other space missions could also aide in decreasing costs, decreasing schedule and increasing innovation.

Desired deliverables include:

- For Earth-based FSW: analysis supported by FSW samples.
- For in-space FSW: analysis, concept to minimize FSW setup in an in-space application.

The expected TRL for this project is 3 to 5.

Specialty Metals - Bulk Metallic Glass (BMG)

In the specialty materials processing area, the focus for this solicitation is on bulk metallic glass (BMG) alloys. Specific areas of interest relate to optimized processing to fabricate these materials while retaining their unique structures and properties.

Of specific interest for BMGs are innovative processing methods for rapid prototyping of net shape bulk metallic glass components. Product forms of interest are uniformly thin walled structures, structures of high dimensional accuracy and precision (from nm to cm scales), and structures with features larger than the critical casting thickness of the BMG alloy but still amorphous. Consideration must be given to the availability of BMG feedstocks or accommodating the raw materials for in-situ alloy fabrication. Any approach must demonstrate control of contaminant elements (e.g. oxygen and carbon) or show an immunity to their presence.

For the 2019 solicitation of specific interest for bulk metallic glasses are innovative processing methods that:

- Rapid prototyping, while maintaining high dimensional accuracy.
- Uniform thin walled structures that again, retain high dimensional accuracy.
- BMG structures with features that are larger than the critical thickness, but still amorphous.

This scope is relevant to the Space Technology Mission Directorate's Game Changing Development (GCD) effort for BMG mechanisms that are being designed for Lunar and Icy World missions. The technology provides alternative manufacturing pathways to reduce fabrication risk for bulk metallic glass gears and similar high precision mechanical components.

Desired deliverables include a test article demonstrating the proposed manufacturing process for a BMG component sized greater than the critical casting thickness and a report discussing the process' suitability for dimensional control including repeatability and reproducibility, processed material properties, and finished product material impurity (oxygen and carbon) levels.

The expected TRL for this project is 3 to 5.

References:

Solid State Joining

- https://blogs.nasa.gov/Rocketology/tag/friction-stir-welding/
- https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20150009520.pdf
- https://www.nasaspaceflight.com/2017/05/sls-core-stage-recovering-weld-pin-change/

Specialty Metals - Bulk Metallic Glass (BMG)

- Hofmann, DC et.al, "Optimizing Bulk Metallic Glasses for Robust, Highly Wear-Resistant Gears," Advanced Engineering Materials, 2016 DOI: 10.1002/adem.201600541
- Hofmann, DC et.al, "Castable Bulk Metallic Glass Strain Wave Gears: Towards Decreasing the Cost of High-Performance Robotics," Scientific Concepts, 2016 DOI: 10.1038/srep37773

Z3.02 Development of Mobile Welding Capabilities for In-Space Manufacturing (SBIR)
Lead Center: MSFC
Participating Center(s): LaRC, MSFC

In-Space Manufacturing/In-Space Material Joining

An in-space welding capability is an important supporting technology for the long duration, long endurance space missions NASA will undertake beyond the International Space Station (ISS). Historically structures in space have been assembled using mechanical fastening techniques and modular assembly. Structural designs for crewed habitats, space telescopes, antennas, and solar array reflectors are primarily driven by launch considerations such as payload faring dimensions and vibrational loads experienced during ascent. An in-space material joining capability can potentially eliminate constraints on the system imposed by launch, enabling the construction of larger, more complex and more optimized structures. Welding is an essential complementary capability to large scale additive manufacturing technologies being developed by NASA and commercial partners. Even without volume additive manufacturing, components will eventually need to be mated to larger structures. Welding is also a critical capability for repair scenarios (ex. repair of damage to a structure from micrometeroid impacts). This subtopic seeks innovative engineering solutions to mobilize joining technology for manufacturing in the external space environment, removing the need for large-scale equipment, specialized operators, and large-footprint manufacturing facilities that may not be available on long duration missions. Technologies developed may be infused into NASA missions and should also have high relevance to earth-based manufacturing applications which require fabrication and repair in the field. Note that concepts for mobile friction stir welding are included in Z3 (solid state joining) and should not be proposed to this subtopic. Priority welding process for external in-space use include electron beam, laser beam welding, gas tungsten arc welding, gas metal arc welding, and plasma arc.

Phase I is a feasibility study and laboratory proof of concept of a mobile, robotic welding process and system for external in-space manufacturing applications. Targeted applications for this technology include joining and repair of habitat modules, trusses, solar arrays, and/or antenna reflectors. The Phase I effort should provide a laboratory demonstration of the mobility of the selected welding process and its applicability to aerospace grade metallic materials, focusing on joint configurations which represent the priority in-space welding applications identified above. A proof of concept for repair capabilities in the scenario where structural material is damaged is also desired. Work under Phase I will inform preliminary design of a mobile welding unit and a concept of operations for how the system would be deployed and operate in the space environment, with a focus on specific scenarios: for example, repair of a metal panel following micrometeroid damage, longitudinal joining of two metal curved panels, and joining of a truss

to an adjacent truss. The Phase I should also provide an assessment of the proposed mobile unit's operational capabilities (for example: classes of materials which can be welded with the process, joint configurations which can be accommodated, and any expected impacts of the microgravity environment on joint efficiency relative to terrestrial system operation), volume, and power budget. A preliminary design and concept of operations are also deliverables under the Phase I. Concepts for teleoperation of the system should also be emphasized-- significant astronaut interaction with the system is not anticipated and proposers must have a maturation path toward teleoperation/remote commanding. The proposed system should thus be capable of remote commanding and evolvable to a self-contained free-flying configuration or a system that is externally mounted on a space station platform. Concepts for ancillary technologies such as post-process inspection, in-situ monitoring, or robotic arms for manipulation of structures to be joined may also be included in the Phase I effort.

Phase I requires a demonstration/proof of concept that:

- The process selected can be mobilized in a manner that enables high-value applications of in-space welding for repair and assembly.
- System shows potential for being operated remotely with very little intervention/setup.

Phase II includes finalization of the mobile welding unit design and demonstration of a ground-based prototype system. Phase III would seek to evolve the technology toward a flight demonstration, either via a system mounted externally on ISS, Gateway, or as a free-flyer.

Expected TRL for this project is 3 to 6.

References:

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Focus Area 15 Materials, Materials Research, Structures, and Assembly

Lead MD: STMD
Participating MD(s): HEOMD, STTR

As NASA strives to explore deeper into space than ever before, lightweight structures and advanced materials have been identified as a critical need. The Lightweight Materials, Structures, Advanced Assembly and Construction focus area seeks innovative technologies and systems that will reduce mass, improve performance, lower cost, be more resilient and extend the life of structural systems. Reliability will become an enabling consideration for deep space travel where frequent and rapid supply and resupply capabilities are not possible.

Improvement in all of these areas is critical to future missions. Applications include structures and materials for launch, in-space and surface systems, deployable and assembled systems, integrated structural health monitoring (SHM) and technologies to accelerate structural certification. Since this focus area covers a broad area of interests, specific topics and subtopics are chosen to enhance and or fill gaps in the space and exploration technology development programs as well as to complement other mission directorate structures and materials needs.

Specific interests include but are not limited to:

- Improved performance and cost from advances in composite, metallic and ceramic material systems as well as nanomaterial and nanostructures.
- Improved performance and mass reduction in innovative lightweight structural systems, extreme environments structures and multifunctional/multipurpose materials and structures.

- Improved cost, launch mass, system resiliency and extended life time by advancing technologies to enable large structures that can be deployed, assembled/constructed, reconfigured and serviced in-space or on planetary surfaces.
- Improved life and risk mitigation to damage of structural systems by advancing technologies that enhance nondestructive evaluation and structural health monitoring.
- Improved approaches that provide the development of extreme reliability technologies.

The specific needs and metrics for this year's focus technology needs are requested in detail in the topic and subtopic descriptions.

H5.01 Lunar Surface Solar Array Structures (SBIR)
Lead Center: LaRC
Participating Center(s): GRC, MSFC
Related Subtopic Pointer(s): Z3.02

NASA intends to start delivering small payloads to the lunar surface in 2019, moving to larger mid-size payloads by 2022 and eventually to human exploration [Ref. 1]. These missions will be powered by some combination of solar arrays with energy storage (e.g., batteries or regenerative fuel cells), radioisotope power converters, and nuclear fission depending on mission location and length, technology maturity, safety, and cost. Power estimates range from hundreds of watts initially to hundreds of kilowatts for an expansive human settlement [Ref. 2]. Mission plans are evolving and are mostly notional today but fault-tolerant power generation systems are critical in all cases.

An identified need for the initial human lander is a vertically deployed, retractable, sun-tracking solar array that generates ~4 kW of average and 6 kW of peak power. This lander would shuttle at least 5 times between the orbiting Lunar Gateway and elevated sites near the poles where the sun shines almost continuously, the temperature remains near -50° C, and frozen water exists in nearby craters [Ref. 3]. The solar array will be deployed and operated in zero gravity on the Gateway and during the initial descent, retract for final descent and landing, deploy again on the lunar surface for a 7-30 day mission, retract again for ascent, and then deploy again for transfer back to the Gateway.

Ideally, the solar array technology developed for this initial human lander can be reconfigured for follow-on habitats, rovers, and in-situ resource processing, and adapted for later use on Mars. For an eventual permanent polar outpost, vertically deployed tracking solar arrays that generate up to 100 kW each (~300 m^2) are imagined. These arrays would be similar in size to the flight-proven solar array wings on the ISS and would provide a centralized and high-efficiency power generation capability.

This subtopic seeks structural and mechanical innovations for lightweight solar arrays that can deploy and retract at least 10 times in both zero-g and lunar gravity from landers, rovers, habitats, and other surface equipment at the lunar poles. Full or possibly partial retraction will minimize rocket plume loads and dust accumulation, and allow valuable solar array hardware to be reused, repurposed, or reconfigured. Because most spacecraft solar arrays do not self-retract, this technology is not well developed. Mechanized, fault-tolerant retraction of lightweight solar arrays would be a valuable design option to have available, and suitable innovations and variations of existing array concepts [e.g., Ref. 4] are of special interest.

Design guidelines for these deployable/retractable solar arrays are:

- Vertical orientation (solar cells pointed at the horizon).
- Sun tracking with dust-resistant mechanisms and motors.
- Deployed area: 30 m^2 initially; up to 300 m^2 eventually per unit.
- Specific mass: >150 W/kg at 30 m^2; >100 W/kg at 300 m^2.
- Specific packing volume: >60 kW/m^3 at 30 m^2; >40 kW/m^3 at 300 m^2.
- Deployment/operation in both zero and lunar gravity.
- Number of mechanized deploy/retract cycles: at least 10; stretch goal >20 (in service).
- Lifetime: >10 years.
- Power generation: State assumptions.

Suggested areas of innovation include:

- Novel packaging, deployment, retraction, and modularity concepts.
- Lightweight, compact components including booms, ribs, substrates, and mechanisms.
- Load-limiting devices to avoid damage during deployment, retraction, and solar tracking.
- Optimized use of advanced lightweight materials (but not materials development).
- Validated modeling, analysis, and simulation techniques.
- High-fidelity, functioning laboratory models and test methods.
- Flight hardware for demonstration on a small or mid-size lander (multiple motorized retractions required).
- Completely new solar array concepts; e.g., thinned "rigid panel" solar arrays.

Proposals should emphasize structural and mechanical innovations, not photovoltaics, electrical, or energy storage innovations, although a complete solar array systems analysis is encouraged. If solar concentrators are proposed, strong arguments must be developed to justify why this approach is better from technical, cost, and risk points of view over unconcentrated planar solar arrays.

In Phase I, contractors should prove the feasibility of proposed innovations using suitable analyses and tests. In Phase II, significant hardware or software capabilities that can be tested at NASA should be developed to advance their Technology Readiness Level (TRL). TRL at the end of Phase II of 4 or higher is desired.

An identified need for the initial lunar human lander is a vertically deployed, retractable, sun-tracking solar array that generates ~3-4 kW of power. This lander would shuttle at least 5 times between the orbiting Lunar Gateway and elevated sites near the poles where the sun shines almost continuously, the temperature remains near -50° C, and frozen water exists in nearby craters. The solar array will be deployed and operated in zero gravity on the Gateway and during the initial descent, retract for final descent and landing, deploy again on the lunar surface for a 7-30 day mission, retract again for ascent, and then deploy again for transfer back to the Gateway.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

References:

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- Petri, D., Cataldo, R., and Bozek, J., "Power System Requirements and Definition for Lunar and Mars Outposts," AIAA Paper 2006-4103, June 2006, https://doi.org/10.2514/6.2006-4103.
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T12.01 Thin-Ply Composite Technology and Applications (STTR) - AMENDED – February 12, 2019
Lead Center: LaRC
Participating Center(s): LaRC
Related Subtopic Pointer(s): Z4.03

The use of thin-ply composites is one area of composites technology that has not yet been fully explored or exploited. Thin-ply composites are those with cured ply thicknesses below 0.0025 in., and commercially available prepregs are now available with ply thicknesses as thin as 0.00075 in. By comparison, a standard-ply-thickness composite would

have a cured ply thickness of approximately 0.0055 in. or greater. Thin-ply composites hold the potential for reducing structural mass and increasing performance due to their unique structural characteristics, which include (when compared to standard-ply-thickness composites):

- Improved damage tolerance.
- Resistance to microcracking (including cryogenic-effects).
- Improved aging and fatigue resistance.
- Reduced minimum-gage thickness.
- Thinner sections capable of sustaining large deformations without damage.
- Increased scalability of structures.

Thin-ply composites are attractive for a number of applications in both aeronautics and space as they have the potential for significant weight savings over the current state-or-the-art standard-ply materials due to improved performance. For example, preliminary analyses show that the notched strength of a hybrid of thin and standard ply layers can increase the notched tensile strength of composite laminates by 30%. Thus, selective incorporation of thin plies into composite aircraft structures may significantly reduce their mass. There are numerous possibilities for space applications. The resistance to microcracking and fatigue makes thin-ply composites an excellent candidate for a deep-space habitation structure where hermeticity is critical. Since the designs of these types of pressurized structures are typically constrained by minimum gage considerations, the ability to reduce that minimum gage thickness also offers the potential for significant mass reductions. For other space applications, the reduction in thickness enables: thin-walled, deployable structural concepts only a few plies thick that can be folded/rolled under high strains for launch (and thus have high packaging efficiencies) and deployed in orbit; and greater freedom in designing lightweight structures for satellite buses, landers, rovers, solar arrays, and antennas. For these reasons, NASA is interested in exploring the use of thin-ply composites for aeronautics and space applications requiring very high structural efficiency, for pressurized structures (such as habitation systems and tanks), for lightweight deep-space exploration systems, and for low-mass high stiffness deployable space structures (such as rollable booms or foldable panels, hinges or reflectors). There are many needs in development, qualification and deployment of composite structures incorporating thin-ply materials – either alone or as a hybrid system with standard ply composite materials. In particular, there is substantial interest in proposals that address manufacturability and production of composite structures utilizing thin-ply composites that at minimum develop the process and plan for the production of one prototype in Phase I and demonstrate reproducibility of prototype manufacturing and key parameter validation of repeated samples in Phase II. Another area requiring development is in new analysis and testing methods adapted for thin-ply flexible composites for folded and rolled structures. The Phase II deliverables will depend on the aspect addressed, but in general will be documentation of the analytical foundation and process, maturing the necessary design/analysis codes, and to validate the approach though design, build, and test of an article representative of the component/application of interest to NASA.

The particular capabilities requested for in a Phase I proposal in this subtopic are:

- New processing methods for making repeatable, consistent, high quality thin-ply carbon-fiber prepreg materials, (i.e., greater than 55% fiber density with low degree of fiber twisting, misalignment and damage, low thickness non-uniformity and minimal gaps in the material across the width) using currently used and commercially available fiber/matrix combinations. Prepreg product forms of interest have fiber areal weights below 70 g/m^2 for unidirectional tape with tape widths between 6 and 100 mm, and below 130 g/m^2 for woven/braided prepreg materials. Dry woven/braided fabrics below 80 g/m^2 are also of interest. Matrices of interest include both toughened epoxy resins for aeronautics applications, and toughened epoxy and cyanate ester resins with a Tg higher than 300 °F qualified for use in space. The intent of this requirement is to provide thin-ply prepreg material with the same quality as the standard-ply material of the same material system in order to facilitate substitution of thin-ply into structural concepts.
- Contributing to the development of the design and qualification database though testing and interrogation of the structural response and damage initiation/progression at multiple scales including evaluation of environmental durability and ageing, from which design recommendations can be formulated.
- Analysis and design tool validation and calibration to ensure appropriate behavior of thin-ply composites is captured, to identify any application-specific shortcomings with suggested improvements, and to certify thin-ply composite components are matured sufficiently to be used for NASA applications.

- Fabrication of very long slender composite structures, such as helicopter blades or deployable booms/beams, are not readily made in autoclaves due to length constraints. Innovative out-of-autoclave processing methods of the thin-ply composite is sought for the fabrication of very long parts to facilitate the use of thin-ply lamina in such structures. Additionally, the method should guarantee the curing process variables (temperature, pressure, etc) are uniform over the long parts to achieve better final products with less process-related defects and part-to-part variability.
- Cured-induced deformation of thin composite structures such as the spring-in effect is a known phenomenon that affects part accuracy during fabrication. Simulation software with general purpose finite element environments such as ABAQUS or ANSYS for the manufacturing process-induced deformations and residual stresses adapted to thin-ply composite structures with a final thickness under 1.5 mm are sought after. The goal is to develop recommendations for geometric tool compensation, as well as cure cycles and tooling that meets cure cycle specifications. In addition, simulation capability of complex hybrid, multi-step processes (co-cure, co-bond and secondary bonding) is of interest.
- Micromechanical models for spread-tow woven/braided lamina, as well as laminates that combine these with spread-tow unidirectional plies, including viscoelastic/viscoplastic and thermo-mechanical response. Such models shall be readily integrated into commercial finite element software packages like ABAQUS for the efficient thermo-mechanical analysis of large structural systems.
- Fracture mechanics models for thin-ply high strain composite shell structures for better prediction or remedy of damage initiation and progression in foldable/rollable structural members. The study of influential parameters such as creep/stress relaxation, fiber sizing, thermal fatigue, radiation dosage, atomic oxygen and ultra-high vacuum exposure, and low-strain resin microcracking as related to environmental ageing and dimensional stability is of special interest as part of a larger goal to qualify these structures for space flight.
- Development of new testing methods adapted for thin-ply flexible composite materials that allow folding/rolling of structures, with particular interest to dedicated large deformation bending relaxation and creep tests. Innovative non-contact techniques for accurately measuring the structure's high surface bending strains, curvatures and overall shape change over the large deformation process are of interest.
- Engineering viscoelastic behavior of thin-ply laminates for controlled deployment of space structures. For bistable thin-shell structures, the study of the viscoelastic/viscoplastic response and how that affects bistability and the ability of the structure to self-deploy after long-term stowage is of special interest.
- Development of novel low creep and low stress relaxation polymer thin-ply composites for inflatable and rollable/foldable space structures. Amongst others, approaches of interest are: designing new molecular structures showing high restriction of distortion of atomic bond angle under stress; controlling cross-linking density by reactive functional groups of molecular chains to keep a good balance between restriction of molecular rearrangement and material brittleness; restricting large scale rearrangements of polymer molecules by second phase of components; and securing strong interfaces between reinforcing fibers and polymer matrix by chemical bonding to prevent fibers and polymer molecules slippage under load. The temperature dependent viscoelastic/viscoplastic properties of the developed thin-ply material shall be characterized to predict the long-term behavior of the system under continuous loading.

Relevance to NASA

The most applicable ARMD program is AAVP, and within that is Advanced Air Transport Tech. (AATT). Additional projects within AAVP that could leverage this technology Commercial Supersonic Tech. (CST), Hypersonic Technology (HT), and Revolutionary Vertical Lift Tech. (RVLT). Projects within TACP could also benefit. That is, any project in need of lightweight structures can benefit from the thin-ply technology development. Within STMD, projects with deployable composite booms, landing struts, space habitats, tanks, and other very lightweight structures can benefit from the thin-ply technology.

References:

- https://www.nasa.gov/aeroresearch/programs/aavp
- https://www.nasa.gov/aeroresearch/programs/tacp
- https://www.nasa.gov/directorates/spacetech/home/index.html
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Z4.01 MISSE Experiments (SBIR)
Lead Center: LaRC
Participating Center(s): LaRC, MSFC

As NASA strives to explore deeper into space than ever before, lightweight structures and advanced materials have been identified as critical needs. The Lightweight Materials, Structures, Advanced Assembly, and Construction focus area seeks innovative technologies and systems that will reduce mass, improve performance, lower cost, be more resilient, and extend the life of structural systems. Reliability will become an enabling consideration for deep space travel, where frequent and rapid supply and resupply capabilities are not possible. Improvement in all these areas is critical to future missions. Applications include structures and materials for launch, in-space and surface systems, deployable and assembled systems, integrated structural health monitoring (SHM), and technologies to accelerate structural certification. Since this focus area covers a broad area of interests, this specific subtopic is chosen to enhance and or fill gaps in the space and exploration technology development programs, as well as to complement other mission directorate structures and materials needs.

Specific interests include:

- Improved performance and cost from advances in composite, metallic, and ceramic materials systems, as well as nanomaterials and nanostructures.
- Improved performance and mass reduction in innovative lightweight structural systems, extreme environments structures, and multifunctional/multipurpose materials and structures.
- Improved cost, launch mass, system resiliency, and extended life time by advancing technologies to enable large structures that can be deployed, assembled, constructed, reconfigured, and serviced in-space or on planetary surfaces.
- Improved life and risk mitigation to damage of structural systems by advancing technologies that enhance nondestructive evaluation and structural health monitoring.
- Improved approaches that provide the development of extreme reliability technologies.

Space technology experiments are solicited to fly on a new space environmental effects platform on the outside of the International Space Station (ISS). The new platform is called the MISSE-FF (Materials International Space Station Experiment - Flight Facility). The MISSE-FF provides experiment accommodations for both active experiments (requires power and/or communications) and passive experiments. The technology can be materials or non-materials (e.g., devices). The physical size of the experiments can vary, depending on the technology being demonstrated (2 inches by 2 inches, up to 7 inches by 14 inches). The depth is a maximum of 3 inches. Of particular interest are space technologies that would mature in TRL (technology readiness level) due to successful demonstration in the space environment. The proposal should justify the need for spaceflight exposure and justify that the ISS environment is adequate to collect the data they need. NASA's commercial partner Alpha Space Test and Research Alliance, LLC (Alpha Space) plans to service the MISSE-FF every 6 months. The MISSE-FF data will be made available to the global community of researchers through the NASA MAPTIS (Materials and Processes Technical Information System) database.

Phase I deliverables could be data from ground-based testing the candidate technology and/or passive samples for flight on the MISSE-FF. Phase II deliverables could include an active technology experiment, packaged and ready for flight on the MISSE-FF. The experiments would fly free of charge with standard services on the NASA surface area allocation of the MISSE-FF. Standard services include the mechanical integration of the experiment/samples with the flight hardware, monthly high-resolution images of the samples on orbit, and space environmental data (ultraviolet radiation, temperature, and contamination), as well as pointing/orientation data.

Any optional services desired from Alpha Space should be included in the proposal budget. Optional services include power, communications, and additional space environmental data (atomic oxygen and ionizing radiation). If an experiment requires a data acquisition unit, then that would need to be an integral part of the proposed experiment.

The award of an SBIR contract does not guarantee that the technology will be flown as a MISSE experiment. The developed technology has to be nominated in a subsequent and separate process and then selected for a MISSE mission by the NASA Flight Opportunities Program (FOP) jointly with the International Space Station Program (ISSP).

In addition to space environment exposure in Low Earth Orbit (LEO) on-board the International Space Station, limited numbers of payloads may be selected for flight aboard commercial lunar landers in the coming years. Thus in addition to the above described MISSE experiments, please consider and justify (need for lunar environment, data acquisition approach) environmental exposure experiments for the lunar environment according to the following opportunity:

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

Relevance to NASA

The Space Technology Mission Directorate (STMD), Human Exploration and Operations Mission Directorate (HEOMD), and Science Mission Directorate (SMD) could use the space technologies resulting from this subtopic. The Flight Opportunities Program (FOP), International Space Station Program (ISSP), Advanced Exploration Systems (AES) Program, and In-space Robotic Manufacturing and Assembly (IRMA) Project would particularly benefit from the technologies developed and tested under this subtopic.

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- Thibeault, Sheila A.; Cooke, Stuart A.; Ashe, Melissa P.; Saucillo, Rudolph J.; Murphy, Douglas G.; de Groh, Kim K.; Jaworske, Donald A.; and Nguyen, Quang-Viet: MISSE-X: An ISS External Platform for Space Environmental Studies in the Post-Shuttle Era. Presented at the 2011 Aerospace Conference, Big Sky, Montana, March 5-12, 2011. Published in the Proceedings of the 2011 Aerospace Conference, published by IEEE, pp.13, 2011.
- https://www.alphaspace.com/

Z4.02 In-Space Sub-Modular Assembly (SBIR)
Lead Center: LaRC
Participating Center(s): MSFC

NASA envisions that persistent (very long duration) assets in space will require modular assembly architectures and interfaces to facilitate routine expansion, upgrade and refurbishment at the module and submodule level. This subtopic seeks novel approaches to three classes of module interface systems.

The first assembly need is autonomous (and highly automated) approaches and hardware concepts that support the interconnection of modules in the 100 – 5,000 kg range using some form of space robotics. (Note: The robotic manipulation systems are not the subject of this solicitation.) The objective of this first subtopic area is to minimize the parasitic mass from the joints and modularity features that are required for inter-module assembly. The lightweight connections between modules must include both electrical (power and data) and structural connections. Joining strategies that support fluid connections are of interest but not necessary to be responsive to this subtopic area. The structural connection should occur at a minimum of 3 discrete locations fixing the rigid body motion of the 2 modules in all 6 degrees of freedom while isolating (minimizing) forces resulting from thermal induced strain between the modules consistent with a LEO orbit. The three (or more) connections do not have to occur simultaneously.

The second assembly need is the development of a lightweight modular, palletizing system to support transport, emplacement and exchange of sub-modules in the 1-100 kg range. The palletizing system must support power and structural connections between the pallet and supporting backbone structure (fluid connections are a plus). Important considerations for the system are structurally efficient approaches that minimize parasitic mass, volume and power

necessary to operate the palletizing system while minimizing forces resulting from temperature induced strain consistent with a LEO orbit.

The third assembly need is the development of assembly strategies that enable individual small spacecraft to support structural connects with suitable rigidity to form superstructures of small spacecraft (for the listed applications and other proposed science and exploration applications). Structural systems formed with assembled dimensions approximately 20x the size of the individual spacecraft are desired. Important consideration for the system are structurally efficient approaches that minimize parasitic mass, volume and power necessary to form assemblies utilizing approaches appropriate for small spacecraft, while minimizing forces resulting from temperature induced strain consistent with a LEO orbit.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

Relevance to NASA

Fundamental to this work is the cross-cutting nature of the technologies to all NASA missions that benefit from multiple visits to an asset, both zero-g and planetary surface assets. The technologies developed under this topic have the potential to radically change the way missions and in-space capabilities are conceived and developed. Rapid emplacement of initial capability followed by systematic upgrade and expansion will increase investment value while simultaneously improving return on investment.

References:

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- Belvin, W. Keith, Dorsey, John T., and Watson, Judith J., "Technology Challenges and Opportunities for Very Large In-Space Structural Systems," Presented at the International Symposium on Solar Energy from Space, Toronto, Canada, Sept. 8 – 10, 2009.
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Z4.03 Lightweight Conformal Structures (SBIR)
Lead Center: LaRC
Participating Center(s): MSFC
Related Subtopic Pointer(s): T12.01

Design and Manufacturing of Conformal Structures

Affordable space exploration beyond LEO will require innovative lightweight structural concepts. Conformal structural designs have the potential to enable not just mass savings, but also packaging efficiency in high center of gravity design concepts through advanced lightweight materials used to allow efficient integration of multifunctional structural elements. Examples of concepts of interest are non-cylindrical, non-spherical pressure vessels including but not limited to toroidal designs, suitable for storing cryogenic liquids or as habitable volumes. Successful demonstration of the manufacturability of such complex shaped pressure vessels can influence spacecraft designs.

Phase I of the award should describe proposed structural design concepts, an assessment of the manufacturability of the proposed structures and a systems benefits study to demonstrate mass and cost savings that can be achieved for Lunar and/or Mars missions. Designs where the pressure vessel is part of the structural load path are of interest. Potential applications anticipated for the successfully demonstrated concepts include lunar landers, Mars landers, habitat modules and ascent vehicles. Phase II will include a manufacturing demonstration of the design proposed in Phase I on a scale that is representative of full scale manufacturing challenges. Advanced materials of interest for the structural design and manufacturing include but are not limited to standard carbon fiber, thin ply laminates, carbon nanotube composites and hybrids of suitable advanced materials. Fabrication approaches such as tow steering and tailoring of hybrid materials to meet design requirements of application are of particular interest. Scaled prototype demonstrations should address manufacturing challenges that are anticipated in the full scale design.

Relevance to NASA

This topic fits under STMD. It is supported by the Lightweight Structures and Materials PT and bridges advanced materials and manufacturing.

Potential users of successful demonstration of the concept include NASA and Commercial Space companies.

References:

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Z11.01 NDE Sensors, Modeling, and Analysis (SBIR)
Lead Center: LaRC
Participating Center(s): ARC, LaRC
Related Subtopic Pointer(s): T11.04, Z4.01

NDE sensors and data analysis

Technologies enabling the ability to perform inspections on large complex structures will be encouraged. Technologies should provide reliable assessments of the location and extent of damage. Methods are desired to perform inspections in areas with difficult access in pressurized habitable compartments and external environments for flight hardware. Many applications require the ability to see through assembled conductive and/or thermal insulating materials without contacting the surface.

Techniques that can dynamically and accurately determine position and orientation of the NDE sensor are needed to automatically register NDE results to precise locations on the structure. Advanced processing and displays are needed to reduce the complexity of operations for astronaut crews who need to make important assessments quickly. NDE inspection sensors are needed for potential use on free-flying inspection platforms. Integration of wireless systems with NDE may be of significant utility. It is strongly encouraged to provide explanation of how proposed techniques and sensors will be applied to a complex structure. Examples of structural components include but are not limited to multi-wall pressure vessels, batteries, tile, thermal blankets, micrometeoroid shielding, International Space Station (ISS) Radiators or aerospace structural components.

Additionally, techniques for quantitative data analysis of sensor data are desired. It is also considered highly desirable to develop tools for automating detection of material Foreign Object Debris (FOD) and/or defects and evaluation of bondline and in-depth integrity for light-weight rigid and/or flexible ablative materials are sought. Typical internal void volume detection requirements for ablative materials are on the order of less than 6mm and bondline defect detection requirements are less than 25mm.

NDE Modeling

Technologies sought under this SBIR include near real-time realistic nondestructive evaluation (NDE) and structural health monitoring (SHM) simulations and automated data reduction/analysis methods for large data sets. Simulation techniques will seek to expand NASA’s use of physics based models to predict inspection coverage for complex aerospace components and structures and to utilize inverse methods for improved defect characterization. Analysis techniques should include optimized automated reduction of NDE/SHM data for enhanced interpretation appropriate for detection/characterization of critical flaws in space flight structures and components, and may involve methods such as machine learning, domain transformation, etc. NASA's interest area is light weight structural materials for space flight such as composites and thin metals. Future purposes will include application to long duration space vehicles, as well as validation of SHM systems.

Techniques sought include advanced material-energy interaction (i.e., NDE) simulations for high-strength lightweight material systems and include energy interaction with realistic damage in complex 3D component geometries (such as bonded/built-up structures). Primary material systems can include metals but it is highly desirable to target composite structures. NDE/SHM techniques for simulation can include ultrasonic, laser, Micro-wave, Terahertz, Infrared, X-ray, X-ray Computed Tomography, Fiber Optic, backscatter X-Ray and eddy current. It is assumed that any data analysis methods will be focused on NDE techniques with high resolution high volume data. Modeling efforts should be physics based and it is desired they can account for material aging characteristics and induced damage, such as micrometeoroid impact. Examples of damage states of interest include delamination, microcracking, porosity, fiber breakage. Techniques sought for data reduction/interpretation will yield automated and accurate results to improve quantitative data interpretation to reduce large amounts of NDE/SHM data into a meaningful characterization of the structure. It is advantageous to use co-processor/accelerator based hardware (e.g., GPUs, FPGAs) for simulation and data reduction. Combined simulation and data reduction/interpretation techniques should demonstrate ability to guide the development of optimized NDE/SHM techniques, lead to improved inspection coverage predictions, and yield quantitative data interpretation for damage characterization.

Phase I Deliverables - For NDE sensors focused proposals, lab prototype and feasibility study or software package including applicable data or observation of a measurable phenomenon on which the prototype will be built. For NDE modeling focused proposals, Feasibility study, including demonstration simulations and data interpretation algorithms, proving the proposed approach to develop a given product (TRL 2-4). Inclusion of a proposed approach to develop a given methodology to Technology Readiness Level (TRL) of 2-4. All Phase I's will include minimum of short description for Phase II prototype/software. It will be highly favorable to include description of how the Phase II prototype or methodology will be applied to structures.

Phase II Deliverables - Working prototype or software of proposed product, along with full report of development, validation, and test results. Prototype or software of proposed product should be of Technology Readiness Level (TRL 5-6). Proposal should include plan of how to apply prototype or software on applicable structure or material system. Opportunities and plans should also be identified and summarized for potential commercialization.

Relevance to NASA

Several missions could benefit from technology developed in the Area of nondestructive evaluation. Currently NASA is returning to manned space flight. The Orion program has continuing to have inspection difficulties and continued development and implementation of NDE tools will serve to keep our missions flying safely. Currently Orion is using several techniques and prototypes that have been produced under the NDE SBIR topic. Space Launch System is NASA's next heavy lift system. Capable of sending hundreds of metric tons into orbit. Inspection of the various systems is on-going and will continue to have challenges such as verification of the friction stir weld on the fuel tanks. As NASA continues to push in deeper space smart structures that are instrumented with structural health monitoring system can provide real time mission critical information of the status if the structure.

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- Burke, E. R.; and Waller, J.: NASA-ESA-JAXA Additive Manufacturing Trilateral Collaboration. Presented at Trilateral Safety and Mission Assurance Conference (TRISMAC), June 4-6, 2018, Kennedy Space Center, Florida.
- Campbell Leckey, C. A.; Juarez, P. D.; Hernando Quintanilla, F.; and Yu, L.: Lessons from Ultrasonic NDE Model Development. Presented at 26th ASNT Research Symposium 2017, March 13-16, 2017, Jacksonville, Florida.
- Campbell Leckey, C. A.: Material State Awareness: Options to Address Challenges with UT. Presented at World Federation of NDE Centers Short Course 2017, July 15-16, 2017, Provo, Utah.
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- Cramer, K. E.; and Klaassen, R.: Developments in Advanced Inspection Methods for Composites Under the NASA Advanced Composites Project. Presented at GE Monthly Seminar Series, April 13, 2017, Cincinatti, Ohio.
- Cramer, K. E.; and Perey, D. F.: Development and Validation of NDE Standards for NASAâ€™s Advanced Composites Project. Presented at ASNT Annual Conference, October 30-November 2, 2017, Nashville, Tennessee.
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- Cramer, K. E.: Research Developments in Non-Invasive Measurement Systems for Aerospace Composite Structures at NASA. Presented at 2018Â International Instrumentation and Measurement Technology Conference, May 14-18, 2018, Houston, Texas.
- Dehaven, S. L.; Wincheski, R. A.; and Burke, E. R.: X-ray transmission through microstructured optical fiber. Presented at QNDE - Review of Progress in Quantitative NDE, July 17-21, 2017, Provo, Utah.
- Dehaven, S. L.; Wincheski, R. A.; and Burke, E. R.: X-ray transmission through microstructured optical fiber. Presented at 45th Annual Review of Progress in Quantitative Nondestructive Evaluation (QNDE), July 15-19, 2018, Burlington, Vermont.

- Frankforter, E.; Campbell Leckey, C. A.; and Schneck, W. C.: Finite Difference Simulation of Ultrasonic Waves for Complex Composite Laminates. Presented at QNDE 2018, July 15-19, 2018, Burlington, Vermont.
- Gregory, E. D.; and Juarez, P. D.: In-situ Thermography of Automated Fiber Placement Parts: Review of Progress in Quantitative Nondestructive Evaluation. Presented at QNDE - Review of Progress in Quantitative NDE, July 17-21, 2017, Provo, Utah.
- Gregory, E. D.; Campbell Leckey, C. A.; and Schneck, W. C.: A Versatile Simulation Framework for Elastodynamic Modeling of Structural Health Monitor.

Focus Area 16 Ground & Launch Processing

Lead MD: HEOMD

Ground processing technology development prepares the agency to test, process and launch the next generation of rockets and spacecraft in support of NASA's exploration objectives by developing the necessary ground systems, infrastructure and operational approaches.

This topic seeks innovative concepts and solutions for both addressing long-term ground processing and test complex operational challenges and driving down the cost of government and commercial access to space. Technology infusion and optimization of existing and future operational programs, while concurrently maintaining continued operations, are paramount for cost effectiveness, safety assurance, and supportability.

A key aspect of NASA's approach to long term sustainability and affordability is to make test, processing and launch infrastructure available to commercial and other government entities, thereby distributing the fixed cost burden among multiple users and reducing the cost of access to space for the United States.

Unlike previous work focusing on a single kind of launch vehicle such as the Saturn V rocket or the Space Shuttle, NASA is preparing common infrastructure to support several different kinds of spacecraft and rockets that are in development. Products and systems devised at a NASA center could be used at other launch sites on earth and eventually on other planets or moons.

Specific emphasis to substantially reduce the costs and improve safety/reliability of NASA's test and launch operations includes development of ground test and launch environment technology components, system level ground test systems for advanced propulsion, autonomous control technologies for fault detection, isolation, and recovery, including autonomous propellant management, and advanced instrumentation technologies including Intelligent wireless sensor systems.

H10.01 Advanced Propulsion Systems Ground Test Technology (SBIR)
Lead Center: SSC
Participating Center(s): KSC
Related Subtopic Pointer(s): T2.02, T13.01, Z10.03

Rocket propulsion development is enabled by rigorous ground testing to mitigate the propulsion system risks that are inherent in spaceflight. This is true for virtually all propulsive devices of a space vehicle including liquid and solid rocket propulsion, chemical and non-chemical propulsion, boost stage and in-space propulsion and so forth. It involves a combination of component and engine-level testing to demonstrate that propulsion devices were designed to meet the specified requirements for a specified operational envelope over robust margins and shown to be sufficiently reliable prior to its first flight.

This subtopic seeks to develop advanced ground test technology components and system level ground test systems that enhance Chemical and Advanced Propulsion technology development and certification. The goal is to advance propulsion ground test technologies in order to enhance environment simulation, minimize test program time, cost and risk and meet existing environmental and safety regulations. It is focused on near-term products that augment and enhance proven, state-of-the-art propulsion test facilities. This project is especially interested in ground test and

launch environment technologies with potential to substantially reduce the costs and improve safety/reliability of NASA's test and launch operations.

In particular, technology needs include designs for stable combustion of propellants in a duct under low pressure, high velocity conditions, developing robust materials, advanced instruments and monitoring systems capable of operating in extreme temperature and harsh environments. This subtopic seeks innovative technologies in the following areas:

- Robust design solutions for liquid oxygen injection and subsequent stable combustion with high temperature (5100°Rankine/2850 K) hydrogen in a duct flowing at low pressure (3-15 psia) and high velocity (~Mach 0.2)
- Devices for measurement of pressure, temperature, strain and radiation in a high temperature and/or harsh environment
- Development of innovative rocket test facility components (e.g., valves, flowmeters, actuators, tanks, etc.) for ultra-high pressure (>8000 psi), high flow rate (>100 lbm/sec) and cryogenic environments
- Robust and reliable component designs which are oxygen compatible and can operate efficiency in high vibro-acoustic, environments
- Advanced materials to accommodate flow and containment high-temperature (<5100°Rankine/2850 K) hydrogen, i.e., materials resistant to hydrogen embrittlement and associated harsh environments generated by high velocity hydrogen flows
- Tools using computational methods to accurately model and predict system performance are required that integrate simple interfaces with detailed design and/or analysis software. SSC is interested in improving capabilities and methods to accurately predict and model the transient fluid structure interaction between cryogenic fluids and immersed components to predict the dynamic loads, frequency response of facilities
- Improved capabilities to predict and model the behavior of components (valves, check valves, chokes, etc.) during the facility design process are needed. This capability is required for modeling components in high pressure (to 12,000 psi), with flow rates up to several thousand lb/sec, in cryogenic environments and must address two-phase flows. Challenges include: accurate, efficient, thermodynamic state models; cavitation models for propellant tanks, valve flows, and run lines; reduction in solution time; improved stability; acoustic interactions; fluid-structure interactions in internal flows.

For all above technologies, research should be conducted to demonstrate technical feasibility during Phase I and show a path towards Phase II hardware/software demonstration with delivery of a demonstration unit or software package for NASA's testing at the completion of the Phase II effort.

This subtopic is relevant to the development of liquid propulsion systems development and verification testing in support of the Human Exploration and Operations Mission Directorate (HEOMD) and supports all test programs at Stennis Space Center and other propulsion system development centers. The expected Technology Readiness Level (TRL) range at completion of this project is 4 to 6.

References:

- J. Schmalzel ; F. Figueroa ; J. Morris ; S. Mandayam ; R. Polikar, "An architecture for intelligent systems based on smart sensors," IEEE Transactions on Instrumentation and Measurement (Volume: 54 , Issue: 4 , Aug. 2005)
- S. Rahman, R. Gilbrech, R. Lightfoot, M. Dawson, "Overview of Rocket Propulsion Testing at NASA Stennis Space Center," NASA Technical Report SE-1999-11-00024-SSC
- David J. Coote, Kevin P. Power, Harold P. Gerrish, and Glen Doughty. "Review of Nuclear Thermal Propulsion Ground Test Options", 51st AIAA/SAE/ASEE Joint Propulsion Conference, AIAA Propulsion and Energy Forum, (AIAA 2015-3773)
- Fernando Figueroa, Randy Holland, David Coote, "NASA Stennis Space Center integrated system health management test bed and development capabilities," Proc. SPIE 6222, Sensors for Propulsion Measurement Applications, 62220K (10 May 2006)
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H10.02 Autonomous Control Technologies (ACT) for Ground Operations (SBIR)
Lead Center: KSC
Participating Center(s): ARC, LaRC, SSC
Related Subtopic Pointer(s): H6.01

Autonomous Control Technologies (ACT) are needed to reduce operations and maintenance (O&M) costs of ground test and payload operations on ground, and to increase systems availability to support mission operations. These technologies are also required for extended surface operations and maintenance on the Moon and Mars. They are also required in activities where human intervention/interaction needs to be minimized, such as in hazardous locations/ operations and in support of remote operations.

ACT performs functions such as systems and components fault prediction and diagnostics, anomaly detection, fault detection and isolation, and enables variable levels of autonomous control and recovery from faults, where recovery may include reconfiguration or repair. ACT are enabled by System Health Management (ISHM) technologies, methodologies, and approaches; command and control architectures; computing architectures; software for decision-making and control; and intelligent components and devices.

ACT will be applied to activities such as rocket engine test facilities, propellant servicing and launch of vehicles and will complement In-Situ Resources Utilization (ISRU) operations by utilizing ISRU-generated commodities to support transportation activities. ACT will also enable surface operations and maintenance, which differs significantly from traditional launch processing operations, due to the required high degree of autonomy and reliability for unattended operations during extended periods of time. ACT enables Autonomous Propellant Management (APM), which requires the unattended or minimally attended storage, transfer, monitoring, and sampling of cryogenic or other propellant use in launch systems. APM includes pre-planned nominal processes, such as vehicle fill and drain, as well as contingency and off-nominal processes, such as emergency safing, venting and system reconfiguration.

ACT capabilities will also enable the autonomous command, monitoring and control of the overall integrated system, resulting from the integration of loading systems and all other associated systems involved in the loading process.

The system autonomy software itself includes both prerequisite control logic (PCL) and reaction control logic (RCL) programming, and may utilize some form of machine learning, neural network, or other form of artificial intelligence to adapt to degraded system components or other forms of off-nominal conditions. In addition to cryogenic and other propellants, propellant management systems may utilize additional commodities to prepare a vehicle for launch, such as high pressure gasses for purges, pressurization, or conditioning, and may include power and data interfaces with the vehicle to configure vehicle valves or other internal systems and utilize on-board instrumentation to gain visibility into the vehicle during loading.

ACT must also support tasks such as systems setup, testing and checkout, troubleshooting, maintenance, upgrades and repair. These additional tasks drive the need for autonomous element to element interface connection and separation, multi-element inspection, and recovery of high value cryogenic propellants and gasses to avoid system losses.

Specifically, this subtopic seeks the application of high-fidelity, physics-based, cryogenic-thermal models and simulations capable of real-time and faster than real-time performance. Along with these simulations, the subtopic seeks the creation of simulation component libraries (generic components), capable to be tailored, to support the rapid prototyping of cryogenic-thermal models. The subtopic also seeks supervisory control software for autonomous control and recovery of propellant loading systems and infrastructure and software development tools to support the rapid prototyping of autonomous control applications. Furthermore, the subtopic seeks architectures that support integration of the above capabilities for integrated autonomous operations.

For all above technologies, research should be conducted to demonstrate technical feasibility during Phase I and show a path toward Phase II demonstration and delivering a demonstration package for NASA testing in operational or analog test environments at the completion of the Phase II effort.

Ideally, Phase I deliverables should include:

Research, identification and evaluation of candidate technologies or concepts for systems and components fault detection, isolation and recovery, fault prediction and diagnosis, and decision-making algorithms for control to enable autonomy of ground systems. Demonstration of the technical feasibility and show a path towards a demonstration. Concept methodology should include the path for adaptation of the technology, infusion strategies (including risk trades), and business model. It should identify improvements over the current state of the art for both operations and systems development and the feasibility of the approach in a multi-customer environment. Bench or lab-level demonstrations are desirable. Deliverables must include a report documenting findings.

Ideally, Phase II deliverables shall:

Place an emphasis on developing, prototyping and demonstrating the technology under simulated operational conditions using analog earth-based systems including dynamic events such as commodity loading, disconnect or engine testing. Deliverables shall include a report outlining the path showing how the technology could be matured and applied to mission-worthy systems, functional and performance test results and other associated documentation. Deliverable of a functional prototype (software and hardware) is expected at the completion of the Phase II effort. The technology concept at the end of Phase II should be at a Technology Readiness Level (TRL) of 6 or higher.

In addition to reducing O&M costs in ground operations, this subtopic provides NASA's Human Exploration & Operations Mission Directorate (HEOMD) with an on-ramp for technologies that enable the unattended setup, operation, and maintenance of ground systems and systems on the surfaces of other planets and moons. These types of technology developments are identified in the NASA Strategic Technology Area (TA) roadmaps, published by the Office of the Chief Technologist (OCT), under TA4: Robotics and Autonomous Systems and TA13: Ground and Launch Systems. This subtopic produces technologies which will also be of use to NASA's Space Technology Mission Directorate (STMD). Autonomous strategies have crosscutting value in other applications and with other mission directorates.

References:

- NASA OCT webpage: https://www.nasa.gov/offices/oct/home/index.html
- 2017 NASA Strategic Space Technology Investment Plan (STIP), found at NASA's OCT webpage: https://www.nasa.gov/offices/oct/feature/nasa-releases-2017-nasa-strategic-technology-investment-plan
- NASA Technology Roadmaps, found within NASA's OCT, are a set of documents that consider a wide range of needed technology candidates and development pathways for the next 20 years (2015-2035). Roadmaps are found at: https://www.nasa.gov/offices/oct/home/roadmaps/index.html. Of special interest are the following Roadmaps:
    - Technology Area TA4: Robotics and Autonomous Systems
    - Technology Area TA13: Ground and Launch Systems
    - Technology Area TA11: Modeling, Simulation, Information Technology and Processing

T13.01 Intelligent Sensor Systems (STTR)
Lead Center: SSC
Participating Center(s):
Related Subtopic Pointer(s): H10.01, S1.09, Z11.01

Advanced Instrumentation for Rocket Propulsion Testing

Rocket propulsion development is enabled by rigorous ground testing in order to mitigate the propulsion system risks that are inherent in spaceflight. Test articles and facilities are highly instrumented to enable a comprehensive analysis of propulsion system performance. This subtopic seeks to develop advanced instrumentation technologies which can be embedded in systems and subsystems. The goal is to provide a highly flexible instrumentation solution capable of

monitoring remote or inaccessible measurement locations, all while eliminating cabling and auxiliary power. It is focused on near-term products that augment and enhance proven, state-of-the-art propulsion test facilities. Rocket propulsion test facilities within NASA provide excellent test beds for testing and using the innovative technologies discussed above. The technologies developed would be capable of addressing multiple mission requirements for remote monitoring such as vehicle health monitoring.

Intelligent wireless sensor systems have the potential for substantial reduction in time and cost of propulsion systems development, with substantially reduced operational costs and evolutionary improvements in ground, launch and flight system operational robustness. Sensor systems should provide an advanced diagnostics capability to monitor test facility parameters including simultaneous heat flux, temperature, pressure, strain, and near-field acoustics. Applications encompass remote monitoring of vacuum lines, gas leaks and fire; where the use of wireless/self-powered sensors to eliminate power and data wires would be beneficial. Nanotechnology enhanced sensors are desired where applicable to provide a reduction in scale, increase in performance, and reduction of power requirements.

Sensor systems should have the ability to provide the following functionality:

- Measure of the quality of the measurement.
- Measure of the health of the sensor.
- Sensor systems should enable the ability to detect anomalies, determine causes and effects, predict future anomalies, and provides an integrated awareness of the health of the system to users (operators, customers, management, etc.).
- Sensors are needed with capability to function reliably in extreme environments, including rapidly changing ranges of environmental conditions, such as those experienced in space. These ranges may be from extremely cold temperatures, such as cryogenic temperatures, to extremely high temperatures, such as those experienced near a rocket engine plume. Collected data must be time stamped to facilitate analysis with other collected data sets.
- Sensor systems should be self-contained to collect information and relay measurements through various means by a sensor-web approach to provide a self-healing, auto-configuring method of collecting data from multiple sensors, and relaying for integration with other acquired data sets.
- The proposed innovative systems must lead to improved safety and reduced test, and space flight costs by allowing real-time analysis of data, information, and knowledge through efficient interfaces to enable integrated awareness of the system condition by users.
- The system provided must interface with existing data acquisition systems and the software used by such systems.
- The system must provide NIST traceable measurements with capability for in-place calibrations.
- The system design should consider an ultimate use of space flight qualified sensor systems, which could be used for multi-vehicle use.

Subtopic is relevant to the development of liquid propulsion systems and verification testing in support of the Human Exploration and Mission Operations Directorate. Supports all test programs at Stennis Space Center (SSC) and other propulsion system development/test and launch facilities. Potential advocates are the Rocket Propulsion Test (RPT) Program Office and all rocket propulsion test programs at SSC.

The expected Technology Readiness Level (TRL) range at completion of the project is 3-6.

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- Fernando Figueroa, Randy Holland, David Coote, "NASA Stennis Space Center integrated system health management test bed and development capabilities," Proc. SPIE 6222, Sensors for Propulsion Measurement Applications, 62220K (10 May 2006);
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Focus Area 17 Thermal Management Systems

Lead MD: STMD
Participating MD(s): SMD

From the smallest satellite to the most complicated human rated spacecraft, thermal is seen as an enabling function to a vehicle. Temperatures must be maintained within design limits, whether those be cryogenic systems for science instruments, or comfortable shirt sleeve operations temperatures for crew missions. As missions evolve and waste energy rejection becomes more of a demand, NASA seeks components for both active and passive thermal systems. Such components complete the thermal cycle which includes waste energy acquisition, transport, rejection/storage, and insulation. The ultimate goal for any thermal control hardware is to minimize mass, volume, and power while maintaining the aforementioned temperature limits on a spacecraft.

S3.06 Thermal Control Systems (SBIR)
Lead Center: GSFC
Participating Center(s): JPL, LaRC, MSFC

Future Spacecraft and instruments for NASA's Science Mission Directorate (SMD) will require increasingly sophisticated thermal control technology. Innovative proposals for the cross-cutting thermal control discipline are sought in the following areas/scopes. Research should be conducted to demonstrate technical feasibility during Phase I and show a path toward a Phase II hardware demonstration. Phase II should deliver a demonstration unit for NASA testing at the completion of the Phase II effort.

Advanced Thermal Devices

Advanced thermal devices capable of maintaining components within their specified temperature ranges are needed for future advanced spacecraft. Some examples are:

- High thermal conductivity, vacuum-compatible interface materials that minimize losses across make/break interfaces
- High flux heat acquisition and transport devices
- High performance, low cost insulation systems for diverse environments
- Durable, radiation stable, electrically dissipative, high emittance, low and high absorptance coatings at cryogenic temperatures below 50K
- Radiator heat rejection turndown devices (e.g., mini heat switches, mini louvers)

These high-performance devices would have widespread applicability to upcoming missions and enable missions that are currently not feasible with present technology.

New generations of electronics used on numerous missions have much higher power densities than in the past. High conductivity, vacuum-compatible interface materials are needed in order to reduce interface temperature gradients and facilitate efficient heat removal. Also needed are high flux heat acquisition and transport devices such as high heat flux heat pipes and loop heat pipes.

Exploration science missions beyond earth orbit require systems which can withstand extreme temperatures ranging from high temperatures on Venus to the cryogenic temperatures of the outer planets. High performance insulation systems, which are more easily fabricated than traditional multi-layer (MLI) systems, are required for both hot and cold environments. Potential applications include traditional vacuum environments, low pressure carbon dioxide atmospheres on Mars, and high pressure atmospheres found on Venus.

There are few options for electrically dissipative, low and high absorptance exterior coatings that exhibit high IR emittance characteristics at cryogenic temperatures at or below 50K and that applied conformally coat complex structures with durability to not become particle contamination or entrapment risks during I&T operations. White and black inorganic silicate-based coatings exhibit durability and particle generation risks. White silicone-based systems lack optical stability for long duration exposures. Other structural solutions such as coated honeycomb are not durable and represent particle entrapment risks.

High radiator heat rejection turn down devices are needed for future NASA missions. A radiator is typically designed to dissipate the maximum heat load at the highest sink temperature. At low sink temperatures, there is a desire to turn down the radiator's heat rejection to reduce heat loss from the instrument to the space and to save the survival heater power.

Flexible Cryogenic Heat Pipe

Heat pipes are efficient and versatile heat transfer devices that can transport a large heat load over a long distance with a small temperature difference. Existing heat pipes are designed mainly for room temperature applications. Some instruments require heat pipes to operate in the cryogenic temperature range. Furthermore, some flight missions call for flexible heat pipes to allow an instrument to track a target during orbit, to allow the deployment of radiator panels, or to minimize mechanical loads and vibration from a cryocooler into an instrument to be cooled. Constant conductance heat pipes (CCHPs) operating below 90 Kelvin with the ability for repeated cycles of pipe flexing are needed.

SMD missions involving cryogenic temperature applications such as Wide Field Infrared Survey Telescope (WFIRST); Plankton, Aerosol, Cloud, ocean Ecosystem (PACE); and SWOT can utilize such devices.
The expected Technology Readiness Level (TRL) range at completion of this project is 5 to 6.

Software Improvements for Integrated Thermal-Structural-Optical Performance Analysis

Sensitive optical components and systems, as are frequently used on space science missions, require structural, thermal, and optical performance (STOP) analysis in their design process to validate performance in expected mission environments. Current STOP analysis codes are tailored to specific optical components or systems, lacking the generality that would make the codes more useful to a range of designs. Improvement in existing STOP analysis codes is needed such that they can be applied to any optical system and integrated with mechanical, structural, thermal, and optical analysis software used at NASA. The improved STOP analysis code should be user-friendly and generate performance predictions based on the optical system mechanical design and structural/thermal material properties.

Any mission/project in which optical components or systems are used will require STOP analyses to be completed. As such, a general, integrated, and easy-to-use STOP software is a common desire among engineers of different disciplines.

The expected Technology Readiness Level (TRL) range at completion of this project is 5 to 6.

Advanced Thermoelectric Converter

Thermoelectric converters (TECs) have advantages of small size, long life, solid state design, and no moving parts or fluid operation. Although TECs can also be used for power generation, this solicitation specifically calls for TECs for thermal cooling applications. TECs are known to have a very low efficiency, and most existing TECs are designed for room temperature applications. Research and development in areas of advanced materials, processes, and designs are

needed in order to improve its efficiency and extend its low temperature (< 90K) capability for space science application.

Many NASA missions have used TECs for localized cooling to remove heat or to achieve low temperatures. Because of the low efficiency of TECs, multi-staged TECs are often used, which adds complexity in the design and integration of this device. A high efficiency TEC will greatly simplify the design and enhance its reliability. A high efficiency, low temperature TEC can also be an enabling technology for many space missions.
The expected Technology Readiness Level (TRL) range at completion of this project is 5 to 6.

Approaches and Techniques for Surface Payload Survival through the Lunar Day/Night Cycle

NASA has plans to purchase services for delivery of small science and technology demonstration payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

The lunar day/night cycle is approximately one earth month. During that time surface temperatures on the lunar surface can reach 400K at local solar noon or drop to below 100K during the lunar night, even colder in permanently shadowed regions. These hot and cold conditions can last several earth days due to the slow rotation of the moon and can pose significant challenges to small, low power, science payloads and supporting systems trying to operate for extended length missions encompassing multiple years. Lunar dust deposited on heat rejection surfaces and coatings may further exacerbate the challenge. As interest in the moon has renewed and potential ways to get there are being developed, updated and new thermal approaches, techniques, components are needed to enable missions to include continuous operation through the day night cycle despite the absence of extended heat sources like radio-isotopes which are typically prohibitive from a cost and safety perspective. As examples: technologies that minimize heat loss might help to survive the lunar night, but need to support heat rejection during the potentially hot lunar day; advanced thermal storage approaches could allow heat from the lunar day to be used to survive the long lunar night but need to consider mass and volume limitations; adding heaters for the lunar night will drive electrical power usage necessitating excessive battery mass. The technologies proposed for this subtopic may address a single aspect of the lunar thermal environment or may offer an approach for accommodating the full range of environmental features over a complete lunation.

SMD lunar surface science investigations will employ small, low power payloads that will require advanced thermal control approaches and techniques to survive and operate for extended durations through extreme thermal environments on the lunar surface.

The expected Technology Readiness Level (TRL) range at completion of this project is 3 to 4.

References:

Flexible Cryogenic Heat Pipe

- Wide Field Infrared Survey Telescope (WFIRST): https://wfirst.gsfc.nasa.gov/
- Plankton, Aerosol, Cloud, ocean Ecosystem (PACE): https://pace.gsfc.nasa.gov/
- Kobel, M., and J. Ku, “Thermal Vacuum Testing of Swift XRT Ethane Heat Pipes,” AIAA 1st IECEC, Paper No. 2003-6080, August 17-21, 2003, Portsmouth, Virginia.

Advanced Thermoelectric Converter

- Swift's X-Ray Telescope (XRT): https://swift.gsfc.nasa.gov/about_swift/xrt_desc.html
- https://www.marlow.com/resources/thermoelectric-technology-guide
- https://www.tellurex.com/services/thermoelectric-cooling-heating/

Approaches and Techniques for Surface Payload Survival through the Lunar Day/Night Cycle

- https://en.wikipedia.org/wiki/Commercial_Lunar_Payload_Services
- https://www.jpl.nasa.gov/news/news.php?release=2007-068
- https://history.nasa.gov/TM-3487/ch2-1.htm
- https://www.lpi.usra.edu/lunar/missions/surveyor/
- https://solarsystem.nasa.gov/missions/lunokhod-01/in-depth/
- https://solarsystem.nasa.gov/missions/lunokhod-02/in-depth/

Z2.01 Spacecraft Thermal Management (SBIR)
Lead Center: JSC
Participating Center(s): GRC, GSFC, JPL, MSFC
Related Subtopic Pointer(s): S3.06

NASA seeks new technologies that will facilitate low mass & highly reliable thermal control systems for exploration vehicles. Of particular interest in this solicitation are thermal control technologies related to heat acquisition, transport, and/or rejection/storage that enable single-loop thermal control systems for long duration human spacecraft, precision thermal control technologies compatible with mechanically pumped two-phase flow thermal systems, and lunar lander technologies that can be matured from small scale vehicles to human class missions. Proposals are expected to provide analytical and/or empirical proof-of-concept at the end of a Phase I effort and result in technology delivery at the end of a Phase II effort. At the culmination of Phase II, deliverables could include thermal math modeling that has been correlated to new technology tests, test data, as well as delivery to NASA.

Single-Loop Enabling Technologies for Thermal Control of Human Spacecraft

Human spacecraft have historically utilized dual fluid loop system architectures that contain a benign internal fluid such as water and a hazardous external fluid such as ammonia. Here, technologies that enable a single-loop thermal control system for human class missions are sought. Technologies should be appropriate for integration into a vehicle that has a nominal load of 6-8 kW during crewed operations (approximately 10% of the year) and a nominal load of 1-2 kW during uncrewed dormancy periods (the remaining 90% of the year). Vehicle external environments can vary between deep space and one sun conditions. Solutions should not pose undue burden on other vehicle subsystems and have a useful life > 10 years of continuous operation. Proposed technologies and associated systems should have a tangible mass, volume, and power benefit over the current state-of-the-art. Currently, single-loop systems are traded against an external pump package's approximately 100kg mass.

Mechanically Pumped Two-Phase Flow Thermal Control System Technology Development

NASA currently has a critical gap in two-phase mechanically pumped thermal management system technologies. NASA plans to extend its traditional single, liquid, phase mechanically pumped architectures to two-phase, liquid/gas systems. Technologies are sought that enable these two-phase mechanically pumped fluid loop architectures and provide high quality precision thermal control.

Of particular interest in this solicitation are two-phase system heat acquisition technologies that can accommodate heat fluxes of up to 5 W/cm^2 and provide instrument isothermalization of < 3°C over 1-m^2 areas with temporal accuracy <0.05° C/minute. Such technologies have significant value as they improve the consistency of scientific data collection. Novel approaches to evaporators, flow boiling cold plates, or other novel technologies are sought to provide this capability. All proposed technologies should provide a useful life of at least 15 years and minimize vehicle level impacts to mass, power, and volume. While not specifically sought in this subtopic, improvements in two-phase mechanical pump efficiency and useful life are desirable. See Subtopic S3.06 Thermal Control Systems for proposals directed to that area.

Lunar Lander Technology Development

Here, NASA is seeking focused efforts to develop small and mid-to-large lunar lander technologies that have the potential to mature into technologies compatible with human lunar exploration. Technologies should address a gap

associated to long duration habitation on the lunar surface where temperatures range from -183° C in shadow regions (including night) to 100° C at the subsolar point. System technologies should be orientation insensitive; for example, lander side mounted radiators must provide their function regardless of lunar surface temperature condition. Additional difficulties include the deposition of dust that will degrade optical properties. Technologies are needed that allow a single mission to operate in all these environments. For example, keeping batteries between the ranges of 0-45° C during the day and night. Technologies should address mass, volume, and power usage relative to current solutions. Adding heaters can add significant vehicle mass to accommodate an additional power source and are not considered a novel architecture approach. Small size landers are around 500kg dissipating around 100W, to large size lander of 6000kg dissipating around 1kW. Human class landers are likely to have a variable heat load with an average of 4-6 kW which must be accommodated in all these environments.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

All of the technology developments listed above have relevance to NASA through the following projects: Lunar Gateway, Europa Clipper/Lander, Venus Landers, Lunar Landers, and long duration habitats (Moon, Mars. etc.).

References:

- Gates, D. W., Harrison, J. K., Jones, B. P., & Watkins, J. R. (1966). Lunar thermal environment. NASA Technical Memorandum, NASA TM X-53499.
- Ochoa, D. A., Miranda, B. M., Conger, B. C., & Trevino, L. A. (2006). Lunar EVA thermal environment challenges. *SAE Transactions*, 492-505.
- Thornton, J., Whittaker, W., Jones, H., Mackin, M., Barsa, R., & Gump, D. (2010). Thermal strategies for long duration mobile lunar surface missions. In *48th AIAA Aerospace Sciences Meeting Including the New Horizons Forum and Aerospace Exposition* (p. 798).
- Swanson, T. D., & Birur, G. C. (2003). NASA thermal control technologies for robotic spacecraft. *Applied thermal engineering*, *23*(9), 1055-1065.
- Bhandari, P., Birur, G. C., & Gram, M. B. (1996). *Mechanical Pumped Cooling Loop for Spacecraft Thermal Control* (No. 961488). SAE Technical Paper.
- Delil, A. A. M., Woering, A. A., & Verlaat, B. (2002). *Development of a Mechanically Pumped Two-Phase CO_2 Cooling Loop for the AMS-2 Tracker Experiment* (No. 2002-01-2465). SAE Technical Paper.
- Crepinsek, M., & Park, C. (2012). Experimental analysis of pump-assisted and capillary-driven dual-evaporators two-phase cooling loop. *Applied Thermal Engineering*, *38*, 133-142

Focus Area 18 Air Vehicle Technology

Lead MD: ARMD
Participating MD(s): STTR

This focus area includes tools and technologies that contribute to both the Advanced Air Vehicles Program (AAVP) and the Transformative Aeronautics Concepts Program (TACP) encompassing technologies in all six Strategic Thrusts within the NASA Aeronautics Mission Directorate (ARMD). AAVP studies, evaluates and develops technologies and capabilities for new aircraft systems, and also explores far-future concepts that hold promise for revolutionary air-travel improvements. Innovative AAVP design concepts for advanced vehicles integrate technologies focus on fuel burn, noise, emissions and intrinsic safety. The goal: to enable new aircraft to fly safer, faster, cleaner, quieter, and use fuel far more efficiently. Partnering with industry, academia, and other government agencies, AAVP pursues mutually beneficial collaborations to leverage opportunities for effective technology transition. TACP encourages

revolutionary concepts, creates the environment for researchers to experiment with new ideas, performs ground and small-scale flight tests, and drives rapid turnover into potential future concepts to enable aviation transformation. Research is organized to aggressively engage both the traditional aeronautics community and non-traditional partners. Although TACP focuses on sharply focused studies, the program provides flexibility for innovators to assess new-technology feasibility and provide the knowledge base for radical aeronautics advances.in noise reduction technology.

A1.01 Aerodynamic and Structural Efficiency - Integration of Flight Control with Aircraft Multidisciplinary Design Optimization (SBIR)
Lead Center: ARC
Participating Center(s): AFRC, LaRC

NASA is conducting fundamental aeronautics research to develop innovative ideas that can lead to next generation aircraft design concepts with improved performance and operation. There is an increasing interest in more integrated aircraft multidisciplinary design optimization (MDO) processes that can bring flight control design into the early stage of an aircraft design cycle. By taking advantage of flight control as an additional discipline for integration with the traditional disciplines of aerodynamics, structures, and propulsion into an aircraft design process, potentially novel aircraft concepts could be identified that may provide better performance and improved safety.

Increasingly, distributed flight control designs are being investigated as future options for next-generation aircraft. A distributed flight control design can have multi-functional capabilities to accomplish multiple vehicle performance and operation objectives. For example, the aircraft industry has developed flight control technologies for drag reduction and load alleviation for modern transports such as B787 using distributed flight control surfaces. Such technologies, when fully integrated into the traditional multidisciplinary aircraft design cycle, could bring benefits to future aircraft designs, such as NASA-funded Truss-Braced Wing aircraft that could reduce interference drag at the strut juncture and lower wing weight using distributed flight control surfaces. Distributed electric propulsion (DEP) has been proposed as a technology solution for future Urban Air Mobility (UAM). The NASA X-57 vehicle utilizes wing-mounted distributed propulsion, which presents a complex aircraft design that could benefit from a more integrated design solution that would leverage DEP as flight control effectors for both operational objectives, such as cruise and roll control, as well as performance objectives, such as drag optimization by altering spanwise lift. Gradually, more advanced aircraft concepts utilize distributed flight control design such as modern Airbus A350 with adaptive drooped hinge flaps and a wide variety of UAM aircraft. The current state-of-the-art does not account for integration of flight control systems into MDO processes. Some low-level effort of addressing flight control surface integration in an MDO process has already begun; however, the investigation does not address the trade study with flight control system actuator and sensor hardware and flight control laws, nor does it address other novel flight control systems such distributed electric propulsion.

Therefore, this subtopic seeks proposals that addresses flight control integration with aircraft MDO processes. Proposed subjects may include, but are not limited to, the following:

- Novel distributed flight control design concepts that can potentially reduce size, weight, and drag relative to the existing state-of-the-art, including concepts that can improve aerodynamic performance by exploring design options with relaxed static stability.
- Flight control integration with aircraft design that results in integrated aero-structural-control optimal design and control layout for optimal L/D, noise reduction, as well as suitable handling and ride quality in all flight phases. This should take into consideration aeroelasticity, propulsion, and flow physics, as necessary.
- Methods and tools for integrating flight control into aero-structural-propulsion design processes that include structural sizing, control layouts, interaction physics with other disciplines, flight control laws as aircraft design parameters, and trade-off between control system weight/power and aircraft performance.
- Novel aircraft design concepts that can demonstrate the benefits of integrated aircraft design solutions with flight control.

The expected Technology Readiness Level (TRL) range at completion of the project is 3 to 4.

Expected deliverables would be methods and tools on how to develop the proposed technologies. In addition, concepts of aircraft designs that demonstrate the benefit of the proposed technologies are desired.

Within the NASA Advanced Air Vehicle Program (AAVP), the Advanced Air Transport Technologies (AATT) Project is conducting research in distributed electric propulsion and adaptive wing technologies. Both of these research elements could benefit from this subtopic. Also, under the NASA AAVP, the Revolutionary Vertical Lift Technologies (RVLT) Project is conducting research in the area of UAM aircraft using distributed electric propulsion for Vertical Take-Off and Landing (VTOL). This subtopic would complement the research in the RVLT project.

References:

- Bret Stanford, "Optimization of an Aeroservoelastic Wing with Distributed Multiple Control Surfaces", 33rd AIAA Applied Aerodynamics Conference, AIAA-2015-2419, 2015.
- Nhan Nguyen, Kevin Reynolds, Eric Ting, Natalia Nguyen, "Distributed Propulsion Aircraft with Aeroelastic Wing Shaping Control for Improved Aerodynamic Efficiency", AIAA Journal of Aircraft, Vol.55, pp. 1122-1140, 2018.

A1.02 Quiet Performance - Airframe Noise Reduction (SBIR)
Lead Center: LaRC
Participating Center(s): GRC, LaRC

Innovative technologies and methods are necessary for the design and development of efficient, environmentally acceptable aircraft. In particular, the impact of aircraft noise on communities around airports is the predominant limiting factor on the growth of the nation's air transportation system. Successful reductions in aircraft noise would lead to wider community acceptance, lower airline operating costs where noise quotas and fees are employed, and increased potential for air traffic growth on a global scale.

In support of the Advanced Air Vehicles, Integrated Aviation Systems, and Transformative Aeronautics Concepts Programs, improvements in technologies and methods for noise prediction, measurement of acoustic and relevant flow field quantities on aircraft, noise propagation modeling, and noise control are needed for subsonic, transonic, and supersonic vehicles. This subtopic is seeking innovations specifically targeting airframe noise sources and noise sources due to the aerodynamic and acoustic interaction of airframe and engines.

State-of-the-art technologies for noise reduction for conventional transport aircraft are generally passive and do not incorporate advanced material systems or adaptive mechanisms that can modify their performance based on the noise state of the vehicle. Advanced material systems for airframe noise control are still in their infancy, especially in the context of robustness and being certifiable. Novel material systems that could be applied to component noise sources on the aircraft are needed, such as shape memory alloy actuators, or active or adaptive systems. In addition, future aircraft designs are envisioned that leverage noise benefits of complex geometrical configurations, such as engine integration with the airframe. However, efficient computational tools that enable rapid-turn evaluations of multiple configurations at the design stage are lacking. Numerical methods to study complex engine/airframe configurations are complex and difficult to leverage at the aircraft design stage where configuration details are not specified. Improvements to numerical methods and models for studying the noise aspects of advanced airframe configurations, including engine integration, would ease consideration of acoustics early in the design cycle, rather than leaving acoustics to the late or post design stages where noise control solutions are costly and less effective.

Therefore, this subtopic is seeking innovations in the following specific areas:

- Fundamental and applied computational fluid dynamics techniques for aeroacoustic analysis that can be adapted for design purposes.
- Prediction of aerodynamic noise sources, including those from the airframe and those that arise from significant interactions between airframe and propulsion systems (i.e., Propulsion Airframe Aeroacoustics).
- Prediction of sound propagation from the aircraft through a complex atmosphere to the ground. This should include interaction between noise sources and the airframe and its flowfield. Thus, acoustic shielding/scattering effects should be incorporated.
- Innovative source identification techniques for airframe noise sources, such as landing gear and high lift systems. This should also include turbulence details related to flow-induced noise typical of separated flow regions, vortices, shear layers, etc.

- Concepts for active and passive control of noise sources for conventional and advanced aircraft configurations, including adaptive flow control technologies, and noise control technology and methods that are enabled by advanced aircraft configurations, which also includes integrated airframe-propulsion control methodologies. Innovative acoustic liner and porous surface concepts for the reduction of airframe noise sources and/or propulsion/airframe interaction are solicited; however, engine nacelle liner applications are specifically excluded.
- Concepts for novel acoustic calibration sources for both open- and closed-wall wind tunnel testing. Such sources should provide well-defined acoustic characteristics—both with and without flow—for typical frequency ranges of interest in scale-model wind tunnel testing for the purposes of magnitude and phase calibration for both single microphones and microphone phased arrays.
- Development of synthesis and auditory display technologies for subjective assessments of aircraft community noise.

The deliverables would be low Technology Readiness Level (TRL) - between 3-5 - technologies that demonstrate a potential for component noise reduction or demonstrate characteristics that could be incorporated into a more sophisticated noise control solution for transport aircraft.

Within the Advanced Air Vehicles Program (AAVP), the Advanced Air Transport Technology (AATT) and Commercial Supersonic Technology (CST) Projects would both benefit from noise reduction technologies that would reduce the aircraft noise footprint at landing and takeoff. Configurations with novel engine placement, such as above the fuselage, can reduce the noise footprint, but technologies are needed to efficiently model the performance and noise impacts of these novel engine locations.

Within the Transformative Aeronautics Concepts Program (TACP), the Transformational Tools and Technologies (TTT) Project would benefit from tool developments to enhance the ability to consider acoustics earlier in the aircraft design process. The TTT project would also benefit from development and demonstration of simple material systems, such as advanced liner concepts with reduced drag or adaptive structures that reduce noise, as these component technologies could have application in numerous vehicle classes in the AAVP portfolio, including subsonic and supersonic transports, as well as vertical lift vehicles.

References:

- AATT: https://www.nasa.gov/aeroresearch/programs/aavp/aatt
- CST: https://www.nasa.gov/aeroresearch/programs/aavp/cst
- TTT: https://www.nasa.gov/aeroresearch/programs/tacp/ttt

A1.03 Low Emissions/Clean Power - Environmentally Responsible Propulsion (SBIR)
Lead Center: GRC
Participating Center(s): LaRC

Environmentally responsible propulsion allows high turbine engine performance with lower pollution and engines that are quiet. Achieving low emissions and finding new pathways to cleaner power are critical for the development of future air vehicles. Vehicles for subsonic and supersonic flight regimes will be required to operate on a variety of certified aircraft fuels and emit extremely low amounts of gaseous and particulate emissions to satisfy increasingly stringent emissions regulations. Future vehicles will be more fuel-efficient, which will result in smaller engine cores operating at higher pressures. Future combustors will also likely employ lean burn concepts that are more susceptible to combustion instabilities. Fundamental combustion research coupled with associated physics-based model development of combustion processes will provide the foundation for technology development critical for these vehicles.

Combustion involves multi-phase, multi-component fuel, turbulent, unsteady, 3-D, and reacting flows where much of the physics of the processes are not completely understood. Computational Fluid Dynamics (CFD) codes used for combustion do not currently have the predictive capability that is typically found for non-reacting flows. Low emissions combustion concepts require very rapid mixing of the fuel and air with a minimum pressure loss to achieve complete combustion in the smallest volume.

Specifically, this subtopic is seeking research that includes:

- Development of laser-based diagnostics for quantitative spatially and temporally resolved measurements of fuel/air ratio in reacting flows at elevated pressure.
- Development of ultra-sensitive instruments for determining the size-dependent mass of combustion generated particle emissions.
- Low emissions combustor concepts for small high-pressure engine cores.
- Development of miniature high-frequency fuel modulation valve for combustion instability control that is able to withstand the surrounding high-temperature air environment.
- Infusion/commercial potential. These developments will impact future aircraft engine combustor designs (e.g., lower emission, control instabilities). In addition, these developments may have commercial applications in other gas-turbine based industries, such as power generation and industrial burners. The modeling and results can and will be employed in current and future hydrocarbon rocket engine designs (e.g., improving combustion efficiency, ignition, stability, etc.).

The expected Technology Readiness Level (TRL) range at completion of the project is between 2-5.

The Transformational Tools and Technologies (TTT) Project is developing computer-based tools and models, as well as scientific knowledge that will lead to significant advances in our ability to understand and predict flight performance for a wide variety of air vehicles.

Examples of this research include the development of new computational tools that are used to predict the flow around vehicles and inside turbine engines. Another area of research that is of benefit to a number of vehicle types is improving the understanding and development of new types of strong and lightweight materials that are important for aviation.

Therefore, several major deliverables will be computer simulation software to predict the best and most effective combustor configuration, prototype flow control devices to control combustor efficiency, and sensor development for monitoring engine emissions and sound levels.

Environmentally responsible propulsion includes all these potential research areas:

- Fuels/Propellants; Thermal; Development Environments; Fluids; Metallics; Nanomaterials
- Actuators & Motors; Exciters/Igniters; Isolation/Protection/Shielding
- Software Tools (Analysis, Design); Operating Systems; Programming Languages; Visible; Infrared; Simulation & Modeling; Active Systems; Heat Exchange; Passive Systems; Diagnostics/Prognostics; Aerodynamics; 3D Imaging; Image Analysis
- Conversion; Generation; Sources (Renewable, Nonrenewable); Characterization; Models & Simulations (see also Testing & Evaluation)
- Detectors (see also Sensors); Lasers (Measuring/Sensing); Analytical Instruments (Solid, Liquid, Gas, Plasma, Energy; see also Sensors)

References:

- https://www.nasa.gov/aeroresearch/programs/tacp/ttt
- https://www.nasa.gov/aeroresearch/programs

A1.04 Electrified Aircraft Propulsion (SBIR)
Lead Center: GRC
Participating Center(s): AFRC, LaRC
Related Subtopic Pointer(s): T15.03

The critical technical need for lightweight, high-efficiency power distribution systems that have flight critical reliability have led to requirements for weight reduction by a factor of 2-3 as well as improved efficiency. Higher

efficiency reduces losses and makes thermal management more achievable in an aircraft. Another need for medium to large aircraft is the ability to operate at voltages above 600V. This capability results in reduced weight, however is called out specifically because it impacts all of the power system components. Technologies that address these gaps enable Electrified Aircraft Propulsion (EAP) which enables new aircraft configurations and capabilities for the point-to-point Urban Air Mobility (UAM) market and a new type of innovation for transport aircraft to reduce fuel consumption and emissions. EAP is an area of strong and growing interest in NASA's Aeronautics Research Mission Directorate (ARMD). There are emerging vehicle level efforts in the area of UAM/On-Demand Mobility like the X-57 electric airplane being built to demonstrate EAP advances applicable to thin haul/short haul aircraft markets, and an ongoing technology development subproject to enable EAP for single aisle aircraft.

NASA Projects working in the vehicle aspects of EAP include: Advanced Air Vehicles Program (AAVP)/Advanced Air Transport Technology (AATT) Projects, Integrated Aviation Systems Program (IASP)/ Flight Demonstrations & Capabilities (FDC) Project, AAVP/Revolutionary Vertical Lift Technology (RVLT) Project, and Transformative Aeronautics Concepts Program (TACP)/Convergent Aeronautics Solutions (CAS) Projects.

Turboelectric, hybrid electric, and all electric power generation as well as distributed propulsive power have been identified as candidate transformative aircraft configurations with reduced fuel consumption/energy use and emissions. However, components and management methods for power generation, distribution, and conversion are not currently available in the high-power ranges with the necessary efficiency, power density, electrical stability and safety required for thin haul/short haul, or transport-class aircraft.

Therefore, technical proposals are sought for the development of enabling power systems, turbofan engines, range extenders, electric machines, batteries, power converters, electrical fault management systems, protective devices (such as circuit breakers), and related materials that will be required for aircraft which use turboelectric, hybrid electric, or all electric power generation as part of the propulsion system.

Specifically, novel developments are sought in these areas:

- Light weight AC and DC electrical fault management systems and protective devices (such as circuit breakers)
- Aircraft power systems operating at or above 600V
- Turbofan engines in which > 20% of power is extracted electrically
- Lightweight multifunctional additively-manufactured heat exchangers/recuperators (using metallics and/or ceramics) which can operate up to 1400° F
- Range extenders which consume fuel and produce electricity with significantly higher efficiency than available turbogenerator or diesel generators
- Electric machines (motors/generators) with efficiency > 98% and specific power > 13 kW/kg
- Magnetic gear systems with gear ratio on the order of 10:1 that connect high speed motors to lower speed propulsors
- Converters (inverters/rectifiers) with efficiency > 99% and specific power > 19kW/kg
- Energy storage systems with specific energy > 400Whr/kg at the system level and cycle life > 10,000 cycles. This SBIR seeks energy storage technologies in the Technology Readiness Level (TRL) range from 2 to 6 for shorter-term infusion into NASA Aeronautics projects. Submissions for technologies in the TRL range 1 to 4 should seek partnering academic institutions and apply to STTR subtopic T15.03 - Electrified Aircraft Propulsion Energy Storage.
- Soft magnetic material with high magnetic saturation and/or lower losses for 100kHz-300kHz operation
- Hard magnetic materials with an energy product greater than neodymium iron boron
- Conductors with a specific resistivity less than copper or aluminum and cable insulation materials with increased dielectric breakdown strength as well as significantly higher thermal conductivity ($\geq$ 1W/m·K) and resistance to ageing effects such as corona, ozone, humidity and dust operating at greater than 3kV
- Advanced material systems for motors

Individual components should target the 15kW-3MW size range and would be combined into power systems in the range of 200kW-10MW total power.

References:

- Electrified Aircraft Propulsion (EAP) is called out as a key part of Thrust 4 in the ARMD strategic plan: https://www.nasa.gov/aeroresearch/strategy
- Overview of NASA's EAP Research for Large Subsonic Aircraft: https://ntrs.nasa.gov/search.jsp?R=20170006235
- NASA X-57 Project: https://www.nasa.gov/aeroresearch/X-57/technical/index.html

A1.05 Computational Tools and Methods (SBIR)
Lead Center: LaRC
Participating Center(s): ARC, GRC

Computational Fluid Dynamics (CFD) plays an important role in the design and development of a vast array of aerospace vehicles, from commercial transports to space systems. With the ever-increasing computational power and the usage of higher fidelity, fast CFD tools and processes will significantly improve the aerodynamic performance of airframe and propulsion systems, as well as greatly reduce non-recurring costs associated with ground-based and flight testing. Historically, the growth of CFD accuracy has allowed NASA and other organizations, including commercial companies, to reduce wind tunnel and single engine component tests. Going forward, increased CFD fidelity for complete vehicle or engine configurations holds the promise of significantly reducing development costs by enabling certification by analysis. Confidence in fast, accurate CFD allows engineers to reach out of their existing design space and accelerate technology maturation schedules.

NASA's CFD Vision 2030 Study (https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20140003093.pdf) highlighted the many shortcomings in the existing technologies used for conducting high-fidelity simulations and made specific recommendations for investments necessary to overcome these challenges. One of the technology areas recognized by the vision study is computational tools for knowledge extraction and visualization of simulation data. Managing the vast amounts of data generated by current and future large-scale simulations will continue to be problematic and is becoming increasingly complex due to changing high performance computing (HPC) hardware. These include effective, intuitive, and interactive visualization of high-resolution simulations, real time analysis and management of large data bases generated by these simulations. Currently, there is no NASA program or project that is directly funding this technology area. However, various programs and projects of NASA missions use CFD for advanced aircraft concepts, launch vehicle design, and planetary entry vehicles. Therefore, the developed technology will enable design decisions by Aeronautics Research Mission Directorate (ARMD) and Human Exploration Operations Mission Directorate (HEOMD).

As HPC systems become faster and more efficient, a single unsteady high-fidelity CFD simulation using more complicated physical models to solve for the flow about a complete aerospace system (e.g., airplane with full engine simulation, aircraft in maneuvering flight, space vehicle launch sequence, etc.), using a much higher number of grid points (~10-100 billion), will become commonplace in the 2030 timeframe. Effective use (visualization and in-situ analysis) of these very large high-fidelity CFD simulations will be paramount. Software and hardware methods to handle data input/output (I/O), memory, and storage for these simulations on emerging HPC systems must improve. Likewise, effective CFD visualization software algorithms and innovative information presentation, particularly for solutions obtained by using high-order methods, are also lacking. Continually increasing HPC capabilities will allow for the rapid and systematic generation of a large number (perhaps, thousands) of CFD simulations for flow physics exploration, trend analysis, experimental test design, design space exploration, etc.

The main goal of this subtopic will be to collect, synthesize, and interrogate this large array of computational data to make engineering decisions in real time. This is complicated by a lack of data standards which makes collection and analysis of results from different codes, researchers, and organizations difficult, time consuming, and prone to error. At the same time, there are no robust and effective techniques for distilling the important information contained in large collections of CFD simulation data into reduced-order models or meta-models that can be used for rapid predictive assessments of operational scenarios, such as the correlation of flow conditions with vehicle performance degradation or engine component failures, or assessments of engineering trade-offs as is required in typical design studies.

Thus, there are a number of technology gaps and impediments that must be overcome to efficiently analyze and utilize CFD simulations in the 2030 timeframe, and this solicitation seeks innovative approaches to overcome the challenges associated with knowledge extraction. Proposers can address one or more technology areas addressed above within the overall knowledge extraction and data analytics topic. The final deliverable will be a software tool that could be used in conjunction with fluid dynamic simulations—including multidisciplinary, such as aeroelastic - to extract information of relevance to NASA missions.

Another focused technology area for which innovative proposals are solicited requires the development of an automated aircraft drag optimization method using knowledge-based design tools. Current aircraft design methods are usually based on computationally intensive optimization approaches that are slow, or faster knowledge-based methods that can improve designs but may not find the optimal solution. The proposer shall develop a fast CFD-based design optimization method that couples NASA's CDISC (https://software.nasa.gov/software/LAR-18693-1) knowledge-based design tools with an optimization driver framework to enable rapid drag-based optimization. The method shall include robust mesh movement capabilities. The new design optimization method shall be demonstrated for configurations of interest to NASA aeronautics programs (http://www.aeronautics.nasa.gov/programs.htm). The deliverable shall be a software tool for computationally efficient design optimization studies.

The expected Technology Readiness Level (TRL) range at completion of the project is between 3-6.

References:

- https://www.nasa.gov/aeroresearch/programs/aavp
- https://www.nasa.gov/aeroresearch/programs/tacp

A1.06 Vertical Lift Technology and Urban Air Mobility (SBIR)
Lead Center: GRC
Participating Center(s):
Related Subtopic Pointer(s): A1.09

Urban Air Mobility (UAM) is a concept for air transportation around metropolitan areas consisting of passenger-carrying operations. An emerging UAM market will require a high density of vertical takeoff and landing (VTOL) operations for on-demand, affordable, quiet, and fast transportation in a scalable and conveniently-accessible "vertiport" network. It is envisioned that UAM will provide increased mobility within a given metropolitan area by flying faster and using shorter and more direct routing as compared to ground vehicles.

The expanding UAM vehicle industry has generated a significant level of enthusiasm amongst aviation designers, manufacturers and researchers. This industry is determined to change the urban transportation paradigm from traditional ground-based vehicles (cars, taxis, buses) to air-based UAM vehicles which can be summoned similar to conventional taxi services. These new UAM vehicles are designed to be small, lightweight and operate autonomously without user interaction.

There are many unknowns as to how the industry will mature. A critical challenge for UAM market growth is to gain public acceptance for being as safe as, or safer than, commercial air travel and automotive transportation. Crash avoidance technologies in autonomous systems are continuously in development. Despite these efforts, crashes will likely occur, and subsidiary crashworthy systems must be in place to mitigate the injurious loads. It is in this area of crash mitigation where much attention is needed.

Crashworthiness involves a system-level approach to account for all energy absorbing contributions from the landing gear/skids, airframe, seat, and restraints. Current landing gear struts on utility helicopters (the category most closely related to UAM) are primarily oleo-pneumatic and limit loads for a variety of sink rates. These legacy struts may be too heavy for UAM concepts, and simpler and more lightweight struts are required. Energy absorbing struts must withstand normal load profiles but activate in the event of a hard landing or crash. Concepts that have been studied to date for crew seats or landing gear include, but are not limited to, tubes with crushable cores, inversion tubes, wire benders, and tube crimpers.

The technology solutions proposed under this subtopic will address the need to demonstrate vehicle safety under a hard landing or crash condition for UAM aircraft. Examples of preliminary missions, requirements, and concepts for UAM aircraft can be found in references. It is conceivable that the capability of a Ballistic Recovery System would be negated if deployed at these proposed low altitudes. Vertical drop testing conducted at NASA Langley Research Center in 2018 has shown that from a height of only 14 ft., which generates an impact velocity of ~29 ft/s, occupant loading limits were exceeded for occupant protection specifications in both General Aviation and rotorcraft. While not directly comparable due to the unknowns in UAM performance and regulations, these tests demonstrate that comparable impacts from even low levels have the capability of causing injury.

The expected technology readiness level (TRL) range at completion of the project is between 3-6.

This subtopic is relevant to NASA's Aeronautics Research Mission Directorate (ARMD) Revolutionary Vertical Lift Technology (RVLT) Project under the Advanced Air Vehicle Program (AAVP). The goal of the RVLT Project is to develop and validate tools, technologies and concepts to overcome key barriers for vertical lift vehicles. The scope encompasses technologies that address noise, speed, mobility, payload, efficiency, environment and safety for both conventional & non-conventional vertical lift configurations. This subtopic directly aligns with the mission, goals and scope in addressing safety of & non-conventional vertical lift configurations.

Phase I of the SBIR should review these concepts and their utilization for UAM applications. Phase II of the SBIR should include the development of novel prototype energy absorbing concepts for VTOL UAM vehicle crashworthiness/survivability.

References:

- Johnson, W., Silva, C., and Solis, E.: Concept Vehicles for Air Taxi Operations. AHS Specialists Conference on Aeromechanics Design for Transformative Vertical Flight, January 16-18, 2018. San Francisco, CA.
- Patterson, M.D., Antcliff, K.R. and Kohlman, L.W.: A Proposed Approach to Studying Urban Air Mobility Missions Including an Initial Exploration. AHS International 74th Annual Forum & Technology Display. May 14-18, 2018. Phoenix, AZ.
- Silva, C., Johnson, W., Antcliff, K.R, and Patterson, M.D.: VTOL Urban Air Mobility Concept Vehicles for Technology Development. AIAA Aviation and Aeronautics Forum and Exposition. June 25–29, 2018. Atlanta, GA.
- Littell, J. D.A.: Discussion of Vehicle Safety in Autonomous Electric Vertical Take-off and Landing (eVTOL) Vehicles. 8th Biennial Autonomous VTOL Technical Meeting and 6th Annual Electric VTOL Symposium. January 29-31, 2019. Mesa, AZ (Forthcoming).
- Littell, J.D. and Annett, M.S. The Evaluation of Anthropomorphic Test Device Response under Vertical Loading. AHS International 74th Annual Forum & Technology Display. May 14-18, 2018. Phoenix, AZ.

A1.07 Propulsion Efficiency - Propulsion Materials and Structures (SBIR)
Lead Center: GRC
Participating Center(s): GRC

Materials and Structures research and development are a key contributor to NASA’s Aeronautics Research Mission Directorate (ARMD), impacting the development of advanced propulsion systems for Next Generation aircraft. Proposals are sought for advanced materials and structures technologies that will be enabling for new propulsion systems for subsonic transport vehicles with high levels of thermal, transmission, and propulsive efficiency.

For future aircraft with hybrid electric or all electric propulsion systems, advanced materials technology is needed for power components including electric machines and power cables. Integrated computational and experimental approaches are needed that can reduce the time necessary for development, testing, and validation of new materials systems and components. Advanced high-pressure-ratio compact gas turbine engines will include components of sufficiently compact size so that new approaches to processing and advanced manufacturing will be needed. Temperature capability, thermo-mechanical performance, environmental durability, reliability and cost-effectiveness are important considerations.

The increased use of various types of modeling to improve R&D effectiveness and enable more rapid and revolutionary materials design has been identified as critical. NASA recently sponsored a study to define a potential 25-year goal for integrated, multiscale modeling of materials and systems to accelerate the pace and reduce the expense of innovation in future aeronautical systems. Through a series of surveys, workshops, and validation exercises, this study identified critical cultural changes and gaps facing the multiscale modeling community. The results of this study were published in a NASA report [Vision 2040: A Roadmap for Integrated, Multiscale Modeling and Simulation of Materials and Systems, NASA/CR-2018-219771]. Some of the critical gaps identified in this report are:

- Under-development of physics-based models that link length and time scales
- Inability to conduct real time characterization at appropriate length and time scales
- Lack of optimization methods that bridge scales
- Lack of models that compute input sensitivities and propagate uncertainties
- Lack of verification and validation methods and data

Proposals emphasizing modeling can address topics which shall advance gaps in the Vision 2040 report. The range of topics could include data management, data analytics, machine learning, linkage and integration across spatiotemporal scales, and characterization of materials over their lifecycle. Proposals may address any material class associated with aeronautics propulsion, multiscale modeling and measurements, multiscale optimization methods, and verification and validation of models and methods. However, approaches should rely on iterative, predictive methods which integrate experiments and simulations to describe the behavior and response of materials at various length and time scales.

Specific technology areas of interest this year include:

- Computational materials and multiscale modeling tools, including methods to predict properties, and/or durability of propulsion materials based upon chemistry and processing for conventional as well as functionally-graded, nanostructured, multifunctional and adaptive materials.
- Robust and efficient methods/tools to design and model advanced propulsion system materials and structures at all scale levels, including approaches that are adaptable for a multi-scale framework.
- Multiscale design tools that integrate novel materials, mechanism design, and structural subcomponent design into system level designs.
- Advancing technology for ceramic matrix composites (CMCs) and their environmental barrier coatings (EBCs) for gas turbine engine components operating at 1482° C (2700° F) or higher. Focus areas include increased thermomechanical durability, increased resistance to environmental interactions, cost-effectiveness of processing and manufacturing, and improved approaches to component fabrication and integration. Computational tools and integrated experimental/computational methods are sought, including models/tools to predict degradation and failure mechanisms.
- Additive Manufacturing and other advanced processing/manufacturing approaches for propulsion system structural components or materials to enable improved engine efficiency through decreasing weight and/or improving component design, properties and performance.
- Soft magnetic material with high magnetic saturation and/or lower losses for 100 – 300 kHz operation, hard magnetic materials with an energy product greater than neodymium iron boron, conductors with a specific resistivity less than copper or aluminum, and cable insulation materials with increased dielectric breakdown strength, and significantly higher thermal conductivity ($\geq$ 1W/m·K) and resistance to ageing effects such as corona, ozone, humidity and dust operating at greater than 3kV.
- Novel materials systems and structures to enable functionality, such as power harvesting, thermal management, self-sensing, and actuation. Approaches may include use of nanotechnology and novel processing to tailor and control properties such as thermal conductivity, electrical conductivity, thermoelectric response, microstructure and porosity, and shape memory behavior.
- Design and development of unique materials such as shape memory alloys and high entropy alloys for aeronautics structures and components.
- Propulsion aeromechanics, damping devices, and analysis and mistuning analysis for turbomachinery rotating blades.

The expected technology readiness level (TRL) range at completion of the project is 2 to 4.

NASA's intent is to select proposals that have the potential to move a critical technology beyond Phase II SBIR funding and transition it to Phase III, where NASA's ARMD programs, Other Government Agencies, or a commercial entity in the aeronautics sector can fund further maturation as needed, leading to actual usage. Each proposed topic area could yield a different deliverable. Some Phase IIs will yield models supported with experimental data, some can yield software related to a model that was developed, some will yield a material system or subcomponent that has been demonstrated to have better properties/performance (ability to operate at a higher temperature, carry more current, modeling tools for incorporation in software, etc.). The goal is to have funded proposals yield products that enable enhanced propulsion systems (with the possibility that additional funding via Phase III required to reach this goal), and the technology is of strong interest to NASA and/or commercial entities.

References:

- Vision 2040: A Roadmap for Integrated, Multiscale Modeling and Simulation of Materials and Systems, NASA/CR-2018-219771 https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20180002010.pdf
- NASA Hybrid Gas-Electric Propulsion (HGEP) Subproject: https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20170004515.pdf
- Hybrid Electric Propulsion:
  - https://www1.grc.nasa.gov/aeronautics/hep/
  - https://www.grc.nasa.gov/www/cdtb/aboutus/workshop2016/HybridElectricPropulsionReportOut-LCCPCDRoadmapWorkshop.pdf
- NASA Transformational Tools and Technologies (TTT) Project: https://www.nasa.gov/aeroresearch/programs/tacp/ttt/description
- NASA Convergent Aeronautics Solutions (CAS) Project:
  - https://www.nasa.gov/aeroresearch/programs/tacp/cas
  - https://nari.arc.nasa.gov/
  - https://www.nasa.gov/aeroresearch/programs/tacp
  - https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20170006909.pdf

A1.08 Aeronautics Ground Test and Measurement Technologies (SBIR)
Lead Center: LaRC
Participating Center(s): ARC, GRC

NASA's aeroscience ground test facilities include wind tunnels, air-breathing engine test facilities, and simulation and loads laboratories. They play an integral role in the design, development, evaluation, and analysis of advanced aerospace technologies and vehicles. These facilities provide critical data and fundamental insight required to understand complex phenomena and support the advancement of computational tools for modeling and simulation. The primary objective of this subtopic is to develop innovative tools and technologies that can be applied in NASA's aeroscience ground test facilities to revolutionize testing and measurement capabilities and improve utilization and efficiency. Of primary interest are technologies which can be applied to NASA's portfolio of large-scale ground test facilities. For this solicitation, NASA seeks proposals for innovative research and development in the following areas:

Flow Diagnostics for High-Speed Flows

Spatially- and temporally-resolved molecular-based diagnostics are sought for high-speed wind-tunnel flows (supersonic, hypersonic), both with and without combustion:

- Improved measurement capabilities are needed for velocity, temperature, density and species concentrations in harsh wind tunnel environments, including (but not limited to) short-duration facilities and luminous flows. Measurement systems should be reliable and robust and able to be implemented in multiple wind tunnel facilities and facility types including blow-down tunnels, shock tubes, shock tunnels and arc-jets. Some of the target facilities have naturally occurring species such as NO, O, and N atoms. Some facilities have thermal nonequilibrium. Combustion applications use both hydrogen and hydrocarbon fuels. Planar or volumetric measurements are preferred but highly-accurate pointwise measurements will also be considered. Ability to measure multiple parameters simultaneously (for example: (i) NO and O concentrations or (ii) temperature and fuel or combustion intermediate concentration) are desirable. The ability to time resolve unsteady flows

so that spectra of the measured phenomena can be obtained is also desirable. Measurement systems should be validated against accepted standards (thermocouples, calibration flames, etc.) to determine measurement accuracy and precision. Proposals should project accuracies and precisions of the proposed measurement system(s) based on prior work.

- Improved measurement capabilities are also needed to obtain high-bandwidth heat flux measurements to complement global techniques like phosphor thermography and provide time-resolved data to investigate boundary layer transition, instability waves, and shock-boundary layer interactions. The application of current approaches is often limited by the sensitivity of the sensor to radiated heat, humidity, and low-tolerance to damage from abrasion. NASA seeks innovative techniques to measure heat flux that that are robust and can easily applied to or integrated into model surfaces without creating steps or significantly changing the surface profile or mold line of the test article.

Current capabilities for measuring velocity, temperature, density, and species concentrations in harsh wind tunnel environments are effective but limited to sample rates of 10 hertz or less. The run times for the facilities where these techniques are used is extremely short (milliseconds to several seconds); therefore, the amount of data that can be acquired in a given amount of time or number of runs is limited. Technology is needed to enable parameters like velocity, temperature, density, and species concentration to be sampled simultaneously and at higher rates to obtain an appropriate amount of data to improve statistical error and provide detailed information about the time-varying nature of these flow fields. Similarly, there are existing technologies for heat flux measurements, but they are often limited by the sensitivity of the sensor to radiated heat, humidity, and low-tolerance to damage from abrasion. Improvements to the existing technology are needed to reduce the sensitivity of existing heat-flux sensors to these factors or entirely new technology and approaches need to be developed to enable heat flux measurements to be made routinely and consistently. The technologies described above are all critical for evaluating and analyzing high-speed vehicle concepts and technologies.

Global Shear Stress Measurements

Shear stress is an important parameter for characterizing the interaction between a fluid and a surface over which it is moving. Quantitative measurements of shear stress provide information about features like flow separation as well as boundary conditions for modeling and simulation tools. Currently, shear stress is measured at discrete locations using different sensors and probes; however, determining the proper measurement locations a priori is a significant challenge, especially for advanced configurations and in regions where the flow is highly unsteady.

To address these known difficulties, NASA seeks innovative techniques to obtain time-averaged, quantitative measurements of the global shear stress field on test articles in ground test facilities. Techniques are needed for each speed regime (subsonic, transonic, supersonic, and hypersonic) with a spatial resolution of 1 mm or less and applicable on complex geometries and vertical surfaces.

Currently, shear stress is measured at discrete locations using different sensors and probes; however, determining the proper measurement locations a priori is a significant challenge, especially for advanced configurations and in regions where the flow is highly unsteady. As such, technology is needed to provide a global picture of the shear-stress field that can be used to validate computational tools and methods and aid in determining where to place discrete shear stress measurement devices and/or make off-body flow field measurements.

The expected Technology Readiness Level (TRL) range at completion of the project is 1 to 4.

This subtopic is cross-cutting and as such, has the potential to improve the measurement capabilities of numerous projects within NASA's Aeronautics Research Mission Directorate (ARMD) that utilize ground-based test facilities for their R&D activities, including:

- Advanced Air Transport Technology (AATT) Project
- Aeronautics Test and Evaluation Project
- Commercial Supersonic Transport (CST) Project
- Hypersonics Technology Project
- Revolutionary Vertical Lift Technology (RVLT) Project

- Transformative Tools and Technologies (TTT) Project

While the main deliverable for this topic will be prototype systems based on the measurement techniques and approaches developed by the company, these systems will inherently include hardware and software that can be used by NASA. In cases where a prototype system cannot be produced, the research itself will provide a wealth of information that will hopefully advance the state-of-the-art and can be used by others.

References:

- https://www.nasa.gov/aeroresearch/programs/aavp/aetc/ground-facilities

A1.09 Vehicle Safety - Internal Situational Awareness and Response (SBIR)
Lead Center: GRC
Participating Center(s): AFRC, ARC, LaRC
Related Subtopic Pointer(s): A3.03, A3.04, A2.02

Achieving a vision for a safer and more efficient National Airspace (NAS) with increasing traffic and the introduction of new vehicle types requires increasingly intelligent vehicle systems able to respond to complex and changing environments in a resilient and trustworthy manner. Future air vehicles, especially autonomous vehicles and those that support Urban Air Mobility (UAM), must operate with a high degree of awareness of their own well-being, and possess the internal intelligence to provide warning and potentially take action in response to off-nominal states. A vehicle's capability to independently assure safety may be the only recourse in some situations and addresses the recurring issue of inappropriate crew response. Further, early warning of impending maintenance conditions reduces maintenance cost and vehicle down-time through improved vehicle availability and throughput. Understanding the vehicle state also has impact on vehicle performance, efficiency, and environmental impact.

It is predominately left to pilots (not the vehicle) to interpret current state and infer future states based on experience and expertise. Commercial Aviation Safety Team (CAST), FAA, NTSB, and the NRC have called for research on systems that can predict the state of the aircraft, including the state of autonomous systems, to provide notifications of trending to unsafe states. In order for there to be trust in autonomy, vehicle situational awareness and response needs to be tailored for independent autonomous systems without human intervention. There has been development in component health management technology with some adoption; integrated subsystem/vehicle system full-field health management is limited. Significant new capabilities are needed to enable safe vehicle operation in the NAS independent of human intervention.

This subtopic seeks technologies to enable intelligent vehicle systems, including subsystems such as airframes, propulsion, and avionics, with an internal situational awareness and ability to respond to off-nominal conditions for piloted vehicles augmented with autonomous capabilities, as well as increasingly autonomous vehicle systems (including On Demand Mobility/Urban Air Mobility and vertical lift vehicles). Specific areas of interest include:

- Networked sensors and algorithms to provide necessary vehicle full-field state information ranging from the component level to the subsystem and system level.
- On-board hardware and software systems that are modular, scalable, redundant, high reliability, and secure with minimal vehicle impact.
- Information fusion technologies to integrate information from multiple, disparate sources and evaluate that information to determine health and operational state.
- Diagnostic and prognostic technologies that inform decision-making functions with critical markers trending to unsafe state.
- Decision-making algorithms and approaches to enable trustworthy real-time operations, take preventive actions as needed in complex uncertain environments, and appropriately communicate status to other components of the NAS.
- Integrated systems technologies that enable the mitigation of multiple hazards, while effectively dealing with uncertainties and unexpected conditions.
- Approaches that combine improved in-flight vehicle state safety awareness with adaptive methods to achieve improved efficiency, performance, and reduced environmental impact.

- Methods that significantly enhance the fidelity and relevance of information provided to ground systems by the vehicle in-flight for use in on-demand maintenance.

The expected Technology Readiness Level (TRL) range at completion of the project is 3 to 6.

This technology development is directly relevant to NASA's Aeronautics Research Mission Directorate (ARMD) Thrust 6 Autonomy Roadmap in order to allow more intelligent vehicle systems and has strong relevance to NASA's autonomy activities. NASA also plans to have an increasing role in the expanding market of ODM/UAM. In these fields, technologies that enable vehicle situational awareness and response will be enabling for NASA to carry out its future missions across a range of ARMD projects. This includes operations not only at the vehicle level, but at the subsystem and component level as well. The approach is to mature technology through this subtopic for ongoing implementation into NASA missions.

References:

- ARMD Strategic Thrust 6: Assured Autonomy for Aviation Transformation, Vision and Roadmap, M.Ballin, June 2016, https://nari.arc.nasa.gov/sites/default/files/attachments/ARMD%20ST%206%20Roadmap%20Webinar%20Briefingv2.pdf
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- M. D. Moore, K. Goodrich, J. Viken, J. Smith, B. Fredericks, T. Trani, J. Barraclough, B. German, and M. Patterson, "High-Speed Mobility through On-Demand Aviation," 2013 Aviation Technology, Integration, and Operations Conference, AIAA AVIATION Forum, AIAA013-4373, Aug. 2013. 10.2514/6.2013-4373
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A1.10 Hypersonic Technology - Innovative Manufacturing for High Temperature Structures (SBIR)
Lead Center: LaRC
Participating Center(s): LaRC
Related Subtopic Pointer(s): H5.02, Z7.03

Hypersonic aircraft have structural challenges that require significant advances in materials and in their manufacturing processes. Because a hypersonic aircraft must be streamlined for low drag, it is volume limited with sharp leading edges and thin wings, and it has tight integration of the propulsion system and the airframe.

Aerodynamic friction produces external structure temperatures above 1,000° C, and heating of the internal surfaces of propulsion structures necessitates their cooling. Therefore, the low mass/high structural efficiency requirement for subsonic aircraft is made more complex by these thermal requirements.

The focus of this subtopic is advanced manufacturing (e.g., additive) that enables cost-effective fabrication of structurally efficient components such as wing, control surfaces, and fuselage structures that operate as hot structures for multiple flights. Also, of interest are innovative manufacturing approaches for engine panels with integral cooling passages that are lightweight and can accommodate thermal growth mismatches with their manifolds and adjacent panels. Manufacturing approaches that can fabricate complex thermal management devices, such as heat-pipe-cooled leading edges—that serve as "heat spreaders" to eliminate hot spots from aerodynamic heating—are also within scope of this subtopic.

Further, the current state-of-the-art for hypersonic hot structures is either nickel-based superalloys or high temperature ceramic-matrix composites. Nickel-based superalloys are heavy and are also limited in their maximum use temperatures. Ceramic-matrix composites have higher use temperatures, but are brittle, are difficult to manufacture, and have limited useful life in service. Hence, an additional goal of this subtopic is to discover new materials that are lighter in weight but capable of repeatedly reaching service temperatures up to 1,000° C through use of advanced manufacturing techniques like additive manufacturing.

Specifically, this subtopic is seeking advanced metallic materials that:

- Can replace current state-of-the-art nickel-based alloys with new alloys that are lower in weight (e.g., gamma titanium aluminides, beryllium-containing alloys, refractory alloys, high temperature metal matrix composites, and oxide dispersion-strengthened alloys).
- Are capable of surviving multiple excursions to temperatures >1,000° C.
- Are resistant to high temperature oxidation.
- Are tough enough to handle thermal shock from sudden changes in temperatures under load.

In combination with these advanced materials and advanced manufacturing methods (including additive manufacturing approaches), this subtopic is seeking capabilities that can: Fabricate complex geometries, such as incorporating cooling passages or integral heat pipes for thermal management; or, providing novel shape controls that enable resiliency and flexibility for seals or connecting hot structures:

- Provide transition from a high-temperature low coefficient of thermal expansion material to a higher coefficient of thermal expansion material, such as connectors between dissimilar materials.
- Provide multi-functionality, incorporating thermal management and structural load capabilities into an integrated component.
- Enable novel materials through novel manufacturing approaches, such as in-situ reactions to form reinforcing phases within the metallic materials during processing; or, functionally graded structures with different materials in different parts of the same component.

The expected Technology Readiness Level (TRL) range at completion of the project is between 2-4.

High temperature materials and manufacturing of relevant shapes with these materials is a challenge that has not been fully exploited. Materials development is a long lead-time research area, and engaging innovation across a wider community through SBIR provides time to develop technologies that can be enabling for future hypersonic vehicles. Accordingly, during Phase I, it is expected that feasibility will be demonstrated to fabricate novel materials using advanced manufacturing methods that meet the environmental, thermal, and structural requirements for high temperature hypersonic vehicles. These would involve simple geometries and material characterizations that show the potential of the material and manufacturing approach selected. During Phase II, the process and materials should scale up to larger, more complex, and realistic components relevant to hypersonic hot structures applications.

Deliverables include understanding of the process and materials used, as well as realistic prototype components relevant to hypersonic hot structures applications.

References:

- https://www.nasa.gov/aeroresearch/programs/aavp/ht

T15.01 Distributed Electric Propulsion (DEP) Vehicles toward Urban Air Mobility (UAM) and Regional Airliners (STTR)
Lead Center: AFRC
Participating Center(s): ARC, GRC, LaRC
Related Subtopic Pointer(s): A1.04, T15.03

Distributed Electric Propulsion (DEP) aircraft employ multiple electric propulsors to achieve unprecedented performances in air vehicles. The propulsors could be ducted/un-ducted fans, propellers, cross-flow fans, etc. Some of the benefits identified using this propulsion system are reductions in fuel burn/energy usage, noise, emissions, and/or field length. A focus on full vehicle performance; stability and control prediction; and safe, efficient operation is considered a high priority. Addressing NASA's Aeronautics Research Mission Directorate (ARMD's) Strategic Thrust #3 (Ultra-Efficient Commercial Vehicles) and #4 (Transition to Low-Carbon Propulsion), innovative approaches in designing and analyzing DEP-enabled Urban Air Mobility (UAM) aircraft are investigated and encouraged. In support of these two Strategic Thrusts, the following DEP aircraft research areas are to be considered under this subtopic:

- *Explore DEP-enabled UAM aircraft concepts and designs* - passenger-carrying UAM vehicles will be required to operate safely and efficiently within an urban airspace setting. The study shall include vehicle system level assessment including feasibility, design, benefits, predicted performance, concept of operations and/or failure assessments.
- *Develop tools and methods to assess DEP-enabled UAM aircraft and its operation* - assessing a feasibility of UAM vehicle concept and operation requires reliable analytical, computational, experimental, and/or simulation tools and methods for safe and efficient operation. The study shall include computational, experimental, and/or simulation tools and methods in addressing safe and efficient operation of DEP-enabled UAM vehicles. The approach to validation of tools and methods should be discussed.
- *Develop low-noise DEP-enabled UAM aircraft* - community noise associated with UAM aircraft operating in an urban setting is very challenging and needs to be addressed from the system and component perspectives. The study shall address the noise problems of the UAM aircraft through vehicle design, noise reduction technologies and vehicle operations strategy. Effectiveness of proposed noise reduction approaches should be validated through reliable noise assessment tools and methods.
- *Develop ride quality and gust load alleviation technologies for safe operation of UAM aircraft* - dynamic gust encounters and wake vortices from neighboring aircraft can pose a very challenging problem for UAM operation. The ride quality of small UAM can suffer during gust or wake encounters. Structural loads on these aircraft could experience large excursions that could cause safety concerns. The study shall address relevant vehicle flight dynamics in the presence of gust and wake encounters and associated flight control technologies that could improve ride quality and gust load alleviation for UAM aircraft.

The expected outcome (Technology Readiness Level range: 2 to 3) of Phase I awards include but are not limited to:

- DEP-enabled UAM aircraft concept definition and system level assessment
- Initial development of analytical/computational/experimental/simulation tools and methods in assessing DEP enabled UAM aircraft and its operation; definition of approach to validate tools and methods

The expected outcome (Technology Readiness Level range: 4 to 6) of Phase II awards include but are not limited to:

- Detailed feasibility study and demonstration of the subscale hardware
- Refinement of tools and methods in assessing DEP-enabled UAM aircraft and its operation; validation of tools and methods developed in Phase I
- Experimental (e.g., wind tunnel, flight demo) results that assess the validity of the DEP aircraft concept

This research area is of particular interest to the following NASA programs:

- ARMD/Advanced Air Vehicles Program (AAVP)
- ARMD/Transformative Aeronautics Concepts Program (TACP)

References:

- NASA Aeronautics Strategic Implementation Plan, 2017 Update: https://www.nasa.gov/sites/default/files/atoms/files/sip-2017-03-23-17-high.pdf
- NASA ARMD – Advanced Air Transport Technology (AATT) Project: https://www.nasa.gov/aeroresearch/programs/aavp/aatt
- NASA ARMD – Revolutionary Vertical Lift Technology (RVLT) Project: https://www.nasa.gov/aeroresearch/programs/aavp
- NASA ARMD – Convergent Aeronautics Solutions (CAS) Project: https://www.nasa.gov/aeroresearch/programs/tacp/cas
- NASA ARMD article on Urban Air Mobility (UAM): https://www.nasa.gov/aero/nasa-embraces-urban-air-mobility

T15.03 Electrified Aircraft Propulsion Energy Storage (STTR)
Lead Center: GRC
Participating Center(s): AFRC, LaRC
Related Subtopic Pointer(s): A1.04

Proposals are sought for the development of enabling rechargeable batteries (or other types of energy storage) for Electrified Aircraft Propulsion (EAP). Two paths to improved battery performance are sought:

- Innovative thermal, structural, and electrical integration that reduce the mass fraction added when scaling from a battery cell to an integrated battery.
- Battery chemistry improvements that substantially enhance useable energy density, cycle life, life cycle cost, and safety.

This subtopic seeks technologies in the Technology Readiness Level (TRL) range 1 to 4 via partnerships between academic institutions and small businesses. Small businesses interested in proposing TRL 3-6 energy storage ideas should apply to SBIR subtopic A1.04 - Electrified Aircraft Propulsion.

Batteries and other energy storage systems with some combination of some or all of the following performance levels at the integrated battery pack level are sought (see below for additional details on these metrics):

- Specific energy > 400Whr/kg at the system level.
- Cycle life > 10,000 cycles.
- Prime flight quality and safety.
- Cost effective enough to close electric air services at a profit.

Battery pack level energy density means the amount of useable energy after derating for depth of discharge, cycle life, C rate limits, thermal constraints, and any other applicable limit to energy that can be used during the mission divided by the mass of the battery package (including the structure, safety devices, battery management system, and thermal management parts that are mounted to the battery). This will typically require cell level energy densities in the range to 600-800 W-hr/kg along with an innovative combination of those cells into a battery system. Alternate electrical energy storage approaches will also be considered.

All-electric conventional and vertical takeoff research vehicles that can carry one or two people have been demonstrated. In order to achieve commercial viability, improvements in batteries are required for the aircraft to have sufficient range, safety, and operational economics for regular service. Markets needs span urban air mobility (UAM), thin/short haul aviation, and commercial air transport vehicles which use EAP. Hybrid electric and all electric power generation as well as distributed propulsive power have been identified as candidate transformative aircraft configurations with reduced fuel consumption/energy use and emissions.

- Specific Energy: Approximately a factor of 2 improvement is needed. Current assessment of battery specific energy requirements for all-electric operations are in the 300-400 Wh/kg at the installed/pack level (installed means after derating for depth of discharge limit, cycle life, battery management, packaging, and thermal environment). This assumes the ability to quickly recharge between flights. Current state of the art (SOA) is about ≈ 160-170 Wh/kg (pack level). Li-ion batteries are nearing practical maximums so new chemistry(s) or energy storage types are likely required to meet all-electric UAM mission needs. All electric helicopters and regional passenger aircraft will likely need 600Wh/kg and 500-700Wh/kg (cell level) respectively. Approximately 30-40% Wh/kg is lost when cells are integrated into packs and installed.
- Cycle Life: A substantial improvement is needed. Current SOA is 1500-3000 cycles which lasts about 3 months for UAM.
- Prime Flight Quality and Safety: The expected reliability of an aviation system is probably a few orders of magnitude higher than an automotive application and safety considerations are a more significant driver – including time needed to get passengers out of danger. Discuss features and plans to ensure reliability.
- Cost: Justify features of the technology and implementation, including comparisons to SOA alternatives, which aid in ensuring that the vehicle concept and overall operations can close profitably.

EAP is an area of strong and growing interest in NASA's Aeronautics Research Mission Directorate (ARMD). Energy storage is an enabling technology for the UAM and Thin Haul segments of the effort. There are emerging vehicle level efforts in UAM/On-Demand Mobility like the X-57 electric airplane being built to demonstrate EAP advances applicable to thin and short haul aircraft markets, and an ongoing technology development subproject to enable EAP for single aisle aircraft.

NASA Projects working in the vehicle aspects of EAP include: Advanced Air Vehicles Program (AAVP)/Advanced Air Transport Technology (AATT) Projects, Integrated Aviation Systems Program (IASP)/ Flight Demonstrations & Capabilities (FDC) Project, AAVP/Revolutionary Vertical Lift Technology (RVLT) Project, and Transformative Aeronautics Concepts Program (TACP)/Convergent Aeronautics Solutions (CAS) Projects.

Key outcomes NASA intends to achieve in this research area are:

- Outcome for 2015-2025: Markets will begin to open for electrified small aircraft.
- Outcome for 2025-2035: Certified small aircraft fleets enabled by EAP will provide new mobility options. The decade may also see initial application of EAP on large aircraft.
- Outcome for > 2035: The prevalence of small-aircraft fleets with EAP will provide improved economics, performance, safety, and environmental impact, while growth in fleet operations of large aircraft with cleaner, more efficient alternative propulsion systems will substantially contribute to carbon reduction.

Deliverables most likely will include prototypes of energy storage units along with research and analysis addressing safety and cost considerations. In some cases, test data for safety may be a deliverable. Ideally, proposals would identify a technology pull area (with a market size estimate), how the proposed idea address the needs of the technology pull area, and then deliver a combination of analysis and prototypes that substantiate the idea's merit.

References:

- Electrified Aircraft Propulsion (EAP) is called out as a key part of Thrust 4 in the ARMD strategic plan: https://www.nasa.gov/aeroresearch/strategy
- NASA Urban Air Mobility (UAM): https://ntrs.nasa.gov/search.jsp?R=20170006235
- NASA X-57 Project: https://www.nasa.gov/aeroresearch/X-57/technical/index.html

Focus Area 19 Integrated Flight Systems

Lead MD: ARMD

This focus area includes technologies that contribute to the Integrated Aviation Systems Program's (IASP) objectives to demonstrate integrated concepts and technologies to a maturity level sufficient to reduce risk of implementation for stakeholders in the aviation community through the rigorous execution of highly complex flight tests and related experiments.

A2.01 Flight Test and Measurement Technologies (SBIR)
Lead Center: AFRC
Participating Center(s): ARC, GRC, LaRC
Related Subtopic Pointer(s): T13.01

NASA continues to use flight research as a critical element in the maturation of technology. This includes developing test techniques that improve the control of in-flight test conditions, expanding measurement and analysis methodologies, and improving test data acquisition and management with sensors and systems that have fast response, low volume, minimal intrusion, and high accuracy and reliability. By using state-of-the-art flight test techniques along with novel measurement and data acquisition technologies, NASA and the aerospace industry will be able to conduct flight research more effectively and also meet the challenges presented by NASA and industry's cutting edge research and development programs.

NASA's Flight Demonstrations and Capabilities Project supports a variety of flight regimes and vehicle types ranging from low speed, sub-sonic applications and electric propulsion, through transonic and high-speed flight regimes. Therefore, this solicitation can cover a wide range of flight conditions and vehicles. NASA also requires improved measurement and analysis techniques for acquisition of real-time, in-flight data used to determine aerodynamic, structural, flight control, and propulsion system performance characteristics. These data will also be used to provide information necessary to safely expand the flight and test envelopes of aerospace vehicles and components. This requirement includes the development of sensors for both in-situ and remote sensing to enhance the monitoring of test aircraft safety and atmospheric conditions during flight testing. This subtopic supports innovative flight platform development for use in hypersonic ground and flight testing, science missions and related subsystems development.

Flight test and measurement technologies proposals should significantly enhance the capabilities of major government and industry flight test facilities comparable to the following NASA aeronautical test facilities:

- Dryden Aeronautical Test Range
- Aero-Structures Flight Loads Laboratory
- Flight Research Simulation Laboratory

Proposals should address innovative methods and technologies to reduce costs and extend the health, maintainability, communication, and test techniques of these types of flight research support facilities.

Areas of interest emphasizing flight test and measurement technologies include:

- High performance, real time reconfigurable software techniques for data acquisition and processing associated with IP based commands and/or IP based data input/output streams.
- High efficiency digital telemetry techniques and/or systems to enable high data rate and high volume IP based telemetry for flight test. This includes Air-to-Air and Air-to-Ground communication.
- Architecture and tools for high integrity data capture and fusion.
- Real-time integration of multiple data sources from on-board, off-board, satellite, and ground-based measurement equipment.
- Innovative cybersecurity protocols for safe transmission of measurement data.
- Improved time-constrained situational awareness and decision support via integrated, secure, cloud-based web services for real-time decision-making.
- Prognostic and intelligent health monitoring for hybrid and/or all-electric propulsion systems using an adaptive embedded control system.
- Methods for accurately estimating and significantly extending the life of electric aircraft propulsion energy source (e.g., batteries, fuel cells, etc.).
- Test techniques, including optical-based measurement methods that capture data in various spectra, for conducting quantitative in-flight boundary layer flow visualization, Schlieren photography, near and far-field sonic boom determination, and atmospheric modeling as well as measurements of global surface pressure and shock wave propagation.
- Measurement technologies for in-flight steady and unsteady aerodynamics, juncture flow measurements, propulsion airframe integration, structural dynamics, stability and control, and propulsion system performance.
- Miniaturized fiber optic-fed measurement systems with low power requirements are desirable for migration to small business class jets or UAS platforms.
- Innovative techniques that enable safer operation of aircraft.
- Wireless sensor/sensing technologies and telecommunication that can be used for flight test instrumentation applications for manned and unmanned aircraft. This includes wireless (non-intrusion) power transferring techniques and/or wirelessly powering remote sensors.
- Innovative measurement methods that exploit autonomous remote sensing measurement technologies for supporting advanced flight testing.
- Fast imaging spectrometry that captures all dimensions (spatial/spectral/temporal) and can be used on UAS platforms.
- Innovative new flight platforms, airframes, and the associated subsystems development for use in all areas of flight tests and missions, e.g., X-planes testing, hypersonic testing, science missions, etc.

The emphasis of this subtopic is on flight test and flight test facility needs.

The expected TRL for this project is 1 to 6.

The technologies developed for this subtopic directly address the technical challenges in the Aeronautics Research Mission Directorate (ARMD) Integrated Aviation Systems Program (IASP) and Flight Demonstrations and Technologies (FDC) projects. The FDC conducts complex flight research demonstration to support different ARMD programs. FDC is seeking to enhance flight research and test capabilities necessary to address and achieve ARMD Strategic plan. Also, they could support Advanced Air Vehicle Program (AAVP) Projects: Commercial Supersonic Technology (CST), and AAVP - Aeronautic Evaluation & Test

References:

- https://sbir.nasa.gov/
- https://www.nasa.gov/mission_pages/lowboom/index.html
- https://www.nasa.gov/centers/armstrong/news/FactSheets/FS-109.html
- https://www.nasa.gov/centers/armstrong/research/X-56/index.html
- https://www.nasa.gov/centers/armstrong/news/FactSheets/FS-108-AFRC.html
- https://www.nasa.gov/centers/armstrong/features/shock_and_awesome.html
- https://technology-afrc.ndc.nasa.gov/featurestory/fiber-optic-sensing

A2.02 Unmanned Aircraft Systems (UAS) Technologies (SBIR)
Lead Center: AFRC
Participating Center(s): ARC, GRC, LaRC
Related Subtopic Pointer(s): A3.01, A3.02, A3.03, A3.04

Unmanned Aircraft Systems (UAS) offer advantages over manned aircraft for applications which are dangerous to humans, long in duration, require great precision, and require quick reaction. Examples of such applications include remote sensing, disaster response, delivery of goods, agricultural support, urban air mobility, and many others that are known or yet to be discovered. The future of UAS promises great economic and operational advantages by requiring less human participation, less human training, an ability to take-off and land at any location, and the ability to react to dynamic situations.

NASA is involved in research that would greatly benefit from breakthroughs in UAS capabilities. Flight research of basic aerodynamics and advanced aero-vehicle concepts would be revolutionized with an ability of UAS teams to cooperate and interact while making real time decisions based upon sensor data with little human oversight. Commercial industry would likewise be revolutionized with such abilities.

There are multiple barriers that are restricting greater use and application of UAS technologies in NASA research and in civil aviation. These barriers include, but are not limited to, the lack of methods, architectures, and tools that enable:

- The verification, validation, and certification of complex and/or nondeterministic systems.
- High level machine perception, cognition, and decision-making.
- Inexpensive secure and reliable communications.

This solicitation is intended to break through these and other barriers with innovative and high-risk research.

The Integrated Aviation Systems Program's work on UAS technology for the FY 2019 NASA SBIR solicitation is focused on breaking through barriers to enable greater use of UAS in NASA research and in civil aviation use. The following three research areas are the primary focus of this solicitation, but other closely related areas will also be considered for award. The primary research areas are:

- Verification, Validation, and Certification - New inexpensive methods of verification, validation, and certification need to be developed which enable application of complex systems to be certified for use in the

National Airspace System (NAS). Proposed research could include novel hardware and software architectures that enable or circumvent traditional verification and validation requirements.
- Sensing, Perception, Cognition, and Decision Making - Technologies need to be developed that provide the ability of UAS to detect and extract internal and external information of the vehicle, transform the raw data into abstract information that can be understood by machines or humans, and recognize patterns and make decisions based on the data and patterns.
- Inexpensive, Reliable, and Secure Communications - Inexpensive methods that ensure reliable and secure communications for increasingly interconnected and complex networks need to be developed that are immune from sophisticated cyber-physical attacks.

This subtopic is relevant to NASA Aeronautics Research Mission Directorate's Strategic Thrusts 5 and 6:

- UAS in the NAS
- Traveler
- ATM-X
- UTM

Phase I deliverables should include, but are not limited to:

- A final report clearly stating the technology challenge addressed, the state of the technology before the work was begun, the state of technology after the work was completed, the innovations that were made during the work period, the remaining barriers in the technology challenge, a plan to overcome the remaining barriers, and a plan to infuse the technology developments into UAS application.
- A technology demonstration in a simulation environment which clearly shows the benefits of the technology developed.
- A written plan to continue the technology development and/or to infuse the technology into the UAS market. This may be part of the final report.

Phase II deliverables should include, but are not limited to:

- A final report clearly stating the technology challenge addressed, the state of the technology before the work was begun, the state of technology after the work was completed, the innovations that were made during the work period, the remaining barriers in the technology challenge, a plan to overcome the remaining barriers, and a plan to infuse the technology developments into UAS application.
- A technology demonstration in a relevant flight environment which clearly shows the benefits of the technology developed. There should be evidence of infusing the technology into the UAS market or a clear written plan for near term infusion of the technology into the UAS market. This may be part of the final report.

The expected TRL for this project is 3 to 6.

References:

- https://www.hq.nasa.gov/office/aero/pdf/armd-strategic-implementation-plan.pdf
- https://www.faa.gov/uas/resources/uas_regulations_policy/media/Second_Edition_Integration_of_Civil_UAS_NAS_Roadmap_July%202018.pdf

Focus Area 20 Airspace Operations and Safety

Lead MD: ARMD

This focus area includes technologies addressing both the Airspace Operations and Safety Program (AOSP), and NASA's ARMD Strategic Thrusts 1, 5, and 6. AOSP is targeting system-wide operational benefits of high impact for NextGen and beyond, both in the areas of airspace operations and safety management. The SBIR Airspace Operations and Safety Topic is focused on research and technology development for enabling a modernized air transportation

system that will achieve much greater capacity and operational efficiency while maintaining or improving safety and other performance measures. This will include the integration of new types of vehicles such as unmanned vehicles, advanced subsonic aircraft, supersonic or commercial space vehicles; new types of business models or operations (i.e., urban air mobility); and new architectures or services for enabling these operations within the NAS

A3.01 Advanced Air Traffic Management System Concepts (SBIR)
Lead Center: ARC
Participating Center(s): LaRC
Related Subtopic Pointer(s): A3.02, A3.03, A3.04, A2.02

This subtopic addresses contributions towards advanced Air Traffic Management (ATM) systems and concepts with potential application in the near- to mid-term National Airspace System (2020-2035), focused on conventional, and mostly commercial, operations. The goals for this system are addressing established ATM challenges of improving efficiency, capacity, and throughput while minimizing negative environmental impact and maintaining or improving safety. The subtopic also seeks to accelerate the implementation of NASA technologies in the current and future National Airspace System (NAS), generally focused on conventional commercial air transportation operations and FAA air traffic management.

NASA's work in this area: Some NASA technologies that are relevant to this subtopic include, but are not limited to, Integrated Arrival, Departure, and Surface (IADS) capabilities, and routing and rerouting around weather from ground-based and cockpit-based systems.

The subtopic is seeking proposals that can apply novel and innovative technologies, concepts, models, algorithms, architectures, and tools towards bridging the gap from NASA's R&D to operational implementation, and should address such nearer-mid-term ATM challenges such as:

- Safe, end-to-end, Trajectory-Based Operations (TBO)
- Enabling and integrating existing independent systems and domains, and increasingly diverse and unconventional operations (gradually enabling the future integration of large Unmanned Vehicles, unconventional commercial airline business models, space traffic management, advanced subsonic and supersonic vehicles)
- Applying elements of the "service-based architecture concept" towards near-future/NextGen and beyond, NAS applications (the approach originally was pioneered for application in the UAS traffic management (UTM) domain).

Expected TRL for this project is 1 to 4.

Relevance to NASA

This technology is applicable to the Airspace Operations and Safety Program (AOSP)

References:

- https://www.nasa.gov/aeroresearch/programs/aosp

A3.02 Increasing Autonomy in the National Airspace System (NAS) (SBIR)
Lead Center: ARC
Participating Center(s): LaRC
Related Subtopic Pointer(s): A3.01, A3.03, A3.04, A2.02

NASA's future concepts for air transportation (2025+) will significantly expand the capabilities of airspace and vehicle management and are anticipated to increasingly rely on autonomy and/or artificial intelligence to ensure safe and equitable operations. Such future concepts propose a seamless, integrated, flexible and robust set of systems that are anticipated to include all vehicle types (traditional as well as novel vehicles); all airspace domains; all mission types; and accommodate changes to a diverse range of environmental and operational conditions while maintaining expected safety levels.

NASA's work in this area related to either transition or end-state autonomous airspace include:

- Data mining, application of machine learning and data science to air transportation data and problems
- Transition of largely human-centric systems to human-autonomy teaming systems
- Autonomy/autonomous technologies and concepts for trajectory management and efficient/safe traffic flows
- Weather and environment-integrated flight planning, rerouting, and execution
- Fleet, crew, and operator management to reduce the total cost of operations
- Graceful, manageable degradation in off-nominal conditions

This subtopic is seeking proposals to advance the future air transportation system (beyond 2025) that will apply novel and innovative techniques, methods, and approaches, to developing tools and/or technologies that will enable the successful transition to, or be an integral component of, the eventual realization of any autonomously operating airspace system in all airspace domains, from one in which human operators and decision-makers play a significant role.

This subtopic is also particularly interested in proposals focused on the application of advanced data science, and unconventional data or information sources, towards air traffic management (ATM) problems while incorporating meaningful ATM domain knowledge for more sophisticated results.

Relevance to NASA

Airspace Operations and Safety Program (AOSP) https://www.nasa.gov/aeroresearch/programs/aosp

Successful technologies in this subtopic have helped to advance the air traffic management/airspace operations objectives of the Program. The technologies also introduce new autonomy/artificial intelligence/data science methods and approaches to air transportation problems for current and near-future application.

A3.03 Future Aviation Systems Safety (SBIR)
Lead Center: ARC
Participating Center(s): LaRC
Related Subtopic Pointer(s): A3.01, A3.02, A3.04, A2.02

Public benefits derived from continued growth in the transport of passengers and cargo are dependent on the improvement of the intrinsic safety attributes of the Nation's and the world's current and future air transportation system. Recent developments to address increasing demand are leading to greater system complexity, including airspace systems with tightly coupled air and ground functions as well as widely distributed and integrated aircraft systems. Current methods of ensuring that designs meet desired safety levels will likely not scale to these levels of complexity (Aeronautics R&D Plan, p. 30). The Airspace Operations and Safety Program (AOSP) is addressing this challenge with a major area of focus on In-time System-wide Safety Assurance (ISSA). A proactive approach to managing system safety requires two abilities:

- The ability to monitor the system continuously and to extract and fuse information from diverse data sources (while identifying emergent anomalous behaviors after new technologies, procedures, and training are introduced).
- The ability to reliably predict probabilities of the occurrence of hazardous events and of their safety risks.

Understanding and predicting system-wide safety concerns of the airspace system and the vehicles as envisioned by the Next Generation Air Transportation System (NextGen) and beyond is paramount. Such a system would include the emergent effects of increased use of automation and autonomy to enhance system capabilities, efficiency, and performance beyond current capability of human-based systems. These systems would advance technology through health monitoring of the system-wide functions that are integrated across distributed ground, air, and space systems. Emerging highly autonomous operations such as those envisioned for Unmanned Aerial Systems (UAS) and Urban Air Mobility (UAM) will play a major role in future airspace systems. In particular operating Beyond the operator's Visual Line-Of-Sight (BVLOS) and near or over populated areas are topics of concern. Safety-critical risks include:

- Flight outside of approved airspace.
- Unsafe proximity to people/property.
- Critical system failure (including loss of C2 link, loss or degraded GPS, loss of power, and engine failure).
- Loss-of-control (i.e., outside the envelope or flight control system failure).

Tools are being sought for use in creating prototypes of ISSA capabilities. The ultimate vision for ISSA is the delivery of a progression of capabilities that accelerate the detection, prognosis, and resolution of system-wide threats.

Proposals under this subtopic are sought, but are not limited to, development and/or demonstration in the following areas (with an emphasis on safety applications):

- Data collection architecture, data exchange model, and data collection mechanism (for example, via UAS Traffic Management (UTM) Technical Capability Level 4 (TCL-4)).
- Data mining tools and techniques to detect and identify anomalies and precursors to safety threats system-wide.
- Tools and techniques to assess and predict safety margins system-wide to ensure airspace safety.
- Prognostic decision support tools and techniques capable of supporting in-time safety assurance.
- Verification and Validation (V&V) tools and techniques for ensuring the safety of air traffic applications during certification and throughout their lifecycles as well as techniques for supporting the in-time monitoring of safety requirements during operation.
- Products to address technologies, simulation capabilities, and procedures for reducing flight risk in areas of attitude and energy aircraft state awareness.
- Decision support tools and automation that will reduce safety risks on the airport surface for normal operations and during severe weather events.
- Alerting strategies/protocols/techniques that consider operational context as well as operator state, traits, and intent.
- Methodologies and tools for integrated prevention, mitigation, and recovery plans with information uncertainty and system dynamics in a UAS and in a Trajectory-Based Operations (TBO) environment.
- Strategies for optimal human-machine coordination for in-time hazard mitigation.
- Methods and technologies enabling transition from a dedicated pilot-in-command or operator for each aircraft (as required per current regulations) to single operators safely and efficiently managing multiple unmanned and UAM aircraft in civil operations.
- Measurement methods and metrics for human-machine team performance and mitigation resolution.
- System-level performance models and metrics that include interdependencies and relationships among human and machine system elements.

Expected TRL for this project is 1 to 3.

Relevance to NASA

This technology is applicable to the Airspace Operations and Safety Program (AOSP).

References:

- https://www.nasa.gov/aeroresearch/programs/aosp

A3.04 Non-Traditional Airspace Operations (SBIR)
Lead Center: ARC
Participating Center(s): LaRC
Related Subtopic Pointer(s): A3.01, A3.02, A3.03, A2.02

NASA is exploring airspace operations incorporating unmanned vehicles and novel operations occurring in all airspaces (controlled and uncontrolled), with a goal to safely and efficiently integrate with existing operations and mission types.

NASA's work in this area has already demonstrated the potential benefits and capabilities of a service-based architecture (such as developed for the Unmanned Aircraft Systems Traffic Management [UTM] R&D evaluations), and has led to new procedures, equipage and operating requirements, and policy recommendations, to enable widespread, harmonized, equitable execution of diverse unmanned missions.

This subtopic is seeking proposals continuing to support and develop the UTM concept, which seeks technologies to enable safe, heterogeneous (manned/unmanned) operations including, but not limited to, the following:

- To demonstrate the scalability of the UTM concept to potentially 10M+ users/operators
- To enable low size, weight, and power sense-and-avoid technologies
- The development of UTM-focused track and locate functions
- Autonomous and safe UAS operations for the last and first 50 feet, under diverse weather conditions.

This subtopic also seeks proposals supporting the Urban Air Mobility (UAM) concept, which seeks technologies including, but not limited to, the following:

- Service-based architecture designs that enable dense urban mobility operations and/or increasingly complex operations at ultra-high altitudes.
- Dynamic route planning that considers changing environmental conditions, vehicle performance and endurance, airspace congestion, and traffic avoidance.
- Dynamic scheduling for on-demand access to constrained resources and interaction between vehicles with starkly different performance and control characteristics.
- Integration of emergent users with legacy users, large commercial transport, including pass-through to and from ultra-high altitudes and interactions around major airports.
- Operational concepts for future vehicle and missions, including vehicle performance, vehicle fleet and network management, market need and growth potential, for future operations and airspace integration.
- Identification of potential certification approaches for new vehicles (such as electric vertical take-off-and landing-VTOL)

The expected TRL for this project is from 1 to 4.

References:

- https://www.nasa.gov/aeroresearch/programs/aosp
- https://www.aviationsystemsdivision.arc.nasa.gov/publications/index.shtml
- https://www.aviationsystemsdivision.arc.nasa.gov/index.shtml

Focus Area 21 Small Spacecraft Technologies

Lead MD: STMD

The concept of distributed spacecraft missions (DSM) involves the use of multiple spacecraft to achieve one or more science mission goals. Small distributed spacecraft acting in cooperation can execute science and exploration missions that would be impossible by traditional large spacecraft operating alone and offer the potential for new concepts in mission design. The goal of this topic is to develop enabling technologies for small spacecraft DSM configurations operating over large distances beyond low Earth orbit (LEO). The term DSM or "swarm" refers to a group of cooperatively distributed spacecraft, scalable up to 100s of spacecraft, in a specific configuration, which has three distinct characteristics. First, as opposed to a constellation, where spacecraft are distributed across multiple orbits, a DSM is comprised of small spacecraft orbiting relatively close to one another, with intersatellite ranges on the order of tens to hundreds of kilometers. Second, the DSM requires inter-spacecraft communications where each spacecraft is capable of sharing data and relative position information so that all swarm members are aware of the overall topology. The swarm topology would be dependent on the spatial and temporal distribution, orbit, ground reference, or other requirements of the science mission. Third, the swarm is commanded from the ground as an entity rather than each spacecraft individually. Thus, the swarm has inherent autonomous capabilities to control individual or complete

swarm topology redistribution depending or requirements or in response to commands. Small spacecraft, for the purpose of this solicitation, are defined as those with a mass of 180 kilograms or less and capable of being launched into space as an auxiliary or secondary payload. Small spacecraft are not limited to Earth orbiting satellites but might also include interplanetary spacecraft, planetary re-entry vehicles, and landing craft. Specific innovations being sought in this solicitation will be outlined in the subtopic descriptions. Proposed research may focus on development of new technologies but there is particular interest in technologies that are approaching readiness for spaceflight testing. NASA's Small Spacecraft Technology Program will consider promising SBIR technologies for spaceflight demonstration missions and seek partnerships to accelerate spaceflight testing and commercial infusion. Some of the features that are desirable for small spacecraft technologies across all system areas are the following:

- Simple design.
- High reliability.
- Tolerant of extreme thermal and/or radiation environments.
- Low cost or short time to develop.
- Low cost to procure flight hardware when technology is mature.
- Small system volume or low mass.
- Low power consumption in operation.
- Suitable for rideshare launch opportunities or storage in habitable volumes (minimum hazards).
- Able to be stored in space for several years prior to use.
- High performance relative to existing system technology.

The following references discuss some of NASA's small spacecraft technology activities:

- www.nasa.gov/smallsats https://www.nasa.gov/smallsat-institute

Another useful reference is the Small Spacecraft Technology State of the Art Report at:

- http://www.nasa.gov/sites/default/files/atoms/files/small_spacecraft_technology_state_of_the_art_2015_tagged.pdf

Z8.01 Chemical Propulsion Systems for Small Satellite Missions (SBIR)
Lead Center: GRC
Participating Center(s): ARC, GSFC, JPL, MSFC
Related Subtopic Pointer(s): S4.03, T2.02, T2.03, Z10.02

NASA is interested in utilizing SmallSats/CubeSats (6U-12U, < 50kg CubeSats targeted) for cislunar, interplanetary and/or deep space missions, including lunar exploration precursor missions and missions identified in recent Planetary Science Deep Space SmallSat Studies (PSDS3). To accomplish this, advances in chemical propulsion systems for these class of spacecraft are sought, to complete maneuvers such as attitude control, trans-orbit injection, orbit changes, and planetary intercept, with a minimum of transit time. Chemical propulsion systems considered here can include cold-gas, warm-gas, monopropellants and/or bi-propellant systems. These propulsion systems are preferentially envisioned as modular, add-on sub-systems to the larger SmallSat/CubeSat payloads, and would be comprised of the sum components of tank(s), valve(s), pressurant, feed system, thrusters and/or controls. Proposers should place emphasis on full propulsion systems offering long life, reliability, and minimalistic use of spacecraft resources (power, energy, volume, and mass). The use of existing component technologies to build a propulsion system is encouraged to minimize overall development, however proposers are also cautioned that experience to date has shown component technologies for larger systems do not necessarily and easily scale down to CubeSat platforms. Since the focus is on complete propulsion systems, proposals will not be considered that focus purely on individual component development (e.g., new thruster designs or propellant formulations) without addressing how the innovative component solution supports improved mission outcomes and clearly identifies how the product will be incorporated into an overall propulsion system solution. Component solutions must clearly demonstrate a willing system infusion customer and/or mission for consideration. Proposals are sought that can deliver a propulsion system hardware prototype at or greater than Technology Readiness Level (TRL) 4 (breadboard validations within a laboratory environment) within Phase II resources.

Propulsion system solutions are sought that provide as many of the following features as possible within a single propulsion system module:

- Volumetric efficient designs (> 50-60% propellant mass fraction), with tank expulsion efficiency of 95-99%. Propellant mass fraction here is defined as the usable mass of propellant divided by the total wet mass of the propulsion system.
- Maximized Delta-V capability, with target capabilities of 200-500 m/s desired. Proposers must clearly delineate how Delta-V capability is defined including anticipated payload mass/volume.
- Thrust levels from 0.2 to 1.0 N to provide rapid orbit insertions (hours vs. days/months of maneuver time)
- Operation on spacecraft bus voltage
- Systems with low/zero pre-launch pressurization needs (< 1.5 atm at launch)
- Thermal regulation of propellant (e.g., Low temp (< 0° C) storage to reduce system power requirements)
- Ability to conduct both translation and attitude control maneuvers
- Restart and pulsed operation capable, with pulse mode and impulse-bit control to meet station keeping and pointing requirements
- Systems presenting reduced ground processing hazards, or reduced risks to primary payloads (i.e., secondary payload safe)
- Ability to tolerate > 12 mo. of loaded storage without degradation or need for servicing
- Ability to drain & flush system of propellant and/or pressurant during ground processing
- System lifetime & reliability > 2 years under flight
- Dual fault tolerance
- Optimized for the rigors of interplanetary/deep space missions (i.e., radiation tolerant > 20 krad, thermal management to minimize heat soak to remainder of spacecraft, etc.)

While electric propulsion (EP) system solutions are recognized as a key enabling advancement for small spacecraft missions, the desire in the current subtopic is to specifically bolster chemical propulsion system capabilities, where high-thrust and short duration maneuvers are required, as a compliment to the growing set of existing EP solutions. Advances in EP technologies for this class of spacecraft are of interest to NASA, and for component EP solutions for CubeSats proposers should consider submitting to STMD subtopic Z10.02 - In-Space Electric Propulsion Component Technologies or STTR subtopic T2.02 - Advanced In-Space Electric Propulsion (EP) Technologies.

This subtopic would provide technologies specifically of interest to NASA Space Technology Mission Directorate (STMD's) Small Spacecraft Technology (SST) Program and the Planetary Exploration Science Technology Office (PESTO). Technology sought would help to develop needed propulsive capabilities for SmallSats for increasingly complex science missions, such as those identified in the PSDS3 studies. Also, the Green Propulsion Working Group (GPWG) is an Agency-level working group seeking to monitor and advise on green propulsion technology development. NASA has a Green Propulsion Technology Development Roadmap, which identifies CubeSats/ SmallSats as a near-term infusion opportunity for advancing green propulsion technology. Proposers are encouraged to consider how they align to the roadmap if proposing green propulsion technologies.

References:

- Proposers are encouraged to review the STMD Small Spacecraft Technology State of the Art Report: https://www.nasa.gov/sites/default/files/atoms/files/small_spacecraft_technology_state_of_the_art_2015_tagged.pdf.
- NASA is seeking advanced propulsive capabilities for SmallSat/CubeSat spacecraft for interplanetary/deep space exploration missions, such as those described in the SIMPLEx solicitation (NNH17ZDA004O-SIMPLEX, available from NSPIRES).
- Additionally, proposers should also review recent Planetary Science mission concept studies (Planetary Science Deep Space SmallSat Studies (PSDS3) program) for sample proposed mission concepts: https://www.hou.usra.edu/meetings/smallsat2018/smallsat_program.pdf. Specifically, chemical propulsion solutions that can meet or exceed metrics identified in several of the mission studies are sought in order to provide greater propulsive capabilities to these class of missions.

- NASA/TP—2018–219861 is the 2018 NASA Green Propulsion Technology Development Roadmap, which identifies CubeSats/SmallSats as a near-term infusion opportunity for advancing green propulsion technology.

Z8.03 Low Cost Radiation Hardened Integrated Circuit Technology (SBIR)
Lead Center: LaRC
Participating Center(s): ARC, GRC, GSFC, JPL
Related Subtopic Pointer(s): H9.07, H6.22, S4.04, S3.08, S1.03, T6.05, Z8.07, Z6.01

Over recent years, the small spacecraft industry has developed into an economic factor for space business. As these small spacecraft develop in capability and developers plan both for longer missions and usage beyond Low Earth Orbit (LEO), they have the need for more advanced electronic solutions beyond what is normally available in existing space-qualified product lines. Designers need these solutions to serve government, commercial, and academic small spacecraft in these higher radiation environments. In keeping with the small spacecraft design philosophy and general mission costing profiles, solutions associated with state-of-the-art Integrated Circuit (IC) devices offer a significant potential for improved functionality in space capabilities and cost reduction, especially by combining complementary functions or providing "smart" capabilities. However, commercial solutions usually have not been certified for use in the space environment and often have issues with the radiation environment found beyond LEO environments.

This subtopic is requesting proposed solutions for supplying this improved functionality while mitigating the radiation environment beyond LEO including, but not limited to:

- Radiation hardened ICs based on commercial, terrestrial applications
- Low-mass/small volume shielding, including shielding with structure or power supply elements, without causing damage from secondary emission
- Affordable low-mass/small volume redundancy with voting and switching mechanisms
- Utilization of Model Based Systems Engineering (MBSE) techniques for modeling the environment, failure modes and effects, and for modeling and evaluating the effectiveness and risks of various proposed mitigation scenarios

Because the entire suite of avionics functions, power systems, and many instrument support functions can benefit from taking commercial IC concepts and making them space-ready, capabilities and requirements are not specified in this subtopic. Proposers should consider functions normally found within spacecraft systems (such as avionics, power systems, etc.) and propose solutions to make available innovative new approaches with added capabilities for new spacecraft designers. In particular, the proposer should:

- Identify the specific function to be advanced and demonstrate how it is not currently available at the full level performance needed.
- Identify the intended application environment(s), e.g., Geosynchronous Earth Orbit (GEO), Cis-lunar space, deep space, etc.
- Explain how the proposed approach exceeds the state of the art in key metrics such as available functionality, power requirements, mass, radiation resistance, temperature range, risk, etc.

For IC development, proposers should identify the technology development necessary to get the proposed IC to the performance needed for the proposed environment and application. The proposal's technology development plan needs to include test and verification to include full environmental testing so that the end customer can readily incorporate the chipsets into new vehicles without additional testing. Low-cost chipsets that are used both for space applications as well as terrestrial applications such as DoD, commercial aircraft, etc. are the primary emphasis for this call.

For redundancy solutions, the proposal should describe how the failure effect mitigations are feasible both economically and within the severe mass budgets typical of small spacecraft.

For shielding solutions, the proposal should provide the necessary initial analysis to demonstrate that the proposed approach in the candidate structural and physical environment will mitigate the radiation effects of the chosen

environment without damage from secondary emissions. Because mass budgets are especially constrained in small spacecraft, supporting evidence should be provided to show how the benefits from shielding outweigh any mass penalty.

For MBSE development, proposers should identify the candidate small spacecraft configurations (including for constellations of cooperative small spacecraft) and the methodology for capturing the intended environmental variables, the mitigation approaches they will incorporate to reduce the radiation risk, and how they will perform the risk assessment. The proposal should specify the modeling elements and tools are required. The development plan must also include methods for validating the model and its results.

General-purpose computing processors are explicitly excluded in this subtopic.

NASA has plans to purchase services for delivery of payloads to the Moon through the Commercial Lunar Payload Services (CLPS) contract. Under this subtopic, proposals may include efforts to develop payloads for flight demonstration of relevant technologies in the lunar environment. The CLPS payload accommodations are yet to be precisely defined, however at least for early missions, proposed payloads should not exceed 15 kilograms in mass and not require more than 8 watts of continuous power. Smaller, simpler, and more self-sufficient payloads are more likely to be accommodated. Commercial payload delivery services may begin as early as 2020 and flight opportunities are expected to continue well into the future. In future years it is expected that payloads of higher mass and with higher power requirements might be accommodated. Selection for award under this solicitation will not guarantee selection for a lunar flight opportunity.

This subtopic is aimed at enhancing the small spacecraft market. This falls within the portfolio domain of the Small Spacecraft Technology program within NASA's STMD (Space Technology Mission Directorate). This small spacecraft portfolio has often supported commercial development to enhance the technology for this class of satellites. Missions longer than one year, at higher orbital altitudes, and missions beyond Earth's orbit will all need improved radiation protection for critical electronics in avionics and instruments. Missions in this class are of strong interest to STMD as well as NASA's Science Mission Directorate (SMD). This technology will also be useful to commercial and academic developers.

The expected Technology Readiness Level (TRL) range at completion of this project is 3 to 5.

Z8.06 DragSails for Spacecraft Deorbit (SBIR)
Lead Center: MSFC
Participating Center(s): ARC
Related Subtopic Pointer(s): S4.03, Z7.03

DragSails are a generic family of drag devices that can:

- Provide coarse, non-propulsive de-orbit capability which can aid in the disposal of end-of-life spacecraft through burnup upon reentry.
- Provide an accurate means of de-orbiting by modulating the ballistic coefficient to guide the system to a desired point at the Von Karman altitude for precision reentry targeting.

Small, lightweight, deployable membranes have been tested and deployed in Earth for both solar sail and drag sail applications. NASA's 10 square meter NanoSail-D2 solar sail and The University of Surrey's InflateSail drag sail are two examples. These systems demonstrated the technical viability of developing a deployable drag device to accelerate the deorbit of satellites to comply with end-of-life regulations and to mitigate the growth of orbital debris. Given the underlining technology similarities between solar sail and drag sail systems there are opportunities for adaptation or cross-use of some system elements. In terms of controlled, targeted de-orbit, the NASA Exo-Brake development effort has yielded promising though nascent results with the development of controllable tension structures. Tension structures don't have the 'beam buckling' issue associated with the more common drag sails at the higher dynamic pressures at atmospheric entry interface. This approach, while not as applicable to larger disposal efforts, can allow for more targeted reentry with potential additional uses in inexpensive entry, descent, and landing (EDL) test-beds or sample return concepts.

Developing systems to actively provide a de-orbit disposal, or targeted de-orbit/re-entry capability, is the next logical step toward such systems becoming widely available for spacecraft manufacturers, NASA and other government agencies as an alternative to conventional propulsion systems. Specific technology development areas of interest include:

- Advancements in sail fabric, boom and alternative deployment technologies that can be used to simplify production, reduce mass, or reduce the stowed volume of mechanically deployed drag sails.
- Concepts designed to augment aerodynamic drag and provide guidance and control during enhanced de-orbit.

Phase I proof of concept and preliminary design efforts that will lead to, or can be integrated into, flight demonstration prototypes in a Phase II effort are of interest. An ideal Phase II deliverable would be a DragSail subsystem suitable for environmental testing.

Desired system-level capabilities include the de-orbit of CubeSats (3U to 12U or larger) and small spacecraft in the 50kg - 200kg mass range (frontal areas on the order of 2000 to 2700 cubic cm) from altitudes between approximately 700km and 2,000km in 25 years or less. Spacecraft flying below 700km will generally meet the 25-year-or-less requirement without augmentation.

Any spacecraft in Earth's orbit must demonstrate how it will be either de-orbited or moved to an orbit that poses no risk to other spacecraft within a set period after its useful life. Therefore, any spacecraft launched by government, universities or industry are potential customers for a DragSail deorbit system.

The expected Technology Readiness Level (TRL) range at completion of this project is 5 to 6.

References:

- Alhorn, Dean, Joseph Casas, Elwood Agasid, Charles Adams, Greg Laue, Christopher Kitts, and Sue O'Brien. "NanoSail-D: The Small Satellite That Could!" (2011), Utah State University Small Satellite Conference (https://digitalcommons.usu.edu/smallsat/2011/all2011/37/)
- Andrew Viquerat, Mark Schenk, Vaios Lappas, and Berry Sanders. "Functional and Qualification Testing of the InflateSail Technology Demonstrator", 2nd AIAA Spacecraft Structures Conference, AIAA SciTech Forum, (AIAA 2015-1627) https://doi.org/10.2514/6.2015-1627

Z8.07 Spacecraft Model-Based Systems Engineering (SBIR)
Lead Center: JPL
Participating Center(s): JPL, LaRC
Related Subtopic Pointer(s): H6.04, S5.05, T11.03, Z8.03

NASA continues to pursue improvements in small spacecraft capabilities and reliability, especially for the deep space environment. Small spacecraft can help NASA achieve science and exploration goals with novel and more affordable mission architectures, including architectures based on ensembles of small spacecraft or that augment larger conventional spacecraft with small spacecraft. NASA seeks innovative model-based systems engineering (MBSE) methods and tools to:

- Define, design, develop, analyze, execute, and validate future small spacecraft missions through development of advanced methods and tools that enable more rapid, comprehensive, deeper, and integrated spacecraft design across the entire project lifecycle from concepts through system operations and end of mission disposal. The capabilities should leverage MBSE approaches being piloted across NASA and enable agile integration of disparate model types and various discipline tools.
- Enable disciplined system analysis for the design of future missions, including modeling of decision and programmatic support for those missions. Such models might also be made useful to evaluate technology alternatives and impacts, science valuation methods, and programmatic and/or architectural trades, including potential mission architectures comprised of multiple spacecraft.

Specific areas of interest are listed below. Approaches that emphasize or address multiple areas are encouraged. Proposers are recommended to be familiar with the state of the art and leverage existing standards where possible:

- Support for rapid trade space evaluations and capabilities for visualization, comprehension, and comparison of results/options.
- User-centered model interaction approaches for engaging domain experts that do not have significant MBSE expertise.
- Automated generation of traditional systems engineering artifacts including requirements and interface documents.
- Simulation and execution of spacecraft missions, leveraging advances in behavior modeling and execution in ALF, FUML, PSSM, PSCS, etc.
- Evaluation of systems performance margins (technical and programmatic).
- Model interchange, traceability, and configuration management between systems and discipline specific models, e.g., integration with CAD, FEM tools, etc.
- Integration of systems models with data analytics platforms for reporting and querying on model information.

As NASA continues its move into greater use of models for formulation and development of NASA projects and programs, there are recurring challenges to address. This subtopic focuses on encouraging solutions to these cross-cutting modeling challenges, including greater modeling breadth (e.g., cost/schedule), depth (scalability), variable fidelity (precision/accuracy vs. computation time), trade space exploration (how to evaluate large numbers of options), interaction (how users interact with the tools) and the processes that link them together. The focus is not on specific tools, but demonstrations of capability and methodologies for achieving the above. While the current focus is on small spacecraft, these tools and techniques will become more and more applicable to NASA's large missions.
The expected Technology Readiness Level (TRL) range at completion of this project is 3 to 5. Desired deliverables at the end of Phase I include new prototypes for tool integrations, visualization and access approaches, best practice patterns, reusable libraries, and model validation frameworks.

References:

- SysML - http://www.omgsysml.org/SysML-2.htm
- Open MBEE (Model-Based Engineering Environment) - http://www.openmbee.org/
- Alf (Action Language for Foundational UML) - https://www.omg.org/spec/ALF/1.1/
- OSLC (Open Services for Lifecycle Collaboration) - https://open-services.net/
- PSSM (Precise Semantics of UML State Machines Specification) - https://www.omg.org/spec/PSSM/About-PSSM/
- PSCS (Precise Semantics of UML Composite Structures) - https://www.omg.org/spec/PSCS/About-PSCS/
- fUML (Foundational UML) - https://www.omg.org/spec/FUML/About-FUML/
- FMI (Functional Mock-Up Interface) - https://fmi-standard.org/
- OpenAPI - https://en.wikipedia.org/wiki/OpenAPI_Specification

Z9.01 Small Launch Vehicle Technologies and Demonstrations (SBIR) – AMENDED: March 4, 2019
Lead Center: MSFC
Participating Center(s): AFRC, KSC, LaRC, MSFC

NASA is recognizing a growing demand for dedicated, responsive small spacecraft launch systems and seeks to facilitate the establishment of a robust launch service provider market sector. The movement toward small spacecraft missions is largely driven by rising development/launch costs associated with conventional spacecraft, and by rapid miniaturization of spacecraft platform capabilities. This topic seeks innovative systems and streamlined processes that will support the development of affordable launch systems having a 5-180kg payload delivery capacity to 350 to 700km at inclinations between 28 to 98.2° to support both CONUS and sun synchronous operations. Affordability objectives are focused on reducing launch costs to below $1.5M/launch for payloads ranging up to 50kg or below $30,000/kg for payloads in excess of 50kg. It is recognized that no single enabling technology is likely to achieve this goal and that a combination of multiple technologies and production practices are likely to be needed. Therefore, it is highly desirable that disparate but complementary technologies formulate and use standardized plug-and-play interfaces to better allow for transition and integration into small spacecraft launch systems.

Technology areas of specific interest are as follows:

- Innovative Propulsion Stages and Launch Systems
- Affordable Guidance, Navigation & Control
- Innovations for Launch Vehicle Structures
- Dual Use Hypersonic Flight Testbeds

Proposers are expected to quantify improvements over relevant SOA technologies and substantiate the performance relative to delivered payload mass and cost. Potential opportunities for technology demonstration and commercialization should be identified along with associated technology gaps. Ideally, proposed technologies would be matured between Technology Readiness Level (TRL) 4 to 6 by the end of the Phase II effort. A brief descriptive summary of desired technical objectives and goals are provided below.

Innovative Propulsion Stages and Launch Systems

Innovative chemical propulsion stages and integrated launch system concepts are sought that can serve as the foundational basis of an affordable, high flight rate launch system architecture. Solutions that directly address vehicle integration, mission profile sensitivities on delivered payload, and projected life cycle effects are desired. This could include main propulsion system design, novel staging concepts, and ground servicing technologies that enable rapid inspection, repair, and refurbishment of stages. Subsystem technologies that enable reusability are of particular interest. Subsystems are expected to demonstrate proof-of-concept by the end of Phase II as a minimum and proposals should include a development roadmap for achieving this goal. Efforts aimed at Phase II delivery of integrated prototype stages that could either be ground tested or flight tested as part of a post Phase II effort are highly encouraged and desired. Design simplicity, reliability, and reduced development and recurring costs are all important factors.

Affordable Guidance, Navigation, & Control

Affordable guidance, navigation & control (GN&C) is a critical enabling capability for achieving small launch vehicle performance and cost goals. Innovative GN&C technologies and concepts are therefore sought to reduce the significant costs associated with avionics hardware, software, sensors, and actuators. The scope of interest includes embedded computing systems, sensors, actuators, algorithms, as well as modeling & design tools. Low cost commercially available components and miniaturized devices that can be repurposed as a basis for low-SWaP GN&C systems are of particular interest. Special needs include sensors that can function during prolonged periods of high-g and high-angular rate (i.e., spin-stabilized) flight, while meeting the stringent launch system environment requirements pertaining to stability and noise. A low-cost GPS receiver capable of maintaining lock, precision, and accuracy during ascent would be broadly beneficial, for example. Sensors that can withstand these conditions might be sourced from industrial and tactical applications, and performance requirements may be achievable by fusing multiple measurements, e.g., inertial and optical (sun, horizon) sensors. Modular actuator systems are also needed that can support de-spin and turn-over maneuvers during ascent. These can include cold-gas or yo-yo type mechanisms. Improved designs are needed to reduce the overall power and volume requirements of these types of actuator systems, while still providing enough physical force to achieve the desired maneuver and enable orbital insertion. Programmable sequencers are required to trigger actuators for events such as stage sequencing, yo-yo and shroud deployment. In addition to hardware, software algorithms for autonomous vehicle control are needed to support in-flight guidance and steering. Robust control laws and health management software are of interest, particularly those that address performance and reliability limitations of affordable hardware. This is especially important in the typical high dynamics (acceleration and angular velocity) conditions of proposed small launch vehicles. Algorithms that are able to merge data from redundant onboard sensors could improve reliability compared to expensive single-string sensors. Similarly, advanced ground-alignment, initialization, and state estimation routines that integrate noisy data are desired to support ascent flight. These algorithms take advantage of improved onboard computational capability in order to process observations from lower accuracy sensors to provide higher fidelity information. Implementations of state-of-the-art Unscented Kalman Filters, and Square-Root-Information Filters with robust noise and sensor models are particularly applicable. Successful technologies should eventually be tested in relevant environments and at relevant flight conditions.

Innovations for Launch Vehicle Structures

The development of more efficient vehicle structures and components are sought to improve small launch vehicle affordability. This may include the adoption and utilization of advanced lightweight materials, including but not limited to carbon fiber composites, nanocomposites, extreme temperature materials, especially as used in combination with advanced manufacturing to enable low cost, reliable, lightweight design innovations. Of interest are systems for actively alleviating launch loads and environments. Approaches for achieving life-cycle cost reductions might also include reduced part count by substitution of multi-functional components; additive and/or combined additive-subtractive manufacturing; re-purposing launch structure for post-launch mission needs; incorporating design features that reduce operating costs. Alternatively, approaches based on the utilization of heavier materials could lead to simpler parts, fewer components, and more robust design margins. Although this could yield a larger rocket and impose performance penalties, significantly reduced life-cycle costs could be realized due to overall lower manufacturing and integration cost.

Dual Use Hypersonic Flight Testbeds

The potential repurposing and dual use applications of small launch vehicles as hypersonic flight technology testbeds is of great interest. If low-cost small launch vehicle concepts can be dual purposed as affordable hypersonic flight testing platforms with a high degree of commonality, it would open up a highly lucrative sector with significant commercial and defense market potential. The scope of interest is on launch vehicle derived concepts that could boost or gravity turn into a cruise altitude in the range of 75-100 Kft and accelerate a hypersonic testbed stage to a speed of Mach 4 or higher. Because small launch vehicle boosters typically undergo stage-1 to stage-2 separation in the Mach 8-10 range, it is conceivable that these vehicles could serve as low-cost boost phase systems for hypersonic flight testbeds equal in weight to the fully loaded orbital upper stage. Testbed concepts adaptable for a wide range of hypersonic technology investigations, including air breathing propulsion systems and thermal protection systems, while also offering payload recovery and partial testbed stage reusability, are strongly encouraged.

Relevance to NASA

The Launch Services Program continues to seek options for small satellite orbital launch capabilities.

Sound rocket capabilities are being improved with options financed through this topic.

References:

- https://ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20160002253.pdf
- https://kscpartnerships.ksc.nasa.gov/Partnering-Opportunities/Launch-Systems/Small-Class-Launch-Vehicles

Focus Area 22 Low Earth Orbit Platform Utilization and Microgravity Research

Lead MD: HEOMD

The Human Exploration and Operations Mission Directorate (HEOMD) provides mission critical space exploration services to both NASA customers and to other partners within the U.S. and throughout the world: operating the International Space Station (ISS); ensuring safe and reliable access to space; maintaining secure and dependable communications between platforms across the solar system; and ensuring the health and safety of astronauts. Additionally, the HEOMD is chartered with the development of the core transportation elements, key systems, and enabling technologies required for beyond-Low Earth Orbit (LEO) human exploration that will provide the foundation for the next half-century of American leadership in space exploration. In this topic area, NASA is seeking technologies that address how to improve and lower costs related to use of flight assets; maximize the utilization of the ISS for in-situ research; and utilize the ISS as a platform for in-space commercial science and technology opportunities. NASA seeks to accomplish these objectives by achieving following goals:

- Investing in the near- and mid-term development of highly-desirable system and technologies that provide innovative ways to leverage existing ISS facilities for new scientific payloads.

- Increasing investments in spaceflight operations and research to prepare for extended duration missions in near Earth space and beyond.
- Enabling U.S. commercial spaceflight opportunities and technology development to support the commercialization of low Earth orbit.

Through the potential projects spurred by this topic, NASA hopes to incorporate SBIR-developed technologies into current and future systems to contribute to the expansion of humanity across the solar system while providing continued cost-effective ISS operations and utilization for its customers, with a high standard of safety, reliability, and affordability.

H8.01 Low Earth Orbit Platform Utilization and Microgravity Research (SBIR)
Lead Center: JSC
Participating Center(s): ARC, GRC, JPL, JSC

Use of Low Earth Orbit Platforms for Commercial Technology and Product Development

NASA continues to invest in the near and mid-term development of highly-desirable systems and technologies that provide innovative ways to leverage existing International Space Station (ISS) facilities for new scientific payloads and to provide on-orbit analysis to enhance capabilities. Additionally, NASA is supporting commercial science, engineering, and technology to provide low earth orbit commercial opportunities utilizing the ISS as a way of enabling an economy in Low-Earth Orbit (LEO). Of particular interest in the ISS SBIR program are technologies and flight projects that can lead to significant terrestrial applications due to development in microgravity, leading to commercial product development within a number of disciplines. These disciplines include but are not limited to biotechnology, medical applications, material sciences, and in-space manufacturing, in areas as cell line development, tissue generation/bio-printing, exotic fiber manufacturing, and advanced materials production.

Research conducted on the ISS and technologies demonstrated on it supports Human Exploration and Operations Mission Directorate (HEOMD) goals for continued human presence in LEO space and for eventual deep space exploration missions. Utilization of the ISS is key in a number of technology areas for both Space Technology Mission Directorate (STMD) and Science Mission Directorate (SMD) as well as for HEOMD. In the case of SMD, the ISS provides the platform, including interfaces and resources (power, cooling, data, etc.) for external observation and measurement instruments. For STMD, the ISS along with emerging LEO commercial providers enable in-space testing of cutting edge technology projects used for the development of next generation spacecraft and that have potentially game changing capabilities. A solid foothold for space product research development can continue to be realized through use of the ISS and other commercial LEO platforms which are expected to lead to further maturation of commercial enterprise ventures in LEO.

The expected Technology Readiness Level (TRL) range at completion of the project is 2-6.

References:

- Space Station Research &Technology: https://www.nasa.gov/mission_pages/station/research/experiments/explorer
- Center for the Advancement of Science in Space: https://www.iss-casis.org/

Focus Area 23 Digital Transformation for Aerospace
Lead MD: STTR

Digital Transformation is the strategic transformation of an organization's processes and capabilities, driven and enabled by rapidly advancing and converging digital technologies, to dramatically enhance the organization's performance and efficiency. These advancing digital technologies include cloud computing, data analytics, artificial intelligence, blockchain, mobile access, Internet of Things, agile software development and processes, social media, and others. Their convergence is producing major transformations across industries — media and entertainment, retail, advertising, software, publishing, health care, travel, transportation, etc. Through digital transformation, organizations

seek to gain or retain their competitive edge by becoming more aware of and responsive to both customer and employee interests, more agile in testing and implementing new approaches, and more innovative and prescient in pioneering the next wave of products and services. Central to the success digital transformation is the pervasive (and often transparent) gathering of data about everything that impacts success--the organization's processes, activities, competencies, products and services, customers, partners, industry, and so on. Organizations can mine this massive, complex, and often unstructured data to develop accurate insights into how to improve organizational performance and efficiency. An organization may also use this data to train machine learning algorithms to automate processes, provide recommendations, or enhance customer experiences. The digital technologies listed above are essential to generate, collect, transform, mine, analyze, and utilize this data across the enterprise. NASA is undertaking a digital transformation journey to enhance mission success and impact. NASA intends to leverage digital transformation to:

- Boost innovation and creation of new knowledge.
- Reduce cost and increase the effectiveness and efficiency of processes for everything from human resources to science and engineering.
- Reduce the time to develop and mature new technologies.
- Facilitate efficient design and development of advanced aerospace vehicles.
- Ensure that increasingly complex missions are both cost-efficient and safe.
- Achieve data-driven insights and decisions.
- Increase autonomy in aerospace vehicles and ground facilities.
- Engage an enthusiastic and talented workforce.
- Maintain worldwide leadership in aerospace.

Through this topic, NASA is seeking to help explore and develop technologies that may be critical to the Agency's successful digital transformation. Specific innovations being sought in this solicitation are:

- Blockchain for aerospace applications, including its use in distributed space missions and in model-based systems engineering.
- Intelligent digital assistants that reduce the cognitive workload of NASA personnel, from scientists and engineers to business and administrative staff.

Details about these applications of digital transformation technologies are in the respective subtopic descriptions.

T11.03 Distributed Digital Ledger for Aerospace Applications (STTR)
Lead Center: ARC
Participating Center(s): GSFC, LaRC
Related Subtopic Pointer(s): H6.04, S5.04, T11.04, Z8.07

Blockchain solutions can benefit all NASA Mission Directorates and functional organizations. NASA activities could be dramatically more efficient and lower risk through Blockchain support of more automated creation, execution, and completion verification of important agreements, such as international, supply chain, or data use.

A Blockchain is a decentralized, online record keeping system, or ledger, maintained by a network of computers that verify and record transactions using established cryptographic techniques. A Blockchain is a data structure that makes it possible to create a consistent, digital ledger of data and share it among a network of independent parties. Blockchain distributed ledger technology may become a key enabler of digital transformation, enabling peer to peer transactions without requiring intermediaries or pre-established trust. Blockchain was originally developed to support digital currency transactions. Now, application of Blockchain is being explored for other financial services, software security, Internet of Things, parts tracking (supply chain), asset management, smart contracts, identify verification, and much more.

NASA is seeking innovative solutions involving Blockchain that would greatly enhance operational efficiency by providing a single, immutable "source of truth", viewable by all authorized parties, and usable by automated reporting and verification systems. In Phase I, expectations are to document a concept study for a Blockchain-based solution to one of the NASA challenges described. This must include a clear explanation of the benefits of a Blockchain solution

over alternative solutions. In Phase II, the goal is to deliver a prototype system. In this call, NASA is seeking Blockchain-based solutions for only the following two NASA-specific challenges:

- *Model Based System Engineering (MBSE)* - A significant challenge in MBSE is knowing that the system model being used is the current (or intended) version, since various aspects evolve through the system development and operations lifecycle. Further, because systems are becoming increasingly complex, tracking the vast number of changes that occur needs to be automated and efficient. Blockchain solutions may enable a single, real-time source of truth for system models, to eliminate several sources of error and inefficiency in MBSE. These issues become more pronounced when considering an ecosystem involving distributed collaboration among multiple entities, a scenario that will emerge more frequently as MBSE becomes the standard of doing business. For example, the government has already begun moving towards model-based acquisition programs (see GBSD and SET references). In any such environment, trust and security, especially relating to intellectual property, become a significant concern. Blockchain technology may be able to play a central role in enabling such a paradigm.
- *Distributed space mission management* - To accomplish complex space mission and Earth observation objectives, constellations of distributed satellites are often the most cost-effective approach. These constellations share key consolidated resources such as ground stations, a space network, communication networks, onboard processes, etc. Schedulers manage the changes to these resources, and may get overloaded when changes occur, especially when a project or agency does not control all of the assets. Users tend to overbook resources to assure they do not run short of communication resources and then release those resources unused at the last minute. These unused resources cannot be reallocated by central planners due to insufficient time. Blockchain could help to solve this problem by the use of smart contracts which rapidly allow other users to claim those resources in a distributed, automated way. Thus, as a preliminary concept, this allows cost-effective federation of resources, even in a federated system in which NASA does not control all resources. There are many other potential examples in which a combination of a distributed and automated management system coupled with a central planning system, with distributed ledgers and smart contracts, can maintain the responsiveness and cost-effectiveness of future distributed spacecraft mission operations. Specifically, a Blockchain solution to managing distributed space missions should enable collaboration in a partially trusted environment and increase responsiveness, reliability, and availability of both spacecraft and ground resources, while enabling strong security that thwarts hacking attempts. The management functions enhance flexibility (e.g., reduce overhead for components to join and leave constellations), and enhance automation (e.g., automate resource outage alerts, facilitate localized re-planning, enable a constellation level model-based diagnostics). To accomplish this, proposed solutions much overcome the slow transaction rate, large file sizes, and concurrency issues of some blockchain implementations.

The expected TRL for this project is 3 to 5.

References:

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T11.04 Digital Assistants for Science and Engineering (STTR)
Lead Center: LaRC
Participating Center(s): ARC
Related Subtopic Pointer(s): H6.03, S5.03, S5.04, S5.02, T11.03

NASA is seeking innovative solutions that combine modern digital technologies (e.g., natural language processing, speech recognition, machine vision, machine learning and artificial intelligence, and virtual reality and augmented reality) to create digital assistants. These digital assistants can range in capability from low-level cognitive tasks (e.g., information search, information categorization and mapping, information surveys, semantic comparisons), to expert systems, to autonomous ideation. NASA is interested in digital assistants that reduce the cognitive workload of its engineers and scientists so that they can concentrate their talents on innovation and discovery. Digital assistant solutions can target tasks characterized as research, engineering, operations, data management and analysis (of science data, ground and flight test data, or simulation data), business or administrative. Digital assistants can fall into one of two categories: productivity multipliers and new capabilities. Productivity multipliers reduce the time that the engineer or scientists spend on tasks defined by NASA policies, procedures, standards and handbooks, on common and best practices in science and engineering domains within the scope of NASA's missions, or on search and transformation of scientific and technical information. Proposals for productivity multipliers should demonstrate an in-depth understanding of NASA science and engineering workflows or NASA's information needs. New capabilities are disruptive transformations of the engineering and science environments that enable technological advances infeasible or too costly under current paradigms. Proposals for new capabilities should show clear applicability to NASA's missions. Examples of useful digital assistants include but are not limited to:

- A digital assistant that can formulate candidate designs (of components or systems) from a concept of operations, a set of high-level requirements, or a performance specification. Such an agent may use a combination of technologies (e.g., reinforcement learning, generative-adversarial networks) to autonomously ideate solutions.
- A digital assistant that uses the semantic, numeric, and graphical content of engineering artifacts (e.g., requirements, design, verification) to automate traces among the artifacts and to assess completeness and consistency of traced content. For example, the digital agent can use semantic comparison to determine whether the full scope of a requirement may be verified based on the description(s) of the test case(s) traced from it. Similarly, the digital assistant can identify from design artifacts any functional, performance, or non-functional attributes of the design that do not trace back to requirements. Currently, this work is performed by project system engineers, quality assurance personnel, and major milestone review teams as defined in NASA governing documents for engineering such as NPR 7123.1 Systems Engineering.
- A digital assistant that can recommend an action in real-time to operators of a facility, vehicle, or other physical asset. Such a system could work from a corpus of system information such as design artifacts, operator manuals, maintenance manuals, and operating procedures to correctly identify the current state of a system given sensor data, telemetry, component outputs, or other real-time data. The digital assistant can then use the same information to autonomously recommend a remedial action to the operator when it detects a failure, to warn the operator when their actions will result in a hazard or loss of a mission objective, or to suggest a course of action to the operator that will achieve a new mission objective given by the operator.
- A digital assistant that can identify current or past work related to an idea by providing a list of related government documents, academic publications, and/or popular publications. This is useful in characterizing the state-of-the-art when proposing or reviewing an idea for government funding. Currently, engineers and

scientists accomplish this by executing multiple searches using different combinations of keywords from the idea text, each on a variety of search engines and databases; then the engineers read dozens of document returns to establish relevance. This example imagines a digital assistant that accomplish a substantial portion of this work given the idea text.

- A digital assistant that can highlight lessons learned, suggest reusable assets, highlight past solutions or suggest collaborators based on the content that the engineer or scientist is currently working on. This example encourages digital solutions that can parse textual and/or graphical information from an in-progress work product and search Agency knowledge bases, project repositories, asset repositories, and other in-progress work products in the Agency to identify relevantly similar information or assets. The digital assistant can then notify the engineer of the relevant information and/or its author (potential collaborator).
- A digital assistant that understands system dependencies and, when presented with a design change, can assist (or autonomously perform) selection, modification, and execution of engineering analyses to be updated.
- A digital assistant that can autonomously subset, transform, analyze, and visualize large science datasets in response to a user query.

This subtopic targets terrestrial uses of digital assistive technologies in science and engineering environments. For application of digital assistive technologies for in-space applications, see subtopic H168-H6.03 Spacecraft Autonomous Agent Cognitive Architectures for Human Exploration.

Further, this subtopic is related to technology investments in the NASA Technology Roadmap, Technical Area 11 Modeling, Simulation, Information Technology, and Processing under sections 11.1.2.6 Cognitive Computer, 11.4.1.4 Onboard Data Capture and Triage Methodologies, and 11 .4 .1 .5 Real-time Data Triage and Data Reduction Methodologies. This subtopic is seeking similar improvements in computer cognition but more generally applied to the activities performed by engineers and scientists and made more easily accessible through technologies like speech recognition.

The expected TRL for this project is 3 to 5.

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Appendices

Appendix A: Technology Readiness Level (TRL) Descriptions

The Technology Readiness Level (TRL) describes the stage of maturity in the development process from observation of basic principles through final product operation. The exit criteria for each level documents that principles, concepts, applications or performance have been satisfactorily demonstrated in the appropriate environment required for that level. A relevant environment is a subset of the operational environment that is expected to have a dominant impact on operational performance. Thus, reduced-gravity may be only one of the operational environments in which the technology must be demonstrated or validated in order to advance to the next TRL.

TRL	Definition	Hardware Description	Software Description	Exit Criteria
1	Basic principles observed and reported.	Scientific knowledge generated underpinning hardware technology concepts/applications.	Scientific knowledge generated underpinning basic properties of software architecture and mathematical formulation.	Peer reviewed publication of research underlying the proposed concept/application.
2	Technology concept and/or application formulated.	Invention begins, practical application is identified but is speculative, no experimental proof or detailed analysis is available to support the conjecture.	Practical application is identified but is speculative, no experimental proof or detailed analysis is available to support the conjecture. Basic properties of algorithms, representations and concepts defined. Basic principles coded. Experiments performed with synthetic data.	Documented description of the application/concept that addresses feasibility and benefit.
3	Analytical and experimental critical function and/or characteristic proof of concept.	Analytical studies place the technology in an appropriate context and laboratory demonstrations, modeling and simulation validate analytical prediction.	Development of limited functionality to validate critical properties and predictions using non-integrated software components.	Documented analytical/experi-mental results validating predictions of key parameters.
4	Component and/or breadboard validation in laboratory environment.	A low fidelity system/component breadboard is built and operated to demonstrate basic functionality and critical test environments, and associated performance predictions are defined relative to the final operating environment.	Key, functionally critical, software components are integrated, and functionally validated, to establish interoperability and begin architecture development. Relevant Environments defined and performance in this environment predicted.	Documented test performance demonstrating agreement with analytical predictions. Documented definition of relevant environment.

5	Component and/or breadboard validation in relevant environment.	A medium fidelity system/component brassboard is built and operated to demonstrate overall performance in a simulated operational environment with realistic support elements that demonstrates overall performance in critical areas. Performance predictions are made for subsequent development phases.	End-to-end software elements implemented and interfaced with existing systems/simulations conforming to target environment. End-to-end software system, tested in relevant environment, meeting predicted performance. Operational environment performance predicted. Prototype implementations developed.	Documented test performance demonstrating agreement with analytical predictions. Documented definition of scaling requirements.
6	System/sub-system model or prototype demonstration in a relevant environment.	A high fidelity system/component prototype that adequately addresses all critical scaling issues is built and operated in a relevant environment to demonstrate operations under critical environmental conditions.	Prototype implementations of the software demonstrated on full-scale realistic problems. Partially integrate with existing hardware/software systems. Limited documentation available. Engineering feasibility fully demonstrated.	Documented test performance demonstrating agreement with analytical predictions.
7	System prototype demonstration in an operational environment.	A high fidelity engineering unit that adequately addresses all critical scaling issues is built and operated in a relevant environment to demonstrate performance in the actual operational environment and platform (ground, airborne, or space).	Prototype software exists having all key functionality available for demonstration and test. Well integrated with operational hardware/software systems demonstrating operational feasibility. Most software bugs removed. Limited documentation available.	Documented test performance demonstrating agreement with analytical predictions.
8	Actual system completed and "flight qualified" through test and demonstration.	The final product in its final configuration is successfully demonstrated through test and analysis for its intended operational environment and platform (ground, airborne, or space).	All software has been thoroughly debugged and fully integrated with all operational hardware and software systems. All user documentation, training documentation, and maintenance documentation completed. All functionality successfully demonstrated in simulated operational scenarios. Verification and Validation (V&V) completed.	Documented test performance verifying analytical predictions.

9	Actual system flight proven through successful mission operations.	The final product is successfully operated in an actual mission.	All software has been thoroughly debugged and fully integrated with all operational hardware/software systems. All documentation has been completed. Sustaining software engineering support is in place. System has been successfully operated in the operational environment.	Documented mission operational results.

Definitions

Proof of Concept: Analytical and experimental demonstration of hardware/software concepts that may or may not be incorporated into subsequent development and/or operational units.

Breadboard: A low fidelity unit that demonstrates function only, without respect to form or fit in the case of hardware, or platform in the case of software. It often uses commercial and/or ad hoc components and is not intended to provide definitive information regarding operational performance.

Brassboard: A medium fidelity functional unit that typically tries to make use of as much operational hardware/software as possible and begins to address scaling issues associated with the operational system. It does not have the engineering pedigree in all aspects, but is structured to be able to operate in simulated operational environments in order to assess performance of critical functions.

Proto-type Unit: The proto-type unit demonstrates form, fit, and function at a scale deemed to be representative of the final product operating in its operational environment. A subscale test article provides fidelity sufficient to permit validation of analytical models capable of predicting the behavior of full-scale systems in an operational environment

Engineering Unit: A high fidelity unit that demonstrates critical aspects of the engineering processes involved in the development of the operational unit. Engineering test units are intended to closely resemble the final product (hardware/software) to the maximum extent possible and are built and tested so as to establish confidence that the design will function in the expected environments. In some cases, the engineering unit will become the final product, assuming proper traceability has been exercised over the components and hardware handling.

Mission Configuration: The final architecture/system design of the product that will be used in the operational environment. If the product is a subsystem/component, then it is embedded in the actual system in the actual configuration used in operation.

Laboratory Environment: An environment that does not address in any manner the environment to be encountered by the system, subsystem, or component (hardware or software) during its intended operation. Tests in a laboratory environment are solely for the purpose of demonstrating the underlying principles of technical performance (functions), without respect to the impact of environment.

Relevant Environment: Not all systems, subsystems, and/or components need to be operated in the operational environment in order to satisfactorily address performance margin requirements. Consequently, the relevant environment is the specific subset of the operational environment that is required to demonstrate critical "at risk" aspects of the final product performance in an operational environment. It is an environment that focuses specifically on "stressing" the technology advance in question.

Operational Environment: The environment in which the final product will be operated. In the case of space flight hardware/software, it is space. In the case of ground-based or airborne systems that are not directed toward space flight, it will be the environments defined by the scope of operations. For software, the environment will be defined by the operational platform.

Appendix B: SBIR/STTR and the Space Technology Roadmaps

NASA's technology development activities expand the frontiers of knowledge and capabilities in aeronautics, science, and space, creating opportunities, markets, and products for U.S. industry and academia. The 2015 NASA Technology Roadmaps are a set of documents that consider a wide range of needed technology candidates and development pathways for the next 20 years (2015-2035). The roadmaps focus on applied research and development activities. Technologies that support NASA's missions may also support science and exploration missions conducted by the commercial space industry and other government agencies. In addition, NASA technology development results in applications for the general population including devices that improve health, medicine, transportation, public safety, and consumer goods. For more information, please see: https://www.nasa.gov/offices/oct/home/roadmaps/index.html.

The research and technology subtopics for the SBIR Program are identified annually by Mission Directorates and Center Programs. The Directorates identify high priority research and technology needs for respective programs and projects. Research and technology subtopics for the STTR Program are aligned with needs associated with the research interest and core competencies across NASA Centers and aligned with the Space Technology Roadmaps. Both programs support a broad range of technologies defined by a list of subtopics that vary in content within each annual solicitation.

The table below relates the current SBIR/STTR subtopics to Technology Areas in the Space Technology Roadmaps:

TA #	Technology Area (TA) Level 1 Title	Technology Area (TA) Level 2 Title	Subtopic Title	Subtopic #
TA02	2.0.0 - In-Space Propulsion Technologies	2.1.0 - Chemical Propulsion	Spacecraft Technology for Sample Return Missions	S4.03
			Chemical Propulsion Systems for Small Satellite Missions	Z8.01
			Small Launch Vehicle Technologies and Demonstrations	Z9.01
		2.2.0 - Non-Chemical Propulsion	Advanced Technologies for In-Space Electric Propulsion (EP)	T2.02
			Nuclear Thermal Propulsion	Z10.03
			In-Space Electric Propulsion Component Technologies	Z10.02
			DragSails for Spacecraft Deorbit	Z8.06
		2.3.0 - Advanced (TRL <3) Propulsion Technologies	Performance Demonstration of High Payoff Propulsion Technology: Rotating Detonation Engine and Dual Mode Ionic Liquid	T2.03
		2.4.0 - Supporting Technologies	Cryogenic Fluid Management	Z10.01
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA03	3.0.0 - Space Power and Energy Storage	3.1.0 - Power Generation	Power Generation and Conversion	S3.01
			Kilowatt-Class Energy Conversion for Small Fission Reactors	Z1.03
		3.2.0 - Energy Storage	Long Duration Lunar Energy Storage and Discharge	Z1.04
		3.3.0 - Power Management and Distribution	Power Electronics and Management, and Energy Storage	S3.03

TA #	Technology Area (TA) Level 1 Title	Technology Area (TA) Level 2 Title	Subtopic Title	Subtopic #
TA04	4.0.0 - Robotics, Telerobotics and Autonomous Systems	4.1.0 - Sensing & Perception	Intelligent Sensor Systems	T13.01
		4.2.0 - Mobility	Extreme Environments Technology	S4.04
			Lunar Rover Technologies for In-situ Resource Utilization and Exploration	Z5.05
		4.3.0 - Manipulation	Robotic Mobility, Manipulation and Sampling	S4.02
			Sample Collection For Life Detection in Outer Solar System Ocean World Plumes	S4.06
		4.5.0 - Autonomy	Flight Test and Measurement Technologies	A2.01
			Spacecraft Autonomous Agent Cognitive Architectures for Human Exploration	H6.03
			Autonomous Control Technologies (ACT) for Ground Operations	H10.02
			Guidance, Navigation and Control	S3.04
			Fault Management Technologies	S5.05
			Information Technologies for Intelligent and Adaptive Space Robotics	T4.01
			Coordination and Control of Swarms of Space Vehicles	T4.03
			Technologies for Intra-Vehicular Activity Robotics	Z5.04
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA05	5.0.0 - Communication and Navigation	5.1.0 - Optical Comm. And Navigation	Long Range Optical Telecommunications	H9.01
		5.2.0 - Radio Frequency Communications	Transformational Communications Technology	H9.05
			Terrestrial Balloons and Planetary Aerial Vehicles	S3.05
			Electric field mapping and prediction methods within spacecraft enclosures	T5.02
		5.4.0 - Position, Navigation, and Timing	Flight Dynamics and Navigation Technology	H9.03
		5.5.0 - Integrated Technologies	Cognitive Communication	H9.07

TA #	Technology Area (TA) Level 1 Title	Technology Area (TA) Level 2 Title	Subtopic Title	Subtopic #
TA06	6.0.0 - Human Health, Life Support and Habitation Systems	6.1.0 - Environmental Control Life Support & Habitation Systems	Microbial Monitoring and Control for Spacecraft Cabins	H3.03
			Spacecraft Solid Waste Management	H3.02
			Integrated Systems Health Management for Sustainable Habitats	H6.01
			Spacecraft Water Sustainability through Nanotechnology	T6.06
		6.2.0 - Extravehicular Activity Systems	Exploration Portable Life Support System (xPLSS) for deep space and surface missions	H4.01
			Exploration Pressure Garment System (xPGS) for deep space and surface missions	H4.04
		6.3.0 - Human Health and Performance	International Space Station (ISS) Utilization and Microgravity Research	H8.01
			Radioprotectors and Mitigators of Space Radiation-induced Health Risks	H12.01
			Reduced Oxygen Usage for Medical Events	H12.05
			Continuous Crew Health Monitoring	H12.06
		6.5.0 - Radiation	Testing of COTS Systems in Space Radiation Environments	T6.05
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA07	7.0.0 - Human Exploration Destination Systems	7.1.0 - In-Situ Resource Utilization	Extraction of Oxygen from Lunar Regolith	Z12.01
			Payloads for Lunar Resources: Volatiles	Z12.02
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA08	8.0.0 - Science Instruments, Observatories & Sensor Systems	8.1.0 - Science Instruments	In Situ Instruments/Technologies for Ocean Worlds Life Detection	S1.11
			Particles and Fields Sensors & Instrument Enabling Technologies	S1.06
			Atomic Interferometry	S1.10
			Technologies for Active Microwave Remote Sensing	S1.02
			Precision Deployable Optical Structures and Metrology	S2.02
			Technologies for Passive Microwave Remote Sensing	S1.03
			In Situ Instruments/Technologies for Lunar and Planetary Science	S1.07
			NDE Sensors, Modeling, and Analysis	Z11.01
		8.2.0 - Observations	Proximity Glare Suppression for Astronomical Direct Detection of Exoplanets	S2.01
			Advanced Optical Systems and Fabrication/Testing/Control Technologies for EUV/Optical and IR Telescope	S2.03

			X-Ray Mirror Systems Technology, Coating Technology for X-Ray-UV-OIR, and Free-Form Optics	S2.04
		8.3.0 - Sensor Systems	Lidar Remote Sensing Technologies	S1.01
			Suborbital Instruments and Sensor Systems for Earth Science Measurements	S1.08
			Sensor and Detector Technologies for Visible, IR, Far-IR, and Submillimeter	S1.04
			Detector Technologies for UV, X-Ray, Gamma-Ray Instruments	S1.05
			Cryogenic Systems for Sensors and Detectors	S1.09
			Photonic Integrated Circuits	T8.02
			Metamaterials and Metasurfaces Technology for Remote Sensing Applications	T8.04
			Low Cost Radiation Hardened Integrated Circuit Technology	Z8.03
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA09	9.0.0 - Entry, Descent and Landing Systems	9.1.0 - Aeroassist & Entry	Entry Descent & Landing Sensors for Environment Characterization, Vehicle Performance, and Guidance, Navigation and Control	Z7.01
		9.4.0 - Vehicle Systems Technology	Deployable Aerodynamic Decelerator and Weave Diagnostic Technology	Z7.03
			Lander Systems Technologies	Z7.04
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA11	11.0.0 - Modeling, Simulation, Information Technology and Processing	11.1.0 - Computing	Deep Neural Net and Neuromorphic Processors for In-Space Autonomy and Cognition	H6.22
			Technologies for Large-Scale Numerical Simulation	S5.01
			Bridging the Gap of Applying Machine Learning to Earth Science	S5.03
			Command, Data Handling, and Electronics	S3.08
			High Performance Space Computing Technology	Z6.01
		11.2.0 - Modeling	Model Based Systems Engineering for Distributed Development	H6.04
			Integrated Science Mission Modeling	S5.04
			Space Weather R2O/O2R Technology Development	S5.06
			Spacecraft Model-Based Systems Engineering	Z8.07
		11.4.0 - Information Processing	Commercial Geospatial Analysis Platforms for Earth Science Applications	S5.02
			Distributed Digital Ledger for Aerospace Applications	T11.03
			Digital Assistants for Science and Engineering	T11.04

TA #	Technology Area (TA) Level 1 Title	Technology Area (TA) Level 2 Title	Subtopic Title	Subtopic #
TA12	12.0.0 - Materials, Structures, Mechanical Systems and Manufacturing	12.1.0 - Materials	Unmanned Aircraft Systems (UAS) Technologies	A2.02
			Thin-Ply Composite Technology and Applications	T12.01
		12.2.0 - Structures	Lunar Surface Solar Array Structures	H5.01
			Hot Structure Technology for Aerospace Vehicles	H5.02
			In-Space Sub-Modular Assembly	Z4.02
		12.4.0 - Manufacturing	In-situ Curing of Thermoset Resin Mixtures	T12.05
			Lightweight Conformal Structures	Z4.03
			Advanced Metallic Materials and Processes Innovation	Z3.01
		12.5.0 - Cross-Cutting	Development of Mobile Welding Capabilities for In-Space Manufacturing	Z3.02
			MISSE Experiments	Z4.01
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA13	13.0.0 - Ground and Launch Systems Processing	13.1.0 - Technologies to Optimize the Operational Life-Cycle	Advanced Propulsion Systems Ground Test Technology	H10.01
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA14	14.0.0 - Thermal Management Systems	14.2.0 - Thermal Control Systems	Spacecraft Thermal Management	Z2.01
			Thermal Control Systems	S3.06
TA #	**Technology Area (TA) Level 1 Title**	**Technology Area (TA) Level 2 Title**	**Subtopic Title**	**Subtopic #**
TA15	15.0.0 - Aeronautics	15.1.0 - Safe, Efficient Growth in Global Aviation	Advanced Air Traffic Management System Concepts	A3.01
			Increasing Autonomy in the National Airspace System (NAS)	A3.02
			Non-Traditional Airspace Operations	A3.04
			Hypersonic Technology - Innovative Manufacturing for High Temperature Structures	A1.10
			Distributed Electric Propulsion (DEP) Vehicles toward Urban Air Mobility (UAM) and Regional Airliners	T15.01
		15.2.0 - Innovation in Commercial Supersonic Aircraft	Aeronautics Ground Test and Measurement Technologies	A1.08
			Quiet Performance - Airframe Noise Reduction	A1.02
		15.3.0 - Ultra-Efficient Commercial Vehicles	Vertical Lift Technology and Urban Air Mobility	A1.06
			Propulsion Efficiency - Propulsion Materials and Structures	A1.07
		15.4.0 - Transition to Low-Carbon Propulsion	Electrified Aircraft Propulsion	A1.04
			Electrified Aircraft Propulsion Energy Storage	T15.03

		15.5.0 - Real-Time System-Wide Safety Assurance	Future Aviation Systems Safety	A3.03
		15.6.0 - Enable Assured Machine Autonomy for Aviation	Vehicle Safety - Internal Situational Awareness and Response	A1.09
		15.7.0 - Other	Aerodynamic and Structural Efficiency - Integration of Flight Control with Aircraft Multidisciplinary Design Optimization	A1.01
			Computational Tools and Methods	A1.05
			Low Emissions/Clean Power - Environmentally Responsible Propulsion	A1.03